REPLICA

JEE MAINS

SOLUTIONS

Detailed Solutions of Replica Test Papers

REPLICA

JEE Mains

Solutions

Researched & Designed

by IITians

VERSUS LEARNING

VERSUS LEARNING

TO THE STUDENT

Dear Student,

This is high time to assess your learning and strategize, how you are going to attempt the JEE Mains.

Now you must be aware of your weaknesses and strengths in all the three subjects. So, now you have to focus on your strengths and not on your weaknesses.

Every exam has its own pattern so we can suggest you that for JEE Mains, you need not to be a master in all the subjects but you should know the basic concepts and their uses in simple problems. So it would be better to focus on the basic theoretical concepts in each subject.

You should learn Inorganic Chemistry, Organic Chemistry, Modern Physics, Errors, Experimental Physics, Semiconductors and many other informative chapters from your 12^{th} boards books.

After a lot of research on pattern, weightage and level of IIT-JEE Mains in past years, we have designed these tests.

We can assure you of your selection in IIT JEE at your highest academic potential if you take these tests seriously and learn to solve these questions given in all the test papers.

Team Replica

Wishing you a great future and All the Best for IIT JEE examination.

ANSWER KEY

R-1

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
B	C	A	C	D	C	D	A	C	A	A	C	D	D	B
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
C	D	C	C	D	C	A	C	A	A	A	B	D	B	D
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
D	C	A	B	B	A	A	A	A	D	A	A	A	A	B
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A	C	C	C	D	A	A	A	B	A	D	B	D	A	C
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
B	C	C	C	B	C	D	A	C	A	C	D	C	C	B
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
C	C	D	A	A	A	B	D	A	B	D	B	A	A	D

R-2

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
C	C	C	C	A	B	D	C	D	B	D	A	D	A	D
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
C	A	A	D	B	C	A	A	B	D	D	C	B	B	A
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
D	A	D	C	A	D	D	B	D	A	D	B	A	B	D
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
C	A	A	B	D	A	D	A	A	A	A	C	C	B	C
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
B	D	B	A	B	B	D	D	C	D	B	B	A	C	A
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A	C	D	A	A	B	A	B	B	C	B	A	A	D	A

R-3

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
B	C	C	C	B	B	A	A	D	A	C	D	C	A	B
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
D	D	C	B	C	D	A	C	A	C	A	C	C	A	C
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
A	A	B	A	B	A	D	D	B	C	C	A	B	A	C
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A	A	A	C	A	C	A	B	C	A	C	B	A	C	A
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
A	D	D	D	C	C	B	D	B	A	D	B	D	B	D
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
B	A	A	D	C	A	C	C	B	A	B	D	C	B	C

R-4

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	D	B	A	D	C	B	B	B	A	A	D	A	D	C
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
C	A	D	A	B	C	D	D	A	C	D	C	A	B	B
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
D	B	A	C	B	B	B	C	B	A	D	D	C	C	A
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B	D	B	C	A	A	C	A	D	B	B	C	B	A	A
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
A	B	B	C	A	D	A	D	D	B	C	A	B	A	C
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
B	C	B	C	C	A	C	C	C	A	C	C	C	B	D

R-5

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	C	C	A	D	A	B	D	D	A	A	B	D	B	D
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
B	B	B	A	B	D	A	C	C	C	C	C	D	B	C
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
A	C	A	B	C	D	B	B	C	A	D	B	A	C	D
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A	B	C	D	B	D	A	B	B	C	A	B	B	C	A
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
C	B	B	B	A	A	A	A	C	D	D	D	B	B	C
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
C	D	A	A	C	D	D	C	C	A	A	B	B	C	C

R-6

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	B	D	D	C	B	B	C	A	B	B	A	C	C	A
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A	C	A	A	B	B	A	D	C	D	D	D	B	A	A
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
B	B	C	A	B	B	B	A	A	B	D	A	D	B	C
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
C	C	D	B	D	D	D	D	C	A	B	A	C	B	C
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
D	A	B	A	C	A	A	C	B	C	A	D	C	D	B
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A	D	A	D	D	C	C	D	B	A	B	D	D	C	A

R-7

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	B	B	A	B	D	D	D	B	D	A	A	B	B	D
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
C	D	D	D	A	D	D	B	A	A	B	B	C	B	B
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
C	B	C	D	C	D	C	B	D	C	A	A	A	C	D
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
B	C	C	A	B	A	B	A	A	D	D	A	A	A	B
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
A	A	D	B	B	B	C	C	B	B	C	B	A	A	B
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
B	B	B	A	A	C	A	D	A	A	A	C	C	C	A

R-8

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
C	A	A	C	D	C	C	D	A	C	B	D	A	D	D
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
D	A	C	C	A	B	D	D	A	B	B	D	B	A	C
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
D	B	D	B	D	A	C	A	B	B	A	A	B	B	A
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
C	C	C	B	C	C	A	B	C	C	B	B	B	A	B
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
B	A	A	B	C	A	D	A	C	B	B	A	A	D	C
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A	A	D	C	A	B	D	A	B	A	B	B	A	D	A

R-9

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
C	B	D	C	A	B	B	C	D	C	D	D	D	D	C
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A	D	B	B	A	A	C	B	A	D	A	D	A	B	B
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
C	D	B	D	A	B	D	C	A	C	C	D	D	B	D
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A	C	D	B	A	A	C	C	B	D	A	D	C	D	D
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
D	A	C	A	D	B	C	D	C	A	A	C	B	A	C
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A	B	B	B	D	D	A	D	C	C	B	A	A	C	B

R-10

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	A	C	C	A	A	D	B	B	A	C	D	D	C	C
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A	D	B	C	A	B	A	C	C	B	A	A	B	A	B
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
A	A	D	C	C	D	B	D	A	B	A	A	C	A	A
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
B	B	D	C	D	B	D	D	A	A	D	B	C	C	B
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
D	C	D	C	D	A	A	C	C	B	D	C	C	C	B
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
D	B	B	C	B	C	B	D	C	C	A	D	C	A	A

R-11

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
C	B	A	B	A	A	C	C	C	A	C	C	D	C	D
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
C	B	A	C	D	B	A	A	B	A	A	A	B	C	B
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
A	A	C	D	C	B	D	C	C	C	D	B	D	C	B
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A	C	B	C	A	B	B	D	C	A	A	C	B	B	D
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
C	D	C	A	D	C	C	A	A	A	A	D	A	A	C
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
C	D	D	D	B	A	B	A	D	A	C	C	A	B	A

R-12

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	C	B	C	C	C	D	A	C	B	C	A	D	C	B
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
C	A	A	D	A	B	D	B	D	C	A	B	C	B	A
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
A	C	A	D	C	B	B	D	A	D	D	B	A	B	A
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A	B	B	B	B	B	A	A	A	C	B	B	D	A	C
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
D	B	C	D	A	A	B	D	B	D	D	C	B	B	B
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
C	D	A	A	A	A	C	A	B	A	B	A	D	C	B

R-13

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
D	A	B	C	D	D	C	A	B	C	C	B	A	C	D
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
C	A	A	C	B	A	A	D	A	C	B	A	C	B	C
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
C	A	D	C	A	C	A	A	B	C	B	C	A	D	A
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
C	B	A	D	A	C	A	C	D	A	A	A	A	B	D
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
C	C	B	A	C	C	B	B	A	D	C	D	B	B	B
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
B	A	A	C	D	D	C	C	D	D	A	A	B	D	D

R-14

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
B	A	B	D	A	B	A	D	A	B	D	C	C	A	C
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A	A	D	A	A	B	D	C	B	B	B	D	A	C	D
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
A	B	C	C	B	A	A	B	C	A	C	C	B	A	B
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
B	A	D	B	B	A	C	A	D	C	A	C	C	C	B
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
C	B	D	A	D	B	C	A	B	A	A	B	A	D	B
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
D	B	A	B	C	D	D	C	B	B	A	A	D	B	D

R-15

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
D	C	A		D	D	B	D	C	A	D	D	D	B	C
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
C	D	B	A	B	B	A	C	A	A	D	A	B	A	B
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
B	A	B	A	B	B	A	B	B	C	A	D	A	B	C
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A	D	A	A	A	D	C	A	D	C	C	B	B	D	C
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
C	A	A	D	C	D	B	B	B	C	A	A	A	B	D
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A	B	D	C	A	A	B	C	B	B	C	B	C	B	B

R-16

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
C	C	B	B	D	B	D	A	D	B	A	B	C	A	A
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A	D	B	A	A	C	B	D	B	B	B	A	B	C	A
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
B	B	A	C	B	C	C	D	A	A	B	B	A	C	C
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
C	B	C	C	D	B	B	D	A	A	C	A	B	B	B
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
B	D	A	A	A	B	B	D	B	C	C	C	B	D	D
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
C	D	D	A	B	C	A	B	B	C	C	D	D	C	A

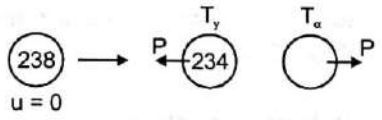
REPLICA 1

P 1 : (B) 25 units

Sol. $I=\left(\sqrt{I_1}+\sqrt{I_2}\right)^2=(2+3)^2=25$

P 2 : (C) Slightly less than T

Sol.



$$K.E.=\frac{P^2}{2m}$$

$$\Rightarrow \quad \frac{T_y}{T_\alpha}=\frac{M_\alpha}{M_y}=\frac{4}{234}$$

and $T_y + T_\alpha = T$

$$\Rightarrow \quad T_\alpha=\frac{234}{238}T$$

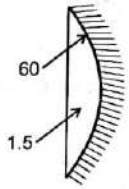
T_α is slightly less than T.

P 3 : (A) 20 cm

Sol. $$\frac{1}{f}=\frac{1}{f_m}-\frac{2}{f_\ell}=\frac{1}{-30}-2(1.5-1)\left[\frac{1}{\infty}-\frac{1}{-60}\right]$$

$$=\frac{1}{-30}-\frac{1}{60}$$

$$\frac{1}{f}=\frac{-3}{60}\ ;\ f=-20$$



P 4 : (C) I and III have the same weight

Sol. Container I and III have same weight because in III container block is floating. It means weight of liquid displaced is equal to weight of block. Since container was fully filled so water equal to weight of block will overflow therefore weight will remain same as container I.

Container II has more weight because block is completely immersed it means weight of water displaced (overflown) is less than that of block so weight of container will be more than that of I.

P 5 : (D) B

Sol. $r^2 = A^2\sin^2\omega t + B^2\cos^2\omega t$;

$$\frac{dr^2}{dt}=2A^2\sin\omega t\cos\omega t\ \omega-2B^2\cos\omega t\sin\omega t\ \omega$$

$$=(\omega\sin2\omega t(A^2-B^2)=0$$

$$\Rightarrow \quad \omega t=0,\frac{\pi}{2},\pi,\frac{3\pi}{2}\ \text{etc.}$$

at $\omega t = 0$; $\frac{d^2r^2}{dt}=2\omega^2\cos\ 2\omega t(A^2-B^2)$

= –ive (so maximum)

r^2 is max at $\omega t = 0$

$$r_{max}=\sqrt{B^2}=B$$

P 6 : (C) 3.2 mm

Sol. For central bright fringe net path difference = 0

$$\frac{2\pi}{\pi}\left[(\pi-1)t-\frac{yd}{D}\right]=0$$

$$(\mu-1)t=\frac{d}{D}\left[30\times\frac{D}{d}\right]$$

$$(\mu-1)t'=\frac{d}{D}\left[20\times\frac{\lambda D}{d}\right]$$

$$\frac{t}{t_1}=\frac{3}{2}\ \Rightarrow\ t' = 3.2\ \text{mm}$$

P 7 : (D) 1.25×10^{-5} cm

Sol. The distance of first diffraction minimum from the central principal maximum,

$x = \Delta\lambda/d$

$$\therefore \quad \sin\theta=\frac{x}{D}=\frac{\lambda}{d}\ \text{ or }\ d=\frac{\lambda}{\sin\theta}$$

$$\therefore \quad d=\frac{5000\times10^{-8}}{\sin30^\circ}=2\times5\times10^{-5}$$

$= 1.0 \times 10^{-4}$ cm

P 8 : (A) $v^2=v_0^2-\frac{2hg}{1+\frac{h}{R}}$

Sol. From work energy theorem,

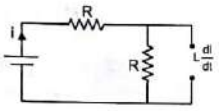
$$\frac{mv^2}{2}-\frac{mv_0^2}{2}=-\left[\left(-\frac{GMm}{R+h}\right)-\left(-\frac{GMm}{R}\right)\right]$$

$$\Rightarrow \quad v^2-v_0^2=\frac{2GM}{R+h}-\frac{2GM}{R}=\frac{2GM(-h)}{(R+h)R}$$

$$\Rightarrow \quad v_0^2-v^2=\frac{2gh}{1+\frac{h}{R}}$$

P 9 : (C) ε/2L

Sol. at t = 0



$i=\frac{\varepsilon}{2R}$, and by KVL,

$$L\frac{di}{dt}=iR=\frac{\varepsilon}{2} \quad \Rightarrow \quad \frac{di}{dt}=\frac{\varepsilon}{2L}$$

P 10 : (A) 0.12 Å

Sol. $V=\frac{P}{i}=\frac{P}{ne}=\frac{1}{6.25\times10^{13}\times1.6\times10^{-19}}$

$= 10^5$ volts.

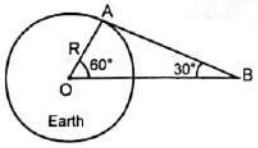
$$\lambda_{min}=\frac{hc}{eV}=\frac{12400}{V}\text{Å}=0.12\text{ Å}$$

P 11 : (A) $\sqrt{\frac{GM}{8R^3}}$

Sol. In Δ AOB : - $\cos 60° = \frac{R}{OB} \Rightarrow OB = 2R$

Here gravitational force will provide the required centripetal force.

Hence $\frac{GMm}{(OB)^2}=m(OB)\omega^2$



$$\Rightarrow \omega=\sqrt{\frac{GM}{(OB)^3}}=\sqrt{\frac{GM}{(2R)^3}} \; ; \; \omega=\sqrt{\frac{GM}{8R^3}}$$

P 12 : (C) $2\sqrt{u}/k$

Sol. Given $t=\sqrt{v}$ or $\frac{dv}{dt}=-k\sqrt{v}$.

Thus $v^{-1/2}\,dv=-k\,dt$

Integrating, we get $2v^{1/2}=-kt+c$.

Using the given initial condition (v = u at t = 0),

we get $c=2\sqrt{u}$.

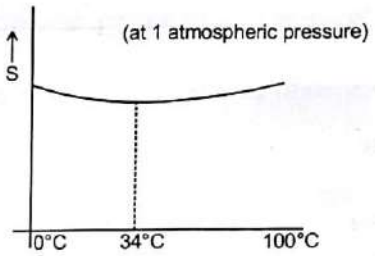
Thus, we have $2(v^{1/2}-u^{1/2})=-kt$

Now, use t = T and v = 0

$\Rightarrow t=2\sqrt{u}/k$

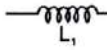
P 13 : (D) has a minimum at about 30°C.

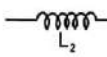
Sol. It is based on information of anamolies of behaviour of water.



So it is not a straight line and minimum at about 30°C.

P 14 : (D) None of the above

Sol.  ⇒ If i current goes $L_1 i$ is flux through it

 ⇒ If i current goes $L_2 i$ is flux through it

⇒ If i current goes flux passing through it will be more than $L_1 i + L_2 i$ as there will be some linkage flux.

⇒ If no linkage flux $L = L_1 + L_2$

⇒ opposite linkage

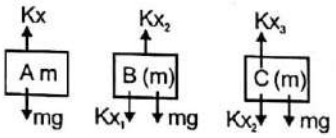
$L < L_1 + L_2$

So L can have any value

P 15 : (B) $\frac{6mg}{K}$

Sol. Displacement of upper block is

$\Delta x_1 + \Delta x_2 + \Delta x_3 = \Delta x$

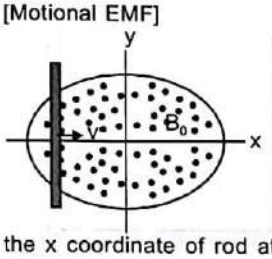


$$\Delta x=\frac{6Mg}{k}$$

$$x_1=\frac{mg}{k} \qquad x_2=\frac{2mg}{k} \qquad x_3=\frac{3mg}{k}$$

P 16 : (C) $2Bbv\sqrt{1-\frac{(vt-a)^2}{a^2}}$

Sol. [Motional EMF]



the x coordinate of rod at t = t will be x = vt – a,
So, its effective length will be 2y that will follow $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.

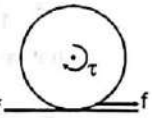
$$y = b\sqrt{1-\frac{(vt-a)^2}{a^2}}$$

So, induced EMF will be :
EMF = 2Byv

$$EMF = 2Bbv\sqrt{1-\frac{(vt-a)^2}{a^2}}$$

P 17 : (D) $\frac{\tau mR}{I+mR^2}$

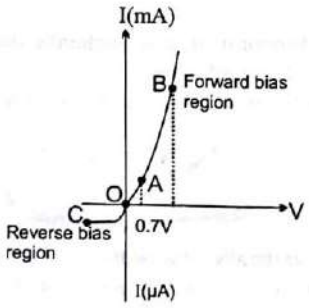
Sol. As engine is giving force but not torque
So f = ma
$\tau - fR = I\alpha$
and for rolling $a = R\alpha$



$$\tau - fR = I\frac{a}{R} \Rightarrow \tau - fR = \frac{If}{mR}$$

$$\Rightarrow f = \frac{\tau mR}{I+mR^2}$$

P 18 : (C) only statements (I) and (III) are correct
Sol. I-V characteristics of silicon diode is



So, statements I and III are correct.

P 19 : (C) 94.4%

Sol. $P_{suppressed} = P_C\left(\frac{m_a}{2}\right)^2 = P_C\left(\frac{(0.5)^2}{4}\right)$
$= 0.0625\ P_C$

$$P_{enhance} = \left(1+\frac{m_a^2}{2}\right)\times P_C$$

$$= \left(1+\frac{(0.5)^2}{2}\right) = 1.125\ P_C$$

$$\therefore \%\ saving = \left[\frac{1.125P_C - 0.0625P_C}{1.125P_C}\right]\times 100$$

$= 94.4\%$

P 20 : (D) 2 μW, 36 MJ

Sol. By definition $SL = 10 \log \frac{I}{I_0}$

$$\therefore \quad 60 = 10 \log \frac{I}{I_0}$$

$$\Rightarrow \quad \frac{I}{I_0} = 10^6$$

$$\Rightarrow \quad I = 10^6 \times 10^{-12} = 10^{-6}\ W/m^2$$
$= 1\ \mu W/m^2$
= Power = Intensity × surface area
$P = IS = 1 \times 10^{-6} \times 2 = 2\ \mu W$
and $E = Pt = 2 \times 10^{-6} \times 5 \times 60 \times 60$
$= 36 \times 10^{-3}\ j = 36\ MJ$

P 21 : (C) $p_0(1 - ah)^{Mg/aRT_0}$
Sol. We know that the variation of pressure with height of a fluid is given by

$$dp = -\rho ghdh \quad ...(i)$$

From gas law

$$pV = \frac{m}{M}RT \text{ or } p = \frac{\rho}{M}RT$$

or $$\rho = \frac{pM}{RT} \quad ...(ii)$$

Putting eq. (ii) in eq. (i)

$$dp = \frac{-\rho Mg}{RT}dh$$

$$\frac{dp}{p} = \frac{Mgdh}{Rt_0(1-ah)}$$

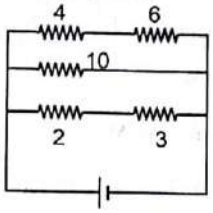
or $$\int_0^p \frac{dp}{p} = \frac{Mg}{RT_0}\int_0^h \frac{dh}{(1-ah)}$$

or $$\ln\frac{p}{p_0} = \frac{Mg}{aRT_0}\ln(1-ah)$$

$$= \ln(1 - ah)^{Mg/aRT_0}$$
$$p = p_0(1 - ah)^{Mg/aRT_0}$$

P 22 : (A) 3Ω

Sol. As it is wheatstone so current in 8 W will be zero so we can reduce the circuit as all 3 branches are in parallel, they will get same potential difference so last branch will have double power in comparision with above two branches. The power in 2Ω and 3Ω will be in the ratio and resistance of 3Ω will have maximum power.



$$P_4 = \frac{4}{10}\left[\frac{E^2}{10}\right] \quad P_6 = \frac{6}{10}\left[\frac{E^2}{10}\right]$$

$$P_{10} = \left[\frac{E^2}{-10}\right] \quad P_2 = \frac{2}{5}\left[\frac{E^2}{5}\right]$$

$$P_3 = \frac{3}{5}\left[\frac{E^2}{5}\right]$$

So P_3 is maximum.

P 23 : (C) $\frac{4\pi}{3}\sqrt{\frac{Re}{g}}$

Sol. Any tunnel in earth will have same angular frequency of particle performing SHM.

$$\omega = \sqrt{\frac{g}{Re}}$$

A to S ⇒ extreme to half of amplitude phase travelled $\pi/3$.

S to P ⇒ extreme to mean phase travelled $= \pi/2$.

P to C ⇒ extreme to mean phase travelled $= \pi/2$.

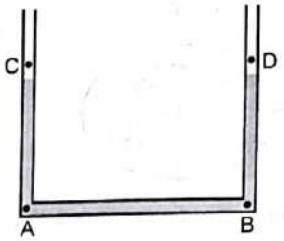
So total phase travelled by particle = $\frac{4\pi}{3}$.

So time taken by particle

$$T = \frac{4\pi}{3\omega} = \frac{4\pi}{3}\sqrt{\frac{Re}{g}}$$

P 24 : (A) Acceleration of u tube is towards left and its magnitude is 10 m/s^2.

Sol. Pressure at C is P_{atm} so



$P_A = P_{atm} + 2 \times 2 \times 10$ {ρgh}

$P_B = P_{atm} + 2 \times 8 \times 10$ {ρgh}

and $P_B > P_A$, it must be travelling towards left. Let acceleration is 'a'.

So $P_B - P_A = 4 \times a \times 3$

$120 = 12\,a \Rightarrow a = 10$ m/s^2

P 25 : (A) 5

Sol. Wavelength of tuning fork

$\lambda = 330/600 = 0.55$m $= 55$ cm

As the one end of tube is closed and other end is open so for resonance,

required length will be, $\ell = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}, \frac{7\lambda}{4} \ldots$

$$\ell = \frac{55}{4}, \frac{3(55)}{4}, \frac{5(55)}{4}, \frac{7(55)}{4}$$

but length cannot be more than 150 cm.

$$(2n + 1)\frac{55}{4} \leq 150$$

$$(2n + 1) \leq \frac{600}{55}$$

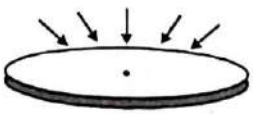
$$(2n + 1) \leq 10.90$$

So $\frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}, \frac{7\lambda}{4}, \frac{9\lambda}{4}$ are possible.

So 5 resonance will be observed.

P 26 : (A) direction of g is vertically downwards only at axis of disc.

Sol. Gravitational field due to disc is as shown



g is vertically downward only at axis, also water accumulated near axis due to high value of g.

P 27 : (B) 10%

Sol. At brewster's angle, intensity of reflected light is $\frac{I_0}{2}$ if reflectivity is 100%.

$\Rightarrow$ if reflectivity is 20% then reflected intensity = $\frac{I_0}{2}\times\frac{20}{100}$

$= \frac{I_0}{10} \Rightarrow$ 10% reflection

P 28 : (D) 9

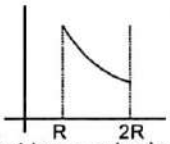
Sol. $\sqrt{\upsilon} = a(Z-1) \Rightarrow \sqrt{\frac{\upsilon_1}{\upsilon_2}} = \frac{(Z_1-1)}{Z_2-1} = \frac{1}{4}$

$Z_2 = 4Z_1 - 3$

$\Rightarrow Z_2 - Z_1 = 3Z_1 - 3 = 3(Z_1 - 1)$

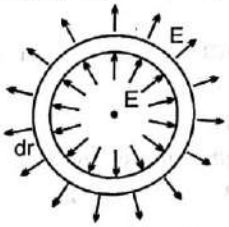
difference in atomic number is multiple of 3.

P 29 : (B)



Sol. Consider a spherical layer of radius r and thickness dr.

So, by Gauss Law



$E\,4\pi(r+dr)^2 - E4\pi r^2 = \frac{\rho}{\epsilon_0}4\pi r^2$

$\Rightarrow E4\pi r^2((1+dr/r)^2-1) = \frac{\rho}{\epsilon_0}4\pi r^2$

$\Rightarrow E4\pi r^2\left(1+\frac{2dr}{r}-1\right) = \frac{\rho}{\epsilon_0}4\pi r^2 dr$

$\Rightarrow \rho = \frac{2E\epsilon_0}{r}$

P 30 : (D) 2

Sol. Electric field is uniform and current is constant so

$E = \rho J \Rightarrow E = \frac{\rho I}{4\pi r^2}$

$\Rightarrow \rho = \frac{4\pi E}{I}r^2$

By comparing it with $\rho = \rho_0 r^n \Rightarrow n = 2$

P 31 : (D) cos 7°

Sol. sin 47° + sin 61° – (sin 11° + sin 25°) =

2sin 54°.cos 7° – 2sin 18° cos 7°

= 2cos 7° (sin 54° – sin 18°)

= 2cos 7° .2cos 36°. sin 18°

$= 4.\cos 7°.\frac{\sqrt{5}+1}{4}.\frac{\sqrt{5}-1}{4} = \cos 7°.$

P 32 : (C) $-\tan\theta + \frac{\tan n\theta}{n}$

Sol. We have $\frac{U_n}{V_n} = \tan n\theta$

And

$\frac{V_n - V_{n-1}}{U_{n-1}} = \frac{\cos n\theta \sec^n\theta - \cos(n-1)\theta\sec^{n-1}\theta}{\sin(n-1)\theta\sec^{n-1}\theta}$

$= \frac{\cos n\theta\sec\theta - \cos(n-1)\theta}{\sin(n-1)\theta}$

$= \frac{\cos n\theta - \cos(n-1)\theta\cos\theta}{\cos\theta\sin(n-1)\theta} =$

$\frac{\cos(n-1)\theta\cos\theta - \sin(n-1)\theta\sin\theta - \cos(n-1)\theta\cos\theta}{\cos\theta\sin(n-1)\theta}$

$= \frac{\cos n\theta - \cos(n-1)\theta\cos\theta}{\cos\theta\sin(n-1)\theta}$

$= -\tan\theta$

So that

$\frac{V_n - V_{n-1}}{U_{n-1}} + \frac{1}{n}\frac{U_n}{V_n} = -\tan\theta + \frac{\tan n\theta}{n} \neq 0$

P 33 : (A) $\frac{\pi}{4}$ or $\frac{3\pi}{4}$

Sol. Equation of plane containing i and i + j is

$[r - i, i, i + j] = 0 \Rightarrow (r - i).[i \times (i + j)] = 0$

$\Rightarrow [(x-1)i + yj + zk]\,k = 0 \Rightarrow z = 0$ ..(1)

Equation of plane containing i – j and i + k is

$\Rightarrow (r - i + j)[(i - j) \times (i + k)] = 0$

$\Rightarrow x + y - z = 0$ ..(2)

Let $a = a_1 i + a_2 j + a_3 k$. Since a is parallel to (1) and (2)

$a_3 = 0,\ a_1 + a_2 - a_3 = 0 \Rightarrow a_1 = -a_2,\ a_3 = 0$

Thus a vector in the direction of a is u = i – j.

If θ is the angle between a and i – 2j + 2k.

Then $\cos\theta = \pm\frac{1(1)+(-1)(-2)}{\sqrt{1+1}\sqrt{1+4+4}} = \pm\frac{3}{\sqrt{2}.3}$

$\Rightarrow \cos\theta = \pm\frac{1}{\sqrt{2}} \Rightarrow \theta = \pi/4$ or $3\pi/4$

P 34 : (B) 1/3

Sol. Given $2b = a + c \Rightarrow 3b = 2s = a + b + c$

$$\tan\frac{A}{2}\tan\frac{C}{2} = \frac{\Delta}{s(s-a)}\cdot\frac{\Delta}{s(s-c)}\cdot\frac{s-b}{s-b}$$

$$= \frac{s-b}{s} = \frac{2s-2b}{2s} = \frac{b}{3b} = \frac{1}{3}$$

P 35 : (B) $m^2 - m(4r + 1) + 4r^2 - 2 = 0$

Sol. Since, the coefficient of given terms are ${}^mC_{r-1}$, ${}^mC_r$, ${}^mC_{r+1}$, respectively and they are in AP

Therefore, ${}^mC_{r-1} + {}^mC_{r+1} = 2\,{}^mC_r$

$$\Rightarrow \frac{m!}{(r-1)!(m-r+1)!} + \frac{m!}{(r+1)!(m-r-1)!} = 2\frac{m!}{r!(m-r)!}$$

$$\Rightarrow \frac{1}{(m-r+1)(m-r)} + \frac{1}{(r+1)r} = \frac{2}{r(m-r)}$$

$$\Rightarrow \frac{r(r+1)+(m-r+1)(m-r)}{r(r+1)(m-r+1)(m-r)} = \frac{2}{r(m-r)}$$

$$\Rightarrow r^2 + r + m^2 + r^2 - 2mr + m - r$$
$$= 2(mr - r^2 + r + m - r + 1)$$

$$\Rightarrow 4r^2 - 4mr - m - 2 + m^2 = 0$$

$$\Rightarrow m^2 - m(4r + 1) + 4r^2 - 2 = 0$$

P 36 : (A) $\frac{1}{n-1}$

Sol. $\int_0^{\pi/4} (\tan^n x + \tan^{n-2} x)\,dx = \int_0^{\pi/4} \tan^{n-2} x(\tan^n x + 1)\,dx$

$$\left[\because\ 0 < x < \frac{\pi}{4}\right]$$
$$\therefore\ [x] = 0$$

$$\int_0^{\pi/4} \tan^{n-2} x.\sec^2 x\,dx$$

$$\left\{\frac{\tan^{n-1} x}{n-1}\right\}_0^{\pi/4} = \frac{1}{(n-1)}$$

P 37 : (A) 8

Sol.
$$\sum_{n=1}^{k} U_n = \begin{vmatrix} \sum_{n=1}^{k} 1 & k & k \\ 2\sum_{n=1}^{k} n & k^2+k+1 & k^2+k \\ 2\sum_{n=1}^{k} n - \sum_{n=1}^{k} 1 & k^2 & k^2+k+1 \end{vmatrix}$$

$$= \begin{vmatrix} k & k & k \\ k(k+1) & k^2+k+1 & k^2+k \\ k^2 & k^2 & k^2+k+1 \end{vmatrix}$$

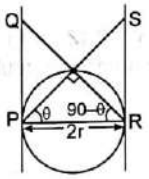
$$= \begin{vmatrix} k & 0 & k \\ k^2+1 & 1 & k^2+k \\ k^2 & 0 & k^2+k+1 \end{vmatrix} \quad [\text{Applying } C_2 \to C_2 - C_1]$$

$= k(k^2 + k + 1) - k^3 = k(k + 1) = 72$

$\Rightarrow k = 8$

P 38 : (A) $\sqrt{PQ \cdot RS}$

Sol.



$$\tan\theta = \frac{RS}{2r} \Rightarrow RS = 2r\tan\theta$$

$$\tan(90-\theta) = \frac{PQ}{2r} \Rightarrow PQ = 2r\cot\theta$$

$$PQ.RS = 4r^2 \Rightarrow 2r = \sqrt{PQ.RS}$$

P 39 : (A) $6, 2\sqrt{5}$

Sol. Length of the string = 6

$a = 3 \qquad b = 2$

$$\therefore\ e = \sqrt{\frac{9-4}{9}} = \sqrt{\frac{5}{9}}$$

$$\therefore\ 2ae = 2 \times 3 \times \frac{\sqrt{5}}{3} = 2\sqrt{5}$$

P 40 : (D) $\frac{y^2}{6} - \frac{x^2}{3} = 1$

Sol. $\because\ 2\frac{a}{e} = 4 \qquad \left\{\because\ e = \sqrt{\frac{3}{2}}\right\}$

$$\therefore a = 2 \times \sqrt{\frac{3}{2}} = \sqrt{6} \quad \Rightarrow b^2 = 6\left(\frac{3}{2} - 1\right) = 3$$

$\therefore$ equation of hyperbola whose transverse axis along y-axis is $\frac{y^2}{6} - \frac{x^2}{3} = 1$

P 41 : (A) $\frac{1}{3}\left\{2(1+x)+\frac{1}{(1+x)}\right\}$

Sol. Given, $2f(x-1)-f\left(\frac{1-x}{x}\right)=x$.

Replacing x by $\frac{1}{x}$, then

$$2f\left(\frac{1}{x}-1\right)-f\left(\frac{1-\frac{1}{x}}{\frac{1}{x}}\right)=\frac{1}{x}$$

$$\Rightarrow \quad 2f\left(\frac{1-x}{x}\right)-f(x-1)=\frac{1}{x}$$

Now, multiplying eq. (i) by 2 and eq. (ii) by 1, then adding, we get

$$3f(x-1)=2x+\frac{1}{x}$$

Now replacing x by x + 1, then

$$f(x)=\frac{1}{3}\left(2(x+1)+\frac{1}{(x+1)}\right)$$

P 42 : (A) 300

Sol. A three digit block from 000 to 999 means 1000 numbers, each number constituting 3 digits. Hence, total digits which we have to write is 3000. Since the total number of digits is 10 (0 to 9) no digit is filled preferentiely.

$\Rightarrow$ number of times we write $3 = \frac{3000}{10} = 30$.

Alternatively any one block can be selected in $^3C_1$ ways and the digit 3 can be filled in it. Now the remaining two blocks can be filled in $9 \times 9 = 81$ ways (excluding the digit 3)

Total ways it can be done = $^3C_1 . 9\ 9 = 243$

Similarly any two blocks can be selected in $^3C_2$ ways and the digit 3 can be put in both of them. Remaining one block can be filled in 9 ways.

Total ways this can be done = $^3C_2 . 2 . 9$

When all the blocks are taken we have $^3C_3 . 3$

Thus the total = $(^3C_1 . 9^2) 1 + (^3C_2 . 9) 2 + ^3C_3 . 3$ = 300

P 43 : (A) 41x – 38y = 0

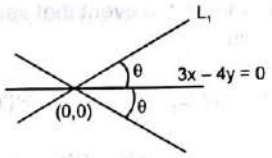
Sol. The bisector are (4x + 3y) (3x – 4y) = 0

choose one bisector 3x – 4y = 0

slope of given line is $\frac{1}{2}$.

Let m be the slope of the other line, then

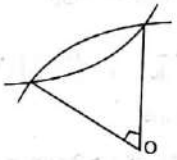
$$\frac{\frac{1}{2}-\frac{3}{4}}{1+\frac{1}{2}\times\frac{3}{4}} = \frac{\frac{3}{4}-m}{1+\frac{3}{4}m} \quad \Rightarrow \quad m = \frac{41}{38}$$



equation of the line L_2 is 41x – 38y = 0

P 44 : (A) g (a′ + b′) – g′ (a + b) = 0

Sol. By making homogeneous second degree equation with the help of given curves we can get pair of lines which are joining the point of intersection of two curves from the origin as follows.



(i) $ax^2g' + 2g'hxy + 2gg'x + by^2g' = 0$

(ii) $a'x^2 g + 2gh'xy + 2gg' x + b'y^2 g = 0$

by subtracting (ii) from (i)

$(ag' – a'g) x^2 + 2(g'h – gh') xy + (bg' – b'g) = 0$

since these lines are right angled to each other hence

$$ag' – a'g + bg' – b'g = 0$$

$$\Rightarrow \quad (a + b)g' – (a' + b') g = 0$$

P 45 : (B) 500

Sol. 2^p when divided by 4 : the remainder is 0 or 2

3^q when divided by 4 : the remainder is 3 or 1

5^r when divided by 4 : the remainder is 1

$\therefore$ $2^p + 3^q + 5^r$ is divisible by 4

if the combinations for the remainder are 0, 3, 1; 2, 1, 1

number of ways for the first set = 5×5×10 = 250

number of ways for the second set = 5 × 5 × 10 = 250

$\therefore$ total number of ways = 500

P 46 : (A) $\frac{20}{41}$

Sol. There are 5W + 3R balls

Let E_1 be the event that first ball drawn is white and E_2 be the event that first ball drawn is red

$$P(E_1) = \frac{5}{8} \qquad P(E_2) = \frac{3}{8}$$

Let W be the event that second ball drawn is white

$$\therefore P(W/E_1) = \frac{4}{9} \text{ and } P(W/E_2) = \frac{7}{9}$$

$$P(E_1/W) = \frac{P(E_1)P(W/E_1)}{P(W)}$$

$$= \frac{P(E_1)P(W/E_1)}{P(E_1)P(W/E_1)+P(E_2)(W/E_2)}$$

$$= \frac{5/8\times 4/9}{5/8\times 4/9+3/8\times 7/9} = \frac{20}{20+21} = \frac{20}{41}$$

P 47 : (C) 2

Sol. We have, $|\log_b a + \log_a b|$

$$= \left|\log_b a + \frac{1}{\log_b a}\right| = \left|x + \frac{1}{x}\right|, \text{ where } x = \log_b a$$

We know that

$x + \frac{1}{x} \geq 2$ for all $x > 0$ and, $x + \frac{1}{x} \leq -2$ for all $x<0$

$$\Rightarrow \left|x + \frac{1}{x}\right| \geq 2 \text{ for all } x \neq 0$$

$\therefore |\log_b a + \log_a b| \geq 2$

Hence, the least value of $|\log_b a + \log_a b|$ is 2.

P 48 : (C) $x^2 + y^2 + (a + b)y + ab = 0$

Sol. Let the variable line be $y = mx + c$

$\therefore$ Equation the lines through the origin

$$= x^2 - (a + b)\, x \left(\frac{y - mx}{c}\right) + ab \left(\frac{y - mx}{c}\right)^2 = 0$$

$\therefore c^2 + (a + b)\, c + ab + abm^2 = 0$

{$\because$ angle between the lines is 90°}

now foot of perpendicular to $y = mx + c$ from (0, 0)

$$\frac{x-0}{m} = \frac{y-0}{-1} = -\frac{c}{1+m^2}$$

$$\text{i.e. } m = -\frac{x}{y} \qquad c = y(1 + m^2) = \frac{x^2+y^2}{y}$$

$\therefore$ the locus is $x^2 + y^2 + (a + b)y + ab = 0$

P 49 : (C) 6

Sol. Given $\sin^3 x \sin 3x = \sum_{m=0}^{n} C_m \cos mx$

$$\Rightarrow \left(\frac{3\sin x - \sin 3x}{4}\right)\sin 3x = \sum_{m=0}^{n} C_m \cos mx$$

$$\Rightarrow \frac{3}{8}(2\sin 3x \sin x) - \frac{1}{8}(2\sin^2 3x) = \sum_{m=0}^{n} C_m \cos mx$$

$$\Rightarrow \frac{3}{8}(\cos 2x - \cos 4x) - \frac{1}{8}(1 - \cos 6x) = \sum_{m=0}^{n} C_m \cos mx$$

$$\Rightarrow \frac{1}{8}\cos 6x + \frac{3}{8}(\cos 2x - \cos 4x) - \frac{1}{8}$$

$\Rightarrow C_0 \cos 0 + C_1 \cos x + C_2 \cos 2x + \ldots + C_n \cos nx$

Therefore n = 6

P 50 : (D) $\alpha < \gamma < \beta$

Sol. We have $a^2\alpha^2 + b\alpha + c = 0$ (1)

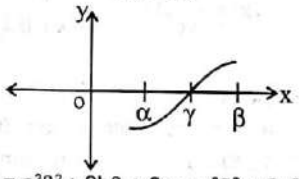
$a^2\beta^2 + b\beta + c = 0$ (2)

Let $f(x) = a^2x^2 + 2bx + 2c$

Therefore $f(\alpha) = a^2\alpha^2 + 2(b\alpha + c)$

$= a^2\alpha^2 + 2(-a^2\alpha^2)$ [from Eq. (1)]

$= -a^2\alpha^2 < 0$



and $f(\beta) = a^2\beta^2 + 2b\beta + 2c = a^2\beta^2 + 2a^2\beta^2 > 0$

Thus $f(\alpha) < 0$ and $f(\beta) > 0$, also $f(x)$ is continuous

Therefore $f(\gamma) = 0$, if $\alpha < \gamma < \beta$

P 51 : (A) $2x - y + 10 = 0$

Sol. Let p (h, k) be the pt of intersection of tangents at A & B to the circle $x^2 + y^2 = 1$

Eqn. of chord of contact AB is

$$xh + y\,k = 1 \qquad \ldots(i)$$

Eqn. of common chord AB is

$$(\lambda + 6)x + (2\lambda - 8)\, y + 2 = 0 \qquad \ldots(ii)$$

Comparing (i) & (ii), we get

$$\frac{\lambda+6}{h} = \frac{2\lambda-8}{k} = \frac{2}{-1}$$

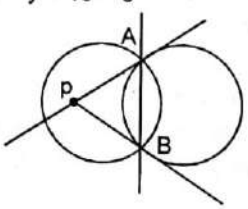
$\lambda + 6 = -2h$

$2\lambda - 8 = -2k$

$\therefore \quad -4h - 12 - 8 = -2k$

$\Rightarrow \quad 4h - 2k + 20$

$\therefore \quad 2x - y + 10 = 0$



P 52 : (A) 2

Sol. $20 = \frac{\pi}{3} \quad \Rightarrow \quad \theta = \frac{\pi}{6}$

$e = \sec\frac{\pi}{6} \quad \text{or} \quad \text{cosec}\frac{\pi}{6}$

$= \frac{2}{\sqrt{3}}, 2$

P 53 : (A) $\frac{1}{x^2} + \frac{1}{y^2} = \frac{1}{a^2}, \quad x > 0, \ y > 0$

Sol. Let the focus be (h, k) then
equation of the tangent at vertex is

$y = -\frac{k}{h}(x - h)$

i.e. $kx + hy - hk = 0$

its distance from (h, k) $= \left|\frac{hk}{\sqrt{h^2 + k^2}}\right| = a$

$\therefore$ the locus is $a^2 (x^2 + y^2) = x^2 y^2$

i.e. $\frac{1}{x^2} + \frac{1}{y^2} = \frac{1}{a^2}, \quad x > 0, \ y > 0$

P 54 : (B) 0

Sol. $\frac{\vec{A}.\vec{B}\times\vec{C}}{\vec{C}\times\vec{A}.\vec{B}} + \frac{\vec{B}.\vec{A}\times\vec{C}}{\vec{C}.\vec{A}\times\vec{B}}$

$= \frac{[\vec{A}\vec{B}\vec{C}]}{[\vec{A}\vec{B}\vec{C}]} + \frac{-[\vec{A}\vec{B}\vec{C}]}{[\vec{A}\vec{B}\vec{C}]} = 0$

P 55 : (A) $\frac{1}{x^2} + \frac{1}{y^2} = \frac{1}{a^2}, \quad x > 0, \ y > 0$

Sol. Let the focus be (h, k) then

equation of the tangent at vertex is $y = -\frac{k}{h}$ (x – h)

i.e. $kx + hy - hk = 0$

its distance from (h, k) $= \left|\frac{hk}{\sqrt{h^2 + k^2}}\right| = a$

$\therefore$ the locus is $a^2 (x^2 + y^2) = x^2 y^2$

i.e. $\frac{1}{x^2} + \frac{1}{y^2} = \frac{1}{a^2}, \quad x > 0, \ y > 0$

P 56 : (D) $\frac{\sqrt{5}}{3}$

Sol. Put $\cot^{-1}\left(\frac{1}{2}\right) = \theta \Rightarrow \cot\theta = \frac{1}{2}$

Therefore, $\sin\theta = \frac{2}{\sqrt{5}}$.

Put $\cos^{-1} x = \phi$ then $x = \cos\phi$

Also $\because \tan\phi = \frac{2}{\sqrt{5}}, \ \therefore x = \cos\phi = \frac{\sqrt{5}}{3}$

P 57 : (B) $\frac{9}{25}$

Sol. D_1 Probability of student (if he misses no test)

$$P(S) = \left(1 - \frac{1}{5}\right) \times \left(1 - \frac{1}{5}\right) = \frac{16}{25}$$

Hence, probability that he will miss at lest one test

$$= 1 - \frac{16}{25} = \frac{9}{25}$$

D_2 The probability that one test is held

$$= 2 \times \frac{1}{5} \times \frac{4}{5} = \frac{8}{25}$$

Probability that test is held on both days = $\frac{1}{5} \times \frac{1}{5} = \frac{1}{25}$

Thus, the probability that the student misses at least one test = $\frac{8}{25} + \frac{1}{25} = \frac{9}{25}$

P 58 : (D) 42

Sol. First select the two men. This can be done in $^4C_2$ ways. Let us say the men are A, B, C and D and their respective wives are a, b, c and d. If we select A and B as the two men when while selecting the women there would be two cases as seen below.

Case 1 :

A : if b is selected to partner A

B : There will be 3 choices for choosing B's partner – viz a, c and d.

Thus, total number of ways in this case $= {}^4C_2 \times 1 \times {}^3C_1 = 18$ ways.

Case 2 :

A : If either c or d is selected to partner A

B : There will be 2 choices for choosing B's partner – viz a and any one of c and d.

Total number of ways of doing this $= {}^4C_2 \times 2 \times {}^2C_1 = 24$ ways.

Hence the required answer is 18 + 24 = 42 ways.

P 59 : (A) $\begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$

Sol. $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = PAP^T$ and $X = P^T Q^{2005} P$

We observe that $Q = PAP^T$

$\Rightarrow \quad Q^2 = (PAP^T)(PAP^T)$

$= PA(P^TP)AP^T$

$= PA(IA)P^T$

$= PA^2P^T$

Proceeding in the same way, we get

$Q^{2005} = PA^{2005}P^T$

Also $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \Rightarrow A^2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

And proceeding in the same way

$$A^{2005} = \begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$$

Now, $X = P^T Q^{2005}$

$= P^T (PA^{2005} P^T) P$

$= (P^T P) A^{2005} (P^T P)$

$= I A^{2005} I$

$= A^{2005}$

$$= \begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$$

P 60 : (C) $\frac{1}{2} \log \left(\frac{e^{2x} - e^x + 1}{e^{2x} + e^x + 1} \right) + C$

Sol. $J - I = \int \frac{(e^{3x} - e^x)\, dx}{e^{4x} + e^{2x} + 1}$

Let $e^x = t$

$$\int \frac{t^2 - 1}{t^4 + t^2 + 1}\, dt = \int \frac{1 - \frac{1}{t^2}}{t^2 + \frac{1}{t^2} + 1}\, dt$$

Let $t + \frac{1}{t} = u$

$$J - I = \int \frac{du}{u^2 - 1} = \frac{1}{2} \ell n \left(\frac{u-1}{u+1} \right) + C$$

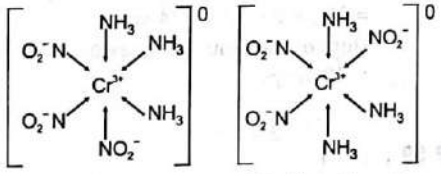
$$= \frac{1}{2} \ell n \left(\frac{t^2 - t + 1}{t^2 + t + 1} \right) + C$$

$$= \frac{1}{2} \ell n \left(\frac{e^{2x} - e^x + 1}{e^{2x} + e^x + 1} \right) + C$$

P 61 : (B) I and II

Sol.

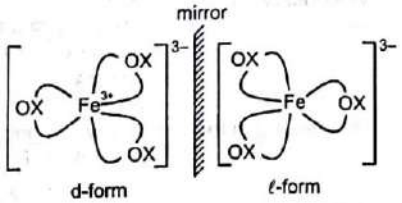
(I)



Facial (or cis) Optically inactive — Meridional (trans) Optically inactive

⇒ Two stereoisomers

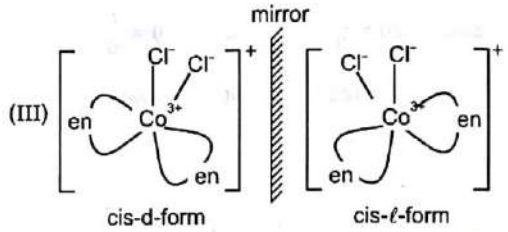
(II)



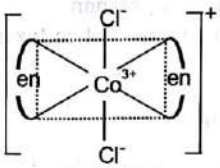
d-form — ℓ-form

⇒ Two stereoisomers

(III)



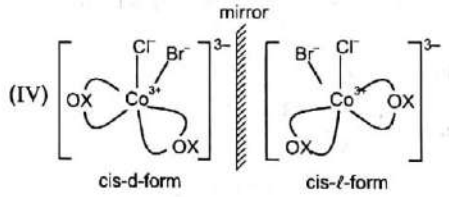
cis-d-form — cis-ℓ-form



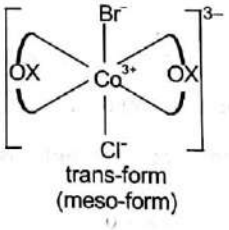
trans-form (meso-form)

⇒ Three stereoisomers

(IV)



cis-d-form — cis-ℓ-form



trans-form (meso-form)

⇒ Three stereoisomers

P 62 : (C) 1,2 and 4

P 63 : (C) $NO_3^- \rightarrow N_2O_5$

P 64 : (C) 2715.63 years

Sol. $\because kt = \ln \frac{N_0}{N}$

$$\Rightarrow \frac{\ln 2}{5730} \times t$$

$$= \ell n \frac{1}{0.72}$$

$\Rightarrow$ t = 2715.63 years.

P 65 : (B) –827 kJ

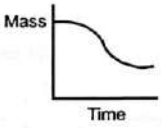
Sol. $\Delta H_f^\circ(KF) = \Delta H_{atomization}(K) + \frac{1}{2}$

$\Delta H_{atomization}(F_2) + I.E(K) - EA(F) + U_0$

$\Rightarrow -567 = 90 + \frac{1}{2} \times 158 + 419 - 328 + U_0$

$\Rightarrow U_0 = -827$ kJ

P 66 : (C)



P 67 : (D) On dilution, molar conductivity increases linearly for strong electrolyte but for weak electrolyte, the increase is gradual in higher concentration range but very rapid in lower concentration range.

Sol. For strong electrolyte, molar conductance increases linearly on dilution. In case of weak electrolyte, dilution first increases molar conductance slowly but very rapidly near to C = 0.

P 68 : (A) 150 g

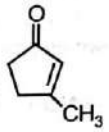
Sol. $\because \frac{P^\circ - P_s}{P_s} = \frac{w_A}{w_B} \times \frac{m_B}{m_A}$

$P_S = \frac{4P^\circ}{5}$, $m_A = 60$, $w_A = ?$, $w_B = 180$ g, $m_B = 18$

$\therefore \frac{P^\circ - \frac{4P^\circ}{5}}{\frac{4P^\circ}{5}} = \frac{w_A \times 18}{60 \times 180}$

$\therefore w = \frac{60 \times 180}{4 \times 18} = 150$ g

P 69 : (C)



Sol. Intramolecular aldol condensation followed by dehydration.

P 70 : (A) II,IV,I,III

Sol. Acid chloride is most reactive followed by acid anhydride then ester and amide is most stable acid derivative.

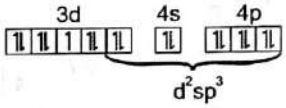
P 71 : (C) (4) (3) (2) (1)

P 72 : (D) 4

Sol. If s-p mixing is not operative than energy order. $\sigma 2p < \pi 2p$

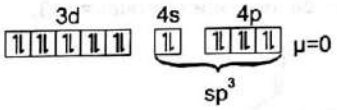
P 73 : (C) I and IV

Sol. Fe^{3+} in $[Fe(CN)_6]^{3-}$

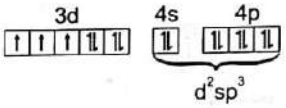


$\Rightarrow$ Low spin complex μ = 1.732 BM

Ni in $[Ni(CO)_4]$

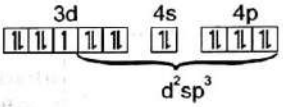


Cr^{3+} in $[Cr(NH_3)_6]^{3+}$



$\Rightarrow$ low and high spin complex is applicable for d^4 to d^7 configuration.

Mn^{2+} in $[Mn(CN)_6]^{4-}$



P 74 : (C) 0.1

Sol. $k_c = \frac{k_f}{k_b} = \frac{2 \times 10^3}{20} = 100$

$k_c = \frac{1}{\sqrt{k_c}} = \frac{1}{\sqrt{100}} = 0.1$

P 75 : (B) less than that of B

Sol. For gas A the temperature remains constant but in case of gas B the temperature increases so the pressure increases. Therefore, the final pressure of 'A' will be less than that of B.

P 76 : (C) +3

Sol. Given,

Current (i) = 5.0 amp

Time period (t) = 4.0 h

Deposition of moles of metal at cathode electrode

= 0.25 mole = $\frac{1}{4}$ mole

Charge in solution = 5.0 × 4 × 60 × 60 C
or charge in Faradays

$$= \frac{5\times4\times60\times60}{96500} = 0.75F$$

$Q = nF$ [n = valency]

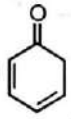
or $n = \frac{Q}{F}$

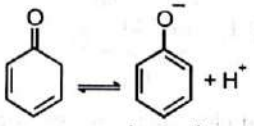
$n = \frac{3F}{4F}$ $\left[0.75 = \frac{3}{4}\right]$

$n = \frac{3}{4}\times 4$ [for 4 hours]

or $n = 3$

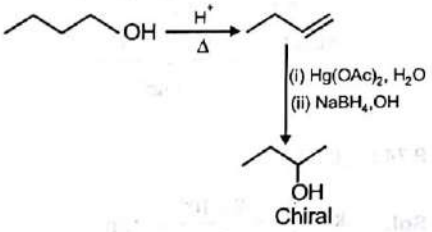
Oxidation state of metal = + 3.

P 77 : (C) 

Sol. 

Aromatic

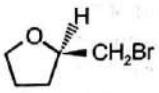
P 78 : (D) $CH_3CH_2CH_2CH_2OH$

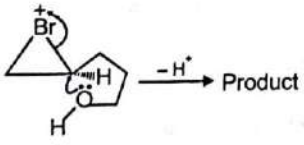
Sol. 

Chiral

P 79 : (A) 1, 2 and 3

Sol. Gamma radiation is more energetic than IR radiation.

P 80 : (A) 

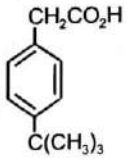
Sol. 

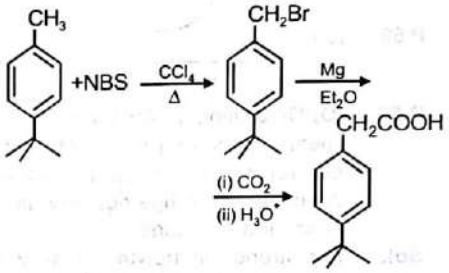
P 81 : (A) $LiAlH_4$, then H_3O^+

Sol. $LiAlH_4$ reduces an acid-chloride to corresponding alcohol.

P 82 : (B) Anisole

Sol. Anisole ($C_6H_5OCH_3$) is activated ring for S_EAr reaction.

P 83 : (D) 

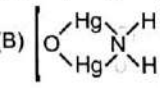
Sol. 

P 84 : (A) The association of two or more peptide chains

Sol. The quaternary structure of protein is the way the individual protein chains are arranged with respect to each other.

P 85 : (B) Mac Arthur–Forest method

P 86 : (D) $Be(OH)_2$, $Zn(OH)_2$

P 87 : (B) 

I–brown iodide of Million's base

P 88 : (A) $[Fe(H_2O)_5(NO)]^{2+}$

Sol. When freshly prepared solution of $FeSO_4$ is added in a solution containing NO_3^- ion, it is leads to formation of a brown coloured complex. This is known as brown ring test of nitrate.

$$NO_3^- + 3Fe^{2+} + 4H^+ \rightarrow NO + 3Fe^{3+} + 2H_2O$$

$$[Fe(H_2O)_6]^{2+} + NO \rightarrow \underset{\text{Brownring}}{[Fe(H_2O)_5(NO)]^{2+}} + H_2O$$

P 89 : (A) Aqueous Ca^{2+} and OH^- ions and gaseous PH_3.

Sol. $Ca_3P_2 + 3H_2O \longrightarrow 3Ca(OH)_2 + 2PH_3\uparrow$

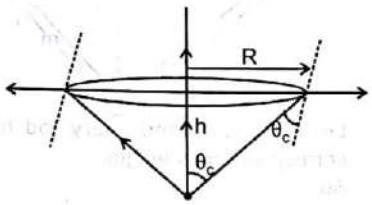
P 90 : (D) Suffix is used wrongly.

Sol. When –COOH is not the part of numbering system of the parent chain, suffix carboxylic acid rather than -oic acid is used.

REPLICA 2

P 1 : (C) $2h \tan \theta_c$

Sol.



Radius of circle is $R = h \tan\theta_c$
diameter $= 2h \tan\theta_c$

P 2 : (C) 9.2×10^{-4} m

Sol. $\frac{\text{size of image}}{\text{size of object}} = \left|\frac{v}{u}\right|$

$$\text{size of the image} = \frac{1.39\times10^{9}\times10^{-1}}{1.5\times10^{11}}$$

$= 0.92 \times 10^{-3}$ m
size of the image $= 9.2 \times 10^{-4}$ m

P 3 : (C) 0.03 mm

Sol. In Young's double slit experiment,
$\sin\theta = \theta = (y/D)$, so $\Delta\theta = (\Delta y/D)$
Hence, angular fringe width $\theta_0 = \Delta\theta$ (with $\Delta y = B$) will be,

$$\theta_0 = \frac{\beta}{D} = \frac{D\lambda}{d}\times\frac{1}{D} = \frac{\lambda}{d}$$

Here, $\theta_0 = 1° = (\theta/180)$ rad. and $\lambda = 6000Å = 6 \times 10^{-7}$ m

$$\text{So, } d = \frac{\lambda}{\theta_0} = \frac{180}{\pi}\times 6 \times 10^{-7} = 3.44 \times 10^{-5}$$

m = 0.0344 mm

P 4 : (C) 500 Hz

Sol. There will be no Doppler effect as the motion of the source is always perpendicular to the direction of propagation of sound.

P 5 : (A) 100

Sol. $L_1 = 10 \log\left(\frac{I_1}{I_0}\right)$; $L2 = 10 \log\left(\frac{I_2}{I_0}\right)$

$$\therefore L_1 - L_2 = 10 \log\left(\frac{I_1}{I_2}\right) \text{ or } 20dB = 10 \log\left(\frac{I_1}{I_2}\right)$$

$$\text{or } \Delta L = 10 \log\left(\frac{I_1}{I_2}\right) \text{ or } 20dB = 10\log\left(\frac{I_1}{I_2}\right)$$

$$\text{or } 10^2 = \frac{I_1}{I_2} \quad \text{or} \quad I_2 = \frac{I_1}{100}$$

P 6 : (B) 98 m

Sol. Motorcycle, $u = 0$, $a = 2$ m/s^2
Observer is in motion and source is at rest

$$\text{or} \quad n' = n\frac{330 - v_a}{330}$$

$$\text{or} \quad 330 - v_a = \frac{330\times94}{100}$$

$$\text{or } v_a = 330 = \frac{94\times33}{10} = \frac{33\times6}{10} \text{ m/s}$$

$$s = \frac{v^2 - v^2}{2a} = \frac{9\times33\times33}{100} = \frac{9\times1089}{100} = 98 \text{ m}$$

P 7 : (D) 1.414

Sol. Given, the maximum electron density in the morning

$N_{max} = 10^{10}$ m^{-3}
$f_e' = 9 (N_{max})^{1/2}$
$= 9 (10)^{1/2}$

Critical frequency, $f_e' = 9 \times 10^5$ Hz
The maximum electron density at noon
$N_{max} = 2 \times 10^{10}$ m^{-3}
Critical frequency, $f_e'' = 9(N_{max})^{1/2}$
$= 9(2 \times 10^{10})^{1/2}$
$= 9 \times \sqrt{2} \times 10^5$

The ratio of critical frequency at noon and the critical frequency in the morning

$$= \frac{f_e''}{f_e'} = \frac{9\times\sqrt{2}\times10^5}{9\times10^5} = \sqrt{2} = 1.414$$

P 8 : (C) Zero

Sol. The magnetic field produced by the current in the loop and the external magnetic field are along the same direction.

P 9 : (D) $40\sqrt{2}$ V

Sol. Given $V_R = V_L = V_C$. Therefore, $R = X_L = X_C$. When the capacitance is short circuited, $X_C = 0$ and the impedence is :

$$Z = \sqrt{R^2 + X_L^2} = \sqrt{R^2 + R^2} = \sqrt{2}R$$

$$(\because R = X_L)$$

the voltage of the a.c. source is given by ($\because V_L = V_C$)

$$V = \sqrt{V_R^2 + (V_L - V_C)^2} = V_R = 80 \text{ V}$$

$\therefore$ Current in the circuit is

$$I = \frac{V}{Z} = \frac{80}{\sqrt{2}R}$$

Hence $V_L = IX_L = \frac{80}{\sqrt{2}R} \times R \quad (\because X_L = R)$

$= 40\sqrt{2}$ volt

P 10 : (B) 62.6 V/m

Sol. (B) Intensity of electromagnetic wave is given by $I = \frac{P_{av}}{4\pi r^2} = \frac{E^2 m}{2\mu_0 c}$

$$E_m = \sqrt{\frac{\mu_0 c P_{av}}{2\pi r^2}}$$

$$= \sqrt{\frac{\left(4\pi \times 10^{-7}\right) \times \left(3 \times 10^8\right) \times 800}{2\pi \times 3.5^2}} = 62.6 \text{ V/m}$$

P 11 : (D) none of the above

Sol. Characteristic X-Ray has wavelengths which only depends on target material and not on potential difference.

P 12 : (A) $\frac{2}{\sqrt{14}}$

Sol. $\vec{A} = 2\hat{i} + 3\hat{j} + 2\hat{k} \Rightarrow |A| = \sqrt{17}$

$\vec{B} = \hat{i} - 2\hat{j} + 3\hat{k} \Rightarrow |B| = \sqrt{17}$

Projection of $\vec{A}$ on $\vec{B}$

$$P = \frac{\vec{A} \cdot \vec{B}}{|\vec{B}|} = \frac{2}{\sqrt{14}}$$

P 13 : (D) is non zero constant.

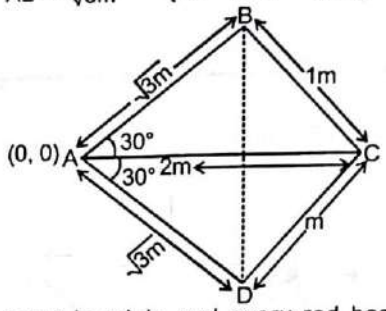
Sol. v_1 = slope of C_1 line = constant

v_2 = slope of C_2 line = constant

$v_1 - v_2 \neq 0$ but constant

P 14 : (A) 1.44 m

Sol. $AB = \sqrt{3}m \quad \{AB^2 + BC^2 = AC^2\}$



Let A is origin and every rod has mass correspond to its length.

So

$$X_{cm} = \frac{(\sqrt{3})\frac{(\sqrt{3}\cos 30)}{2} \times 2 + (2)(1) + 2(1)\left(2 - \frac{1\cos 60}{2}\right)}{2\sqrt{3} + 2 + 2}$$

$$X_{cm} = \frac{3\sqrt{3} + 2 + 7/2}{2\sqrt{3} + 2 + 2} = 1.44 \text{ m}$$

P 15 : (D) $\frac{3}{4}$

Sol. At given point, path difference = $\lambda/6$

$$\Rightarrow \Delta Q = \frac{2\pi}{\lambda}(\lambda/6) = \frac{\pi}{3}$$

So $I = I_0 \cos^2 \frac{\Delta\theta}{2} = I_0(\cos\pi/6)^2 = \frac{3I_0}{4}$

$$\Rightarrow \frac{I}{I_0} = \frac{3}{4}$$

P 16 : (C) 80

Sol. Every oxygen O^{-2} have charge $-2e$

and no. of oxygen atoms = $\frac{1}{16} \times 6.02 \times 10^{23}$

Now $i = \frac{q}{t}$

$$= \frac{\frac{1}{16} \times 6.02 \times 10^{23} \times 1.6 \times 10^{-19} \times 2}{t \times 60} = 1A$$

(Current is half due to +ive ions and half due to –ive ions) $\Rightarrow$ t = 210 min.

P 17 : (A) 300 volt

Sol. $X_c = 350\ \Omega$ and $R = 180\ \Omega$

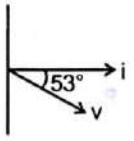
$$\tan 53° = \frac{X_L - X_C}{R} = \frac{4}{3}$$

$$X_L - X_C = \frac{4}{3} \times 180 = 240\Omega$$

$$Z = \sqrt{(240)^2 + (180)^2} = 300\Omega$$

$$P = i^2 R \Rightarrow i = \sqrt{\frac{180}{180}} = 1A = \frac{V}{Z}$$

V = 300 volt



P 18 : (A) $\sqrt{\frac{2d}{R}}$

Sol. Power received = Power radiated

$$\left(\frac{\sigma T_S^4 4\pi R_S^2}{4\pi d^2}\right)\pi r^2 = \sigma T_e^4 4\pi r^2$$

$$T_S^4 = \frac{T_e^4 4}{R_S^2} \quad \Rightarrow \quad T_S = \left[\frac{4d^2}{R^2}\right]^{1/4} T_e$$

$$x = \sqrt{\frac{2d}{R}}$$

P 19 : (D) NAND gate

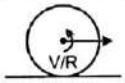
Sol. Truth table of Nand gate is

A	B	y
0	0	1
1	0	1
0	1	1
1	1	0

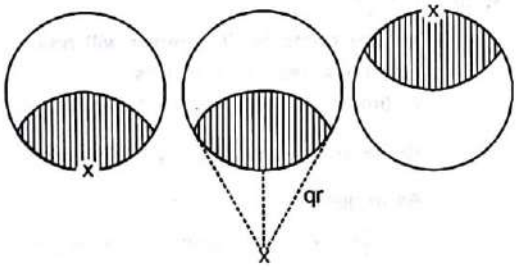
It implies that for both, high signal output will be low.

P 20 : (B) disc 1 = disc 3 < disc 2

Sol. Instantaneous Axis of Rotation passes through the point whose velocity is zero

disc 1	disc 2	 disc 3
$V_p = V_{cm} + \omega r$ $V_p = 0$; $r = R$ $r = R$ below centre	$V_p = V_{cm} + \omega r$ $V_p = 0$; $r = 2R$ $r = 2R$ below centre	$V_p = V_{cm} + \omega r$ $V_p = 0$; $r = R$ $r = R$ above centre

And points at distance less than distance of C.M. from I.O.R. will have less velocity.



As radius of curvature for 2 is 2r, it is relatively flat so area will be more than 1 and 3.

$A_1 = A_3 < A_2$

P 21 : (C) $\frac{r\mu_0 n I_0 \omega \cos\omega t}{2}$

Sol. Magnetic field in solenoid = $\mu_0 ni$

$B = \mu_0 n I_0 \sin\omega t$

So induced electric field at ($r < R$)

$$E = \frac{r}{2}\frac{dB}{dt} \qquad \left\{\oint \vec{E}d\vec{l} = \frac{d\phi}{dt}\right\}$$

$$E = \frac{r\mu_0 n I_0 \omega \cos\omega t}{2}$$

P 22 : (A) $Re^{\left[\frac{2\pi mv}{\mu_0 ei}\right]}$

Sol. At point P, it is farthest from the axis and its velocity is parallel to the wire but its speed will not change as magnetic field cannot do any work.

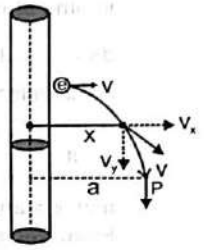
So now impulse in y direction

$$\int eBv_x dt = mv$$

$$e\int_R^a \frac{\mu_0 i}{2\pi x} dx = mv$$

$$\frac{\mu_0 ei}{2\pi}\ln\left(\frac{a}{R}\right) = mv$$

$$a = Re^{\left[\frac{2\pi mv}{\mu_0 ei}\right]}$$



P 23 : (A) 2 MHz

Sol. The critical frequency for reflection

$f_c = 9(N_{max})^{1/2} = 9(10^{11})^{1/2} = 2.7$ MHz

So, all the waves having frequency less than 2.7 MHz will be reflected back.

P 24 : (B) $\frac{E^2A}{8\pi}\left(\frac{1}{d_2} - \frac{1}{d_1}\right)$

Sol. Due to motion of the plates, the charge on the charge on the capacitor increases by

$$\Delta Q = Q_2 - Q_1 = \frac{EA}{4\pi}\left(\frac{1}{d_2} - \frac{1}{d_1}\right)$$

Work done by the battery is given as

$$W = E\Delta Q = \frac{E^2A}{4\pi}\left(\frac{1}{d_2} - \frac{1}{d_1}\right)$$

$\therefore$ Energy of the capacitor is increased by

$$\Delta W_e = W_{e_2} = \frac{EQ_2}{2} - \frac{EQ_1}{2} = \frac{E^2A}{8\pi}\left(\frac{1}{d_2} - \frac{1}{d_1}\right)$$

Now W_1 = work done when plates are moved closer to each other Hence, according to law of conservation of energy.

$W = W_1 + \Delta W_e$

$W_1 = W - \Delta W_e$

$= \frac{E^2A}{8\pi}\left(\frac{1}{d_2} - \frac{1}{d_1}\right)$

P 25 : (D) $\frac{\mu_0 I^2}{4\pi}\log_e\frac{b}{a}$

Sol. In the region between the 2 cylinders

$B = \mu_0 I/2\pi r$

and $W_2 = \frac{1}{2\mu_0}\int_a^b\left(\frac{\mu_0 I}{2\pi\rho}\right)2\pi r$

$= \frac{\mu_0 I^2}{4\pi}\log_e\frac{b}{a}$

P 26 : (D) 4

Sol. Area covered by line joining planet and sun in time dt is,

$ds = \frac{1}{2}x^2 d\theta$

Areal velocity

$= \frac{ds}{dt} = \frac{1}{2}x^2\frac{d\theta}{dt} = \frac{1}{2}x^2\omega$

Where x = distance between planet and sun and ω = angular speed of planet about sun. From Kepler's second law, areal velocity of planet is constant.

At farthest position,

$A = \frac{ds}{dt} = \frac{1}{2}(2R-r)^2\omega$

$= \frac{1}{2}(2R-r)[(2R-r)\omega]$

$= \frac{1}{2}(2R-r)v_B$

or $v_B = \frac{2A}{2R-r}$ = least speed.

Using value v_B = 40 km/sec, thus K = 4

P 27 : (C) $3m_A = 2m_B$

Sol. For gas in A →

$P_1 = \left(\frac{RT}{M}\right)\frac{m_A}{V_1}$ and $P_2 = \left(\frac{RT}{M}\right)\frac{m_A}{V_2}$

$\therefore \Delta P = P_1 - P_2 = \left(\frac{RT}{M}\right)m_A\left[\frac{1}{V_1} - \frac{1}{V_2}\right]$

Putting $V_1 = V$ and $V_2 = 2V$ we get;

$\Delta P = \left(\frac{RT}{M}\right)\frac{m_A}{2V}$ ...(i)

Similarly, for gas B,

$1.5\ \Delta P = \left(\frac{RT}{M}\right)\frac{m_B}{2V}$ ...(ii)

From equations (i) and (ii), $3m_A = 2m_B$.

P 28 : (B) 339.97 m/s

Sol. As the two resonance occurs at 15.4cm and 48.6 cm so their must be end correction as they are not in 1 : 3.

So $\frac{\lambda}{4} = 15.4 + e$

$\frac{3\lambda}{4} = 48.6 + e$

$\Rightarrow \frac{\lambda}{2} = 33.2$ cm

$\Rightarrow \lambda = 66.4$ cm

So wave velocity

$\upsilon = f\lambda$

$\upsilon = 512 \times 66.4$ cm/s

$\upsilon = 339.97$ m/s

P 29 : (B) 10

Sol. $n = \frac{1}{2\ell}\sqrt{\frac{T}{m}}$ (where m is mass per unit length)

$\therefore m_1 = \frac{M}{\ell_1} = \frac{\rho_1 \times \pi r_1^2 \ell_1}{\ell_1} = \pi r_1^2 \rho_1$

and $m_2 = \pi r_2^2 \rho_2$

Therefore, $\frac{n_1}{n_2} = \frac{\ell_2}{\ell_1}\sqrt{\frac{T_1}{T_2}\cdot\frac{r_2^2\rho_2}{r_1^2\rho_1}}$

$= \frac{35}{36}\sqrt{\frac{8}{1}\times\frac{1}{16}\times\frac{2}{1}} = \frac{35}{36} = 10$

P 30 : (A) $\sqrt{5}$

Sol. As the bomb is dropped it will have only horizontal velocity = 20 m/s

At time t, its vertical velocity = 10t

So its speed $v = \sqrt{(20)^2 + (10t)^2}$

As in question,

$\sqrt{(20)^2 + (10t_1)^2} = 2(20) \Rightarrow 10t_1 = 20\sqrt{3}$

$\sqrt{(20)^2 + (10t_2)^2} = 4(20) \Rightarrow 10t_2 = 20\sqrt{15}$

$\frac{t_2}{t_1} = \sqrt{5}$

P 31 : (D) 2.76×10^{-3}

Sol. Subtraction of second equation from first will give the required equation. Hence

$\Delta G = -41.0 - (-55.6)$

$= 14.6 \text{ kJ} = -RT \log K$

$$\Rightarrow \quad \log K = -\frac{14.6 \times 1000}{8.3 \times 298 \times 2303}$$

$$= -2.56$$

$$\Rightarrow \quad K = 2.7 \times 10^{-3}$$

P 32 : (A) 133.33 mL

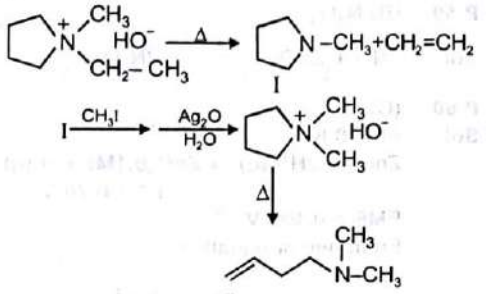
Sol. The half-life of H^3 = 12.3 yrs indicates that the decay of H^3 causing circulatory mixing is negligibly small and can be ignored.

$\Rightarrow 1 \times 2 \times 10^6 = 1.5 \times 10^4 \text{ V}$

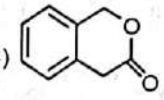
$\Rightarrow$ V = 133.33 mL

P 33 : (D) $H_2C = CH_2CH_2 - N(CH_3)_2$

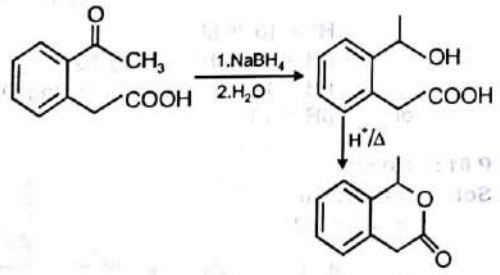
Sol.



P 34 : (C)



Sol.



P 35 : (A) 0.98 bar

Sol. Calculate mole fraction (x_{O_2}):

Mass of 1 L of solution = 1000 g

$\therefore$ Mass of solvent (water) = 1000 g – 0.05 g = 1000 g [d = 1 g mL^{-1}]

$$n_{H_2O} = \frac{1000 \text{g}}{18 \text{gmol}^{-1}} = 1.56 \times 10^{-3} \text{ mole}$$

$$n_{O_2} = \frac{n_{O_2}}{n_{O_2} + n_{H_2O}} = \frac{n_{O_2}}{n_{H_2O}} = \frac{1.56 \times 10^{-3}}{55.5}$$

$= 2.81 \times 10^{-5}$

Calculation of partial pressure :

Applying Henry's law,

$p_{O_2} = K_H \times x_{O_2}$

$= 34.86 \times 10^3 \text{ bar} \times (2.81 \times 10^{-5}) = 0.98 \text{ bar}$

P 36 : (D) 40

Sol. $Z = 4 \times 0.8 = 3.2$ $\frac{a}{2} = 100$

a = 200 pm

$$Cl = \frac{Z \times M}{N_A \times a^3} = \frac{3.2 \times 60}{6 \times 10^{23} \times (2 \times 10^{-8})^3}$$

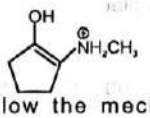
P 37 : (D) 1, 2 and 4

Sol. IR radiation has greater frequency than radiowave.

P 38 : (B) A will be faster both in S_N1 and S_N2 reaction

Sol. Allylic halide gives both S_N1 and S_N2 reaction.

P 39 : (D)



Sol. Follow the mechanism of reaction of ammonia and amines with aldehydes and ketones.

P 40 : (A) 50

Sol. Millimoles of NaCl = molarity × volume (ml)

$= 10 \times 0.5 = 5$

$\therefore$ 100 ml colloidal solution requires 5 millimoles

$\therefore$ 1000 ml colloidal solution requires =

$$\frac{5 \times 1000}{100} = 50 \text{ millimoles}$$

$\therefore$ Flocculation value = 50

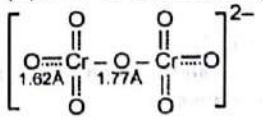
P 41 : (D) All of the above

P 42 : (B) $BaCl_2$

P 43 : (A) I, II

P 44 : (B) 6 Cr–O bonds are equivalent

Sol.



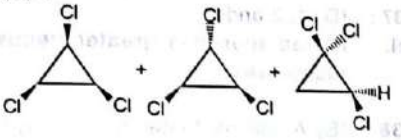
P 45 : (D) NH_2^-, NH_3, NH_4^+
Sol. Number of lone pair ↑ bond angle ↓.

P 46 : (C) Z 195 500
Sol. For alkali metals
$(IE)_1 << (IE)_2$

P 47 : (A) EDTA
Sol. Form complex with EDTA.

P 48 : (A) (i)
Sol. This hydrogen is bonded to electronegative nitrogen as well as conjugate base will be stabilized by resonance by the adjacent carbonyl.

P 49 : (B) 3
Sol.



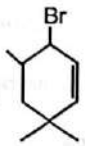
P 50 : (D) BH_3, THF followed by H_2O_2, OH^-
Sol. hydroboration oxidation of alkene occur according to anti-Markownikoff's rule. Syn hydration takes place.

P 51 : (A) 5 kJ
Sol. $q = ms\Delta T = 2 \times 200 \times 0.5 = 200J$

$$DH = 200 \times \frac{50}{2} = 5000\ J = 5\ kJ$$

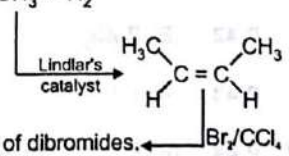
P 52 : (D) (4) (3) (1) (2)

P 53 : (A)



Sol. Allylic bromination takes place by NBS

P 54 : (A) I and III in equal amount.
Sol. $CH_3 - C \equiv C - CH_3 + H_2$



Racemic mixture of dibromides.

P 55 : (A) $[Co(en)_2Cl_2]^+$
Sol. $[Co(en)_2Cl_2]^+$ shows geometrical as well as optical isomerism.

P 56 : (A) The benzaldehyde carbonyl is more reactive than acetone carbonyl
Sol. Carbonyl carbon of an aldehyde has less steric hindrance than that of a ketone, undergo faster nucleophilic addition reaction.

P 57 : (C) van der Waal's force
Sol. Natural rubber is a pure hydrocarbons therefore the only force of attraction between the adjacent chain is van der Waals' force.

P 58 : (C) 8
Sol. 1.0 mol of protein will contain at least 1.0 mol of Fe.

$$\Rightarrow \quad M \times 10^4 \times \frac{0.07}{100} = 56$$

$$\therefore \quad M = 8$$

P 59 : (B) N_2O_3
Sol. $2NO + 2NO_2 \xrightarrow{250K} 2N_2O_3$

P 60 : (C) 10
Sol. At 298 K,
$Zn(s) + 2H^+(aq) \rightarrow Zn^{2+}(0.1M) + H_2(g)$; $E^\circ = 0.76$ V
EMF = 0.199 V
From nernst equation,

$$E_{cell} = E^0_{cell} + \frac{0.0591}{n}\log\frac{[H^+]^2}{[Zn^{2+}]} \quad [n = 2e^-]$$

$$0.199V - 0.76\ V = \frac{0.0591}{2}\log\frac{[H^+]^2}{[0.1]}$$

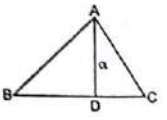
$$H^+ = 10^{-10}\ M$$

$$\therefore \quad pH = -\log H^+ = -\log 10^{-10}$$

$$pH = 10 \log 10 \qquad [\because \log 10 = 1]$$

or $pH = 10$

P 61 : (B) s/Δ
Sol. $\Delta = (1/2)a\alpha$
$\Rightarrow \alpha^{-1} = a/2\Delta$



$$\Sigma\alpha^{-1} = \frac{a+b+c}{2\Delta} = \frac{2s}{2\Delta} = \frac{s}{\Delta}$$

P 62 : (D) cos 7°
Sol. $\sin 47° + \sin 61° - (\sin 11° + \sin 25°) =$
$2\sin 54°.\cos 7° - 2\sin 18° \cos 7°$
$= 2\cos 7° (\sin 54° - \sin 18°) = 2\cos 7° .2\cos 36°. \sin 18°$

$$= 4.\cos 7°.\frac{\sqrt{5}+1}{4}.\frac{\sqrt{5}-1}{4} = \cos 7°.$$

P 63 : (B) $\frac{1}{6}n(n-1)(4n+1)$

Sol. $\int_0^{n^2}[\sqrt{x}]dx = \int_0^{1^2}[\sqrt{x}]dx + \int_0^{2^2}[\sqrt{x}]dx + \int_{2^2}^{3^2}[\sqrt{x}]dx$

$+.... + \int_{(n-1)^2}^{n^2}[\sqrt{x}]dx$

$= 0 + 1\cdot(2^2 - 1^2) + 2\cdot(3^2 - 2^2) + ...+(n-1)$
$[n^2 - (n-1)^2]$

$= -1^2 - 2^2 - 3^2 - ..... - (n-1)^2 + (n-1)n^2$

$= -\frac{(n-1)n(2n-1)}{6} + (n-1)n^2$

$= \frac{1}{6}n(n-1)(4n+1)$

P 64 : (A) $5\sqrt{2}$

Sol. Given that $|\vec{A}| = 3$; $|\vec{B}| = 4$; $|\vec{C}| = 5$

$\vec{A} \perp (\vec{B}+\vec{C}) \Rightarrow \vec{A}\cdot(\vec{B}+\vec{C}) = 0$

$\Rightarrow \quad \vec{A}\cdot\vec{B} + \vec{A}\cdot\vec{C} = 0 \quad ....(1)$

$\vec{B} \perp (\vec{C}+\vec{A}) \Rightarrow \vec{B}\cdot(\vec{C}+\vec{A}) = 0$

$\Rightarrow \quad \vec{B}\cdot\vec{C} + \vec{B}\cdot\vec{A} = 0 \quad ....(2)$

$\vec{C} \perp (\vec{A}+\vec{B}) \Rightarrow \vec{C} \perp (\vec{A}+\vec{B}) = 0$

$\vec{C}\cdot\vec{A} + \vec{C}\cdot\vec{B} = 0 \quad ....(3)$

Adding (1), (2) and (3), wet get

$2(\vec{A}.\vec{B} + \vec{B}.\vec{C} + \vec{C}.\vec{A}) = 0 \quad ....(4)$

Now, $[\vec{A}+\vec{B}+\vec{C}]^2$

$= (\vec{A}+\vec{B}+\vec{C})\cdot(\vec{A}+\vec{B}+\vec{C})$

$= |\vec{A}|^2 + |\vec{B}|^2 + |\vec{C}|^2 + 2(\vec{A}.\vec{B} + \vec{B}.\vec{C} + \vec{C}.\vec{A})$

$= 9 + 16 + 25 + 0 = 50$

$= |\vec{A}+\vec{B}+\vec{C}| = 5\sqrt{2}$

P 65 : (B) a

Sol. Let $a = \sin 4\theta$

$\Rightarrow \sqrt{1+a} = \cos 2\theta + \sin 2\theta$

and $\sqrt{1-a} = \cos 2\theta - \sin 2\theta$

$(1+\sqrt{1+a})\tan\alpha = (1+\sqrt{1-a})$

$\Rightarrow (1 + \cos 2\theta + \sin 2\theta)\tan\alpha$
$= 1 + \cos 2\theta - \sin 2\theta$

$\Rightarrow \quad \frac{2\cos\theta(\cos\theta+\sin\theta)}{2\cos\theta(\cos\theta-\sin\theta)} = \cot\alpha$

$\Rightarrow \quad \frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta} = \cot\alpha$

$\Rightarrow \quad \frac{1+\tan\theta}{1-\tan\theta} = -\cot\alpha$

$\Rightarrow \quad \tan\left(\frac{\pi}{4}+\theta\right) = \tan\left(\frac{\pi}{2}+\alpha\right)$

$\Rightarrow \quad \theta = \left(\frac{\pi}{4}-\alpha\right)$

$\Rightarrow \quad a = \sin 4\theta = \sin(\pi - 4\alpha) = \sin 4\alpha$

P 66 : (B) $\frac{1}{1-\log_a z}$

Sol. We have $y = a^{\frac{1}{1-\log_a x}}$ and $z = a^{\frac{1}{1-\log_a y}}$

$\Rightarrow \log_a y = \frac{1}{1-\log_a x}$ and $\log_a z = \frac{1}{1-\log_a y}$

$\Rightarrow \log_a x = \frac{\log_a y - 1}{\log_a y}$ and $1 - \log_a z = \frac{\log_a y}{\log_a y - 1}$

$\Rightarrow \log_a x = \frac{1}{1-\log_a z}$

$\Rightarrow k = \frac{1}{1-\log_a z}$ [$\because x = a^k \therefore \log_a x = k$]

P 67 : (D) None

Sol. $\frac{b}{2}(1 + \cos A) + \frac{a}{2}(1 + \cos B) = \frac{3c}{2}$

$\Rightarrow a + b + c = 3c \Rightarrow a + b = 2c$

$\Rightarrow$ a,c,b are in A.P.

P 68 : (D) 84 cm^2

Sol. $\Delta = rs$

$\tan\frac{B}{2} = \frac{4}{6} = \frac{2}{3}$, $\tan\frac{C}{2} = \frac{4}{8} = \frac{1}{2}$

and $\Sigma \tan\frac{A}{2}\tan\frac{B}{2} = 1$

$\therefore \quad \frac{2}{3}\tan\frac{A}{2} + \frac{1}{2}\tan\frac{A}{2} + \frac{2}{3}\cdot\frac{1}{2} = 1$

i.e. $\tan\frac{A}{2} = \frac{4}{7}$

$\therefore \quad \frac{4}{7} = \frac{4}{x}$

i.e. $x = 7$

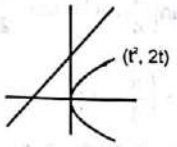
$\therefore \quad s = 6 + 8 + 7 = 21$

$\therefore \quad \Delta = 4 \times 21 = 84$ cm^2

P 69 : (C) $4(y-2) = (x+2)^2$

Sol. $\frac{x-t^2}{1} = \frac{y-2t}{-1} = -2\,\frac{(t^2-2t+2)}{2}$

$\Rightarrow \quad x = 2t-2 \quad \Rightarrow t = \frac{x+2}{2}$



and $\quad y = t^2+2 \quad \Rightarrow y-2 = \frac{(x+2)^2}{4}$

$\Rightarrow \quad 4(y-2) = (x+2)^2$

P 70 : (D) $\{x|x>0|\} \cup \{-1\}$

Sol. $\frac{|x+1|}{|x|} + |x+1| = \frac{|x+1|^2}{|x|}$

$\Rightarrow |x+1|\left\{\frac{1}{|x|}+1-\frac{|x+1|}{|x|}\right\} = 0$

Therefore, $\quad |x+1| = 0$

or $\quad 1+|x|-|x+1| = 0$

$\Rightarrow \quad x = -1 \; |x+1| = 0$

If $\quad x < -1,\ 1+|x|-|x+1| = 0$

$\Rightarrow \quad 1-x+x+1 = 0$

$\Rightarrow \quad 2 = 0$

If $\quad -1 \le x < 0$

$\quad 1+|x|-|x+1| = 0$

$\Rightarrow \quad 1-x-(x+1) = 0$

$\Rightarrow \quad x = 0$ (Not possible)

If $\quad x \ge 0,\ 1+x-(x+1) = 0$

$\Rightarrow \quad 0 = 0$

$\Rightarrow \quad x$ can have any value in the interval

Therefore $x = -1,\ x > 0\ (\because x \ne 0)$

P 71 : (B) $1+\sin^2 1$

Sol. Let $= f(\theta) = \cos^2(\cos\theta) + \sin^2(\sin\theta)$

$-1 \le \cos\theta \le 1$ and $-1 \le \sin\theta \le 1$

$\cos 1 \le \cos(\cos\theta) \le 1$ and

$\quad -\sin 1 \le \sin(\sin\theta) \le \sin 1$

$\cos^2 1 \le \cos^2(\cos\theta) \le 1$ and

$\quad 0 \le \sin^2(\sin\theta) \le \sin^2 1.$

Maximum value of $f(\theta) = 1+\sin^2 1$

P 72 : (B) 0

Sol. Let $\Delta = \begin{vmatrix} 1 & \sin3\theta & \sin^3\theta \\ 2\cos\theta & \sin6\theta & \sin^3 2\theta \\ 4\cos^2\theta - 1 & \sin9\theta & \sin^3 3\theta \end{vmatrix}$

Multiplying C_1 by $\sin\theta$, determinant can be written as

$$\Delta = \frac{1}{\sin\theta}\begin{vmatrix} 1 & \sin3\theta & \sin^3\theta \\ 2\cos\theta\sin\theta & \sin6\theta & \sin^3 2\theta \\ \sin\theta(4\cos^2\theta-1) & \sin9\theta & \sin^3 3\theta \end{vmatrix}$$

$$= \frac{1}{\sin\theta}\begin{vmatrix} \sin\theta & \sin3\theta & \sin^3\theta \\ \sin2\theta & \sin6\theta & \sin^3 2\theta \\ \sin3\theta & \sin9\theta & \sin^3 3\theta \end{vmatrix}$$

Now, applying $C_1 \to 3C_1 - 4C_3$

$$\Delta = \frac{1}{\sin\theta}\begin{vmatrix} 3\sin\theta-\sin^3\theta & \sin3\theta & \sin^3\theta \\ 3\sin2\theta-4\sin^3 2\theta & \sin6\theta & \sin^3 2\theta \\ 3\sin3\theta-4\sin^3 3\theta & \sin9\theta & \sin^3 3\theta \end{vmatrix}$$

$$\Delta = \frac{1}{\sin\theta}\begin{vmatrix} \sin3\theta & \sin3\theta & \sin^3\theta \\ \sin6\theta & \sin6\theta & \sin^3 2\theta \\ \sin9\theta & \sin9\theta & \sin^3 3\theta \end{vmatrix} = 0$$

P 73 : (A) 2

Sol. $\quad 4x^2+9y^2+24x-6y+33 = 0$

i.e. $\quad 4x^2+24x+36+9y^2-6y+1 = 4$

$\quad (2x+6)^2+(3y-1)^2 = 4$

$$4(x+3)^2 + 9\left(y-\frac{1}{2}\right)^2 = 4$$

i.e. $\quad \frac{(x+3)^2}{1} + \frac{\left(y-\frac{1}{2}\right)^2}{\frac{4}{9}} = 1 \qquad \therefore$

$a = 1 \quad b = \frac{2}{3}$

$\therefore$ greatest distance between two points on the ellipse is = 2

P 74 : (C) $\sqrt{2}$ & 2

Sol. $\frac{y}{\sin60^\circ} = \frac{x}{\sin30^\circ} \Rightarrow \quad y = \sqrt{3}\,x$

and $\quad x\left(\sqrt{3}+1\right) = \sqrt{3}+1 \Rightarrow \quad x = 1$

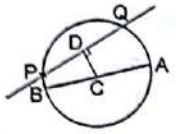
$\Rightarrow \quad r_1 = x\sqrt{2} = \sqrt{2}$ and $r_2 = 2$

P 75 : (A) $(-\infty, -\sqrt{3}) \cup (\sqrt{3}, \infty)$

Sol. Since $\angle APB = \angle AQB = \frac{\pi}{2}$ so $y = mx+8$ intersect the circle whose diameter is AB.

Eq. of circle is $x^2+y^2 = 16$

$\quad CD < 4$



$\Rightarrow \frac{8}{\sqrt{1+m^2}} < 4 \qquad \Rightarrow 1 + m^2 > 4$

$m \in [-\infty, -\sqrt{3}] \cup (\sqrt{3}, \infty)$

P 76 : (A) $(2x + 1) = 0$ & $(y - 2) = 0$

Sol. $\because$ Joint equation of asymptotes is : $2xy - 4x + y + \lambda = 0$ ... (1)

(1) will present two asymptotes if

$abc + 2hgf - af^2 - bg^2 - ch^2 = 0$

$\Rightarrow \quad \lambda = -2$ put in (1), we get.

$2xy - 4x + y - 2 = 0$

$\Rightarrow \quad 2x(y - 2) + 1(y - 2) = 0$

$\Rightarrow \quad (2x + 1) = 0$ and $y = 2$

P 77 : (C) 6

Sol. Applying $C_1 \to C_1 + C_2$

$$f(x) = \begin{vmatrix} 2 & \cos^2 x & 4\sin 2x \\ 2 & 1+\cos^2 x & 4\sin 2x \\ 1 & \cos^2 x & 1+4\sin 2x \end{vmatrix}$$

Applying $R_1 \to R_2 - R_1$, we have

$$f(x) = \begin{vmatrix} 2 & \cos^2 x & 4\sin 2x \\ 0 & 1 & 0 \\ 1 & \cos^2 x & 1+4\sin 2x \end{vmatrix}$$

$f(x) = 2 + 4\sin 2x$. The value of $f(x)$ is maximum when $\sin 2x = 1$

Therefore, Maximum value of $f(x) = 6$

P 78 : (D) $\left(-\frac{1}{5}, -\frac{1}{5}\right)$ if $|PA - PB|$ is minimum

Sol. P lies on the line $2x + 3y + 1 = 0$ ....(1)

$|PA - PB|$ is maximum if P, A, B are collinear.

Equation of the equation AB is $x + y = 2$...(2)

P is the point of intersection of (1) and (2) which is $(7, -5)$

Equation of perpendicular bisector of AB is $y = x$ ....(3)

For minimum value of $|PA - PB|$

P is the point of intersection of (1) and (3)

which is $\left(-\frac{1}{5}, -\frac{1}{5}\right)$

P 79 : (A) isosceles

Sol. Let the given lines be represented by AB, BC and CA, respectively, Then A $(-2, 2)$, B$(2, -2)$ and C$(1, 1)$ are the vertices of a triangle ABC. Also, note that $AB = 4\sqrt{2}$, $AC = \sqrt{10}$ and $BC = \sqrt{10}$. This shows that ΔABC is isosceles; it is clearly not right-angled or equilateral. Since

$$\cos C = \frac{BC^2 + AC^2 - AB^2}{2(BC)(AC)} = \frac{10+10-32}{2\sqrt{10}\sqrt{10}} < 0$$

$\Rightarrow$ ABC is obtuse-angled

P 80 : (A) $\frac{1}{15}$

Sol. Since 3 balls drawn are red

$\therefore$ three hypothesis

(i) E_1: the bag contains 3 red and 2 white balls

E_2 : the bag contains 4 red and 1 white ball

E_3 : the bag contains 5 red balls

Let us assume that $P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}$

further let A be the event that 3 balls drawn are red.

$$P\left(\frac{A}{E_1}\right) = \frac{{}^3C_3}{{}^5C_3} = \frac{1}{10},$$

$$P\left(\frac{A}{E_2}\right) = \frac{{}^4C_3}{{}^5C_3} = \frac{4}{10},$$

$$P\left(\frac{A}{E_3}\right) = \frac{{}^5C_3}{{}^5C_3} = 1$$

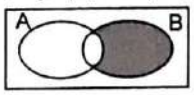
$$\therefore P(A) = \sum P(E_1)\, P\left(\frac{A}{E_1}\right)$$

$$= \frac{1}{3}\cdot\frac{1}{10} + \frac{1}{3}\cdot\frac{4}{10} + \frac{1}{3}\times 1 = \frac{1}{2}$$

$$\text{Now } P\left(\frac{E_1}{A}\right) = \frac{P(E_1)P(A/E_1)}{P(A)} = \frac{\frac{1}{3}\cdot\frac{1}{10}}{\frac{1}{2}} = \frac{1}{15}$$

P 81 : (B) $\frac{13}{21}$

Sol. $P(A \cup B^c) = 1 - [P(B) + P(A \cap B)]$



$$= 1 - \left\{\frac{2K}{4K} + \frac{3K}{21K} + \frac{5K}{21K}\right\} + \left\{\frac{2K}{21K}\right\}$$

$$= 1 - \frac{8}{21} = \frac{13}{21}$$

P 82 : (A) (1, 0, 7)

Sol. Let P(1, 6, 3) be the given point, and let L be the foot of the perpendicular from P to the given line. The coordinates of a general point on the given line are given by

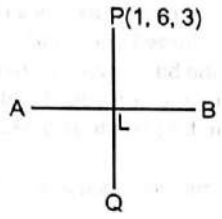
$$\frac{x-0}{1}=\frac{y-1}{2}=\frac{z-2}{3}=\lambda$$

i.e., $x=\lambda,\ y=2\lambda+1,\ z=3\lambda+2$

Let the coordinates of L be

$$(\lambda,\ 2\lambda+1,\ 3\lambda+2) \qquad (1)$$

So, direction ratios of PL are $\lambda-1,\ 2\lambda+1-6,\ 3\lambda+2-3$ i.e. $\lambda-1,\ 2\lambda-5,\ 3\lambda-1$.



Direction ratios of the given line are 1, 2, 3 which is perpendicular to PL.

Therefore, $(\lambda-1).1+(2\lambda-5)2+(3\lambda-1)3=0$

$\Rightarrow 14\lambda-14=0 \Rightarrow \lambda=1$

So, coordinates of L are (1, 3, 5). Let $Q(x_1, y_1, z_1)$ be the image of P(1, 6, 3) in the given line.

Then L is the mid-point of PQ.

Therefore, $\frac{x_1+1}{2}=1, \frac{y_1+6}{2}=3$ and $\frac{z_1+3}{2}=5$

$\Rightarrow x_1=1,\ y_1=0$ and $z_1=7$.

Hence the image of P(1, 6, 3) in the given line is (1, 0, 7).

P 83 : (B) 6 m = 13 n

Sol. $m=3^4-3=78$

$$n=3^4-\left[3\left(2^4-2\right)+3\right]=81-45=36$$

Hence $\frac{m}{n}=\frac{78}{36}=\frac{13}{6} \Rightarrow 6m=13n$

P 84 : (B) 156

Sol. 11

6 (no 4 lies on the same circle)

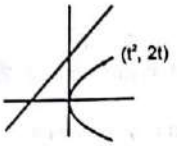
$\Rightarrow {}^5C_2 \cdot {}^6C_1 + {}^6C_2\ {}^5C_1 + {}^6C_3 + 1 = 156$

alternatively ${}^{11}C_3 - {}^5C_3 + 1$

P 85 : (C) $4(y-2)=(x+2)^2$

Sol. $$\frac{x-t^2}{1}=\frac{y-2t}{-1}=-2\,\frac{(t^2-2t+2)}{2}$$

$$\Rightarrow x=2t-2 \quad \Rightarrow t=\frac{x+2}{2}$$



and $y=t^2+2 \quad \Rightarrow y-2=\frac{(x+2)^2}{4}$

$\Rightarrow \quad 4(y-2)=(x+2)^2$

P 86 : (B) $-\frac{1}{2}$

Sol. $$\lim_{n\to\infty}\left[\frac{1}{1-n^2}+\frac{2}{1-n^2}+\ldots.+\frac{n}{1-n^2}\right]$$

$$\lim_{n\to\infty}\frac{\Sigma n}{1-n^2}=\frac{1}{2}\lim_{n\to\infty}\frac{n^2+n}{1-n^2}=-\frac{1}{2}.$$

P 87 : (A) $\left(3, \frac{7}{3}\right)$

Sol. Centroid of

$$\Delta ABC = G\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$$

$$G=\left(\frac{(\text{Sum of abscissa's of vertices})}{3},\right.$$

$$\left.\frac{(\text{Sum of ordinates of vertices})}{3}\right)$$

$$G=\left(\frac{2\frac{(x_1+x_2+x_3)}{2}}{3}, \frac{2\frac{(y_1+y_2+y_3)}{2}}{3}\right)$$

$$G=\left(\frac{\frac{x_1+x_2}{2}+\frac{x_2+x_3}{2}+\frac{x_3+x_1}{2}}{3}\right.$$

$$\left.\frac{\frac{y_1+y_2}{2}+\frac{y_2+y_3}{2}+\frac{y_3+y_1}{2}}{3}\right)$$

The centroid of the triangle coincides with the centroid of the triangle formed by joining the middle points of the triangle. Hence, the coordinates of the centroid of the given triangle are

$$\left(\frac{4+3+2}{3}, \frac{2+3+2}{3}\right) \text{i.e, } \left(3, \frac{7}{3}\right).$$

P 88 : (A) Lie on a straight line

Sol. Let $x_2 = x_1 r$, x_3 and so is $y_2 = y_1 r$, $y_3 = y_1 r_2$ where r is the common ratio. Therefore

$$\Delta = \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \begin{vmatrix} x_1 & y_1 & 1 \\ rx_2 & ry_2 & 1 \\ r^2x_3 & r^2y_3 & 1 \end{vmatrix}$$

$$= r \times r^2 \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$$

Hence the points are collinear.

P 89 : (D) 164

Sol. In order to solve this question you should count the digit 6 appearing in units digit, separately from the instances of the digit 6 appearing in the tens place and appearing in the hundreds place.

When you want to find out the number of times 6 apears in the unit digit, you will have to make a series as follows : 6, 16, 26, 36,....636.

It should be evident to you that the above series has 64 terms because it starts from 06, 16, 26..... and continues till 636. The digit 6 will appear once in the unit digit for each of these 64 numbers.

Next you need to look at how many times the digit 6 appears in the ten's place.

In order to do this we will need to look at instances when 6 appears in the tens place. These will be in 6 different ranges 60s, 160s, 260s, 360s, 460s, 560s and in each of these ranges there are 10 numbers each with exactly one instance of the digit 6 in the tens place, a total of 60 times.

Lastly, we need to look at the number of instances where 6 appears in the hundreds place. For this we need to form the series 600, 601, 602, 603....639. This series will have 40 numbers each with exactly one instance of the digit 6 appearing in the hundreds place.

therefore the required answer would be 64 + 60 + 40 = 164.

P 90 : (A) Statement-1 is True, statement-2 is True ; Statement-2 is a correct explanation for statement-1

Sol. $f'(x) = g(x) \cos x + g'(x) \sin x$

$f'(0) = g(0) = 0$

$$f''(0) = \lim_{x \to 0} \frac{f'(x) - f'(0)}{x}$$

$$= \lim_{x \to 0} \frac{g(x)\cos x + g'(x)\sin x - g(0)}{x}$$

$$= \lim_{x \to 0} \frac{g(x)\cos x - g(0)}{x} + \lim_{x \to 0} \frac{g'(x)\sin x}{x}$$

$$= \lim_{x \to 0} \frac{g(x)\cos x - g(0)}{x}$$

$$= \lim_{x \to 0} \frac{g(x)\cos x - g(0)}{\sin x}$$

$$= \lim_{x \to 0} (g(x) \cot x - g(0) \operatorname{cosec} x)$$

REPLICA 3

P 1 : (B) $R_2 \exp \lambda t_2 = R_1 \exp \lambda t_1$

Sol. $R_1 = \lambda N_0 e^{-\lambda t_1}$

$R_2 = \lambda N_0 e^{-\lambda t_2}$

$R_1 e^{-\lambda t_2} = R_2 e^{-\lambda t_1}$

$R_1 e^{-\lambda t_1} = R_2 e^{-\lambda t_2}$

P 2 : (C) 0.33 sec

Sol. $5\cos\theta_2 \Rightarrow 2$ $\quad 5\cos\theta_1 \Rightarrow 4$

$\cos\theta_2 \Rightarrow 0.4$ $\quad \theta_1 \Rightarrow 37°$

$\theta_2 \Rightarrow 67°$

$$t = \frac{\theta_2 - \theta_1}{2\pi/T} \Rightarrow \frac{\pi/6}{2\pi/T} \Rightarrow \frac{T}{12} \Rightarrow \frac{4}{12} \Rightarrow \frac{1}{3} \text{ sec}$$

P 3 : (C) 1.525 cm

Sol. Given that ; Separation between marks = d; Distance between paper and observer (D) = 50 m ; Aperture of eye lens (a) = 2 mm = 2×10^{-3} m and mean wavelength of light (λ) = 5000 Å = 5000×10^{-10} m. We know that the least distance between the marks to be seen separate,

$$d = \frac{1.22\lambda}{a} \times D = \frac{1.22 \times (5000 \times 10^{-10})}{2 \times 10^{-3}} \times 50$$

$= 15.25 \times 10^{-3}$ m = 1.525 cm

P 4 : (C) both of the above

Sol. Unpolarised light contains electric field vector in two planes. Due to superposition of two polarised lights in different planes, we get unpolarised light hence no interference pattern observed.

P 5 : (B) diffraction is due to interaction of light from same wavefront, whereas the interference is the interaction of two waves derived from the same source

Sol. By defination of diffraction and interference: Diffraction is due to interaction of light from same wavefront, whereas the interference is the interaction of two waves derived from the same source

P 6 : (B) less than n

Sol. Less than n(the frequency will be higher when the source moves towards the stationary observer)

P 7 : (A) decreases

Sol. Here, $v_s = 0$ and v_L is negative, where v_s is velocity of source and v_L is velocity of listener (aeroplane)

P 8 : (A) The initial potential difference across the capacitor is 200 volt.

Sol. $V_0 = I_0 R = 10 \times 10 = 100$ volts (since, I_0 = 10amp from figure)

Also : $I = I_0 e^{-t/RC}$

Taking log ; $\log\left(\frac{I_0}{I}\right) = \frac{t}{RC}$

$$\Rightarrow \quad C = \frac{t}{R\log(I_0/I)}$$

At ; t = 2 sec, $I = 2.5$ A

$$C = \frac{2}{10\log\left(\frac{10}{2.5}\right)}$$

$$C = \frac{2}{10\log 4} = \frac{2}{10 \times 2\log 2} = \frac{1}{10\ell n2}$$

$C = \frac{1}{10\ell n2}$. Heat produced $= \frac{1}{2}CV^2$

$$= \frac{1}{2}\left(\frac{1}{10\ell n2}\right)(100^2) = \frac{500}{\ell n2} \text{ joules.}$$

P 9 : (D) 1.26×10^{-23}

Sol. $M = \pi\, evr^2 = 3.14 \times (1.6 \times 10^{-19}) \times 10^{16} \times (0.5 \times 10^{-10})^2 = 1.26 \times 10^{-23}$ Am^2.

P 10 : (A) 2.2A

Sol. Displacement current and conduction current are equal.

$$\therefore \quad I_D = \frac{E}{Z} = \frac{220}{100} = 2.2\text{A}$$

The conduction current and the displacement current are always equal, i.e., $I_C = I_D$

Like conduction current, the displacement current is also the source of magnetic field.

P 11 : (C) $\vec{E} \times \vec{B}$

Sol. Electric field and magnetic field, both will be perpendicular to each other and also, they will be perpendicular to the propagation of light.

P 12 : (D) $4N_0$, 1 hr

Sol. Total no of decays $\Delta N = N_0 - N(t)$

$= N_0\left(1-e^{-\lambda t}\right) = N_0\left(1-e^{-\ln 2.t/t_{1/2}}\right)$

$= N_0\left(1-2^{-t/t_{1/2}}\right)$

This quantity is maximum for option (D).

P 13 : (C) $\frac{\sqrt{2}+\sqrt{6}}{4}$

Sol. Since the vector u is a unit vector, its x- and y – components are $\cos(45°) = \frac{\sqrt{2}}{2}$ and $\sin(45°) = \frac{\sqrt{2}}{2}$.

So, the component form of the vector u is

$u = (\cos(45°), \sin(45°)) = (\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$.

Similarly, the component form of the vector v is $v = (\cos(60°), \sin(60°)) = (\frac{1}{2}, \frac{\sqrt{3}}{2})$.

cosines of the angle between the vectors u and v

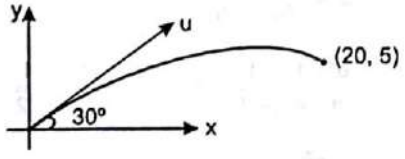
$$\cos(\alpha) = \frac{\left(\frac{\sqrt{2}}{2}\right)\cdot\left(\frac{1}{2}\right)+\left(\frac{\sqrt{2}}{2}\right)\cdot\left(\frac{\sqrt{3}}{2}\right)}{|u|\cdot|v|}$$

$|u| = 1$ and $|v| = 1$, because the vectors u and v are unit vectors,

$$\Rightarrow \quad \cos(15°) = \frac{\sqrt{2}+\sqrt{6}}{4}$$

P 14 : (A) $u = 40\sqrt{\frac{(4+\sqrt{3})}{13\sqrt{3}}}$ m/s

Sol.



$$y = x\tan\theta - \frac{1}{2}\, g\, \frac{x^2}{u^2\cos^2\theta}$$

$$5 = 20\tan 30° - \frac{1}{2}\times\frac{10\times 20^2}{u^2\cos^2 30°}$$

$$\Rightarrow \quad u^2 = \frac{1600}{\sqrt{3}(4-\sqrt{3})} = \frac{1600}{13\sqrt{3}}\,(4+\sqrt{3})$$

P 15 : (B) 4.8×10^{-3} N

Sol. Magnetic force $f = il_{eq}B$

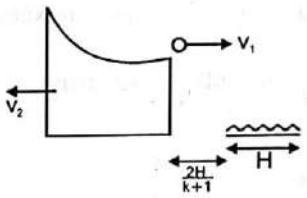
$f = (1.2)(4 \times 10^{-2})(0.1)$

$= 4.8 \times 10^{-3}$ N

P 16 : (D) $0.21 < k < 1.57$

Sol. Let mass of wedge = km

let when they are separating, velocity of particle is v,



$mv_1 = kmv_2 \Rightarrow v_1 = kv_2$

By work energy theorem

$$mgH = \frac{1}{2}mv_1^2 + \frac{1}{2}kmv_2^2$$

$$mgH = \frac{1}{2}mv_1^2 + \frac{1}{2}km\left(\frac{v_1}{k}\right)^2$$

$$2gH = v_1^2\left(\frac{k+1}{k}\right)$$

$$\Rightarrow \quad v_1 = \sqrt{\frac{2gHk}{k+1}}$$

Now range of ball

$$R = \sqrt{\frac{2H}{g}}\sqrt{\frac{2gHk}{k+1}}$$

It would be more than $\frac{2H}{k+1}$ and less than $\frac{2H}{k+1}+H$

$$\Rightarrow \quad \frac{2H}{k+1} \le 2H\sqrt{\frac{k}{k+1}} \le \frac{2H}{(k+1)}(k+3)$$

$$\Rightarrow \quad 1 \le 4k(k+1) \le (k+3)^2$$

$$\Rightarrow \quad \frac{\sqrt{2}-1}{2} < k < \frac{\sqrt{109}-1}{6}$$

$$\Rightarrow \quad 0.21 < k < 1.57$$

$\Rightarrow$ mass of wedge is between 0.21 m and 1.57 m

P 17 : (D) 130 cm

Sol. The only external force on system F = 6N.
So, acceleration of COM will be $a_{cm}=2m/s^2$
So, displacement of COM will be

$$s_{cm} = \frac{1}{2}at^2 = 1\text{meter}$$

$$\text{So, } 100 = \frac{1\times S_1 + 2\times S_2}{1+2} \Rightarrow 300 = S_1 + 170$$

$$\Rightarrow S_1 = 130 \text{ cm}$$

P 18 : (C) 0.3 m/s

Sol. For central Maxima & $\phi = 0$

$$\therefore \quad \Delta x = 0$$

$$(\mu - 1)t + \frac{dy}{D} = 0 \text{ [say thickness = a]}$$

$$y = \frac{(\mu-1)D}{d}(a) = \frac{(2-t)Da}{d}$$

$$\text{as} \quad \frac{dy}{dt} = \frac{aD}{d}$$

$$v_y = \frac{3\times10^{-4}\times1}{1\times10^{-3}} = 0.3 \text{ m/s}$$

P 19 : (B) $1-\frac{1}{\sqrt{2}}$

Sol.

	Q	W	Δu
AB	$nC_v\Delta T$	0	$nC_v\Delta T$
BC	0	$\frac{nR[T_C - T_B]}{1-\gamma}$	$\frac{nR[T_C - T_B]}{1-\gamma}$
CD	$nC_v\Delta T$	0	$nC_v\Delta T$
DA	0	$\frac{nR[T_A - T_D]}{1-\gamma}$	$\frac{nR[T_A - T_D]}{1-\gamma}$

$$n = \frac{W_{net}}{Q_{absorbed}} = \frac{\frac{nR}{1-r}[T_C - T_B + T_A - T_D]}{nC_v[T_B - T_A]}$$

$$\text{as} \quad c_v = \frac{R}{r-1}$$

$$n = \frac{T_B - T_A - T_C + T_D}{T_B - T_A}$$

For process BC and DA

$$T_B V_B^{\gamma-1} = T_C V_C^{\gamma-1} \quad | \quad 2v_B = v_c$$

$$T_D V_D^{\gamma-1} = T_A V_A^{\gamma-1}$$

$$T_C = T_B\left(\frac{V_B}{V_C}\right)^{\gamma-1}$$

$$T_D = T_A\left(\frac{V_A}{V_D}\right)^{\gamma-1}$$

$$n = 1 - \frac{T_B 2^{\gamma-1} + T_A 2^{\gamma-1}}{T_B - T_A}$$

$$n = 1 - 2^{\gamma-1} = 1 - \frac{1}{\sqrt{2}}$$

P 20 : (C) 1.15×10^{-7} Tesla

Sol. We know that, $H = \frac{1}{2}\frac{B_0^2}{\mu_0}C$

$$\frac{40}{4\pi(1)^2} = \frac{B^2\times3\times10^8}{2\times4\pi\times10^{-7}} \quad \left\{I = \frac{\rho}{4\pi r^2}\right\}$$

$$\Rightarrow \quad B_0 = \sqrt{\frac{8}{3}}\times10^{-7} \text{ Tesla}$$

$$\Rightarrow \quad B_{RMS} = \sqrt{\frac{4}{3}}\times10^{-7} \text{ Tesla}$$

$$B_{RMS} = 1.15 \times 10^{-7} \text{ Tesla}$$

P 21 : (D) $\frac{q^2}{a}(1-2\sqrt{2})$

Sol. The electrostatic energy required to compose such a system is given as follows

$$U = \frac{1}{2}\sum_{i\neq j}\frac{q_i q_j}{r_{ij}}$$

where $r_{ij} = |r_i - r_j|$
Inserting data from the figure to eq.(i), we get,

$$U = \frac{q_1q_2}{r_{12}} + \frac{q_1q_3}{r_{13}} + \frac{q_1q_4}{r_{14}} + \frac{q_2q_4}{r_{24}} + \frac{q_2q_3}{r_{23}} + \frac{q_3q_4}{r_{34}}$$

$$\frac{q^2}{a}\left(\frac{1}{2} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right)$$

$$= \frac{q^2}{a}(1-2\sqrt{2})$$

P 22 : (A) $\sqrt{v_0^2 + \frac{2qEy}{m}}$

Sol. Work done by electric field = change in KE

$$qEy = \frac{1}{2}mv^2 + \frac{1}{2}mv_0^2$$

$$\Rightarrow v = \sqrt{v_0^2 + \frac{2qEy}{m}}$$

P 23 : (C) 2160 Hz

Sol. When the air passes through hole and stops by disc it makes one wave pulse so in one revolution 72 wave pulses will be formed. So 72 × 30 wave pulses/sec

So frequency of wave 2160 Hz

P 24 : (A) $300\left(\frac{2\rho-1}{2\rho}\right)^{1/2}$

Sol. Fundamental frequency = $\frac{\upsilon}{2\ell}$ (As both ends of string are fixed)

$$\upsilon = \sqrt{\frac{T}{\mu}}$$

$$\text{So } f = \frac{1}{2\ell}\sqrt{\frac{T}{\mu}}$$

initially $T = \rho vg$

$$\text{so } 300 = \frac{1}{2\ell}\sqrt{\frac{\rho vg}{\mu}} \qquad ..(i)$$

later, $T = \rho vg - F_B = \rho vg - \frac{1}{2}vg$

$$\text{So } f = \frac{1}{2\ell}\sqrt{\frac{\rho vg - \frac{1}{2}vg}{\mu}}$$

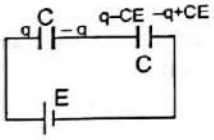
$$\Rightarrow \frac{f}{300} = \sqrt{\frac{(2\rho-1)}{\rho}}$$

$$\Rightarrow f = 300\sqrt{\frac{(2\rho-1)}{\rho}}$$

P 25 : (C) CE^2

Sol. Let q charge flows from battery, so by loop law

$$\frac{q}{c} + \frac{q-CE}{C} = E \Rightarrow q = CE$$



$w_B = CE^2$ (by $w_B = q(EnF)$)

$\Delta U = 0$

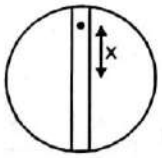
(as one capacitor will have charge CE and other zero)

$H = w_B - \Delta U = CE^2$

P 26 : (A) both particles excute S.H.M with equal time period ($T_1 = T_2$).

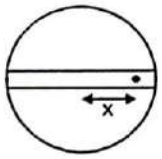
Sol. $F = \frac{GMmx}{R^3}$

$$T_1 = 2\pi\sqrt{\frac{R^3}{GM}}$$



$$F = \frac{GMmx}{R^3} - m\omega^2 x$$

$$T_2 = 2\pi\sqrt{\frac{R^3}{GM - \omega^2 R^3}}$$



$\Rightarrow T_1 < T_2$

P 27 : (C) 2E/3

Sol. It can be treated as combination of three batteries connected parallel between A and B. (E, R) (E, R) and (0, R) are their specification

$$E_{eq} = \frac{R}{3}\left[\frac{E}{R} + \frac{0}{R} + \frac{E}{R}\right] = \frac{2E}{3}$$

P 28 : (C) 1 cm

Sol. Let length of organ pipe is ℓ and end correction is e. So

$$\text{For 1st resonance, } f = \frac{\upsilon}{4(30+e)}$$

$$\text{for 2nd resonance, } f = \frac{3\upsilon}{4(92+e)}$$

$$\Rightarrow \frac{\upsilon}{4(30+e)} = \frac{3\upsilon}{4(92+e)} \Rightarrow e = 1 \text{ cm}$$

P 29 : (A) 12 cm

Sol. Linear magnification, $m = \frac{f}{f-u}$

Given that u = 16 cm

$\therefore \frac{f}{f-16}$

When object is moved towards the lens by 8 cm, then

$$m = \frac{f}{f-16+8} = \frac{f}{f-8}$$

According to question

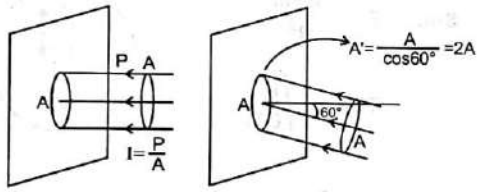
$$\frac{f}{f-16} = \frac{-f}{f-8}$$

or, $f - 8 = -f + 16$ or $2f = 24$

$\therefore$ $f = 12$ cm

P 30 : (C) $I_0/2$

Sol.



$P \rightarrow$ Power incident on surface

New intensity = $\frac{P}{2A} = \frac{I_0}{2}$

P 31 : (A) B_1 is acute angle bisector

Sol. Let d_1 & d_2 be the distance of point (1, 2) from the bisector B_1 & B_2.

$$d_1 = \frac{|3+8-7|}{5} = \frac{4}{5}$$

$$d_2 = \frac{|4-6-14|}{5} = \frac{16}{5}$$

$\because d_1 < d_2$

$\therefore B_1$ is an acute angle bisector

P 32 : (A) $\frac{2\hat{i}+\hat{j}+\hat{k}}{\sqrt{6}}$

Sol. Required unit vector

$$\hat{a} = \frac{\overrightarrow{PQ} \times \overrightarrow{PR}}{|\overrightarrow{PQ} \times \overrightarrow{PR}|}$$

$\overrightarrow{PQ} = \hat{i} + \hat{j} - 3\hat{k}$; $\overrightarrow{PR} = -\hat{i} + 3\hat{j} - \hat{k}$

Therefore, $\overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -3 \\ -1 & 3 & -1 \end{vmatrix}$

$= 8\hat{i} + 4\hat{j} + 4\hat{k}$

Therefore, $|\overrightarrow{PQ} \times \overrightarrow{PR}| = \sqrt{64+16+16}$

$= \sqrt{96} = 4\sqrt{6}$

Therefore, $\hat{n} = \frac{8\hat{i}+4\hat{j}+4\hat{k}}{4\sqrt{6}} = \frac{2\hat{i}+\hat{j}+\hat{k}}{\sqrt{6}}$

P 33 : (B) tan(A – B)

Sol. $2\tan A = 3\tan B \Rightarrow \tan A = \frac{3}{2}\tan B = \frac{3}{2}t$

(Let $\tan B = t$) $\Rightarrow \sin 2B = \frac{2t}{1+t^2}$, $\cos 2B = \frac{1-t^2}{1+t^2}$

Therefore, $\frac{\sin 2B}{5-\cos B} = \frac{\left(\frac{2t}{1+t^2}\right)}{5-\left(\frac{1-t^2}{1+t^2}\right)} = \frac{2t}{4+6t^2} =$

$\frac{t}{2+3t^2} = \tan(A-B)$

P 34 : (A) AP

Sol. $\cos^4\theta\sec^2\alpha$, $\frac{1}{2}$ and $\sin^4\theta\operatorname{cosec}^2\alpha$, are in AP

$1 = \cos^4\theta\sec^2\alpha + \sin^4\theta\operatorname{cosec}^2\alpha$,

$$1 = \frac{\cos^4\theta}{\cos^2\alpha} + \frac{\sin^4\theta}{\sin^2\alpha}$$

$$(\sin^2\theta + \cos^2\theta)^2 = \frac{\cos^4\theta}{\cos^2\alpha} + \frac{\sin^4\theta}{\sin^2\alpha}$$

$$\cos^4\theta\left(\frac{1}{\cos^2\alpha} - 1\right) + \sin^4\theta\left(\frac{1}{\sin^2\alpha} - 1\right) -$$

$2\sin^2\theta + \cos^2\theta = 0$

$\Rightarrow \sin^4\alpha\cos^4\theta + \sin^4\theta\cos^4\alpha -$

$2\sin^2\theta\cos^2\theta\sin^2\alpha\cos^2\alpha = 0$

$\Rightarrow \tan^2\theta = \tan^2\alpha$

$\theta = n\pi \pm a$, $n \in I$

Now, $\cos^8\theta\sec^6\alpha = \cos^8\theta\sec^6\alpha = \cos^2\alpha$

And $\sin^8\theta\operatorname{cosec}^6\alpha = \sin^8\theta\operatorname{cosec}^6\alpha = \sin^2\alpha$

Hence $\cos^8\theta$, $\sec^6\alpha$, $\frac{1}{2}$, $\sin^8\theta\operatorname{cosec}^6\alpha$

i.e. $\cos^2\alpha$, $\frac{1}{2}$, $\sec^6\alpha$, are in AP.

P 35 : (B) $\frac{1}{\sqrt{6}}$

Sol. $SD = \frac{\begin{vmatrix} 2-1 & 4-2 & 5-3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix}}{\sqrt{(15-16)^2+(12-10)^2+(8-9)^2}}$

$= \frac{\begin{vmatrix} 1 & 2 & 2 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix}}{\sqrt{1+1+4}} = \frac{1}{\sqrt{6}}$.

P 36 : (A) $\frac{AB+BC-AC}{2}$

Sol. $\tan\frac{B}{2} = 1 = \frac{r}{s-b} \Rightarrow r = s-b$

$\Rightarrow r = \frac{a+c-b}{2}$

P 37 : (D) $210c^{12}$

Sol. In the expansion of $\left(y+\frac{c^3}{y^2}\right)^{10}$,

the r^{th} term is given by $T_r = {}^{10}C_{r-1}y^{10-r+1}\left(\frac{c^3}{y^2}\right)^{r-1}$

$= {}^{10}C_{r-1}y^{11-r}\frac{c^{3r-3}}{y^{2r-2}}$

$= {}^{10}C_{r-1}y^{11-r-2r+2}c^{3r-3}$

Thus, $T_r = {}^{10}C_{r-1}y^{13-3r}c^{3r-3}$

Let the r^{th} term contains y^{-2}

Therefore, $13-3r = -2 \Rightarrow r = \frac{15}{3} = 5$

Therefore, 5^{th} term will contain y^{-2}

From Equation (1), we get,

$T_5 = {}^{10}C_{5-1}c^{15-3}y^{-2} = {}^{10}C_4c^{12}y^{-2}$

Thus, coefficient of $y^{-2} = {}^{10}C_4c^{12}$

$= \frac{10!c^{12}}{4!(10-4)!} = \frac{10!c^{12}}{4!6!} = \frac{10.9.8.7}{4.3.2.1}.c^{12} = 210c^{12}$

P 38 : (D) none of these

Sol. If $C = 2\cos\theta$

Expanding along with R_1

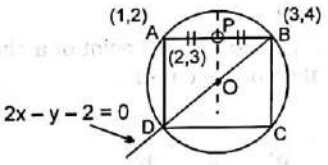
$\Delta = C(C^2-1) - 1(C-6) + 0(1-6C)$

$= C^3 - 2C + 6$

$= 8\cos^3\theta - 4\cos\theta + 6$

P 39 : (B) $\left(\frac{7}{3}, \frac{8}{3}\right)$

Sol. $\because$ OP $\perp$ to given lies.



$\because$ Eq. of OP is $y-3 = -1(x-2)$

$y-3 = -x+2$

$x+y-5 = 0$ ... (1)

$2x-y-2 = 0$ ... (2)

$3x = 7 \Rightarrow x = 7/3$ put in (1)

$\frac{7}{3}+y-5 = 0 \Rightarrow y = 5-\frac{7}{3} = \frac{8}{3}$

$\therefore \left(\frac{7}{3}, \frac{8}{3}\right)$

P 40 : (C) (5, 5), (–3, –1)

Sol. Any circle which touches the line $4x + 3y = 10$ at (1, 2) is

$(x-1)^2 + (y-2)^2 + \lambda(4x+3y-10) = 0$

i.e. $x^2 + y^2 + (4\lambda-2)x + (3\lambda-4)y + 5 - 10\lambda = 0$

its centre is $\left(1-2\lambda, 2-\frac{3\lambda}{2}\right)$ and radius,

$r = \sqrt{(1-2\lambda)^2 + \left(2-\frac{3\lambda}{2}\right)^2 - 5 + 10\lambda} = 5$

$\therefore 1 + 4\lambda^2 - 4\lambda + 4 + \frac{9\lambda^2}{4} - 6\lambda - 5 + 10\lambda = 25$

$\frac{25\lambda^2}{4} = 25 \quad \lambda = \pm 2$

$\therefore$ the centre are (– 3, – 1), (5, 5)

P 41 : (C) 336

Sol. Total = 6 !

n (A) = when A_1A_2 together = 5 ! 2 !

n (B) = when B_1B_2 together = 5 ! 2 !

So n (A $\cap$ B) = 4 ! 2 ! 2 ! = 96

Hence $n\left(\bar{A} \cap \bar{B}\right)$ = 6 ! [2 . 5 ! 2 ! - 96]

= 720 - 384 = 336

P 42 : (A) a hyperbola similar to the given hyperbola if $\frac{h^2}{a^2} > \frac{k^2}{b^2}$

Sol. $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Let (α, β) be the mid point of a chord. Then equation of the chord is

$$\frac{x\alpha}{a^2} - \frac{y\beta}{b^2} = \frac{\alpha^2}{a^2} - \frac{\beta^2}{b^2}$$

$$\frac{2h\alpha}{a^2} - \frac{2k\beta}{b^2} = \frac{\alpha^2}{a^2} - \frac{\beta^2}{b^2}$$

$$\Rightarrow \frac{(\alpha-h)^2}{a^2} - \frac{(\beta-k)^2}{b^2} = \frac{h^2}{a^2} - \frac{k^2}{b^2}$$

P 43 : (B) TFF

Sol. (i) Let angle between asymptotes be 2θ, then

$$\cos\theta = \frac{a}{\sqrt{a^2+b^2}} = \frac{a}{\sqrt{a^2+b^2}} = \frac{1}{e}$$

$$\therefore \theta = \cos^{-1}\left(\frac{1}{e}\right)$$

$$\therefore 2\theta = 2\cos^{-1}\frac{1}{e}$$ Statement is true

(ii) $c = \pm\sqrt{15}$ $\therefore$ false

(iii) director circle does not exist

$\therefore$ false

P 44 : (A) $\frac{1}{2}$

Sol. $\lim\limits_{x\to 0}\frac{\tan x - \sin x}{x^3}$

$$= \lim_{x\to 0}\frac{\sin x - \sin x\cos x}{x^3\cos x}$$

$$= \lim_{x\to 0}\frac{\sin x\left(2\sin^2\frac{x}{2}\right)}{x^3\cos x}$$

$$= \lim_{x\to 0}\left[\frac{\sin x}{x}\cdot\frac{2}{\cos x}\cdot\frac{\sin^2\frac{x}{2}}{\left(\frac{x}{2}\right)^2}\cdot\frac{1}{4}\right] = \frac{1}{2}$$

P 45 : (C) 141

Sol. 0 0 0 0 0 0 $\Rightarrow$ 1 way; 0 0 0 0 1, – 1

$$\Rightarrow \frac{6!}{4!} = 30 \text{ ways;}$$

$$0\,0\,1\,1\,-1\,-1 \Rightarrow \frac{6!}{2!2!2!} = 90 \text{ ways;}$$

$$1\,1\,1\,-1\,-1\,-1 \Rightarrow \frac{6!}{3!3!} = 20 \text{ ways}$$

$\Rightarrow$ Total = 141

P 46 : (A) $(-9, 5) - \left\{-\frac{3}{5}\right\}$

Sol. Lines $3x + by + 4 = 0$ and $5x - y + 2 = 0$ are intersecting if $b \neq -\frac{3}{5}$ ... (i)

Equation of a straight line passing through the intersection of lines
$3x + by + 4 = 0$ and $5x - y + 2 = 0$
is given by
$3x + by + 4 + \lambda(5x - y + 2) = 0$
since it passes through (0, 0)
$\therefore \quad 4 + 2\lambda = 0 \quad \lambda = -2$
$\therefore$ The line is
$3x + by + 4 - 2(5x - y + 2) = 0$
or $-7x + (b + 2)y = 0$

its slope $= \frac{7}{b+2}$

Since this line makes an angle more than 45° with the axis of x.

$$\therefore \left|\frac{\frac{7}{b+2} - 0}{1+0}\right| > 1 \qquad \text{i.e. } b \in (-9, 5) \quad \text{... (ii)}$$

from (i) and (ii) $b \in (-9, 5) - \left\{-\frac{3}{5}\right\}$

P 47 : (A) $\frac{1}{17}$

Sol. There are two hypothesis
E_1 : there are 3 kings
E_2 : there are 4 kings

$$P(E_1) = \frac{1}{13}; \quad P(E_2) = \frac{12}{13}$$

Let A be the event "card drawn is a king"

$$P\left(\frac{A}{E_1}\right) = \frac{3}{51}; \; P\left(\frac{A}{E_2}\right) = \frac{4}{51}$$

$$\therefore P(A) = P(E_1)\,P\left(\frac{A}{E_1}\right) + P(E_2)\,P\left(\frac{A}{E_2}\right)$$

$$= \frac{1}{13} \times \frac{3}{51} + \frac{12}{13} \times \frac{4}{51} = \frac{1}{13}$$

$$\therefore P\left(\frac{E_1}{A}\right) = \frac{P(E_1)P\left(\frac{A}{E_1}\right)}{P(A)}$$

$$= \frac{3}{13 \times 51} \times 13 = \frac{1}{17}$$

P 48 : (A) (– 2, –8)

Sol. Any point on the parabola is
$(\alpha, \alpha^2 + 7\alpha + 2)$
slope of tangent = $2\alpha + 7$
tangent is parallel to the line $y = 3x - 3$
$\therefore 2\alpha + 7 = 3 \quad \therefore \alpha = -2$
$\therefore$ The point is (–2, –8)
(Note that the line does not intersect the parabola)

P 49 : (C) $\log_2 a^2$

Sol. We have,

$$\frac{1}{\log_2 a} + \frac{1}{\log_4 a} + \frac{1}{\log_8 a} + \ldots \frac{1}{\log_{2^n} a} = \frac{n(n+1)}{\lambda}$$

$$\Rightarrow \log_a 2 + \log_a 2^2 + \log_a 2^3 + \ldots + \log_a 2^n = \frac{n(n+1)}{\lambda}$$

$$\Rightarrow \log_a 2 + 2\log_a 2 + 3\log_a 2 + \ldots + n\log_a 2 = \frac{n(n+1)}{\lambda}$$

$$\Rightarrow (1 + 2 + \ldots + n)\log_a 2 = \frac{n(n+1)}{\lambda}$$

$$\Rightarrow \frac{n(n+1)}{2}\log_a 2 = \frac{n(n+1)}{\lambda}$$

$$\Rightarrow \frac{1}{2}\log_a 2 = \frac{1}{\lambda} \Rightarrow \lambda = 2\log_2 a = \log_2 a^2$$

P 50 : (A) $\theta = n\pi \pm \frac{\pi}{3}, \phi = n\pi \pm \frac{\pi}{6}$

Sol. $\left(\frac{\sin\theta}{\sin\phi}\right)^2 = \frac{\tan\theta}{\tan\phi}$

$\Rightarrow \sin\theta\cos\theta = \sin\phi\cos\phi$
$\Rightarrow \sin 2\theta = \sin 2\phi$

$2\theta = \pi - 2\phi \Rightarrow \theta = \frac{\pi}{2} - \phi$

But $\frac{\tan\theta}{\tan\phi} = 3 \Rightarrow \frac{\tan\theta}{\cot\theta} = 3 \Rightarrow \tan^2\theta = 3$

$\Rightarrow \theta = n\pi \pm \frac{\pi}{3}$ so that $\phi = n\pi \pm \frac{\pi}{6}$.

P 51 : (C) $ax^2 - 2bx + c < 0$ for all real x

Sol. Since a, b, c are in HP

Therefore, $b = \frac{2ac}{a+c}$

Also, $D = 4b^2 - 4ac$

Putting $b = \frac{2ac}{a+c}$ and solving, we get,

$$D = 4\left(\frac{2ac}{a+c}\right)^2 - 4ac = -4ac\left(\frac{a-c}{a+c}\right)^2 = -ve$$

Therefore, $ax^2 - 2bx + c < 0$ for all

P 52 : (A) (0, b)

Sol. Equation of tangent at P (a cosθ, b sinθ) is

$$\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$$

$\therefore$ Equation of the lines OA and OB

is $$x^2 + y^2 - a^2\left(\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b}\right)^2 = 0$$

since the lines OA and OB are perpendicular

$$\therefore \quad 1 - \cos^2\theta + 1 - \frac{a^2}{b^2}\sin^2\theta = 0$$

i.e. $$\sin^2\theta + 1 = \frac{a^2}{b^2}\sin^2\theta$$

i.e. $$\frac{b^2}{a^2} = \frac{\sin^2\theta}{1+\sin^2\theta}$$

$$\therefore e^2 = 1 - \frac{b^2}{a^2} = 1 - \frac{\sin^2\theta}{1+\sin^2\theta} = \frac{1}{1+\sin^2\theta}$$

$\therefore$ e is least if $\theta = \pm\frac{\pi}{2}$

$\therefore$ the point P is $(0, \pm b)$

P 53 : (B) (–1, –1)

Sol. Let $P(t^2, 2t)$ and $Q\left(\frac{1}{t^2}, -\frac{2}{t}\right)$

circle with OP as diameter is
$S_1 \equiv x^2 + y^2 - t^2x - 2ty = 0$
circle with OQ as diameter is

$$S_2 \equiv x^2 + y^2 - \frac{1}{t^2}x + \frac{2}{t}y = 0$$

R lies on $S_1 - S_2 = 0$

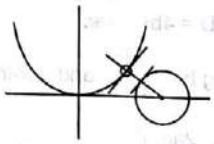
i.e. $$2y = \left(\frac{1}{t} - t\right)x \quad \therefore t - \frac{1}{t} = -1$$

Point of intersection of tangents is $\left(-1,\ t - \frac{1}{t}\right)$

i.e. (–1, –1)

P 54 : (C) (4a, 4a)

Sol. The line of shortest distance will be normal to the parabola and also be normal to the circle.
Let point of contact of normal is $(2at, at^2)$



Equation of normal is $(y - at^2) = -\frac{1}{t}(x - 2at)$

It passes through the centre of circle (12a, 0)

hence $(0 - at^2) = -\frac{1}{t}(12a - 2at)$

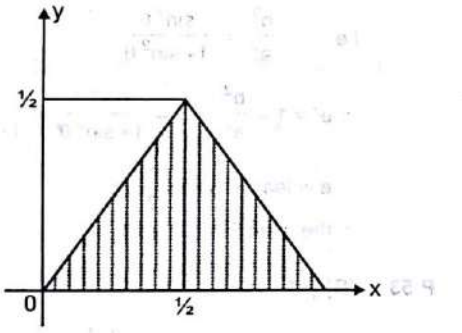
$\Rightarrow \quad t = 2$

$\Rightarrow$ (4a, 4a) is the nearest point.

P 55 : (A) 50

Sol. $\int_{-100}^{100} f(x)dx = \int_{-100}^{100} \min(\{x\},\{-x\})dx$

$= 2\int_{-100}^{100} \min(\{x\},\{-x\})dx$ (by property)



$= 200\int_{0}^{1} \min(\{x\},\{-x\})dx$ (by property)

$= 200 \times \frac{1}{2} \times 1 \times \frac{1}{2} = 50$

P 56 : (C) $\left(\frac{7}{2},\frac{3}{2}\right)$

Sol. As C lies on the line y = x + 3, let the coordinates of C be $(\lambda, \lambda+3)$.
Also A ≡ (2,1), B ≡ (3,–2), then the area of ΔABC is given by :

$$\frac{1}{2}\begin{vmatrix} 2 & 1 & 1 \\ 3 & -2 & 1 \\ \lambda & \lambda+3 & 1 \end{vmatrix} = 5$$

or $|4\lambda - 4| = 10$
i.e., $4\lambda - 4 = 10$ or $4\lambda - 4 = -10$
i.e., $4\lambda = 14$ or $4\lambda = -6$

i.e., $\lambda = \frac{7}{2}, \lambda = \frac{3}{2}$.

Hence the coordinates of C are (7/2, 13/2) or (–3/2, 3/2)

P 57 : (B) 5

Sol. We have
$\sin^{-1}(\sin 12) + \cos^{-1}(\cos 12)$
$= \sin^{-1}(\sin(4\pi - (4\pi - 12)) +$
$\cos^{-1}(\cos(4\pi - (4\pi - 12))$.
$= -(4\pi - 12) + (4\pi - 12) = 0$
so that, $(n-2)x^2 + 8x + n + 4 > 0 \ \forall \ x \in R$
$\Rightarrow n - 2 > 0 \Rightarrow n \geq 3$
and $8^2 - 4(n-2)(n+4) < 0$ or $n^2 + 2n - 24 > 0$
$\Rightarrow n > 4$,
$\Rightarrow n \geq 5 \Rightarrow n_{min} = 5$

P 58 : (A) 70821

Sol. In order to solve this question you need to visualise the series of numbers, which satisfy this condition of the remainder 3 when divided by 7.
the first number in 3 digits for this condition is 101 and the next will be 108, followed by 115, 122....
This series would have its highest value in 3 digits as 997.
The number of terms in this series would be 129 (using the logic that for any AP, the number of terms is given by $\frac{D}{d}+1$)
Also the average value of this series is the average of the first and the last term i.e., the average of 101 and 997. Hence the required sum = 549 × 129 = 70821.

P 59 : (C) $\frac{\vec{B}\times\vec{A}+c\vec{A}}{(\vec{A}\cdot\vec{A})}$

Sol. $\vec{A}\times\vec{X} = \vec{B}$

$\Rightarrow (\vec{A}\times\vec{X})\times\vec{A} = \vec{B}\times\vec{A}$

$\Rightarrow (\vec{A}\times\vec{A})\vec{X} - (\vec{X}\cdot\vec{A})\vec{A} = \vec{B}\times\vec{A}$

$\Rightarrow (\vec{A}\cdot\vec{A})\vec{X} - c\vec{A} = \vec{B}\times\vec{A}$

$\Rightarrow \vec{X} = \frac{\vec{B}\times\vec{A}+c\vec{A}}{(\vec{A}\cdot\vec{A})}$

P 60 : (A) $-4\left\{1+\frac{1}{9}+\frac{1}{25}+....+\frac{1}{(2N-1)^2}\right\}$

Sol. $g'(x)=\frac{f'(x)}{f(x)}$

and $g''(x)=\frac{f(x)\,f''(x)-(f'(x))^2}{[f(x)]^2}$

$f'(x+1)=f(x)+x\,f'(x)$

$f''(x+1)=2f'(x)+x\,f''(x)$

$g''(x+1)$

$=\frac{x\,f(x)[2f'(x)+xf''(x)]-[f(x)+xf'(x)]^2}{x^2f(x)^2}$

$\Rightarrow \quad g''(x+1)$

$=\frac{2x\,f(x)\,f'(x)+x^2f(x)\,f''(x)-f^2(x)-x^2(f'(x))^2-2x\,f(x)f'(x)}{x^2f^2(x)}$

$\Rightarrow \quad g''(x+1)=\frac{f(x)\,f''(x)-(f'(x))^2}{f^2(x)}-\frac{1}{x^2}$

$\therefore \quad g''(x+1)-g''(x)=-\frac{1}{x^2}$

$\Rightarrow \quad g''\left(N+\frac{1}{2}\right)-g''\left(N-\frac{1}{2}\right)=\frac{-1}{\left(N-\frac{1}{2}\right)^2}$

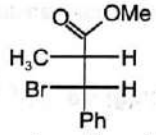
$g''\left(N-\frac{1}{2}\right)-g''\left(N-\frac{3}{2}\right)=\frac{-1}{\left(N-\frac{3}{2}\right)^2}$ ⋮

$g''\left(\frac{3}{2}\right)-g''\left(\frac{1}{2}\right)=\frac{-1}{1/4}$

On adding,

$g''\left(N+\frac{1}{2}\right)-g''\left(\frac{1}{2}\right)=-4\left(1+\frac{1}{9}+\frac{1}{25}+.....+\frac{1}{(2N-1)^2}\right)$

P 61 : (A)



Sol. Inversion of configuration takes place at β-carbon.

P 62 : (D) Only 4

Sol. Energy of electrons in an atom is quantised.

P 63 : (D) 4

Sol. $NO + NO_3 \ddagger \uparrow \uparrow 2NO_2$

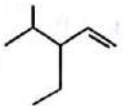
$1-\frac{x}{2} \quad 3-\frac{x}{2} \quad x$

$3-x \quad 3-x \quad 2x$

$K=\frac{x^2}{\left(1-\frac{x}{2}\right)\left(3-\frac{x}{2}\right)} \Rightarrow K=\frac{4x^2}{(3-x)^2}$

Equating the two gives $x=\frac{3}{2}$, then K = 4

P 64 : (D)



Sol. With bulky base, hofmann's elimination occur to give the major elimination product.

P 65 : (C) 0.094 bar

Sol. $\alpha=\frac{i-1}{n-1}$

$0.9=\frac{i-1}{2-1}$; i = 1.9

$\pi = iCRT$

$= 1.9 \times 0.002 \times 0.082 \times 300$

$= 0.094$ bar

P 66 : (C) x = 3, y = 1

Sol. $\frac{1.9}{12+x+80y}(1+x+y)=\frac{0.6}{18}\times 3$ ...(i)

$x+y=4$ ...(ii)

Solving (i) and (ii) gives x = 3, y = 1.

P 67 : (B) 25

Sol. For 200 ml, gold sol starch required = 0.50 g

For 10 ml, gold sol starch required = $\frac{0.5\times 10}{200}$

= 0.025 g = 25 mg

Gold number = 25

P 68 : (D) 9

Sol. Number of moles of Mg

$=\frac{12\times 10^{-3}}{24}=0.5\times 10^{-3}$

Number of Mg atoms

$= 0.5 \times 10^{-3} \times 6 \times 10^{23} = 3 \times 10^{20}$ atoms

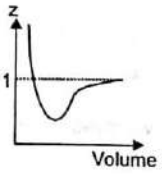
In HCP unit cell z = 6 & number of tetrahedral void = 12 & number of octahedral void = 6

Total void = 18

Total number of voids

$=\frac{3\times 10^{20}}{6}\times 18 = 9\times 10^{20}$

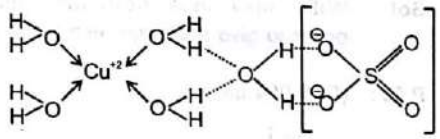
P 69 : (B)



P 70 : (A) $Ni(CO_4)_4$, $[Fe(CN)_6]^{4-}$

P 71 : (D) I, II, III

Sol.



P 72 : (B) 3 1 4 2

P 73 : (D) 210.7 g/mol

Sol. $-\frac{dN}{dt} = kN = 2.9 \times 104$ dpm

$\Rightarrow N = \frac{2.9\times10^4}{k}$

$= \frac{2.9\times10^4\times1.3\times10^9\times364\times24\times60}{\ln 2}$

$= 2.85 \times 10^{10}$

$\Rightarrow$ Molar mass $= \frac{0.01}{2.85\times10^{19}} \times 6.023 \times 10^{23}$

$= 210.7$ g/mol

P 74 : (B) $Na^+ > Li^+ > Mg^{2+} > Be^{2+}$

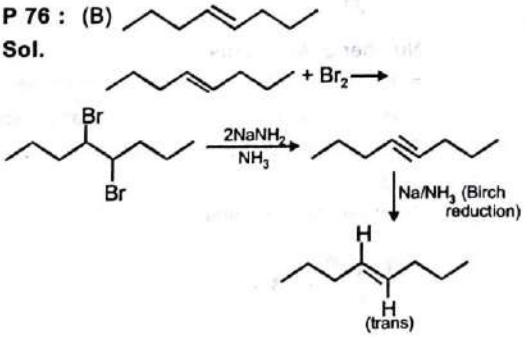
Sol. Li and Na are first and second members of alkali group, therefore, $Na^+ > Li^+$.

P 75 : (D) Under the kinetically controlled conditions, A and B can easily interconvert into each other.

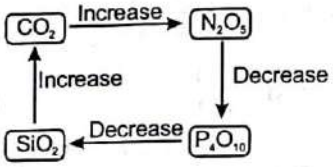
Sol. A is kinetically controlled product and B is thermodynamically controlled.

P 76 : (B)

Sol.

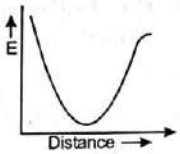


P 77 : (A)



Sol. Acidic strength = $N_2O_5 > P_4O_{10}$
Acidic strength = $CO_2 > SiO_2$
Acidic strength = $N_2O_5 > CO_2$

P 78 : (A)



Sol. As two H atoms approach each other, there is stable bond formation where P.E. is minimum as shown in (a).

P 79 : (D) decreases, increases

Sol.

	Electrons	Bond order	
N_2	14	3.0	
N_2^+	13	2.5	decreases
O_2	16	2.0	
O_2^+	15	2.5	increases

P 80 : (C) CaO

Sol.
$Ca_2N_2 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2H_3\uparrow$
$CaC_2 + 2H_2O \longrightarrow Ca(OH)_2 + H-C\equiv C-H$
$Ca_3P_2 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2PH_3\uparrow$
$CaO + H_2O \longrightarrow Ca(OH)_2$

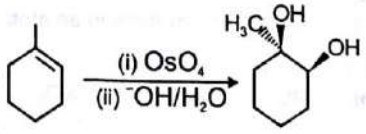
P 81 : (A) $Cl^-(aq) \rightarrow \frac{1}{2}Cl_2(g) + e^-$; $E^\ominus_{cell} = 1.36$ V

Sol. Brine is electrolysed by using inert electrodes. The possible reactions occurring at anode are

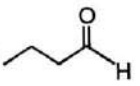
$Cl^-(aq.) \rightarrow \frac{1}{2}Cl_2(g) = e^-$; $E^\ominus_{cell} = 1.36$ V

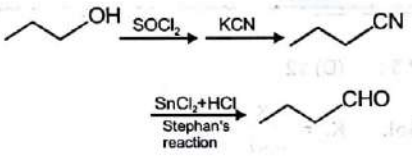
$2H_2O(l) \rightarrow O_2(g) + 4H^+ + 4e^-$; $E^\ominus_{cell} = 1.23$V

P 82 : (C)



Sol. Vicinal diol is formed with OsO_4 by syn addition.

P 83 : (C) 

Sol. 

P 84 : (B) 14.18 min

Sol. Volume of Ag required = 5×1×0.1×2
=1.00 cm^3

Mass of Ag required = $\frac{1}{10.5}$ = 0.0952 g

Gram-equivalent of Ag = $\frac{0.0952}{108}$

$\Rightarrow t = \frac{Q}{I} = \frac{0.0952 \times 96500}{108 \times 0.10 \times 60} = 14.18$ min

P 85 : (A) 5 Amp

Sol. $q = ms\Delta T = 100 \times 4.18 \times 1 = 418 = QV$

$\Rightarrow Q = \frac{418}{20.9} = 20C = x \times 4$

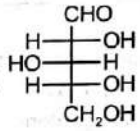
$\Rightarrow x = 5$ Amp

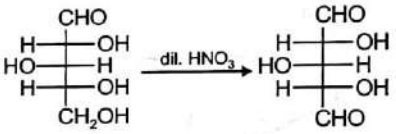
P 86 : (B) SO_2 gas is passed into $Cr_2O_7^{2-}/H^+$

Sol. $\underset{+6}{Cr_2}O_7^{2-} + \underset{+4}{SO_2} \longrightarrow \underset{+6}{SO_4^{2-}} + \underset{+3}{Cr^{3+}}$

P 87 : (D)

Sol. Phenol is activated and chlorobenzene is deactivating in S_EAr reaction.

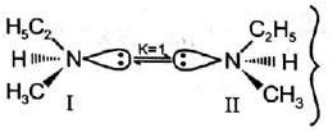
P 88 : (C) 

Sol. 

P 89 : (B) 1 > 3 > 2

Sol. 1 has most acidic hydrogen as the conjugate base will be stabilized by –R effect of $-NO_2$. 3 has more acidic hydrogen (ketone) than α – H of ester 2.

P 90 : (C) The nitrogen atom rapidly inverts its configuration leading to a racemic mixture.

Sol. 

Rapid interconversion between I and II leads to recemic mixture.

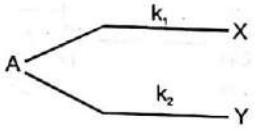
REPLICA 4

P 1 : (A) Goldstein discovered neutrons in his experiments on "canal rays."

Sol. Goldstein discovered proton.

P 2 : (D) 9

Sol.



$$\frac{n(Y)}{n(X)} = \frac{10^{10}}{2.5\times10^{9}} = 4 \Rightarrow n(Y) = 4n(X)$$

$\because n(Y) = 70 \Rightarrow n(X) = 17.5 ; n(X + Y) = 87.5$

Also, overall decay constant

$$(k) = k_1 + k_2 = \ln 2\left(\frac{1}{10} + \frac{1}{2.5\times10^{9}}\right) = \frac{\ln 2}{2\times10^{9}}$$

$$\left(\frac{\ln 2}{2\times10^{9}}\right)t = \ln\frac{100}{100-87.5} = \ln 8 = 3\ln 2$$

$\Rightarrow t = 6 \times 10^9$ yr

P 3 : (B) 3 g

Sol. $(\Delta p)_{glucose} = (\Delta p)_{urea}$

$(x_B)_{glucose} = (x_B)_{urea}$

$$\text{i.e., } \left(\frac{n_B}{n_A}\right)_{glucose} = \left(\frac{n_B}{n_A}\right)_{urea}$$

$$\frac{w_B}{50}\times\frac{18}{180} = \frac{1\times18}{50\times60}$$

$w_B = 3$ g

P 4 : (A) 14.26 hour

Sol. $C_6H_5OH + Br_2 \rightarrow C_6H_4Br(OH) + HBr$

Moles of Br_2 required

$$= \frac{500\times1000}{94} = 5319.15 \text{ mole}$$

$\Rightarrow$ g equivalent of $Br_2 = 5319.15 \times 2$

$$\Rightarrow t = \frac{Q}{I} = \frac{5319.15\times2\times96500}{20000} = 51329.8 \text{ s}$$

$= 14.26$ hr

P 5 : (D) 12

Sol. $K_C = \dfrac{x}{2(2)^2} = \dfrac{1}{2}$

$x = 4$ mol AB_2 present.

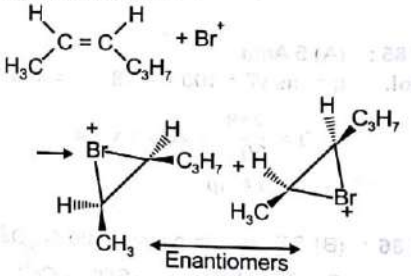
Now let y mol of AB_2 is added.

$$K_C = \frac{1}{2} = \frac{4+y}{2(2+2)^2}$$

$\Rightarrow y = 12$

P 6 : (C) The enantiomeric bromonium ions are formed in equal amounts.

Sol. Enantiomeric bromonium ions are formed in equal amounts.



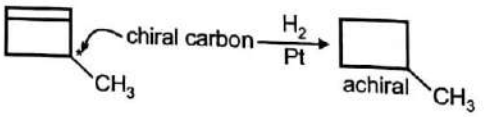
P 7 : (B) 6

Sol. $\Delta H = \Delta U + \Delta(pV)$

$= 14 + (2 - 10) = 4$ L atm.

P 8 : (B) 3-methylcyclobutene

Sol.



P 9 : (B) Li^+ (aq.) > Na^+ (aq.) > K^+ (aq.) > Rb^+(aq.) > Cs^+ (aq.)

Sol. Hydrated radii $\propto \dfrac{\text{charge}}{\text{ionic radii}}$

$Li^+_{(aq)} > Na^+_{(aq)} > Rb^+_{(aq)} > Cs^+_{(aq)}$ Hydrated radii

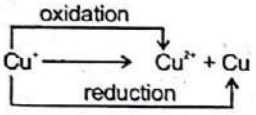
P 10 : (A) $B(OH)_3$ is a Lewis acid

Sol. $B(OH)_3 + H_2O \longrightarrow [B(OH)_4]^- + H^+$

$\bar{e}$ acceptance $\longrightarrow$ Lewis acid

P 11 : (A) (i)

Sol. The reaction in which oxidation as well as reduction occur upon same atom simultaneously is known as disproportionation reaction.

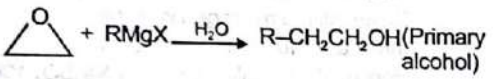


P 12 : (D) $C_2O_4^{2-} > CO > CO^+$

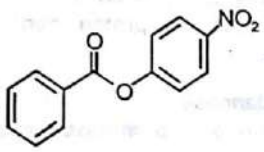
Sol. Bond order of C–O in $C_2O_4^{-2}$ is $\Rightarrow$ 1.5
Bond order of C–O in CO is $\Rightarrow$ 3
Bond order of C–O in CO^+ is$\Rightarrow$more than 3

P 13 : (A) a primary alcohol

Sol.



P 14 : (D)



Sol. Nitration occurs in phenoxy ring at para position mainly, due to steric reason and due to +M effect of oxygen atom of ester. .

P 15 : (C) Boiling occur's

Sol. (A) At critical point, $d_L = d_{vapour}$
(B) Density of vapours increase due to increase in amount of vapours.
(C) Boiling does not occur
(D) Density of liquid decreases due to thermal expansion of liquids

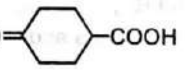
P 16 : (C) During polymerization, a linear, unbranched, crystalline polymer is usually obtained.

Sol. Chain branching always occur during olefinic polymerization due to abstraction of some H-radical from the growing polymer chain.

P 17 : (A) $3d^1_{xy}, 3d^1_{yz}, 3d^1_{zx}$

Sol. Magnetic moment indicates that there are three unpaired electrons present in chromium. These must be present in lower energy orbitals which are $3d^1_{xy}, 3d^1_{yz}, 3d^1_{zx}$

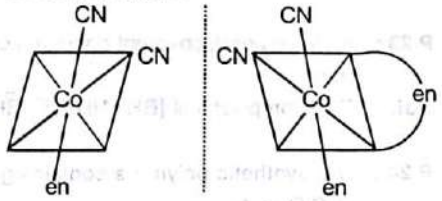
P 18 : (D)



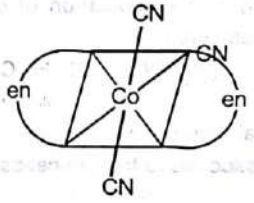
Sol. The primary alcohol group will be oxidised to acid while secondary alcohol will be oxidised to ketone.

P 19 : (A) Trans dicyano bis(ethylenediamine) cobalt (III) nitrate

Sol.



Enantiomorphic pair
d and l cis $[Co(CN)_2(en)_2]NO_3$



trans $[CO(CN)_2(en)_2]NO_3$
Optically inactive

P 20 : (B) Mercuric chloride, $HgCl_2$

Sol. Chromyl chloride test is used for the conformation of chloride ion. Formation of red vapour, yellow solution, yellow ppt.
[Chloride of Sb^{+3}, Sn^{+4}, Ag^{+1}, Pb^{+2}, Hg^{+2}, Hg^+, Sb^{+5} does not give chromyl chloride test bebause these does not dissolve significantly in water.]

Chromyl test :

- $KCl + H_2SO_4 \xrightarrow{\Delta} KHSO_4 + HCl$
- $K_2Cr_2O_7 + 2H_2SO_4 \xrightarrow[(conc.)]{} 2KHSO_4 + CrO_3 + H_2O$

  $2HCl + CrO_3 \longrightarrow \underset{\text{Chromyl chloride}}{CrO_2Cl_2} \uparrow + H_2O$
- $CrO_2Cl_2 + 4NaOH \longrightarrow \underset{\text{yellow}}{Na_2CrO_4} + 2NaCl + 2H_2O$
- $Na_2CrO_4 + Pb(CH_3COO)_2 \rightarrow \underset{\text{yellow}}{PbCrO_4} \downarrow + 2CH_3COONa$

Note : This test is carried out in dry otherwise CrO_2Cl_2 will dissolve in water and form H_2CrO_4 (Chromic acid).

$$CrO_2Cl_2 + H_2O \rightarrow \underset{\text{yellow}}{H_2CrO_4} + HCl$$

P 21 : (C) XeF_4 and HCl_4^-

Sol. Both have 4 σ bonds and two lone pair.

P 22 : (D) H_2O_2 gets decomposed during chemical reaction

Sol. H_2O_2 reacts with the compounds dissolved in it due to its oxidation, reduction, acidic and forming addition compound properties.

P 23 : (D) Ionic bond, covalent bond and coordinate bond

Sol. The compound is $[BH_2(NH_3)_2]^+ [BH_4]^-$

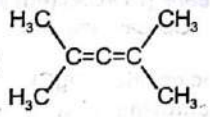
P 24 : (A) Synthetic polymers containing repeated R_2SiO units.

P 25 : (C) Self dissociation of sodium palmitate molecules

Sol. $C_{15}H_{31}COONa \rightarrow \underset{\text{Palmitation}}{C_{15}H_{31}COO^-} + Na^+$

Palmitate ions being of colloidal sizes also associate to form micelles.

P 26 : (D)



Sol. Cumulated diene is least stable, evolve maximum heat in the hydrogenation reaction.

P 27 : (C) 1.736×10^{24} atoms

Sol. Density = $\frac{z \times M}{N \times a^3}$ gram/cm^3

Number of atoms in unit cell (z) = 1 + 2 = 3,
M = ?, N = 6.02×10^{23}, $a^3 = 24 \times 10^{-24}$ cm^3,
d = 7.2 g/cm^3

$$7.2 = \frac{3 \times M}{6.02 \times 10^{23} \times 24 \times 10^{-24}} \text{ or } 34.69$$

(atomic mass)

34.69 gram of element contains atoms = 6.02×10^{23}

100 gram of element contains atoms

$$= \frac{6.02 \times 10^{23} \times 100}{34.69} = 1.736 \times 10^{24} \text{ atoms}$$

P 28 : (A) $Na_2SO_4.10H_2O$

Sol.

Element	%	Atomic ratio	Least ratio	Empirical formula
Na	14.31	$\frac{14.31}{23} = 0.622$	$\frac{0.622}{0.311} = 2$	$Na_2SH_{20}O_{14}$
S	9.97	$\frac{9.97}{32} = 0.311$	$\frac{0.311}{0.311} = 20$	
H	6.22	$\frac{6.22}{1} = 6.22$	$\frac{6.22}{0.311} = 20$	
O	69.50	$\frac{69.50}{16} = 4.34$	$\frac{4.34}{0.311} = 14$	

Empirical formula mass = 46 + 32 + 20 + 224 = 322

Also, molar mass = 322 (given)

$$n = \frac{\text{Molecular weight}}{\text{Empirical formula weight}} = \frac{322}{322} = 1$$

Hence, molecular formula = $Na_2SH_{20}O_{14}$

Since, all the hydrogen is in the form of water, there are thus $10H_2O$ molecules.

Hence, of molecular formula is $Na_2SO_4.10H_2O$

P 29 : (B) His acting as base

Sol. His accepting proton, hence acting as a base.

P 30 : (B) Mannose

Sol. Mannose is a monosaccharide, a C-2 epimer of glucose.

P 31 : (D) none of the above

Sol. $f_1 = \frac{(2n+1)}{4\ell} = \frac{85}{2}\text{Hz}; \frac{255}{2}\text{Hz}; \frac{425}{2}\text{Hz}.....$

$f_2 = \frac{nv}{2\ell}$ = 85 Hz, 170 Hz, 255 Hz, 340 Hz...

in given option, no frequency is common in both options.

P 32 : (B) 5 cm

Sol. The radius of curvature of silvered face is given by :

$$\frac{1}{f} = (\mu - 1) \text{ or } \left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\text{or } \frac{1}{10} = (1.5 - 1)\left(\frac{2}{R}\right) \text{ or } R = 10 \text{ cm}$$

P 33 : (A) 6×10^{-6} m

Sol. 11th bright fringe from Q represents 10th fringe from O.

Phase difference, 10th bright fringe = 20π

$$\Rightarrow \frac{2\pi}{\lambda}(\Delta x) = 20\pi \Rightarrow \Delta x = 10\lambda$$

= 60000 Å = 6×10^{-6} m

$S_1B = 6 \times 10^{-6}$ m

P 34 : (C) all images C, B and A will shift towards O

Sol. Position of first maxima is $\sin\theta = \frac{\lambda}{a}$

λ decreases its angular position decreases therefore pattern shifts towards 'O'

P 35 : (B) varies periodically with time

Sol. In linearly polarised light electric field varies sinusoidally.

P 36 : (B) cn/(c – 2v)

Sol. Here, we can assume that both the source and observer are moving towards each other with velocity v. If c be the velocity of signal, then

$$n' = \left(\frac{c+v}{c-v}\right)n = \frac{c^2-v^2}{(c-v)^2}\times n$$

$$= \frac{c^2-v^2}{c^2-v^2-2cv}\times n \;\square\; \frac{c^2}{c^2-2cv}\times n = \frac{c}{c-2v}\times n$$

P 37 : (B) 5 : 4

Sol. When an observer is approaching the stationary source,

$$v_{observed} = v_{emitted}\,\frac{(v_{sound}+v_{observer})}{v_{sound}}$$

$$\therefore \quad \frac{v_{observer}}{v_{emitted}} = \frac{v+v/4}{v} = \frac{5}{4}$$

or $\quad v_{observed} = v_{emitted} = 5 : 4$

P 38 : (C) $\frac{v}{3}$

Sol. If v_s be the velocity of car, then frequency of sound striking the cliff

$$n' = \frac{v\times n}{v-v_s}$$

The frequency of sound heard on reflection (listener moving towards source),

$$n'' = \frac{(v+v_s)n'}{v} = \frac{(v+v_s)}{v}\times\frac{v\times n}{(v-v_s)}$$

or $\quad \frac{n''}{n} = \frac{(v+v_s)}{(v-v_s)} = 2$

or $\quad v + v_s = 2v - 2v_s$

$\therefore \quad 3v_s = v$

or $\quad v_s = v/3$

P 39 : (B) $\sqrt{\frac{a}{2b}}$

Sol. At origin $a = a_c = v^2/r$ where r is radius of curvature given by :

$$r = \frac{\left(1+\left(\frac{dy}{dx}\right)^2\right)^{3/2}}{\frac{d^2y}{dx^2}} = \frac{(1+2bx)^{1/2}}{2b} = 1/2b$$

at x = 0

P 40 : (A) 13.86 sec

Sol. The growth of voltage in a C-R circuit is given by $V = V_0[1 - e^{-t/RC}]$

For the given circuit, as per condition given :

$$V_C(t) = \frac{3}{4}V_0$$

$$\Rightarrow \frac{3}{4}V_0 = V_0[1 - e^{-t/RC}]$$

$\Rightarrow e^{-t/RC} = 1/4 \Rightarrow t = 2RC \log_e 2$

Solving, we get $t = 13.86$ sec

P 41 : (D) All of the above

Sol. Magnetic lines due to any current carrying line segment are circular.

P 42 : (D) 55.6nT

Sol. $\frac{\mu_0\varepsilon_0 r}{2}\frac{dE}{dt} = \frac{1}{2\times 9\times 10^{16}}\times 10^{10} = 5.56\times 10^{-8}\,T$

P 43 : (C) an alpha particle

Sol. The mass number of x must be 238 – 234 = 4 and atomic number of x must be 92 – 90 = 2.

P 44 : (C) 60°

Sol. Given $(\vec{a}+3\vec{b})\cdot(7\vec{a}-5\vec{b}) = 0$

$$\Rightarrow 7a^2 - 15b^2 + 16\,\vec{a}\cdot\vec{b} = 0 \qquad ...(1)$$

Also, $\quad (\vec{a}-4\vec{b})\cdot(7\vec{a}-2\vec{b}) = 0$

$$\Rightarrow 7a^2 + 8b^2 - 30\,\vec{a}\cdot\vec{b} = 0 \qquad ...(2)$$

Subtracting, $\quad -23b^2 + 46\,\vec{a}\cdot\vec{b} = 0$

$$\Rightarrow \vec{a}\cdot\vec{b} = \frac{b^2}{2}$$

Putting this in (1),

$7a^2 - 7b^2 = \Rightarrow |\vec{a}| = |\vec{b}|$.

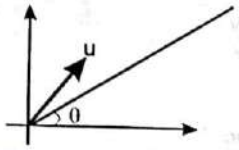
Thus $\vec{a}\cdot\vec{b} = ab\cos\theta$

$$\Rightarrow \frac{b^2}{2} = b^2\cos\theta \Rightarrow \cos\theta = \frac{1}{2} \text{ or } \theta = 60°.$$

P 45 : (A) the amplitude of a vector $(3\hat{i}+4\hat{j})$ is 5

P 46 : (B) $\frac{u^2}{g(1+\sin\theta)}$

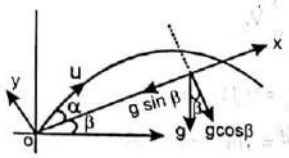
Sol.



Range max = ?

Let $\theta = \beta$

And angle of projection from the inclined plane $= \alpha$



$u_y = u \sin\alpha$ $\quad u_x = \cos\alpha$

$a_x = -g \sin\beta$ $\quad a_y = -g \cos\beta$

Range $= s_x = u_x T + \frac{1}{2} a_x T^2$

(on the inclined plane) where $T = \frac{2u_y}{g_y}$

$$\Rightarrow s_x = (u\cos\alpha)\left[\frac{2u\sin\alpha}{g\cos\beta}\right] + \frac{1}{2}[-g\sin\beta]\left[\frac{2u\sin\alpha}{2\cos\beta}\right]^2$$

$$= \frac{2u^2\sin\alpha}{g\cos^2\beta}[\cos\alpha\cos\beta - \sin\alpha\sin\beta]$$

$$s_x = \frac{2u^2\sin\alpha}{g\cos^2\beta}[\cos(\alpha+\beta)]$$

$$= \frac{u^2}{g\cos^2\beta}[2\sin\alpha\cos(\alpha+\beta)]$$

$$= \frac{u^2}{g\cos^2\beta}[\sin(2\alpha+\beta)+\sin(-\beta)]$$

$$s_x = \frac{u^2}{g\cos^2\beta}[\sin(2\alpha+\beta)-\sin\beta]$$

Now s_x is max

when $\sin(2\alpha+\beta)$ is max

($\because \beta =$ constt.)

$$\Rightarrow \quad 2\alpha+\beta = \frac{\pi}{2} \quad \Rightarrow \quad \alpha = \frac{\left(\frac{\pi}{2}-\beta\right)}{2}$$

i.e., when ball is projected at the angle bisector of angle formed by inclined plane and direction of net acceleration reversed.

$$\& \ (s_x)_{max} = \frac{u^2}{g\cos^2\beta}[1-\sin\beta]$$

$$= \frac{u^2(1-\sin\beta)}{g(1-\sin\beta)(1-\sin\beta)}$$

Max. Range on an inclined plane $= \frac{u^2}{g(1+\sin\beta)}$

Here $\beta = \theta \quad \Rightarrow \quad R_{max} = \frac{u^2}{g(1+\sin\theta)}$

P 47 : (D) They have high degree of parallelism.

Sol. By definition, they have high degree of parallelism.

P 48 : (B) $\frac{1}{\sqrt{2}}$

Sol. Intensity reduces to half of its value and

$I_0 \propto E^2$

$E = \frac{E_0}{\sqrt{2}}$

P 49 : (C) 3m

Sol. Distance travelled by A before coming to rest is $x_A = 8$m, towards right. Distance travelled by B before coming to rest is $x_B = 2$m, towards left.

$\therefore$ Distance moved by centre of mass is,

$$x = \frac{1\times 8 - 1\times 2}{1+1} = 3\text{m}$$

P 50 : (A) 0.6 cal/gm-°C

Sol. Let V_{cc} be the volume of either liquid.

Thermal capacity of each calorimeter is V cal per °C.

Mass of water $V \times 1 = V$ g

Mass of alcohol $= V \times 0.8 = 0.8V$ g

Rate of cooling of water,

$$= \frac{1}{100}[V(50-40) + V\times 1(50-40)]$$

$$= \frac{V}{5} \text{ cal/sec}$$

Rate of cooling of alcohol

$$= \frac{1}{74}[V(50-40) + s\times 0.8V\times 10]$$

$$= \frac{1}{74}[10V + 8V\times s]$$

Now, $\frac{1}{74}[10V + 8V\times s] = \frac{V}{5}$

$\therefore$ s = 0.6 cal/g-°C.

P 51 : (B) $\frac{2}{7}$

Sol. It is about to slip as given in question so

$$\frac{2mg}{3}\sin 37° - \frac{2\mu mg}{3}\cos 37°$$

$$= \frac{mg}{3}\sin 53° + \frac{\mu mg}{3}\cos 53°$$

$$\Rightarrow \frac{6}{5} - \frac{4\mu}{5} = \frac{4}{5} + \frac{3\mu}{5}$$

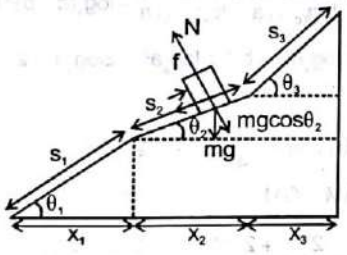
$$\Rightarrow \frac{2}{5} = \frac{7\mu}{5} \Rightarrow \mu = \frac{2}{7}$$

P 52 : (C) intensity of central maxima will decrease

Sol. As D increases, fringe width $\left(\beta = \frac{\lambda D}{d}\right)$ also increases so intensity of central maxima decreases.

P 53 : (A) $\mu\, mg(x_1+x_2+x_3)$, always

Sol.



$w_{f_r} = \bar{f}.\bar{s}$

$f_r = \mu N$

$\Rightarrow \mu mg\cos\theta = \mu mg\cos\theta_1 s_1 + \mu mg\cos\theta_2 s_2 + \mu mg\cos\theta_3 s_3$

$= \mu mgx_1 + \mu mgx_2 + \mu mgx_3$

$= \mu mg\,(x_1 + x_2 + x_3)$

Always independent of angle of inclined

P 54 : (D) $\sqrt{\frac{2\lambda D}{3}}$

Sol. Path difference = $[SS_1 + S_10] - [SS_20]$

$= (D^2 + d^2)^{1/2} + (d^2 + 4D^2)^{1/2} - 3D$

$$= D\left[\left[1+\frac{d^2}{D^2}\right]^{1/2} + 2D\left[1+\frac{d^2}{4D^2}\right]^{1/2}\right] - 3D$$

by B.T $(1 + x)^n = 1 + nx$

$$\Delta x = \frac{3d^2}{4D}$$

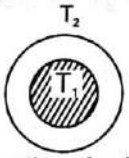
for minima $\Rightarrow \frac{2z}{\lambda}(\Delta x) = \pi$

$$\frac{2\pi}{\lambda}\left(\frac{3d^2}{4D}\right) = \pi$$

$$\Rightarrow \frac{3d^2}{4D} = \frac{\lambda}{D} \Rightarrow d = \sqrt{\frac{2\lambda D}{3}}$$

P 55 : (B) $\left[\frac{H}{\sigma 4\pi R_2^2}\right]^{1/4}$

Sol. Let Temperature of solid sphere is T_1 and of shell is T_2



Steady state equation of sphere

$H + \sigma(4\pi R_1^2)T_2^4 = \sigma(4\pi R_1^2)T_1^4$ ...(i)

Steady state equation of shell

$(\sigma 4\pi R_1^2)T_1^4 + \sigma(4\pi R_2^2 - 4\pi R_1^2)T_2^4 = 2\sigma(4\pi R_2^2)T_2^4$ ...(ii)

by equation (i) and (ii)

$\sigma(4\pi R_2^2)T_2^4 = 4$

$$T_2 = \left(\frac{H}{4\pi\sigma R_2^2}\right)^{\frac{1}{4}}$$

P 56 : (B) kinetic energy of voyager-1 is not enough to escape from sun's gravitational field.

Sol. (A) $V_{escape} = \sqrt{\frac{2GM}{r}} = \sqrt{\frac{2\times 6.67\times 10^{-11}\times 2\times 10^{30}}{21\times 10^{12}}}$

$V_{escape} = 12.7 \times 10^3$ m/s

$V_{Probe} = 62000 \times \frac{5}{18}$ m/s $= 17.2 \times 10^3$ m/s

$V_{Probe} > V_{escape} \Rightarrow$ Probe escapes

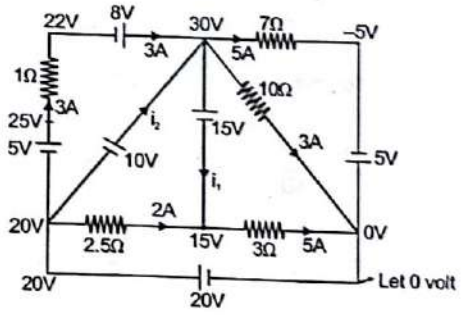
(C) $t = \frac{s}{v} = \frac{10^9 km}{62000 km/h} = \frac{10^6}{62}$ hour

$= \frac{10^6}{24\times 62}$ days $\simeq$ 670 days

(D) $t = \frac{21\times 10^{12}}{3\times 10^8} \times \frac{1}{3600} \times \frac{1}{24}$ days $\approx$ 0.8 days

P 57 : (C) 5A

Sol.



by KCL $i_1 = 5 - 2 = 3A$

$i_2 = 5 + 3 - 3 = 8A$

P 58 : (B) 1000

Sol. For single slit experiment, $x = \frac{2\lambda D}{a}$

For double slit experiment, $y = \frac{\lambda D}{d}$

$$\frac{x}{y} = \frac{2d}{a} = \frac{2\times10^{-3}}{2\times10^{-6}} = 1000$$

P 59 : (A) $\frac{I_0\pi R^2}{c}$

Sol. For absoptivity = 1

per sec incoming momentum = $\frac{I\pi R^2}{c}$

per sec outgoing momentum = 0

so force = $\frac{I\pi R^2}{c}$

for reflectivity = 1

per sec incoming momentum = $\frac{I\pi R^2}{c}$

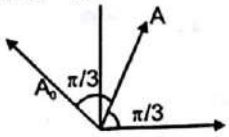
per sec outgoing momentum is zero because after relfection wave will go in all

possible directions so force = $\frac{I\pi R^2}{c}$

that means for absorption or reflection force would be same so any value of 'a' or 'r' answer would be the same.

P 60 : (B) 4I

Sol. If we draw the phasor diagram so the resultant amplitude is '2A'. So resultant intensity will be '4I'



P 61 : (A) More than 90%

Sol. it is diffifult to calculate directly the chance of at least two matching birthdays, because you have to allow for so many possibilities : just two matching, three matching, two pairs matching, etc, the COMPLEMENT of this event is, however, quite simple. If there are NOT at least two matching birthdays, then there are NONE!

$$P(\text{all different}) = \frac{365}{365}\frac{364}{365}\frac{363}{365}\cdots\cdots\frac{336}{365}$$

Therefore, the probability that at least two birth days match is over 70%!.

P 62 : (B) 2

Sol. Since, a, b, c are the sides of a right angled triangle with c as hypotenuse.

Therefore, $c^2 = a^2 + b^2$

Now, $\frac{\log_{c+b} a + \log_{c-b} a}{\log_{c+b} a \cdot \log_{c-b} a}$

$$= \frac{1}{\log_{(c-b)} a} + \frac{1}{\log_{(c+b)} a} = \log_a(c-b)+\log_a(c+b)$$

$$= \log_a(c^2 - b^2) = \log_a a^2 = 2\log_a a = 2$$

P 63 : (B) $2n\pi + \frac{7\pi}{4}, n \in I,\ 2^{1-1/\sqrt{2}}$

Sol. AM ≥ GM

$$\Rightarrow \frac{2^{\sin\theta} + 2^{-\cos\theta}}{2} \geq \sqrt{2^{\sin\theta}.2^{-\cos\theta}}$$

$$\Rightarrow 2^{\sin\theta} + 2^{-\cos\theta} \geq 2^{1+\frac{1}{2}(\sin\theta-\cos\theta)}$$

$$\geq 2^{1+(\sqrt{2}/2)\sin\left(\theta-\frac{\pi}{4}\right)}$$

$$\geq 2^{1+(1/\sqrt{2})\sin\left(\theta-\frac{\pi}{4}\right)}$$

For minimum value, $\sin\left(\theta-\frac{\pi}{4}\right) = -1$

$$\theta - \frac{\pi}{4} = 2n\pi + \frac{3\pi}{4}$$

$$\theta = 2n\pi + \frac{7\pi}{4}, n \in I$$

and minimum value = $2^{1-1/\sqrt{2}}$

P 64 : (C) $(-1/\sqrt{2}, 7\sqrt{2})$

Sol. Reflection about the line y = x, changes the point (4, 1) to (1, 4). On the translation of (1, 4) through a distance of 2 units along the positive direction of the x-axis, the point becomes (1 + 2, 4)
i.e., (3, 4).
On rotation about the origin through an angle $\pi/4$, point P takes the position P' such that OP = OP'. Also, OP = 5 = OP' and $\cos\theta = 3/5$, $\sin\theta = 4/5$. Now

$$x = OP' \cos\left(\frac{\pi}{4}+\theta\right)$$

$$= 5\left(\cos\frac{\pi}{4}\cos\theta - \sin\theta\right)$$

$$5\left(\frac{3}{5\sqrt{2}} - \frac{4}{5\sqrt{2}}\right) = -\frac{1}{\sqrt{2}}$$

$$Y = OP' \sin\left(\frac{\pi}{4}+\theta\right)$$

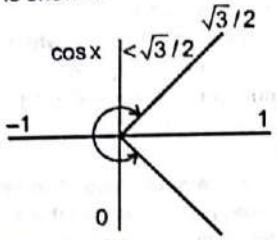
$$= 5\left(\sin\frac{\pi}{4}\cos\theta + \cos\frac{\pi}{4}\sin\theta\right)$$

$$= \frac{3}{5\sqrt{2}} + \frac{4}{5\sqrt{2}} = \frac{7}{\sqrt{2}}$$

Therefore, $P' = \left(\frac{-1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$

P 65 : (A) $\left[-\pi, \frac{-\pi}{6}\right) \cup \left(\frac{\pi}{6}, \pi\right]$

Sol. Here, $\cos x < \frac{\sqrt{3}}{2}$. The value scheme for this is shown :



From the figure,

$$-\pi \le x < -\frac{\pi}{6} \text{ or } \frac{\pi}{6} < x \le \pi$$

Therefore, $x \in \left[-\pi, \frac{-\pi}{6}\right) \cup \left(\frac{\pi}{6}, \pi\right]$

P 66 : (D) −2

Sol. We have $\frac{x-1}{1} = \frac{y+3}{-\lambda} = \frac{z-1}{\lambda} = s$ and

$$\frac{2x}{1} = \frac{y-1}{1} = \frac{z-2}{-1} = t$$

i.e., $\frac{x-0}{1} = \frac{y-1}{2} = \frac{z-2}{-2} = \frac{t}{2}$

Since, lines are coplanar,

Then, $\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0$

$$\Rightarrow \begin{vmatrix} -1 & 4 & 1 \\ 1 & -\lambda & \lambda \\ 1 & 2 & -2 \end{vmatrix} = 0$$

On solving, $\lambda = -2$.

P 67 : (A) $2ac = ab + b^2$

Sol. $\alpha + \beta = \alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$

$$\frac{-b}{a} = \left(\frac{-b}{a}\right)^2 - 2\left(\frac{c}{a}\right)$$

$$\Rightarrow \frac{-b}{a} = \frac{b^2}{a^2} - \frac{2c}{a} \qquad \Rightarrow -b = b^2/a - 2c$$

$$\Rightarrow b^2 = 2ac - ab \qquad \Rightarrow 2ac = ab + b^2$$

P 68 : (D) $\frac{xy}{1-x-y+2xy}$

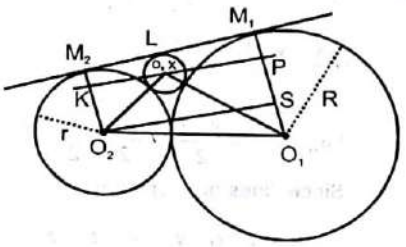
Sol. A and B will agree in a certain statement if both speak truth or both tell a lie. We define the following events:
E_1 = A and B both speak truth $\Rightarrow P(E_1) = xy$
E_2 = A and B both tell a lie
$\Rightarrow P(E_2) = (1-x)(1-y)$
E = A and B agree in a certain statement
Clearly, $P(E/E_1) = 1$ and $P(E/E_2) = 1$
The required probability is $P(E_1/E)$.
Using Baye's theorem :

$$P(E_1/E) = \frac{P(E_1)P(E/E_1)}{P(E_1)P(E/E_1) + P(E_2)P(E/E_2)}$$

$$= \frac{xy \times 1}{xy \times 1 + (1-x)(1-y)1} = \frac{xy}{1-x-y+2xy}$$

P 69 : (D) $\dfrac{Rr}{\left(\sqrt{R}+\sqrt{r}\right)^2}$

Sol. $O_3K = \sqrt{(r+x)^2-(r-x)^2} = 2\sqrt{rx}$ and



similarly $O_3P = \sqrt{(R+x)^2-(R-x)^2} = 2\sqrt{Rx}$

Now $O_3K + O_3P = O_3S = \sqrt{(R+r)^2-(R-r)^2}$

$= 2\sqrt{Rr}$

Hence $2\left(\sqrt{r}+\sqrt{R}\right)\sqrt{x} = 2\sqrt{Rr}$

$\Rightarrow x = \dfrac{Rr}{\left(\sqrt{R}+\sqrt{r}\right)^2}$

P 70 : (B) $\frac{5}{3}\hat{i}+\frac{2}{3}\hat{j}+\frac{2}{3}\hat{k}$

Sol. Given $\vec{A} = \hat{i}+\hat{j}+\hat{k}$ and $\vec{C} = \hat{j}-\hat{k}$

Let $\vec{B} = x\hat{i}+y\hat{j}+z\hat{k}$

Given that $\vec{A}\times\vec{B} = \vec{C} \Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ x & y & z \end{vmatrix} = \hat{j}-\hat{k}$

or $(z-y)\hat{i}+(x-z)\hat{j}+(y-x)\hat{k} = \hat{j}-\hat{k}$

$z - y = 0$, $x - z = 1$ and $y - x = -1$ ...(1)

Also, $\vec{A}.\vec{B} = 3 \Rightarrow x + y + z = 3$ ...(2)

Using (1) and (2), we get

$y = 2/3$, $x = 5/3$, $z = 2/3$

Therefore, $\vec{B} = \frac{5}{3}\hat{i}+\frac{2}{3}\hat{j}+\frac{2}{3}\hat{k}$.

P 71 : (C) $4x + 3y = 12$

Sol. Mid point of $(-3, 0)$ and $(3, 8)$ is $(0, 4)$. It lies on the ellipse.

$\therefore$ the ray strikes the ellipse at $(0, 4)$

Since $(-3, 0)$ is one of the fociii,

$\therefore$ the reflected ray will pass through the other focus i.e. $(3, 0)$

Equation of the reflected ray i.e. the line passing through $(0, 4)$ and $(3, 0)$ is $4x + 3y = 12$

P 72 : (A) $\left(\dfrac{x^2}{a^2}-\dfrac{y^2}{b^2}\right)^2 = c^2\left(\dfrac{x^2}{a^4}+\dfrac{y^2}{b^4}\right)$

Sol. Let (x_1, y_1) be the middle point of a variable chord of hyperbola $\dfrac{x^2}{a^2}-\dfrac{y^2}{b^2} = 1$

Equation of the chord is

$$\frac{xx_1}{a^2}-\frac{yy_1}{b^2} = \frac{x_1^2}{a^2}-\frac{y_1^2}{b^2}$$

If the chord touches the circle $x^2 + y^2 = c^2$ then length of perpendicular from the centre $(0, 0)$ is equal to radius c

$$\Rightarrow \left|\frac{\frac{x_1^2}{a^2}-\frac{y_1^2}{b^2}}{\sqrt{\frac{x_1^2}{a^4}+\frac{y_1^2}{b^4}}}\right| = c$$

$$\Rightarrow \left(\frac{x^2}{a^2}-\frac{y^2}{b^2}\right)^2 = c^2\left(\frac{x^2}{a^4}+\frac{y^2}{b^4}\right)$$

P 73 : (B) 24

Sol. If none of the boxes is empty,

Number of ways = 3! = 6

If one of the boxes is empty,

The number of ways in which 3 balls can be divided into the groups containing 1 and 2 balls respectively = ${}^3C_1 = 3$

Now the number of ways end selection can be put in boxes = 3! = 6

$\therefore$ the number of ways in which one box remains empty = $6 \times 3 = 18$

$\therefore$ total number of ways = 6 + 18 = 24

P 74 : (A) 126

Sol. Any interior intersection point corresponds to 4 of the fixed points, namely the 4 end points of the intersecting segments. Conversely, any 4 labled points determine a quadrilateral, the diagonals of which intersect once within the circle.

P 75 : (C) (– 1/2, 3/2)

Sol. $a^2 + 9b^2 - 4c^2 = 6ab$

$\Rightarrow (a - 3b)^2 - (2c)^2 = 0$

$\Rightarrow (a - 3b + 2c)(a - 3b - 2c) = 0$

when $a - 3b + 2c = 0$, the family of lines is

$(3b - 2c)x + by + c = 0$

i.e. $(3x + y)b + (-2x + 1)c = 0$

Each member of this family passes through the point $(1/2, -3/2)$ .....(i)

when $a - 3b - 2c = 0$, the family of lines is

$(3b + 2c)x + by + c = 0$

i.e. $(3x + y)b + (2x + 1)c = 0$

Each member of this family passes through the point $(-1/2, 3/2)$ .....(ii)

P 76 : (B) $a^2 + ac + bd + d^2 = 0$

Sol. Let $y = mx$ is any line represented by the equation $ax^3 + bx^2y + cxy^2 + dy^3 = 0$

$\Rightarrow ax^3 + b^2(mx) + cx(m^2x^2) + dm^3x^3 = 0$

$\Rightarrow a + bm + cm^2 + dm^3 = 0$ ...(i)

which is a cubic equation. This equation will give three roots as three lines out of which two are perpendicular hence

$m_1m_2 = -1$

and $m_1m_2m_3 = -\frac{a}{d}$

$\Rightarrow m_3 = \frac{a}{d}$

and m_3 is the root of the given equation

hence $a + b\left(\frac{a}{d}\right) + c\left(\frac{a}{d}\right)^2 + d\left(\frac{a}{d}\right)^3 = 0$

$\Rightarrow d^2 + bd + ca + a^2 = 0$

P 77 : (C) $\frac{35}{2^{12}}$

Sol. Let coin is tossed n times

$\therefore {}^nC_6 \cdot \left(\frac{1}{2}\right)^n = {}^nC_{10} \cdot \left(\frac{1}{2}\right)^n \Rightarrow n = 16$

Required probability $= {}^{16}C_3 \left(\frac{1}{2}\right)^3 \cdot \left(\frac{1}{2}\right)^{13}$

$= \frac{16 \times 15 \times 14}{6} \times \frac{1}{2^{16}} = \frac{35}{2^{12}}$

P 78 : (B) $\frac{5}{8}$

Sol. Given, $\sin 2\theta + \sin 2\phi = \frac{1}{2}$ (1)

and $\cos 2\theta + \cos 2\phi = \frac{3}{2}$ (2)

Squaring and adding, we get

$\therefore (\sin^2 2\theta + \cos^2 2\theta) + (\sin^2 2\phi + \cos^2 2\phi) +$

$2[\sin 2\theta.\sin 2\phi + \cos 2\theta.\cos 2\phi] = \frac{1}{4} + \frac{9}{4}$

$\Rightarrow \cos 2\theta.\cos 2\phi + \sin 2\theta.\sin 2\phi = \frac{1}{4}$

$\Rightarrow \cos(2\theta - 2\phi) = \frac{1}{4} \Rightarrow \cos^2(\theta - \phi) = \frac{5}{8}$

P 79 : (C) 6

Sol. Let $a + b + c = 2S$

than we have to find minimum value of

$$\frac{a}{S-a} + \frac{b}{S-b} + \frac{c}{S-c} =$$

$$-3 + \frac{S}{S-a} + \frac{S}{S-b} + \frac{S}{S-c}$$

Also,

$$\frac{\frac{S}{S-a} + \frac{S}{S-b} + \frac{S}{S-c}}{3} \geq \frac{3}{\frac{S-a}{S} + \frac{S-b}{S} + \frac{S-c}{S}}$$

$$\because \frac{S-a}{S} + \frac{S-b}{S} + \frac{S-c}{S} = 1$$

$$\Rightarrow \frac{S}{S-a} + \frac{S}{S-b} + \frac{S}{S-c} \geq 9.$$

Thus minimum value of the expression is 6.

P 80 : (C) only one point (1, 2)

Sol. The given hyperbola is a rectangular hyperbola so centre is the only point from where two perpendicular asymptotes can be drawn.

P 81 : (A) $\alpha < -3$ or $\alpha > 1$

Sol. Since (α, α^2) & (1, 2) lie in the same angular region

$\therefore (3\alpha + 4\alpha^2 - 1) > 0$ ($\because 3 + 4 - 1 > 0$ & $2 + 2 - 3 > 0$) & $2\alpha + \alpha^2 - 3 > 0$

$\therefore 4\alpha^2 + 3\alpha - 1 > 0$ & $\alpha^2 + 2\alpha - 3 > 0$

$(4\alpha - 1)(\alpha + 1) > 0$ & $(\alpha - 1)(\alpha + 3) > 0$

$\alpha \in (-\infty, -1) \cup \left(\frac{1}{4}, \infty\right)$ & $\alpha \in (-\infty, -3) \cup (1, \infty)$

$\therefore \alpha \in (-\infty, -3) \cup (1, \infty)$

P 82 : (C) (1, 2)

Sol. A(– 2, 0), B(1, 3) P(t^2, 2t)

$$\text{Area} = \frac{1}{2}\begin{vmatrix} -2 & 0 & 1 \\ 1 & 3 & 1 \\ t^2 & 2t & 1 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} -2 & 0 & 1 \\ 3 & 3 & 0 \\ t^2+2 & 2t & 0 \end{vmatrix}$$

$$= \frac{1}{2}|6t - 3t^2 - 6|$$

$$= \frac{1}{2}|3t^2 - 6t + 6|$$

$$= \frac{3}{2}|t^2 - 2t + 1 + 1|$$

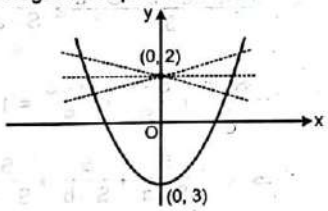
$$= \frac{3}{2}|(t-1)^2 + 1|$$

$\therefore$ t = 1

$\therefore$ the point is (1, 2)

P 83 : (C) 0

Sol. The lines y = ax + 2 are concurrent at (0, 2). Thus bounded area is minimum provided a = 0. Because the area tend to increase if we take 'a' to be negative or positive.



P 84 : (C) $\frac{2}{3}$

Sol. $$\lim_{x\to 0}\frac{\log(3+x)-\log(3-x)}{x} = \lim_{x\to 0}\frac{\log\left(\frac{3+x}{3-x}\right)}{x}$$

$$= \lim_{x\to 0}\frac{\log\left(\frac{1+(x/3)}{1-(x/3)}\right)}{x}$$

$$= \lim_{x\to 0}\frac{\log(1+(x/3))}{x} - \lim_{x\to 0}\frac{\log(1-(x/3))}{x}$$

$$= \frac{1}{3} - \left(-\frac{1}{3}\right) = \frac{2}{3}.$$

P 85 : (A) $\left[0, \frac{1}{17}\right]$

Sol. We have $(1-0)\left(\frac{0}{0^3+16}\right) \le \int_0^1 \frac{x\,dx}{(x^3+16)}$

$$\le (1-0)\left(\frac{1}{1^3+16}\right)$$

$$0 \le \int_0^1 \frac{x\,dx}{x^3+16} \le \frac{1}{17}$$

$$\int_0^1 \frac{x\,dx}{x^3+16} \in \left[0, \frac{1}{17}\right]$$

P 86 : (C) $0 \le p \le 1, q = \frac{1}{2}$

Sol. $\cos^{-1}\sqrt{p} + \cos^{-1}\sqrt{1-p}$

$$= \frac{3\pi}{4} - \cos^{-1}\sqrt{1-q}$$

$$\Rightarrow \cos^{-1}\sqrt{p} + \cos^{-1}\sqrt{1-p}$$

$$= \cos^{-1}\left(\frac{-1}{\sqrt{2}}\right) - \cos^{-1}\sqrt{1-q}$$

$$\Rightarrow \sqrt{p}\sqrt{1-p} - \sqrt{1-p}\sqrt{p}$$

$$= -\left[\frac{1}{\sqrt{2}}\sqrt{1-q} - \frac{1}{\sqrt{2}}\sqrt{q}\right]$$

$$\Rightarrow 0 = \sqrt{1-q} - \sqrt{q} \Rightarrow q = \frac{1}{2}.$$

P 87 : (C) vertices of a rectangle, which is not a square

Sol. It is a better idea to first of all plot the points approximately on the coordinate plane, this will give a basic idea of the geometrical figure involved.

If the points approximately seem to lie along a line, we will find the slopes.

If the points approximately seem to form a quadrilateral, then the next thing we must check is whether that quadrilateral is a square, rectangle, rhombus or something else.

From the rough sketch of the points on the coordinate plane, it is clear that the points A, B, C and D are not collinear.

Therefore, to check whether the quadrilateral thus formed will be a square or rectangle or rhombus or something else, we must find out the lengths of sides AB, BC, CD and DA along with their slopes and justify accordingly.

So, $|AB| = 2\sqrt{5}$; $|BC| = 4\sqrt{5}$; $|CD| = 2\sqrt{5}$

$|DA| = 4\sqrt{5}$

since, the opposite sides are equal in length, therefore the points A, B, C and D are vertices of a parallelogram.

Also, m_1 = slope of AB = –2

m_2 = slope of AD = $\frac{4}{8}$

$\therefore m_1m_2 = -1 \Rightarrow AB \perp AD$

Hence, ABCD is a rectangle.

P 88 : (C) odd and is strictly increasing in $(-\infty, \infty)$

Sol. $g(u) = 2\tan^{-1}(e^u) - \frac{\pi}{2}$

$g(u) = \tan^{-1}(e^u) - \cot^{-1}(e^u)$

$g(u) = \tan^{-1}(e^u) - \tan^{-1}(e^{-u})$ odd function.

$g'(u) = \frac{e^u}{1+e^{2u}} + \frac{e^{-u}}{1+e^{-2u}} > 0$

Strictly increasing function

P 89 : (B) Statement -1 is True, statement-2 is True ; Statement-2 is NOT a correct explanation for statement-1

Sol. Since the two equations have exactly one root common, α and β should be real

Hence $p^2 - q \geq 0$ and $b^2 - ac \geq 0$. Therefore statement '1' is correct

Also, since $\beta \neq 1, -1, 0$

$\alpha + \frac{1}{\beta} \neq \alpha + \beta \Rightarrow \frac{-2b}{a} \neq -2p \Rightarrow b \neq pa.$

Similarly $\frac{\alpha}{\beta} \neq \alpha\beta \Rightarrow \frac{c}{a} \neq q \Rightarrow c \neq qa$

Hence both statement are correct but '2' does not explain '1'

P 90 : (D) $\frac{2}{9}(2p-q)(2q-p)$

Sol. Let α, β be roots of equation $x^2 - px + r = 0$

So $\alpha + \beta = p, \alpha\beta = r$ .....(i)

$\frac{\alpha}{2}, 2\beta$ be the roots of equation $x^2 - qx + r = 0$

then $\frac{\alpha}{2} + 2\beta = q, \alpha\beta = r$ .....(ii)

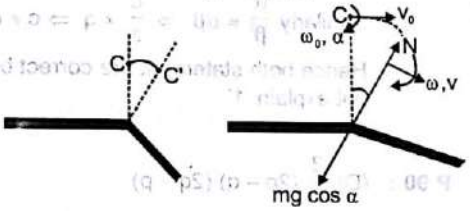
by (1) and (2) $\beta = \frac{2q-p}{3}$ and $\alpha = \frac{2(2p-q)}{3}$

So $r = \alpha\beta = \frac{2}{9}(2p-q)(2q-p).$

REPLICA 5

P 1: (A) $\sqrt{\frac{7gR}{3}\cos\theta - \frac{4}{3}g}$

Sol. The situation (not complete view) along with FBD is shown clearly in figure.



Initial energy $E_1 = \frac{1}{2}mv_0^2 + \frac{1}{2}I_{c.m.}\omega^2 + mgR$

[Considering horizontal plane as the reference for potential energy]

For pure rolling, $\frac{v_0}{R} = \omega_0$

$\Rightarrow E_1 = \frac{1}{2}mv_0^2 + \frac{1}{2}\cdot\frac{1}{2}mR^2\frac{v_0^2}{R^2} + mgR$

$= \frac{3}{4}mv_0^2 + mgR$

$E_2 = \frac{1}{2}mv^2 + \frac{1}{2}I_{c.m.}\omega'^2 + mgR\cos\alpha$

$= \frac{3}{4}mv^2 + mgR\cos\alpha \quad [\because v = R\omega]$

From energy conservation

$\frac{3}{4}mv^2 + mgR\cos\alpha = \frac{3}{4}mv_0^2 + mgR$

$\Rightarrow mv^2 = mv_0^2 + \frac{4}{3}mgR(1 - \cos\alpha)$

Centre of mass of the cylinder describes circular motion about P.

Hence $mg\cos\alpha - N = mv^2/R$

$\Rightarrow N = mg\cos\alpha - mv^2/R$

$= mg\cos\alpha - \frac{mv_0^2}{R} - \frac{4}{3}mg + \frac{4}{3}mg\cos\alpha$

For no jumping, $N \geq 0$

$\Rightarrow \frac{7}{3}mg\cos\alpha - \frac{4}{3}mg - \frac{mv_0^2}{R} \geq 0$

$\Rightarrow v_0 \leq \sqrt{\frac{7gR}{3}\cos\theta - \frac{4}{3}g}$

P 2: (C) $[(x - 1)/(x + 1)]^2$

Sol. $\frac{A + A_r}{A - A_r} = x$

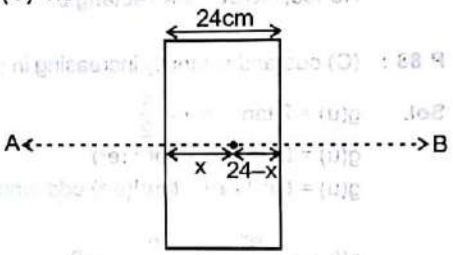
$A - Ar = Ax - A_rx$

$A_r = \frac{A(x-1)}{x+1}$

$\frac{A_r^2}{A^2} = \left(\frac{x-1}{x+1}\right)^2$

P 3: (C) 1.5

Sol.



W.r.t A, apparent depth = $10 = \frac{x}{\mu}.1$

W.r.t. B, apparent depth = $6 = \frac{24-x}{\mu}.1$

$\Rightarrow \mu = \frac{24}{16} = 1.5$

P 4: (A) the increase in the resolving power by 2.54 times for the same λ

Sol. RP of telescope = $\frac{D}{1.22\lambda}$

where, D is the diameter of the objective lens and λ is the wavelength of light used.

$\therefore$ RD $\propto$ D

P 5: (D) all of the above

Sol. $I_{max} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2$

$I_{min} = \left(\sqrt{I_1} - \sqrt{I_2}\right)^2$

as sheet absorbs half the intensity

$I_1 = I_0$ and $I_2 = I_0/2$

Intensity of both I_1 and I_2 increases

P 6 : (A) It reduces the light intensity to half on account of polarisation

Sol. Intensity is reduced to half due to polarization.

P 7 : (B) 2.5 m/s

Sol. Apparent frequency due to source A.

$$n = \frac{v - v_a}{v} = \frac{v-u}{v} \times n$$

Apparent frequency due to source B.

$$n'' = \frac{v-u}{v} \times n$$

$$\therefore\ n'' - n' = \frac{2u}{v} \times n = 10$$

$$\therefore\ u = \frac{10v}{2n} = \frac{10 \times 340}{2 \times 680} = 2.5 \text{ m/s}$$

P 8 : (D) 0.63 m, 0.75 m

Sol. In front of locomotive, $\lambda' = \frac{\lambda(v - v_s)}{500} = \frac{v - v_s}{n}$

$$\therefore \quad \lambda' = \frac{345-30}{500} = \frac{315}{500} = 0.63 \text{ m}$$

Behind locomotive, $\lambda'' = \frac{\lambda}{v}(v + v_s)\frac{v + v_s}{n}$

$$\lambda'' = \frac{345+30}{500} = \frac{375}{500} = 0.75 \text{ m.}$$

P 9 : (D) v

Sol. By using, $v' = v\left[\frac{v - v_a}{v - v_s}\right]$

$$\text{or} \quad 2v = v\left[\frac{v - v_a}{v - 0}\right] \text{ or } v_0 = -v$$

Negative sign indicates that observer is moving opposite to the direction of velocity of second.

P 10 : (A) 300 μJ

Sol. Energy, $E_1 = \frac{1}{2}C_1V_1^2 = \frac{1}{2} 1\times10^{-6} \times (30)^2$

$= 450 \times 10^{-6}$ J

Common potential, $V = \frac{q_1 + q_2}{C_1 + C_2}$

$= \frac{1 \times 30 + 0}{1+2} = 10$ volt

$E_2 = \frac{1}{2}(C_1 + C_2)V^2 = \frac{1}{2}(1 + 2) \times 10^{-6} \times (10)^2$

$= 1.5 \times 100 \times 10^{-6} = 150 \times 10^{-6}$ J

Loss of energy = $E_2 - E_1 = 300$ μJ

P 11 : (A) A current carrying wire produces magnetic field but not electric field.

Sol. A current carrying wire does not produce electric field because it is electrically neutral.

A proton beam is positively charged. Hence a moving proton beam produces both electric and magnetic field.

P 12 : (B) $^{A-4}_{Z-2}Y$

Sol. Alpha particle has mass number 4 and atomic number 2. Thus A decreases to A – 4 and Z decreases to Z – 2.

P 13 : (D) $50\sqrt{2}$ km/hr south-west

Sol. $v_1 = 50\hat{j}$, $v_2 = -50\hat{i}$

$\Delta v = -50\hat{i} - 50\hat{j} \Rightarrow \Delta v = 50\sqrt{2}$

P 14 : (B) $10\sqrt{3}$ m/s

Sol. $\vec{V}_r = 10\hat{j}$

$\vec{V}_c = v\ \hat{i}$

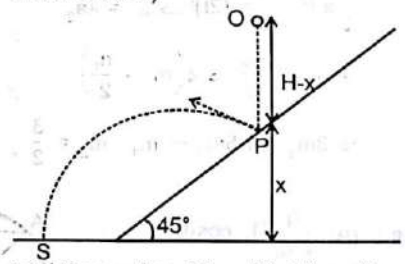
$\vec{V}_r - \vec{V}_c = 10\ \hat{j} - v\hat{i}$

$|\vec{V}_r - \vec{V}_c| = \sqrt{10^2 + v^2} = 20$

$v = 10\sqrt{3}$

P 15 : (D) $\frac{H}{2}$

Sol. As body collides elastically and goes horizontally after collision, the angle of inclination must be 45°. (but this information is not needed)



total time = time (O → P) + time (P → S)

$$t = \sqrt{\frac{2(H-x)}{g}} + \sqrt{\frac{2x}{g}}$$

So for maximum time $\frac{dt}{dx} = 0$

$$\Rightarrow \quad x = \frac{H}{2}$$

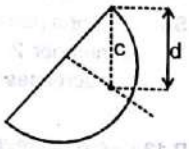
P 16 : (B) $\frac{1}{2\pi}\sqrt{\frac{g\sqrt{\pi^2+4}}{2\ell}}$

Sol. Let R is the radius of semicircular ring $\pi R = \ell$. It is compound pendulum which has

time period, $T = 2\pi\sqrt{\frac{I}{mgd}}$

{d is COM distance from point of suspension and His moment of inertia about point of suspension}

So $H = 2mR^2$



this is equilibrium

$d = \left[R^2 + \left(\frac{2R}{\pi}\right)^2\right]^{1/2}$

So $T = 2\pi\sqrt{\frac{2R\pi}{g(4+\pi^2)^{1/2}}} = 2\pi\sqrt{\frac{2\ell}{g(4+\pi^2)^{1/2}}}$

$\Rightarrow f = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{g(4+\pi^2)^{1/2}}{2\ell}}$

P 17 : (B) $\frac{3}{2}$

Sol. When m_1 is hanging and m_2 is at inclined plane, acceleration will be

$a_1 = \frac{m_1 g - m_2 g \sin 30}{m_1 + m_2}$

When m_2 is hanging and m_1 is at inclined plane, acceleration will be

$a_2 = \frac{m_2 g - m_1 g \sin 30}{m_1 + m_2}$

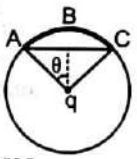
So according to the question

$\frac{1}{2}a_1 t^2 = \frac{1}{2}a_2 (2t)^2 \Rightarrow a_1 = 4a_2$

$\Rightarrow m_1 - \frac{m_2}{2} = 4\left(m_2 - \frac{m_1}{2}\right)$

$\Rightarrow 3m_1 = 4.5m_2 \Rightarrow m_1 : m_2 = \frac{3}{2}$

P 18 : (B) $\frac{q}{2\epsilon_0}(1-\cos\theta)$



Sol. The area of cap ABC,

$A = 2\pi r^2(1 - \cos\theta)$

So flux passing through this area

$\phi = 2\pi r^2(1 - \cos\theta)\frac{1}{4\pi\epsilon_0}\frac{q}{r^2}$ $\left(\text{as } \vec{E} \parallel \overrightarrow{dA}\right)$

$\phi = \frac{q}{2\epsilon_0}(1 - \cos\theta)$

P 19 : (A) 1

Sol. For motion of particle along CD

$u = 0, s = CD, a = g\cos\beta, t_1 = ?$

by $s = ut + \frac{1}{2}at^2 \Rightarrow CD = \frac{1}{2}g\cos\beta\, t_1^2$

$t_1^2 = \frac{2CD}{g\cos\beta}$

From right angled triangle ACD

$\cos\beta = \frac{CD}{AD} \Rightarrow t_1^2 = \frac{2AD}{g} = \frac{2d}{g} \Rightarrow t_1 = \sqrt{\frac{2d}{g}}$

For motion of particle along CD

$u = 0, a = g\cos\alpha, s = AP$

by $s = ut + \frac{1}{2}at^2 \Rightarrow AB = \frac{1}{2}g\cos\alpha\, t_1^2$

$t_2^2 = \frac{2AB}{g\cos\alpha}$

From right angled triangle APB

$\cos\alpha = \frac{AB}{AD} \Rightarrow t_2^2 = \frac{2AD}{g} = \frac{2d}{g} \Rightarrow t_1 = \sqrt{\frac{2d}{g}}$

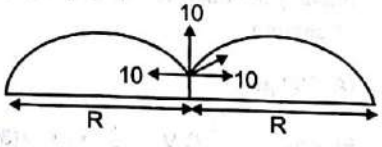
So, $t_1/t_2 = 1$

P 20 : (B) Common emitter is commonly used because current gain is maximum.

Sol. By definition, current gain is maximum in common emitter that is why it is commonly used.

P 21 : (D) New area covered by sprinkler is circle whose centre is at distance of 20m from sprinkler.

Sol.



It would be circle of radius equal to range.

$R = (2)(10) = 20$ meter

because of wind horizontal velocity will change but vertical component will not change, so time of flight will not change so every element of water sprinkler will shift by 20 m in the direction of wind. So still a circle.

P 22 : (A) In photodiode P–N junction is used in forward bias.

Sol. In voltage stabilizer P–N junction zener diode is reverse biasing.

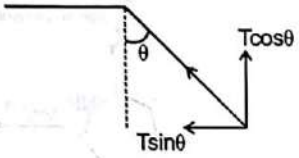
In L.E.D P–N junction is forward biased.

In Photodiode P–N junction is in reverse biased.

P 23 : (C) $\sqrt{\frac{3}{2}}$ rad/sec

Sol. $T\cos\theta = mg$

$T\sin\theta = m\omega^2 r$



$\tan\theta = \frac{\omega^2 5}{g}$

$\omega = \sqrt{\frac{3}{2}}$ rad/sec

P 24 : (C) $\frac{R}{\sqrt{2}}$

Sol. Motion of particle is S.H.M. before and after collision

$v_0 = R\omega$

after collision, speed of particle

$= v_0 \cos 60° = \frac{R\omega}{2}$

Let new amplitude is A

then $A\omega = \frac{R\omega}{2} \Rightarrow A = \frac{R}{2}$

P 25 : (C) saturation current will be 4 times and stopping potential will be less than half.

Sol. Saturation current $i_s = \eta\left(\frac{IA\lambda}{hc}\right)e$

$\Rightarrow$ Current becomes 4 times

Stopping potential $V_s = \frac{hc}{e\lambda} - \frac{\phi}{e}$

$\Rightarrow V_s$ become less than half

P 26 : (C) 7%

Sol. $T = 2\pi\sqrt{\frac{\ell}{g}}$

By differentiation

$$1\frac{\Delta T}{T} = \frac{1}{2}\frac{\Delta \ell}{\ell} - \frac{1}{2}\frac{\Delta g}{g}$$

$$2\frac{\Delta T}{T} + \frac{\Delta \ell}{\ell} = \frac{\Delta g}{g}$$

As we have to report maximum error we will always use positive sign in calculating error.

$$2[3\%] + [1\%] = \frac{\Delta g}{g} = 7\%$$

P 27 : (C) 1 : 2

Sol. For any sphere rate of cooling will be

$$-\frac{dT}{dt} = \frac{e\sigma A\left(T^4 - T_s^4\right)}{ms} = \frac{e}{r}\text{ (constant)}$$

as both material have same T, T_s, S density (ρ) their rate of cooling is same

So, $\frac{e_1}{r_1} = \frac{e_2}{r_2} \Rightarrow \frac{r_1}{r_2} = \frac{e_1}{e_2} = \frac{1}{2}$

So, $\frac{e_1}{e_2} = \frac{1}{2} \Rightarrow \frac{a_1}{a_2} = \frac{1}{2}$ as

e = a for any material.

P 28 : (D) $\frac{8}{15}\pi\rho R^5$

Sol. Charge index, $A = \int r^2 dq$ and we know

moment of inertia, $H = \int r^2 dm$

So, for uniform distribution, $A = \frac{Iq}{m}$

So for solid sphere

$$A = \frac{2}{5}\frac{mR^2 q}{m} = \frac{2qR^3}{5} \Rightarrow q = \rho\frac{4}{3}\pi R^3$$

So, $A = \frac{8}{15}\pi\rho R^5$

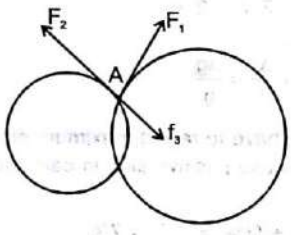
P 29 : (B) Pulse code modulation [PCD]

Sol. By theory given in NCERT, Pulse code modulation [PCD] is digital.

Note : Study NCERT carefully because it is very important for JEE Mains.

P 30 : (C) $\alpha'=120°$ always

Sol.



Net force an A must be zero for equilibrium and as surface tension is same $F_1 = F_2 = f_3 = (2\pi r)T$ and if equal magnitude force acts for equilibrium, angle between them is

$\frac{360}{3}$ is 120°.

P 31 : (A) $\frac{4\times63}{3}$

Sol. In this case only 2 mol of $NO_3^{\ominus}$ undergo reduction

$3e^- + NO_3^{\ominus} \longrightarrow NO$ ('n' = 3) (reduction)

$x - 6 = -1$ $\quad x - 2 = 0$

$x = 5$ $\quad x = 2$

6 mol of HNO_3 are not changing so $6NO_3^{\ominus}$ are added in the reaction to get 3 mol of $Cu(NO_3)_2$.

$\therefore Ew = M + \frac{M}{3} = \frac{4M}{3} = \frac{4\times63}{3}$

P 32 : (C) All the molecules of heavier gas will move at a slower speed as compared to any molecule of a lighter gas.

P 33 : (A) 2.5 g/g adsorbent

Sol. $\frac{x}{m} = k.P^{1/n}$ since, log k = 0.699,

hence, k = 5 ;

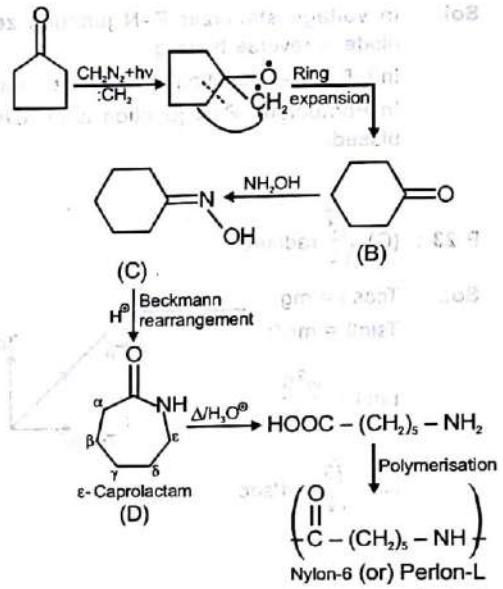
slope $= \frac{1}{n} = \tan 45° = 1$

Thus, $\frac{x}{m} = 5 \times 0.5 = 2.5$ g/g adsorbent

P 34 : (B) Analgesic

P 35 : (C) Nylon-6

Sol.



P 36 : (D) – 0.15 V

Sol. For the reaction

$AgI \rightleftharpoons Ag^+ + I^-$

$$\ln K_{sp} = \frac{nE^0F}{RT}$$

$$\Rightarrow \quad E^\circ = \frac{RT\ln K_{sp}}{nF} = -0.95V$$

the above reaction is combination of :

$Ag \rightleftharpoons Ag^+ + e^-$ $\quad E°=-0.80V$

$AgI + e^- \rightleftharpoons Ag + I^-$ $\quad E°=?$

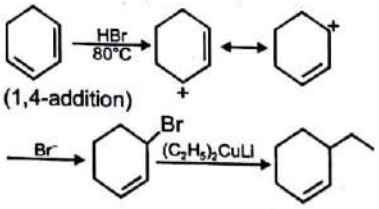
Adding $AgI + e^- \rightleftharpoons Ag^+ + I^-$

$E° = -0.95 = -0.80 + E°(AgHAg, H)$

$\Rightarrow \quad E°(AgHAg, H) = -0.15$ V

P 37 : (B) (i) HBr, 80°C (ii) $(CH_3CH_2)_2CuLi$, ether

Sol.



P 38 : (B) 8

Sol. Initial pH = $-\log \sqrt{K_a.C} = 2$

Required pH = 4 $\Rightarrow [H^+] = c\alpha = 10^{-4}$

$$K_a = 10^{-4} = \frac{c\alpha.\alpha}{1-\alpha} = \frac{10^{-4}\alpha}{1-\alpha}$$

$$\alpha = 0.5 \Rightarrow C = \frac{10^{-4}}{0.5} = 2 \times 10^{-4}\ M$$

$$\Rightarrow M_1V_1 = M_2V_2$$

$$\text{or } V_2 = \frac{1 \times 1.6 \times 10^{-3}}{2 \times 10^{-4}} = 8L$$

P 39 : (C) 269.07 K

Sol. $$\Delta T = K_f \times \frac{w_B \times 100}{m_B \times w_A}$$

$$2.15 = K_f \times \frac{5 \times 1000}{342 \times 95} \quad ...(i) \qquad \text{(for sucrose)}$$

$$\Delta T = K_f \times \frac{5 \times 1000}{180 \times 95} \quad ...(ii) \qquad \text{(for glucose)}$$

Dividing eq. (i) by eq. (ii), we get,

$\Delta T = 4.18$ K

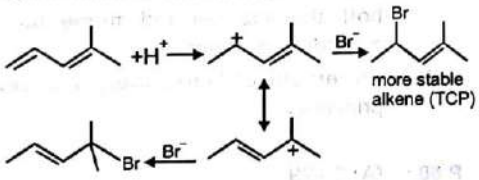
$T = T_0 - 4.18$

$= 273.15 - 4.18$

$= 269.07$ K

P 40 : (A) A

Sol.



P 41 : (D) 8

Sol. $\gamma = 1.5$ and $T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$

$$T_2 = T_1\left(\frac{V_1}{V_2}\right)^{\gamma-1} = 20\sqrt{\frac{4}{25}}$$

$= 8K$

P 42 : (B) H_3BO_3 combines with OH^-

Sol.

$$\underset{sp^3}{\ddot{N}H_3} + H^+ \longrightarrow \underset{sp^3}{NH_4^+}$$

$$\underset{sp^3}{2NH_3} \rightleftharpoons \underset{sp^3}{NH_2^-} + \underset{sp^3}{NH_4^+}$$

$$H_3BO_3 + OH^- \longrightarrow \underset{sp^3}{[B(OH)_4]^-}$$

$$\underset{sp^3}{H_2O} + H^+ \rightleftharpoons \underset{sp^3}{H_3O^+}$$

P 43 : (A) NH_4Cl

Sol. $$\underset{}{NH_4Cl} \xrightarrow{\Delta} \underset{Y}{NH_3} + \underset{Z}{HCl}$$

$$NH_3 + 2K_2[HgI_4] + 3KOH$$

Nessler's reagent

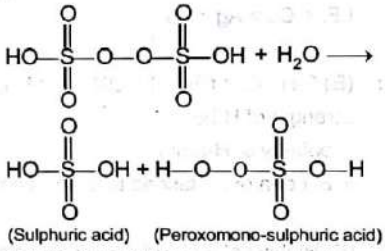
$$\rightarrow H_2NHgO.HgH + 7KH + H_2O$$

brown ppt.

iodide of Millon's base

P 44 : (C) one mole of sulphuric acid, one mole of peroxomono-sulphuric acid

Sol.



(Sulphuric acid) (Peroxomono-sulphuric acid)

P 45 : (D) $[Cr(H_2O)_6]SO_4$

Sol. The electronic configurations of Ni, Fe, Co and Cr in the given complexes are $(3d)^8$, $(3d)^6$, $(3d)^6$ and $(3d)^4$, respectively. These involves dsp^2, d^2sp^3, d^2sp^3 and sp^3d^2-hybridization respectively. The complex $[Cr(H_2O)_6]SO_4$ will be paramagnetic as Cr contains four unpaired electrons.

P 46 : (A) Oxidation of NH_3

Sol. Catalytic oxidation of NH_3 by O_2.

P 47 : (B) 0.042 Ms^{-1}

Sol. $$-\frac{1}{5}\frac{d[Br^-]}{dt} = \frac{1}{3}\frac{d[Br_2]}{dt}$$

$$\Rightarrow \frac{-d[Br^-]}{dt} = \frac{5}{3}\frac{d[Br_2]}{dt}$$

$$= \frac{5}{3} \times 0.025 = 0.042 \text{ mol } L^{-1}s^{-1}$$

P 48 : (C) Na_2O_2

Sol. $2Na_2O_2 + 2CO_2 \longrightarrow 2Na_2CO_3 + O_2$

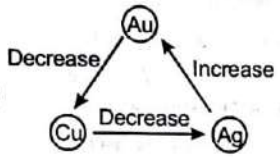
P 49 : (D) Reduction in the use and production of hazardous chemicals

Sol. It is a fact.

P 50 : (B) $KCl + K_2Cr_2O_7 + \text{conc. } H_2SO_4 \xrightarrow{\Delta}$

Sol. In this case $Cr_2O_2Cl_2$

P 51 : (D)



Sol.

Cu	Ag	Au
$[Ar]3d^{10}4s^1$	$[Kr]4d^{10}5s^1$	$[Xe]4f^{14}5d^{10}6s^1$

(Small size and higher HE. due to the effect of lanthanide contraction)

Size = Cu < Ag > Au

I.E. = Cu > Ag < Au

P 52 : (B) ClH...Cl < NH...H < OH...O < FH...F

Sol. Strength of H-bond

$\propto$ polarity of H-atom

$\propto$ EN of atom attached to the H-atom.

P 53 : (B) Highly delocalized nature of π-electrons

Sol. In graphite there is sp^2 hybridized carbon, thus due to highly delocalized π electrons it possesses electrical conductivity.

P 54 : (B) 78% dextro and 21.5% laevo

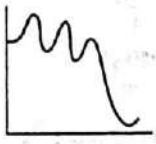
Sol. Let fraction of laevo be x.

$\Rightarrow -21x + (1 - x) \times 21 = 12$

$\Rightarrow x = \frac{9}{42} = 0.215$

$\Rightarrow$ % laevo = 21.5, % dextro = 78.5

P 55 : (C)



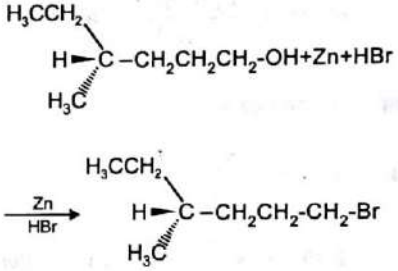
Sol. Reaction involve two reactive intermediates and three transition states. Every subsequent step is easier than previous step and product is more stable (exothermic) than reactant.

P 56 : (A) an electron deficient atom

Sol. It has only three pairs of electrons, hence electron deficient atom.

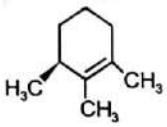
P 57 : (B) single chiral compound without its enantiomer

Sol.



Retention of configuration because reaction doesn't involve chiral carbon.

P 58 : (B)



Sol. Carbocation formation followed by rearrangement gives the major product.

P 59 (C) it is impossible to determine accurately both the position and momentum of an electron simultaneously.

Sol. Statement of Heisenberg's uncertainty principle.

P 60 : (A) 0.729

Sol. Number of Cs^+ in one cell = $8 \times \frac{1}{8} = 1$

Number of Cl^- in one unit cell = $1 \times 1 = 1$

Number of CsCl units in one unit cell = 1

Density = $\frac{z \times M}{N \times a^3}$ gram/cm^3

z = 1, M = 168.36, d = 3.97, N = 6.02×10^{23}, a = ?

$$3.97 = \frac{1\times 168.36}{6.02\times 10^{23}\times a^3} \text{ grams/cm}^3$$

$a = 4.13 \times 10^{-8}$ cm or 4.13 Å (edge length)

Interionic distance ($r^+ + r^-$)

$$= \frac{\sqrt{3}\times a}{2} = \frac{\sqrt{3}\times 4.13}{2} = 3.57 \text{ Å}$$

Now if we assume that two Cl^- ions touch each other,

Length of unit cell = $2r^- = 4.13$ Å

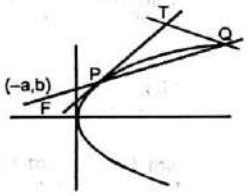
or $r^- = 2.065$ Å

$r^+ = 3.57 - 2.065 = 1.505$ Å

$$\frac{r^+}{r^-} = \frac{1.505}{2.065} = 0.729$$

P 61 : (C) $by = 2a(x - a)$

Sol. Chord of contact of (h, k)



$ky = 2a(x + h)$. It passes through $(-a, b)$

$\Rightarrow$ $bk = 2a(-a + h)$

$\Rightarrow$ Locus is $by = 2a(x - a)$

P 62 : (B) $\frac{\pi}{6}$

Sol. $$\int_0^{\pi/2} \frac{\cos\theta}{\sqrt{4-\sin^2\theta}}d\theta = \left[\sin^{-1}\left(\frac{\sin\theta}{2}\right)\right]_0^{\pi/2}$$

$$= \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$$

P 63 : (B) 1

Sol. We have, $\frac{\log x}{2a+3b-5c} = \frac{\log y}{2b+3c-5a} = \frac{\log z}{2c+3a-5b}$

$$\Rightarrow \frac{\log x}{2a+3b-5c} = \frac{\log y}{2b+3c-5a} = \frac{\log z}{2c+3a-5b}$$

$$= \frac{\log x+\log y+\log z}{0}$$

$\Rightarrow \log x + \log y + \log z = 0 \Rightarrow 0\log(xyz) = 0$

$\Rightarrow xyz = 1$

P 64 : (B) $\frac{2\cos x}{\sqrt{\cos 2x}}$

Sol. Given $\tan x = \frac{b}{a} \Rightarrow \sqrt{\frac{a+b}{a-b}} + \sqrt{\frac{a-b}{a+b}} =$

$$\sqrt{\frac{1+b/a}{1-b/a}} + \sqrt{\frac{1-b/a}{1+b/a}} = \sqrt{\frac{1+\tan x}{1-\tan x}} + \sqrt{\frac{1-\tan x}{1+\tan x}}$$

$$= \frac{2}{\sqrt{1-\tan^2 x}}$$

Now, multiplying by $\sqrt{1+\tan^2 x}$ in N'r and D'r =

$$\frac{2}{\frac{\sqrt{1-\tan^2 x}}{\sqrt{1+\tan^2 x}}.\sqrt{1+\tan^2 x}} = \frac{2}{\sqrt{\cos 2x}.\sqrt{\sec^2 x}} = \frac{2\cos x}{\sqrt{\cos 2x}}$$

P 65 : (A) (–4, –7)

Sol. We have $AH \perp BC$

$m_{AH} \times m_{BC} = -1$

or $\frac{k}{h}\times\frac{3+1}{-2-5} = -1$

or $4k - 7h = 0$

Also, $BH \perp AC$

$$\frac{-1}{5}\times\frac{3-k}{-2-h} = -1$$

Or $3 - k = -10 - 5h$

or $5h - k + 13 = 0$

Solving equations (1) and (2) we get,

$h = -4$, $k = -7$,

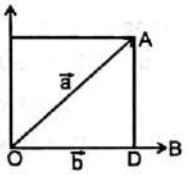
Hence the third vertex is (–4, –7)

P 66 : (A) $\vec{a} - \left(\frac{\vec{a}.\vec{b}}{|\vec{b}|^2}\right)\vec{b}$

Sol. Component of $\vec{a}$ along $\vec{b}$

$$= \overline{OD} = OA\cos\theta.\frac{\vec{b}}{|\vec{b}|}$$

$$= \left(\frac{\vec{a}.\vec{b}}{|\vec{b}|}\right)\frac{\vec{b}}{|\vec{b}|} = \left(\frac{\vec{a}.\vec{b}}{|\vec{b}|^2}\right)\vec{b}$$



Component of $\vec{a}$ perpendicular to $\vec{b}$

$$= \overline{DA} = \vec{a} - \overline{OD}$$

$$= \vec{a} - \left(\frac{\vec{a}.\vec{b}}{|\vec{b}|^2}\right)\vec{b}$$

P 67 : (A) $\left(\frac{5}{12}, \frac{5}{12}\right)$

Sol. Equation of the given circle is $x^2 + y^2 + 8x + 4y - 5 = 0$ .....(i)

Equation of any circle which passes through (0, 0) and touches the line $x - y = 0$ is

$x^2 + y^2 + \lambda (x - y) = 0$ .....(ii)

$\therefore$ equation of common chord of (i) and (ii) is

$8x + 4y - 5 - \lambda (x - y) = 0$ ...(iii)

system of lines represented by (iii) pass through the point $\left(\frac{5}{12}, \frac{5}{12}\right)$

P 68 : (A) $2n\pi \pm \frac{\pi}{4} + \frac{\pi}{12}$

Sol. $(\sqrt{3}-1)\sin\theta + (\sqrt{3}+1)\cos\theta = 2$

Dividing by $\sqrt{(\sqrt{3}-1)^2 + (\sqrt{3}+1)^2} = 2\sqrt{2}$ on both sides,

We get $\frac{(\sqrt{3}-1)}{2\sqrt{2}}\sin\theta + \frac{(\sqrt{3}+1)}{2\sqrt{2}}\cos\theta = \frac{2}{2\sqrt{2}}$

$\sin\theta \sin 15° + \cos\theta \cos 15° = \frac{1}{\sqrt{2}}$

$\Rightarrow \sin\theta \sin\frac{\pi}{12} + \cos\theta \cos\frac{\pi}{12} = \cos\frac{\pi}{4}$

$\cos\left(\theta - \frac{\pi}{12}\right) = \cos\frac{\pi}{4}$

$\Rightarrow \theta - \frac{\pi}{12} = 2n\pi \pm \frac{\pi}{4}$

$\Rightarrow \theta = 2n\pi \pm \frac{\pi}{4} + \frac{\pi}{12}$

P 69 : (C) 13

Sol. Centre of sphere is (–2, 1, 3)

Radius of sphere is $\sqrt{4+1+9+155} = 13$

Distance of centre from plane

$= \frac{-24+4+9-327}{\sqrt{144+16+9}} = \frac{338}{13}$

Therefore, plane cuts the sphere and hence

$SD = \frac{338}{13} - 13 = \frac{169}{13} = 13.$

P 70 : (D) none of these

Sol. Let

$$f(x) = \frac{\sqrt{\frac{1}{2}(1-\cos 2x)}}{x} = \frac{\sqrt{\frac{1}{2}(2\sin^2 x)}}{x} = \frac{|\sin x|}{x}$$

Then, $\lim_{x\to 0^-} f(x) = \lim_{h\to 0} f(0-h) = \lim_{x\to 0} f(-h)$

$$= \lim_{h\to 0} \frac{|\sin(-h)|}{(-h)} = \lim_{h\to 0} \frac{|-\sin h|}{-h}$$

$$= \lim_{h\to 0} \frac{\sin h}{h} = -1.$$

And $\lim_{x\to 0^+} f(x) = \lim_{h\to 0} f(0+h)$

$$= \lim_{h\to 0} f(h) = \lim_{h\to 0} \frac{|\sin h|}{h}$$

$$= \lim_{h\to 0} \frac{\sin h}{h} = 1.$$

$\Rightarrow \lim_{x\to 0^-} f(x) \neq \lim_{x\to 0^+} f(x)$

Hence, $\lim_{x\to 0} f(x)$, i.e., $\lim_{x\to 0} \frac{\sqrt{\frac{1}{2}(1-\cos 2x)}}{x}$ does not exist.

P 71 : (D) At least one of the ex-radii is smaller than the inradius of the triangle

Sol. (A) Δ must be obtuse, because, in an acute angled triangle all the four points are interior to the triangle

(B) Circumcentre and orthocentre will lie outside the triangle

(C) If the triangle is isosceles obtuse angled, then incentre may be collinear with the other three.

(D) Since $r = \frac{\Delta}{s}$, $r_1 = \frac{\Delta}{s-a}$, $r_2 = \frac{\Delta}{s-b}$ &

$r_3 = \frac{\Delta}{s-c}$, $\therefore$ $r < r_1, r_2, r_3$.

P 72 : (D) none of these

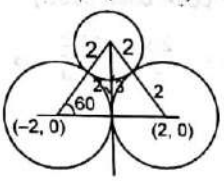
Sol. The given circles are $x^2 + y^2 - 4x = 0$ and $x^2 + y^2 + 4x = 0$

their centres and (2, 0) and (–2, 0)

$\therefore$ the circle which touches these circles and whose radius = 2 has its centre at $(0, 2\sqrt{3})$ or $(0, -2\sqrt{3})$

$\therefore$ equations of the required circles are

$(x - 0)^2 + \left(y \pm 2\sqrt{3}\right)^2 = 4$



i.e. $x^2 + y^2 \pm 4\sqrt{3}\, y + 8 = 0$

P 73 : (B) $\left(-\infty, -\frac{2}{3}\right) \cup \left(-\frac{2}{3}, 2\right)$

Sol. $(2 + \lambda)x^2 - 2\lambda xy + (\lambda - 1)y^2 - 4x - 2 = 0$

represents a hyperbola if

$\Delta \neq 0, \quad h^2 > ab$

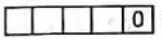
$\Rightarrow \quad \lambda \neq \frac{4}{3}$ and $\lambda < 2$

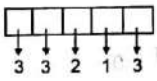
$\Rightarrow \quad \lambda \in \left(-\infty, \frac{4}{3}\right) \cup \left(\frac{4}{3}, 2\right)$

P 74 : (B) 150

Sol. I. Excluding zero $= {}^3C_1 \times 41 = 72$

II. Excluding three

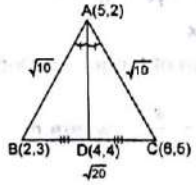
(i)  $= 4! = 24$

(ii)  $= 3.3.31 = 54$

Total = 72 + 24 + 54 = 150

P 75 : (C) 2x + y – 12 = 0

Sol.



$\frac{BD}{DC} = \frac{AB}{AC} = \frac{\sqrt{10}}{\sqrt{10}} = \frac{1}{1} \Rightarrow$ D is midpoint of BC

$\therefore \quad$ D(4, 4)

AD is 2x + y – 12 = 0.

P 76 : (C) infinitely many points

Sol. Equation of line segment by (– 2, 7), (– 5, – 3) is :

$(y - 7) = \frac{10}{3}(x - z)$

$\Rightarrow 3y - 10x = 41$

Equation of line segment by (–8, – 13), (1, 17) is :

$(y - 17) = \frac{10}{3}(x - 1)$

$\Rightarrow 3y - 10x = 41$

both the equations are same. It means infinite common points.

P 77 : (D) $\frac{80}{243}$

Sol. $P = {}^5C_2 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^3$

$= \frac{80}{243}$

P 78 : (A) (–1, 2), (–1, – 6)

Sol. $25x^2 + 9y^2 + 50x + 36y - 164 = 0$

$25(x + 1)^2 + 9(y + 2)^2 = 225$

$\frac{(x+1)^2}{9} + \frac{(y+2)^2}{25} = 1$

$\therefore \quad a = 5, b = 3 \quad \therefore e = \sqrt{\frac{25-9}{25}} = \frac{4}{5}$

$\therefore \quad ae = 4$

$\therefore \quad$ the focii are

$x + 1 = 0 \qquad y + 2 = \pm 4$

$x = -1 \qquad y = 2, -6$

(– 1, 2), (– 1, – 6)

P 79 : (A) $^{103}C_2 - 3^{52}C_2$

Sol. Let the amount received by the sons be Rs. x , Rs. y and Rs. z respectively, then

$x \le y + z = 101 - x$

i.e., $2x \le 101 \quad \therefore \; x \le 50, y \le 50, z \le 50$

$x + y + z = 101$

The corresponding mutlinomial is $(1+x+.... x^{50})^3$

$\therefore$ coefficient of x^{101} in the expansion of $(1 + x + ... x^{50})^3$

$= 1 \times {}^{103}C_{101} - 3. {}^{52}C_{50}$

$= {}^{103}C_2 - 3. {}^{52}C_2$

P 80 : (C) $\frac{1}{4}$

Sol. $\sin\left(\frac{\pi}{14}\right) \sin\left(\frac{3\pi}{14}\right) \sin\left(\frac{5\pi}{14}\right)$

$= \cos\left(\frac{\pi}{2} - \frac{\pi}{14}\right) \cos\left(\frac{\pi}{2} - \frac{3\pi}{14}\right) \cos\left(\frac{\pi}{2} - \frac{5\pi}{14}\right)$

$= \cos\left(\frac{3\pi}{7}\right) \cos\left(\frac{2\pi}{7}\right) \cos\left(\frac{\pi}{7}\right)$

$= \cos\left(\pi - \frac{4\pi}{7}\right) \cos\frac{2\pi}{7} \cos\left(\frac{\pi}{7}\right)$

$= \cos\left(\frac{\pi}{7}\right) \cos\frac{2\pi}{7} \cos\left(\frac{10\pi}{7}\right)$

$- \sin\left(\frac{\pi}{14}\right) \sin\left(\frac{3\pi}{14}\right) \sin\left(\frac{5\pi}{14}\right)$

$= \cos\left(\frac{\pi}{7}\right) \cos\left(\frac{2\pi}{7}\right) \cos\left(2\pi - \frac{10\pi}{7}\right)$

$+ \cos\left(\frac{\pi}{7}\right) \cos\left(\frac{2\pi}{7}\right) \cos\left(\frac{4\pi}{7}\right)$

$= 2\cos\left(\frac{\pi}{7}\right) \cos\left(\frac{2\pi}{7}\right) \cos\left(\frac{4\pi}{7}\right)$

$= \frac{2\sin\left(\frac{\pi}{7}\right) \cos\left(\frac{\pi}{7}\right) \cos\left(\frac{2\pi}{7}\right) \cos\left(\frac{4\pi}{7}\right)}{\sin\left(\frac{\pi}{7}\right)}$

$= \frac{2\sin\left(\frac{2\pi}{7}\right) \cos\left(\frac{2\pi}{7}\right) \cos\left(\frac{4\pi}{7}\right)}{2\sin\left(\frac{\pi}{7}\right)}$

$= \frac{2\sin\left(\frac{4\pi}{7}\right) \cos\left(\frac{4\pi}{7}\right)}{2 \times 2\sin\left(\frac{\pi}{7}\right)}$

$= \frac{\sin\left(\frac{8\pi}{7}\right)}{4\sin\left(\frac{\pi}{7}\right)} = \frac{\sin\left(\pi + \frac{\pi}{7}\right)}{4\sin\left(\frac{\pi}{7}\right)}$

$= \frac{\sin\left(\frac{\pi}{7}\right)}{4\sin\left(\frac{\pi}{7}\right)} = -\frac{1}{4}$

P 81 : (D) 4 R (1 + cosA cosB cosC)

Sol. $a\ell = 2\Delta \Rightarrow \ell = \frac{2\Delta}{a} = \frac{abc}{2Ra} \Rightarrow \frac{bl}{c} = \frac{bc}{2R}$

$\therefore \frac{b}{c} = \frac{b^2}{2R}$

Hence LHS $= \frac{1}{2R}(a^2 + b^2 + c^2)$

$\Rightarrow \quad 2R(\sin^2 A + \sin^2 B + \sin^2 C)$

$= 2R(2 + 2\cos A \cos B \cos C)$

$= 4R(1 + \cos A \cos B \cos C)$

P 82 : (D) a, b, c are irrational

Sol. Let $f(x) = ax^2 + bx + c$

$f\left(\frac{1}{3}\right) = \frac{a}{9} + \frac{b}{3} + c = \frac{1}{9}(a + 3b + 9c) = 0$

Therefore, $\frac{1}{3}$ is the root of the given equation

Also, product of the roots $= \frac{c}{a} = \frac{ac}{a^2} < 0$, therefore another root must be negative hence, it will be $-\frac{1}{\sqrt{3}}$ and required equation is

$\left(x - \frac{1}{3}\right)\left(x + \frac{1}{\sqrt{3}}\right) = 0$

$\Rightarrow \quad 3\sqrt{3}x^2 + (3 - \sqrt{3})x - 1 = 0$

P 83 : (C) $xy_1^2 - yy_1 + a = 0$

Sol. $y^2 = 4ax$

Equation of tangent in slope form is

$y = mx + \frac{a}{m}$ where $m = \frac{dy}{dx} = y_1$

$\therefore$ the differential equation is $y = y_1 x + \frac{a}{y_1}$

$\Rightarrow xy_1^2 - 4y_1 + a = 0$

which represents differential equation of all tangents of the parabola $y^2 = 4ax$.

P 84 : (C) $\frac{\pi}{4}$

Sol. $4\tan^{-1}\frac{1}{5} - \tan^{-1}\frac{1}{70} + \tan^{-1}\frac{1}{99} =$

$$2\tan^{-1}\left[\frac{\frac{2}{5}}{1-\frac{1}{25}}\right] - \tan^{-1}\frac{1}{70} + \tan^{-1}\frac{1}{99}$$

$$= 2\tan^{-1}\left(\frac{5}{12}\right) - \tan^{-1}\frac{1}{70} + \tan^{-1}\frac{1}{99}$$

$$= \tan^{-1}\left[\frac{\frac{5}{6}}{1-\frac{25}{144}}\right] - \tan^{-1}\frac{1}{70} + \tan^{-1}\frac{1}{99}$$

$$= \tan^{-1}\left(\frac{120}{119}\right) - \tan^{-1}\frac{1}{70} + \tan^{-1}\frac{1}{99}$$

$$= \tan^{-1}\frac{120}{119} + \tan^{-1}\left[\frac{\frac{1}{99}-\frac{1}{70}}{1+\frac{1}{99}\cdot\frac{1}{70}}\right]$$

$$= \tan^{-1}\frac{120}{119} + \tan^{-1}\left(-\frac{29}{6931}\right)$$

$$= \tan^{-1}\frac{120}{119} - \tan^{-1}\left(\frac{29}{6931}\right)$$

$$= \tan^{-1}\frac{120}{119} - \tan^{-1}\left(\frac{1}{239}\right)$$

$$= \tan^{-1}\left[\frac{\frac{120}{119}-\frac{1}{239}}{1+\frac{120}{119}\times\frac{1}{239}}\right] = \tan^{-1}(1) = \frac{\pi}{4}$$

P 85 : (A) $\frac{1}{40}$

Sol. Let A, B and C produce 2n, n and n products

Let E_1 = Event of a product being produced at A

E_2 = Event of a product being produced at B.

E_3 = Event of a product being produced at C.

$$P(E_1) = \frac{2n}{2n+n+n} = \frac{1}{2} \quad P(E_2) = \frac{n}{2n+n+n} = \frac{1}{4}$$

$$P(E_3) = \frac{1}{4}$$

Also, $P(D/E_1)$ = Probability of a product being defective when produced at AP(D/E_1)

$$= \frac{2}{100} = \frac{1}{50}$$

Similarly, $P(D/E_2) = \frac{4}{100} = \frac{1}{25}$

$$P(D/E_3) = \frac{2}{100} = \frac{1}{50}$$

As $P(E_1) + P(E_2) + P(E_3) = \frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1$

events, E_1, E_2, E_3 are exhaustive.

Also, they are mutually exclusive. The probability that the product is defective

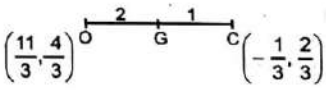
$= P(E_1).P(D/E_1) + P(E_2).P(D/E_2) + P(E_3).P(D/E_3)$

$$= \frac{1}{2}\cdot\frac{1}{50} + \frac{1}{4}\cdot\frac{1}{25} + \frac{1}{4}\cdot\frac{1}{50}$$

$$= \frac{1}{100} + \frac{1}{100} + \frac{1}{200} = \frac{1}{40}$$

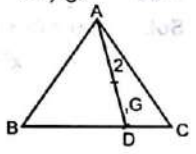
P 86 : (A) (1, – 11/3)

Sol.

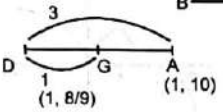


$$G = \left(\frac{-\frac{2}{3}+\frac{11}{3}}{3}, \frac{\frac{4}{3}+\frac{4}{3}}{3}\right) = \left(1, \frac{8}{9}\right)$$

$$\frac{DA}{DG} = \frac{3}{1}$$



$\therefore$



$$D \equiv \left(\frac{3-1}{3-1}, \frac{8/3-10}{3-1}\right)$$

$$D \equiv \left(1, \frac{-11}{3}\right)$$

P 87 : (B) $(xy)^n$

Sol. Solve this for a sample GP. Let us say we take the GP as 2, 6, 18, 54. x, the first term is 2, let n=3 then the 3rd term y = 18 and the product of 3 terms p = 2 × 6 × 18 = 216 = 6^3. The value of p^2 = 216 × 216 = 6^6.

Putting these values in the options we have Option (a) gives us $(xy)^{n-1} = 36^2$ which is not equal to the value of p^2 we have from the options

Option (b) gives us $(xy)^n = 36^3 = 6^6$ which is equal to the value of p^2 we have from the options

It can be experimentally verified that the other options yield values of p^2 which are different from 6^6 and hence we can conclude that option (b) is correct.

P 88 : (B) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$

Sol. Given $\vec{a}+\vec{b}+\vec{c}$ (by triangle law). Therefore,

$$\vec{a} \times (\vec{a}+\vec{b}+\vec{c}) = \vec{a} \times \vec{0}$$

$$\vec{a} \times \vec{a} + \vec{a} \times \vec{b} + \vec{a} \times \vec{c} = \vec{0}$$

$$\vec{a} \times \vec{b} = \vec{c} \times \vec{a}$$

Similarly by taking cross product with $\vec{b}$, we get

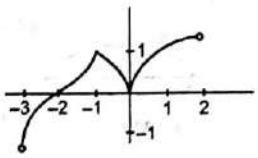
$$\vec{a} \times \vec{b} = \vec{b} \times \vec{c}$$

$$\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$$

P 89 : (C) 2

Sol. $f(x) = (2+x)^3 \quad -3 < x \le -1$

$x^{2/3} \quad -1 < x < 2$



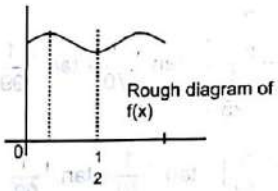
There is one point of local maxima (i.e. x = – 1) and one point of local minima (i.e. x = 0)

P 90 : (C) $\int_{-1/2}^{1/2} f\left(x+\frac{1}{2}\right) \sin x \, dx = -1$

Sol. $f(x) = f(1-x) \Rightarrow f\left(x+\frac{1}{2}\right) = f\left(\frac{1}{2}-x\right)$

f(x) is symmetric about $x = \frac{1}{2}$.

Hence $f'\left(\frac{1}{2}\right) = 0$



also $f'\left(\frac{1}{4}\right) = 0$ & $f'\left(\frac{3}{4}\right) = 0$.

Applying Rolles theorem for

$f'(x)$ in $\left[\frac{1}{4}, \frac{1}{2}\right]$ & $\left[\frac{1}{2}, \frac{3}{4}\right]$

we can say that $f''(x) = 0$ at least twice in [0, 1]
Hence 'A' is correct, 'B' is also correct.
In option 'C' applying property

$$\int_{-1/2}^{1/2} f\left(x+\frac{1}{2}\right)\sin x \, dx = \int_{-1/2}^{1/2} f\left(\frac{1}{2}-x\right)\sin(-x)\, dx$$

$$= -\int_{-1/2}^{1/2} f\left(x+\frac{1}{2}\right)\sin x \, dx$$

Hence $\int_{-1/2}^{1/2} f\left(x+\frac{1}{2}\right)\sin x \, dx = 0$

Hence C is false.
In option 'D'
Put $t = 1 - z$

$$\int_{1/2}^{1} f(1-t)e^{\sin \pi t}\, dt = -\int_{1/2}^{0} f(z)e^{\sin \pi z}\, dz$$

$$= \int_{0}^{1/2} f(z)e^{\sin \pi z}\, dz \text{ which is same as L.H.S.}$$

REPLICA 6

P 1 : (A) $4x^2 - b^2y^2 = 4a^2$, $y^2 - 4x > 0$

Sol. Let the point be (h, k). Clearly (h, k) lies to the exterior of parabola.

$\therefore k^2 - 4h > 0$ .....(1)

and equation of the chord of contact is

$ky = 2(x + h)$

it touches the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\therefore$ comparing $y = \frac{2}{k}x + \frac{2h}{k}$ with

$$y = mx \pm \sqrt{a^2m^2 + b^2}$$

$a^2 \cdot \frac{4}{k^2} + b^2 = \frac{4h^2}{k^2}$ i.e. $4a^2 + b^2k^2 = 4h^2$

$\therefore$ the locus of (h, k) is

$4x^2 - b^2y^2 = 4a^2$, $y^2 - 4x > 0$

P 2 : (B) $\sqrt{2}\pi$

Sol. We can write $\ell = \ell_1 + \ell_2$

where $\ell = \int_{-\pi}^{\pi} \frac{2x(1+\sin x)}{1+\cos^2 x}dx$,

$\ell_1 = \int_{-\pi}^{\pi} \frac{2x}{1+\cos^2 x}dx = 0$, $\ell_2 = 4\int_0^{\pi} \frac{x\sin x}{1+\cos^2 x}$

$= 4\int_0^{\pi} \frac{(\pi - x)\sin(\pi - x)}{1+(\cos(\pi - x))^2}dx$

$= 4\pi\int_0^{\pi} \frac{dx}{1+\cos^2 x} - \ell_2$

$\Rightarrow \ell_2 = 2\pi\int_0^{\pi} \frac{\operatorname{cosec}^2 x}{\operatorname{cosec}^2 x + \cot^2 x}dx$

$= 2\pi\int_0^{\pi} \frac{\operatorname{cosec}^2 x dx}{1 + 2\cot^2 x}$

$= \frac{-2\pi}{\sqrt{2}}\tan^{-1}\left[(\sqrt{2}\cot x)\right]_0^{\pi}$

$= -\sqrt{2}\left(-\frac{\pi}{2} - \frac{\pi}{2}\right) = \sqrt{2}\pi$.

P 3 : (D) Infinitely many

Sol. We have, $1 + \log_5(x^2 + 1) \geq \log_5(x^2 + 4x + 1)$

$\log_5 5 + \log_5(x^2 + 1) \geq \log_5(x^2 + 4x + 1)$

$5(x^2 + 1) \geq x^2 + 4x + 1$

$4x^2 - 4x + 4 \geq 0$

$x^2 - x + 1 \geq 0$ Which is true for all $x \in R$.

Also, $(x^2 + 4x + 1)$ is defined for $x^2 + 4x + 1$

i.e., $x \in \left(-\infty, -2, \sqrt{3}\right) \cup \left(-2, +\sqrt{3}, \infty\right)$

Hence, the solution set of the given inequation is $\left(-\infty, -2, \sqrt{3}\right) \cup \left(-2, +\sqrt{3}, \infty\right)$

P 4 : (D) $\frac{51}{101}$

Sol. P of heads showing on 50 coins

$= {}^{100}C_{50} \times P^{50}(1 - p)^{50}$

P of head showing on 51 coins

$= {}^{100}C_{51} \times P^{51}(1 - p)^{49}$

Both are equal

${}^{100}C_{50} \times P^{50}(1 - p)^{50} = {}^{100}C_{51} \times P^{51}(1 - P)^{49}$

or $\frac{100 \times 99 \times \ldots \times 52 \times 51}{1 \times 2 \times \ldots \times 49 \times 50} \times (1 - P) =$

$\frac{100 \times 99 \times \ldots \times 53 \times 52}{1 \times 2 \times \ldots \times 48 \times 49} \times P$

or $\frac{51}{50} \times (1 - P) = P$

or $51 \times (1 - P) = P$

or $51 - 51P = 50P$

or $51 = 101P$ $\therefore P = \frac{51}{101}$.

P 5 : (C) 0

Sol. Centre and radius of the circle $x^2 + y^2 - 4x - 32 = 0$ are (2, 0) and 6

$mx - y + 4 = 0$

distance from (2, 0) is $\left|\frac{2m + 4}{\sqrt{m^2 + 1}}\right|$

$\therefore \left|\frac{2m + 4}{\sqrt{m^2 + 1}}\right| \leq 6$

i.e. $(m+2)^2 \le 9(m^2+1)$

i.e. $8m^2 - 4m + 5 \ge 0$

$\therefore$ $m \in R$ ...(i)

again centre and radius of the circle $x^2 + y^2 = 4$ are (0, 0), 2

distance of my $- y + 4 = 0$ from (0, 0) is

$$\frac{4}{\sqrt{m^2+1}}$$

$$\therefore \quad \frac{4}{\sqrt{m^2+1}} > 2$$

$\therefore$ $4 > m^2 + 1$ i.e. $m^2 < 3$

$\therefore$ $-\sqrt{3} < m < \sqrt{3}$ ...(ii)

from (i) and (ii) $m \in \left(-\sqrt{3}, \sqrt{3}\right)$

$\therefore$ integral values of m are $-1, 0, 1$

$\therefore$ sum = 0

P 6 : (B) 0

Sol. $$\frac{x}{\cos\alpha} = \frac{y}{\cos\left(\alpha - \frac{2\pi}{3}\right)} = \frac{z}{\cos\left(\alpha + \frac{2\pi}{3}\right)} = k(\text{say})$$

Therefore, $x + y + z$

$$= k\left\{\cos\alpha + \cos\left(\alpha - \frac{2\pi}{3}\right) + \cos\left(\alpha + \frac{2\pi}{3}\right)\right\}$$

$$= k\left\{\cos\alpha + 2\cos\alpha\cos\frac{2\pi}{3}\right\}$$

$$= k\left\{\cos\alpha + 2\cos\alpha \times -\frac{1}{2}\right\} = 0$$

P 7 : (B) $\left(\frac{2}{a}\right)^{2/3}$

Sol. $$\cos^2\theta = \frac{a^2-1}{3}, \tan^2\frac{\theta}{2} = \tan^{2/3}\alpha$$

$$\tan^3\frac{\theta}{2} = \tan\alpha \Rightarrow \frac{\sin^3\frac{\theta}{2}}{\sin^3\frac{\theta}{2}} = \frac{\sin\alpha}{\cos\alpha}$$

$$= \frac{\sin^3\frac{\theta}{2}}{\sin\alpha} = \frac{\sin^3\frac{\theta}{2}}{\cos\alpha} = k$$

$$\sin^3\frac{\theta}{2} = k\sin\alpha \qquad (1)$$

$$\cos^3\frac{\theta}{2} = k\cos\alpha \qquad (2)$$

$$k^{2/3}\sin^{2/3}\alpha + k^{2/3}\cos\alpha = 1$$

$$\sin^{2/3}\alpha + \cos^{2/3}\alpha = \frac{1}{k^{2/3}}$$

Squaring and adding (1) and (2)

$$k^2(\sin^2\alpha + \cos^2\alpha) = \sin^6\frac{\theta}{2} + \cos^6\frac{\theta}{2} =$$

$$\left(\sin^2\frac{\theta}{2} + \cos^2\frac{\theta}{2}\right)^3 - 3\sin^2\frac{\theta}{2}\cos^2\frac{\theta}{2}\left(\sin^2\frac{\theta}{2}\cos^2\frac{\theta}{2}\right)$$

$$k^2 = 1 - \frac{3}{4}\sin^2\theta = 1 - \frac{3}{4} + \frac{3}{4} + \cos^2\theta$$

$$k^2 = \frac{a^2}{4} \qquad \Rightarrow k = \frac{a}{2}$$

$$\sin^{2/3}\alpha + \cos^{2/3}\alpha = \left(\frac{2}{a}\right)^{2/3}$$

P 8 : (C) $\frac{\pi}{2}, \frac{3\pi}{8}, \frac{\pi}{8}$

Sol. In ΔABM $\frac{x}{\cos\theta} = \frac{a}{\sin 3\theta}$ — (1)

and in ΔAMC $\frac{a}{\sin\theta} = \frac{x}{\cos 3\theta}$ — (2)

From (1) and (2)

$\sin\theta \cos\theta = \sin 3\theta \cos 3\theta$

$\Rightarrow \sin 2\theta = \sin 6\theta \Rightarrow 8\theta = \pi$

$\Rightarrow \theta = \pi/8$

P 9 : (A) $\frac{3b}{2a}$

Sol. $\alpha + \beta + \gamma = 0$ as coefficient of $x^2 = 0$

Therefore, $\alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma = -3b$

$\alpha^2 + \beta^2 + \gamma^2 = (\Sigma\alpha)^2 - 2\Sigma\alpha\beta = 0 - 2a = -2a$

Thus, $\frac{\Sigma\alpha^3}{\Sigma\alpha^2} = \frac{-3b}{-2a} = \frac{3b}{2a}$

P 10 : (B) 53

Sol. The number of matches in the first round
$= {}^6C_2 + {}^6C_2 = 15 + 15 = 30$

The number of matches in the next round
$= {}^6C_2 = 15$

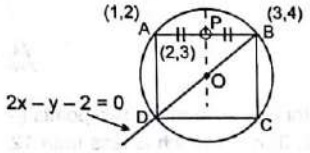
The number of matches in the semi-final round
$= {}^4C_2 = 6$

Therefore, the required number of matches
$= 30 + 15 + 6 + 2 = 53$

($\because$ for best of three at least two matches are played)

P 11 : (B) $\left(\frac{7}{3}, \frac{8}{3}\right)$

Sol. $\because$ OP $\perp$ to given lies.



$\because$ Eq. of OP is $y - 3 = -1\,(x - 2)$

$y - 3 = -x + 2$

$x + y - 5 = 0$ ... (1)

$2x - y - 2 = 0$ ... (2)

$3x = 7 \Rightarrow x = 7/3$ put in (1)

$\frac{7}{3} + y - 5 = 0 \Rightarrow y = 5 - \frac{7}{3} = \frac{8}{3}$

$\therefore \left(\frac{7}{3}, \frac{8}{3}\right)$

P 12 : (A) $3x - 3y - c = 0$

Sol. Centre is $\equiv \left(-\frac{c}{3}, \frac{-2c}{3}\right)$

other assympote is perpendicular to the first

$\therefore \quad x - y - \frac{c}{3} = 0$

$3x - 3y - c = 0$

P 13 : (C) 288

Sol. Since the chairs are numbered, it is equivalent to linear arrangements. However one case will come if persons to be seated together occupied '1' & '6' chair.

Hence No. of ways to arrange in a row if two persons sit together is 5! × 2! (when 2 persons are bundled together.)

Also if '1' & '6' are occupied,

than numbers of ways of sitting is 2! × 4!

Thus total ways 288.

P 14 : (C) 54

Sol. If interior angle = θ then exterior angle = $\pi - \theta$

Sum of all the exterior angles = 2π

$\therefore$ number of sides $= \frac{2\pi}{\pi - \theta} = \frac{360°}{180° - \theta°}$

$= 360°/30° = 12$

and number of diagonals $= \frac{n(n-3)}{2}$

$= 12(12-3)/2 = 54$

P 15 : (A) $u = 0$

Sol. Note that the lines are perpendicular.

Assume the coordinate area to be directed along $u = 0$ & $v = 0$. Now the lines $k_1u - k_2 v = 0$ & $k_1u + k_2v = 0$ are equally inclined with uv axes. Hence the bisectors are $u = 0$ & $v = 0$.

P 16 : (A) $x^2 + y^2 + 2gx + 2fy + k$

Sol. Let P(h, k) be such points whose distance from C(–g, –f) is always at a constant distance a.

$\Rightarrow |PC| = a$

Therefore, $\sqrt{(h+g)^2 + (k+f)^2} = a$

$\Rightarrow (h + g)^2 + (k + f)^2 = a^2$

$\Rightarrow h^2 + k^2 + 2gh + 2fk + g^2 + f^2 = a^2$

$\Rightarrow h^2 + k^2 + 2gh + 2fk + g^2 + f^2 - a^2 = 0$

$\Rightarrow h^2 + k^2 + 2gh + 2fk + k = 0$

$\{k = g^2 + f^2 - a^2\}$

Replace h by x and k by y, the locus of point P is :

$x^2 + y^2 + 2gh + 2fy + k = 0$

Which is the locus of general equation of a circle centered at C(–g, –f) and radius

$r = \sqrt{g^2 + f^2 - k}$

P 17 : (C) $\frac{112}{625}$

Sol. $P = P(3) + P(4)$

$= {}^4C_3 \left(\frac{2}{5}\right)^3 \left(\frac{3}{5}\right) + {}^4C_4 \left(\frac{2}{5}\right)^4$

$= \frac{96 + 16}{625} = \frac{112}{625}$

P 18 : (A) 420

Sol. $p(\text{no girl}) = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$

$\therefore p(\text{at least one girl}) = 1 - \frac{1}{8} = \frac{7}{8}$

$\therefore$ total number of families expected to have at least one girl child $= \frac{7}{8} \times 480 = 420$

P 19 : (A) $\pm\dfrac{(a^2+b^2)^4}{\sqrt{a^2+b^2}}\left[\dfrac{a}{b^8}+\dfrac{b}{a^8}\right]$

Sol. Given, $\tan\theta = a/b$

$$\Rightarrow \cos 2\theta = \frac{1-\tan^2\theta}{1+\tan^2\theta} = \frac{b^2-a^2}{b^2+a^2}$$

$$\sin\theta = \pm\frac{a}{\sqrt{a^2+b^2}};\ \cos\theta = \pm\frac{b}{\sqrt{a^2+b^2}}$$

Therefore, $\dfrac{\sin\theta}{\cos^8\theta}+\dfrac{\cos\theta}{\sin^8\theta}$

$$\frac{\left(\dfrac{a}{\sqrt{a^2+b^2}}\right)}{\left(\dfrac{b}{\sqrt{a^2+b^2}}\right)^8}+\frac{\left(\dfrac{b}{\sqrt{a^2+b^2}}\right)}{\left(\dfrac{a}{\sqrt{a^2+b^2}}\right)^8}$$

$$=\frac{a(a^2+b^2)^4}{b^8(a^2+b^2)^{1/2}}+\frac{b(a^2+b^2)^4}{a^8(a^2+b^2)^{1/2}}$$

$$=\pm\frac{(a^2+b^2)^4}{\sqrt{a^2+b^2}}\left[\frac{a}{b^8}+\frac{b}{a^8}\right]$$

P 20 : (B) $\left(3,\dfrac{5}{3}\right)$

Sol. O(4, 2) . H (1, 1)

$$x = \frac{2\times4+1\times1}{1+2} = 3,\ y = \frac{2\times2+1\times1}{3} = \frac{5}{3}$$

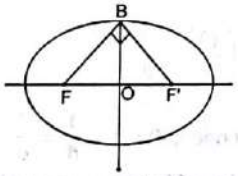
$$\therefore G = \left(3, \frac{5}{3}\right)$$

P 21 : (B) $\dfrac{1}{\sqrt{2}}$

Sol. By the given condition

$b = ae$

$\therefore \quad b^2 = a^2(1-e^2)$

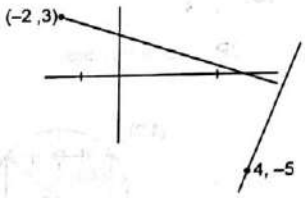


$\Rightarrow \quad a^2e^2 = a^2 - a^2e^2$

$\Rightarrow \quad 2e^2 = 1 \qquad e = \dfrac{1}{\sqrt{2}}$

P 22 : (A) 0

Sol.



Distance between the two points (4, –5) and (– 2, 3) is 10 which is less than 12. Hence no such line can be drawn.

P 23 : (D) hyperbola

Sol. $yy_1 = 2a(x + x_1)$;

$x^2 = 4by = 4b[(2a/y_1)(x + x_1)]$

$\Rightarrow y_1x^2 - 8abx - 8abx_1 = 0$;

$D = 0$ gives $xy = -2ab$

P 24 : (C) 1/2

Sol. The given series is

$$S_n = \frac{1}{1.3}+\frac{1}{3.5}+\frac{1}{5.7}+\ldots.+\frac{1}{(2n-1).(2n+1)}$$

Its n^{th} term, $t_n = \dfrac{1}{(2n-1)\cdot(2n+1)}$

$$=\frac{1}{2}\left[\frac{1}{2n-1}-\frac{1}{2n+1}\right]$$

$$\therefore S_n = \frac{1}{2}\left[\left(1-\frac{1}{3}\right)+\left(\frac{1}{3}-\frac{1}{5}\right)+\left(\frac{1}{5}-\frac{1}{7}\right)+\ldots+\left(\frac{1}{2n-1}-\frac{1}{2n+1}\right)\right]$$

$$=\frac{1}{2}\left(1-\frac{1}{2n+1}\right)$$

$$\therefore \text{Given limit} = \lim_{n\to\infty} S_n = \lim_{n\to\infty}\frac{1}{2}\left(1-\frac{1}{2n+1}\right)$$

$$=\frac{1}{2}\lim_{n\to\infty}\left(1-\frac{1}{2n+1}\right)=\frac{1}{2}(1-0)=\frac{1}{2}$$

P 25 : (D) $\dfrac{-\hat{j}+\hat{k}}{\sqrt{2}}$

Sol. Let $x\hat{i}+y\hat{j}+z\hat{k}$ be a unit vector coplanar with $\hat{i}+\hat{j}+2\hat{k}$ and $\hat{i}+2\hat{j}+\hat{k}$ and also perpendicular to $\hat{i}+\hat{j}+\hat{k}$. Then

$$\begin{vmatrix} x & y & z \\ 1 & 1 & 2 \\ 1 & 2 & 1 \end{vmatrix} = 0$$

Or $-3y + y + z = 0$ ...(1)

And $x + y + z = 0$ ...(2)

Solving the above by cross-product method, we get

$$\frac{x}{0} = \frac{y}{4} = \frac{z}{-4} \text{ or } \frac{x}{0} = \frac{y}{1} = \frac{z}{-1} = \lambda \text{ (Say)}$$

$\Rightarrow \quad x = 0, y = \lambda, z = -\lambda$

As $x\hat{i} + y\hat{j} + z\hat{k}$ is a unit vector, we have

$0 + \lambda^2 + \lambda^2 = 1$

or $\lambda^2 = \frac{1}{2}$ or $\lambda = \pm\frac{1}{2}$

Required vector $= \frac{\hat{j} - \hat{k}}{\sqrt{2}}$ or $\frac{-\hat{j} + \hat{k}}{\sqrt{2}}$.

P 26 : (D) $x = 2n\pi, 2k\pi - \frac{3\pi}{4}$

Sol. Clearly the equation has solution for $\sin x \le 0$.

Now, $-\sin x = \sin\frac{\pi}{8}\cdot\sqrt{(1-\cos x)^2 + \sin^2 x}$. Upon squaring we have,

$$\sin^2 x = \sin^2\frac{\pi}{8}.(1 - 2\cos x + \cos^2 x + \sin^2 x)$$

$$= 2\sin^2\frac{\pi}{8}.(1 - \cos x)$$

$$\text{or } 1 - \cos^2 x = \left(1 - \cos\frac{\pi}{4}\right)(1 - \cos x)$$

$$\text{or } \cos^2 x - \left(1 - \frac{1}{\sqrt{2}}\right)\cos x - \frac{1}{\sqrt{2}} = 0$$

$$\text{or } (\cos x - 1)\left(\cos x + \frac{1}{\sqrt{2}}\right) = 0$$

$$\Rightarrow \cos x = 1 \text{ or } \cos x = -\frac{1}{\sqrt{2}}$$

$$\text{i.e., } x = 2n\pi \text{ or } x = 2k\pi \pm \frac{3\pi}{4}\ ;\ n, k \in I$$

But $\sin x \le 0 \Rightarrow x = 2n\pi$ and $x = 2k\pi - \frac{3\pi}{4}$ are the required solutions.

P 27 : (D) Statement-1 is False, statement-2 is True

Sol. $$\vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} = 2\hat{j} + 2\hat{k}$$

$$\vec{n}_2 \times \vec{n}_3 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 1 & -3 & 3 \end{vmatrix} = -4\hat{j} - 4\hat{k}$$

$$\vec{n}_3 \times \vec{n}_1 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -3 & 3 \\ 1 & -1 & 1 \end{vmatrix} = 2\hat{j} + 2\hat{k}$$

The three lines are respectively parallel to $\vec{n}_1 \times \vec{n}_2$, $\vec{n}_2 \times \vec{n}_3$, $\vec{n}_3 \times \vec{n}_1$. Also $\vec{n}_1 \times \vec{n}_2$, $\vec{n}_2 \times \vec{n}_3$, $\vec{n}_3 \times \vec{n}_1$ are parallel

$\therefore$ Thus the lines are parallel.

$\Rightarrow$ Statement 1 is wrong

P 28 : (B) 1210

Sol. Given that, roots of equation

$x^2 - 10ax - 11b = 0$ are c, d

So $c + d = 10a$ and $cd = -11b$

and a, b are the roots of equation

$x^2 - 10cx - 11d = 0$

$\therefore \quad a + b = 10c, ab = -11d$

So $a + b + c + d = 10(a + c)$

and $(c + d) - (a + b) = 10(a - c)$

$(c - a) - (b - d) + 10(c - a) = 0$

$b + d = 9(a + c)$ ....(i)

$abcd = 121\ bd$

$ac = 121$ ....(ii)

$b - d = 11(c - a)$ ....(iii)

c & a satisfies the equation $x^2 - 10ax - 11b = 0$ and $x^2 - 10cx - 11d = 0$ respectively

$\therefore \quad c^2 - 10ac - 11b = 0$

$a^2 - 10ca - 11\,d = 0$

$(c^2 - a^2) - 11(b - d) = 0$ by eq. (iii)

$(c - a)(c + a) = 11(b - d) = 11.\ 11\ (c - a)$

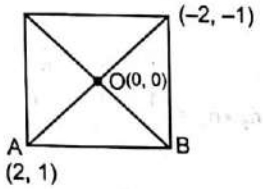
$c + a = 121$

$a + b + c + d = 10(c + a)$

$10.121 = 1210$

P 29 : (A) (1, –2)

Sol. The diagonals are $y = \frac{1}{2}x$ and $y = -2x$



$AO = \sqrt{5}$

The point B lies on the diagonal $y = -2x$

Let the coordinates of B be $(\alpha, -2\alpha)$

$\therefore OB = \alpha^2 + 4\alpha^2 = 5$

$\therefore \alpha^2 = 1$ i.e. $\alpha = \pm 1$

$\therefore$ the reqd. points are (1, – 2) and (– 1, 2)

P 30 : (A) $\frac{a}{b} = \frac{c}{d}$

Sol. The given equation is ,

$ac\, z_1 z_2 + ad\, z_1 \bar{z}_2 + bc\, \bar{z}_1 z_2 + bd\, \bar{z}_1 \bar{z}_2$

$= ac\, z_1 z_2 + ad\, \bar{z}_1 z_2 + bc\, z_1 \bar{z}_2 + bd\, \bar{z}_1 \bar{z}_2$

i.e. $(ad - bc)(z_1 \bar{z}_2 - \bar{z}_1 z_2) = 0$

$ad - bc = 0$ gives $\frac{a}{b} = \frac{c}{d}$.

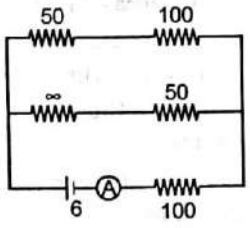
If θ_1 & θ_2 are the arguments of z_1, z_2

$z_1 \bar{z}_2 - \bar{z}_1 z_2 = 0$ gives

$$\frac{|z_1| e^{i\theta_1}}{|z_1| e^{-i\theta_1}} = \frac{|z_2| e^{i\theta_2}}{|z_2| e^{-i\theta_2}} \quad \text{or} \quad \theta_1 = \theta_2$$

P 31 : (B) 0.02 A

Sol. $i = \frac{6}{50 + 150 + 100} = 0.02\ A$



P 32 : (B) $f_A = 4f_B$

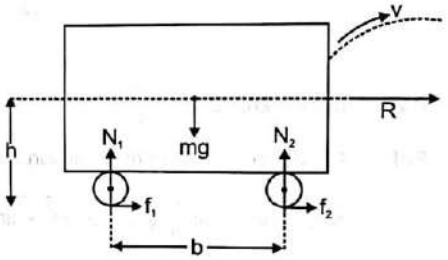
Sol. $f_A = \frac{1}{2\ell}\sqrt{\frac{T}{\rho\pi r^2}}$

$f_B = \frac{1}{4\ell}\sqrt{\frac{2T}{2\rho\pi 4r^2}}$

$\frac{f_A}{f_B} = \frac{4}{2}\sqrt{4} = 4$

P 33 : (C) $\sqrt{\frac{bgr}{2h}}$

Sol. The FBD of the car is as shown in figure (Rear view). The resultant of friction forces acting on inner and outer wheels are providing the necessary centripetal force.



$f_1 + f_2 = \frac{mv^2}{R}$ where m is mass of vehicle

For vertical equilibrium, $N_1 + N_2 = mg$

For rotational equilibrium use $\Sigma\tau = 0$ about centre of mass.

$(f_1 + f_2)h + N_2 \times \frac{b}{2} = N_1 \times \frac{b}{2}$

Solving above equations, we get

$N_1 = \frac{mg}{2} + \frac{mv^2}{r} \times \frac{h}{b}$

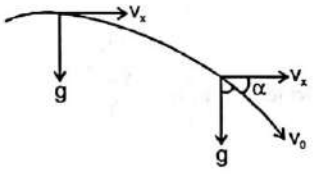
$N_2 = \frac{mg}{2} - \frac{mv^2}{r} \times \frac{h}{b}$

For N_2 to be zero, $\frac{mg}{2} = \frac{mv^2}{r} \times \frac{h}{b} \Rightarrow v = \sqrt{\frac{bgr}{2h}}$

Here as $N_1 < N_2$ it means as v increases N_2 will become zero first and car will tip over outer wheels. We can solve this question in car frame of reference also.

P 34 : (A) $\rho = \frac{\rho_{min}}{\cos^3\alpha}$

Sol. $\frac{v_x^2}{g} \Rightarrow R_{min}$



$$R = \frac{v_0^2}{g\cos\alpha} \Rightarrow \frac{\left(\frac{v_x}{\cos\alpha}\right)^2}{g\cos\alpha} \Rightarrow \frac{v_x^2}{g}\left(\frac{1}{\cos^3\alpha}\right)$$

$$R = \frac{R_{min}}{\cos^3\alpha}$$

P 35 : (B) two points propagating in parallel directions

Sol. Net deviation by parallel slab is zero so both rays emerge parallel to each other, but lateral shift of blue is more in comparison to red, so emerge out from different points.

P 36 : (B) 0.1 mm

Sol. We know that ; $\beta = \frac{D\lambda}{2d}$

Here, $\beta = 5 \times 10^{-3}$ m, $\lambda = 5000 \times 10^{-10}$ m and D = 1m

$\therefore$ Separation between two coherent sources

$$d = \frac{D\lambda}{\beta} = \frac{1\times5000\times10^{-10}}{5\times10^{-3}} = 0.1 \text{ mm}$$

P 37 : (B) the reflected ray will be completely polarised and the transmitted ray will be partially polarised

Sol. Reflected light is completely polarized while transmitted light is partially polarized.

P 38 : (A) 667 Hz

Sol. The apparent frequency heard by the stationary listener is given by :

$$v' = \left(\frac{v}{v+v_s}\right)v = \frac{v}{v+0.2v} \times 800$$

$$= \frac{800}{1.2} = 667 \text{ Hz}$$

P 39 : (A) 24 m

Sol. $s = ut + \frac{1}{2}at^2 \Rightarrow s = 10\times6 + \frac{1}{2}\times(-2)\times6^2$

So, s = 24m

P 40 : (B) 4.1×10^4

Sol. By simple mathematical calculation answer is 41301.2208 but the least significant digit is 2 in the expression (in number 21) so answer cannot have more than 2 significant digits.

So answer is 4.1×10^4

P 41 : (D) a circular disc

Sol. A rolling body will give

$$K.E. = \frac{1}{2}mv^2 + I\frac{1}{2}\left(\frac{V}{R}\right)^2.$$

If it reaches upto height 'h' that means its speed is zero at that height.

So by mechanical energy conservation

$$\frac{1}{2}mv^2 + \frac{1}{2}I\left(\frac{V}{R}\right)^2 = mg\left(\frac{3V^2}{4g}\right)$$

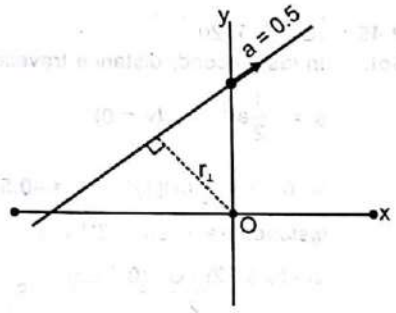
$$\Rightarrow \frac{1}{2}\frac{IV^2}{R^2} = \frac{mv^2}{4} \Rightarrow I = \frac{mR^2}{2}$$

So it would be disc

P 42 : (A) $(0.02)\sqrt{3}(-\hat{k})$

Sol. We know $\vec{L} = \vec{r} \times m\vec{v}$

$\vec{L} = mvr_\perp(-\hat{k})$



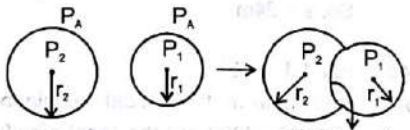
$r_\perp = 2\sqrt{3}$ and v = at = 1m/s

So $\vec{L} = (10\times10^{-3}) \times (2\sqrt{3})(1)(-\hat{k})$

$\vec{L} = 20\sqrt{3}(-\hat{k})\times10^{-3} = (0.02)\sqrt{3}(-\hat{k})$

P 43 : (D) $\frac{r_1r_2}{r_2-r_1}$

Sol.



We know $P_1 - P_A = \frac{4T}{r_1}$ ....(1)

$P_2 - P_A = \frac{4T}{r_2}$ ....(2)

Now when they come in contact, we assumed pressure of gas in any bubble does not change so

$P_1 - P_2 = \frac{4T}{r}$ ....(3)

$\{P_1 > P_2$ and $r_1 < r_2\}$

So by equation (1) and (2)

$P_1 - P_2 = \frac{4T}{r_1} - \frac{4T}{r_2}$

$\Rightarrow \quad \frac{1}{r_1} - \frac{1}{r_2} = \frac{1}{r} \quad \Rightarrow r = \frac{r_1r_2}{r_2-r_1}$

P 44 : (B) 3

Sol. $x = 4\cos^2(50t)\sin(500t)$

$x = 2[2\cos^2(50t)]\sin(500t)$

$x = 2[1 + \cos(100t)]\sin(500t)$

$x = 2\sin(500t) + 2\cos(100t)\sin(500t)$

$x = 2\sin(500t) + \sin(600t) + \sin(400t)$

So it is combination of 3 SHM.

P 45 : (C) –0.5, 20

Sol. In last second, distance travelled :

$s = -\frac{1}{2}at^2 \qquad \{v = 0\}$

so $0.25 = \frac{1}{2}(a)(1)^2 \Rightarrow a = -0.5$ m/s^2

distance travelled in 3rd sec

$\frac{U-(0.5\times2)+U-(0.5\times3)}{2} = S_3$

distance travelled in 8th sec

$\frac{U-(0.5\times7)+U-(0.5\times8)}{2} = S_8$

as $S_3/S_8 = \frac{15}{13} \Rightarrow u = 20$ m/s

P 46 : (C) $\frac{1}{16}$

Sol. Refractive index $n = \frac{v_2}{v_d}$

So for string $n = \frac{\sqrt{\frac{T_r}{\mu_r}}}{\sqrt{\frac{T_d}{\mu_d}}}$

$n = \frac{\sqrt{\frac{1}{9}}}{\sqrt{\frac{1}{25}}} = \frac{5}{3}$ as $\mu\uparrow \Rightarrow v\downarrow$

and tesion is same

So reflection coefficent $= \left(\frac{\frac{5}{3}-1}{\frac{5}{3}+1}\right)^2 = \frac{1}{16}$

P 47 : (C) monatomic

Sol. Sudden expansion can be treated as an adiabatic expansion as there is no sufficent time for heat exchange so

$P_1V_1^\gamma = P_2V_2^\gamma$

$(80)(1000)^\gamma = 60(1190)^\gamma$

$\frac{4}{3} = (1.19)^\gamma$

so $\gamma = 5/3$ that means gas is monoatomic.

P 48 : (D) $\frac{\ell_2\alpha_2 + \ell_2\alpha_1}{\ell_1+\ell_2}$

Sol. We know for composite rod $\ell = \ell_1 + \ell_2$

So $\frac{d\ell}{dT} = \frac{d\ell_1}{dT} + \frac{d\ell_2}{dT}$

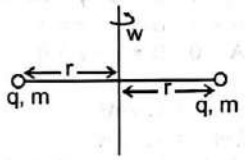
$\Rightarrow \left(\frac{1}{\ell}\frac{d\ell}{dT}\right)\ell = \ell_1\left(\frac{1}{\ell_1}\frac{d\ell_1}{dT}\right) + \ell_2\left(\frac{1}{\ell_2}\frac{d\ell_2}{dT}\right)$

$\alpha_{com}(\ell_1 + \ell_2) = \ell_1\alpha_1 + \ell_2\alpha_2$

$\alpha_{com} = \frac{\ell_1\alpha_1 + \ell_2\alpha_2}{\ell_1+\ell_2}$

P 49 : (B) $\frac{q}{2m}$

Sol. Angular momentum of
Rod $L = mr^2\omega \times 2$



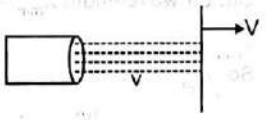
contributed current $i = \frac{2q}{\frac{2\pi}{\omega}} = \frac{q\omega}{\pi}$

Area of loop $= \pi r^2$
So magnetic dipole moment $\mu = iA = q\omega r^2$

So $\frac{\mu}{L} = \frac{q\omega r^2}{2mr^2\omega} = \frac{q}{2m}$

P 50 : (D) $(v - V)^2$

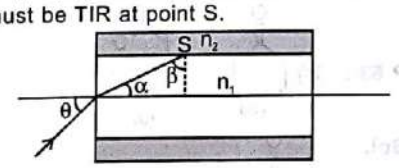
Sol.



The relative velocity of jet $= v - V$
So colliding mass per unit time $= \rho A(v - V)$
So colliding momentum per unit time
$= \rho A(v - V)^2$
This would be the force so force is proportional to $(v - V)^2$

P 51 : (D) $\sin^{-1}\sqrt{n_1^2 - n_2^2}$

Sol. To propogate down the fibre lens, there must be TIR at point S.



So $\beta \geq \theta_c \Rightarrow \beta \geq \sin^{-1}\left(\frac{n_2}{n_1}\right)$

$\Rightarrow 90 - \alpha \geq \sin^{-1}\left(\frac{n_2}{n_1}\right)$

$\Rightarrow \alpha \leq 90 - \sin^{-1}\left(\frac{n_2}{n_1}\right)$

$\Rightarrow \sin^{-1}\left(\frac{1}{n_1}\sin\theta\right) \leq 90 - \sin^{-1}\left(\frac{n_2}{n_1}\right)$

$\Rightarrow \frac{1}{n_1}\sin\theta \leq \sin\left(90 - \sin^{-1}\left(\frac{n_2}{n_1}\right)\right)$

$\Rightarrow \frac{1}{n_1}\sin\theta \leq \cos\left(\sin^{-1}\left(\frac{n_2}{n_1}\right)\right)$

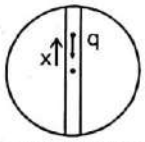
$\Rightarrow \frac{1}{n_1}\sin\theta \leq \frac{\sqrt{n_1^2 - n_2^2}}{n_1}$

$\Rightarrow \sin\theta \leq \sqrt{n_1^2 - n_2^2}$

$\Rightarrow \theta \leq \sin^{-1}\sqrt{n_1^2 - n_2^2}$

P 52 : (D) $\sqrt{\frac{1}{4\pi\epsilon_0}\frac{qQ}{mR^3}}$

Sol. The force at a distance x from centre will be $F = -\frac{1}{4\pi\epsilon_0}\frac{Qxq}{R^3}$



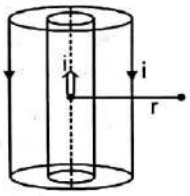
(towards centre)

So motion is SHM

$\Rightarrow \omega = \sqrt{\frac{Qq}{4\pi\epsilon_0 mR^2}}$

P 53 : (D) $\frac{\mu_0}{2\pi}\ln\left(\frac{b}{a}\right)$

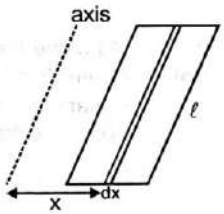
Sol. $B = 0$
$r < a$ and $r > b$



$B = \frac{\mu_0 i}{2\pi r}$

for $a < r < b$
Magnetic flux between r = a and r = b

$\phi = \int_a^b \frac{\mu_0 i}{2\pi x}.\ell dx$



$\phi = \frac{\mu_0 i\ell}{2\pi}\ln\left(\frac{b}{a}\right) = Li$

$\frac{L}{\ell} = \frac{\mu_0}{2\pi}\ln\left(\frac{b}{a}\right)$

Also by energy $\int_a^b \frac{\left(\frac{\mu_0 i}{2\pi x}\right)^2}{2\mu_0}.2\pi x dx.\ell = \frac{1}{2}Li^2$

P 54 : (C) $\left(\frac{C_1}{C_1+C_2}\right)I$

Sol. $C_{eq} = C_1 + C_2$

So $i = \frac{E}{R}e^{-\frac{t}{R(C_1+C_2)}}$

$\Rightarrow i_1 = \frac{C_1}{C_1+C_2}i \quad \left\{\frac{q_1}{C_1} = \frac{q_2}{C_2} \Rightarrow \frac{i_1}{C_1} = \frac{i_2}{C_2}\right\}$

P 55 : (A) $\frac{1}{2R}$

Sol. $E \propto m$

For $e^- \frac{1}{\lambda} = R\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

For $e^- \lambda = \frac{1}{R}$ as $n_1 = 1$ and $n_2 = \infty$

As E depends on mass and hyopothetical e^- is double the mass of other.

$\frac{\lambda_e}{\lambda_{Hp}} = \frac{M_{He}}{Me} \quad \Rightarrow \quad \lambda_{Hp} = \frac{1}{2R}$

P 56 : (B) 3.02×10^{11}

Sol. Given, wavelength of ultraviolet light, $(\lambda) = 150 \text{ nm} = 1.5 \times 10^{-7} \text{ m}$
Intensity of ultraviolet light (I) = 4 W/m^2
Area (A) = 1 cm^2 = 1×10^{-4} m^2
According to question, number of photo electrons will be

$n = \frac{IA}{(hc/\lambda)}\left(\frac{0.1}{100}\right)$ [Here, h = planck constant]

$= \frac{4\times1\times10^{-4}\times1.5\times10^{-7}}{(6.626\times10^{-34})(3\times10^{8})} \times \frac{0.1}{100}$

$= 0.03018 \times 10^{13} = 3.02 \times 10^{11}$

P 57 : (A) move the lens out by about 1 cm

Sol. Given, focal length of lens (f) = 5cm
Distance of object (u) = – 30 cm
We know that,

$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{1}{5} = \frac{1}{v} - \frac{1}{(-30)}$

$\Rightarrow v = \frac{30}{5} = 6 \text{ cm}$

Thus, lens has to be moved out by 1cm as it was focussed for object at infinity so screen will be at 5 cm from the lens.

P 58 : (C) XNOR

Sol. $y = A\cdot B + \bar{A}\cdot\bar{B}$ for XNOR gate
A = 0 ; B = 0 ; y = 1
A = 1 ; B = 0 ; y = 0
A = 1 ; B = 1 ; y = 1
A = 0 ; B = 1 ; y = 0

P 59 : (B) $L = k\rho v^2 A^z$

Sol. Let $L = k\rho^x v^y A^z$
So by dimensional analysis
$[MLT^{-2}] = [ML^{-3}]^x[LT^{-1}]^y [L^2]^z$
$\Rightarrow 1 = x$
$1 = -3x + y + 2z$
$-2 = -y$
$\Rightarrow x = 1, y = 2, z = 1$
So $L = k\rho v^2 A^z$

P 60 : (C) 6.2 eV

Sol. Cut off wavelength $\lambda_{min} = \frac{hc}{eV} = \frac{1240}{12.4}$ nm

$\lambda_{min} = 100$ nm
So, $\lambda_{K\alpha} = 200$ nm

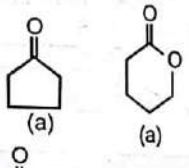
So, $E_2 - E_1 = \frac{hc}{\lambda_{K\alpha}} = \frac{1240}{200} = 6.2$ eV

P 61 : (D) All the above are correct statements.

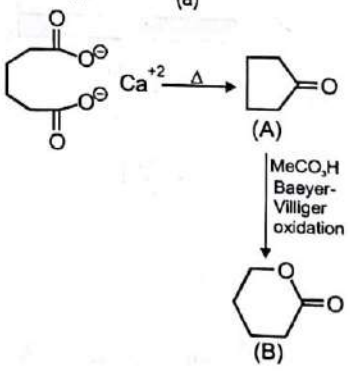
P 62 : (A) The sols will coagulate each other mutually

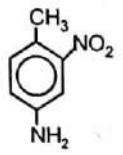
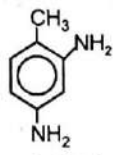
Sol. The sols obtained in the two cases will be oppositely charged so coagulate each other.

P 63 : (B)



Sol.



P 64 : (A) (a)  (b) 

Sol. NH_4HS selectively reduces ($-NO_2$) group *para* to the (Me(EDG)) group, whereas ($SnCl_2$ + HCl) reduces both $-NO_2$ group.

P 65 (C) no two electrons in the same atom can have all four of their quantum numbers the same

Sol. It is a statement of Pauli exclusion principle.

P 66 : (A) 210.0 g mol^{-1}

Sol. For isotonic solutions ;

$\pi_1 = \pi_2$ $\quad \pi = CRT$

$C_1 = C_2$

$$\frac{1.5/60}{V} = \frac{5.25/m}{V}$$

$\therefore \quad m = 210$

P 67 : (A) 2.66 g/cc

Sol. Density crystal decreases due to Schottky defects. Effective no. of atoms per unit cell in given lattice

$$= 4\left(1-\frac{0.25}{100}\right) = 3.99$$

$$p = \frac{50\times 3.99}{6\times 10^{23}(0.50\times 10^{-7})^3}$$

$$= \frac{50\times 3.99}{6\times 125\times 0.1}$$

$p = 2.66\ g/cm^3$

P 68 : (C) 5

Sol. 5 mmol H_2SO_4 would require 10 mmol (40 mL) NH_3 solution. mmol of NH_4^+ formed = 10 in 50 mL

$$[NH_4^+] = \frac{10}{50} = 0.2\ M,\ K_h = \frac{K_w}{K_b} = \frac{10^{-14}}{2\times 10^{-5}}$$

$= 5 \times 10^{-10}$

$[H^+] = \sqrt{K_hC} = \sqrt{5\times 10^{-10}\times 0.2} = 10^{-5}$, pH = 5

P 69 : (B) Step 2

Sol. Step 2 is wrong because Friedel - Crafts reaction will not take place in the presence of e^--withdrawing group ($-COO^-$ and $\overset{\oplus}{N}H_3$) in another ring.

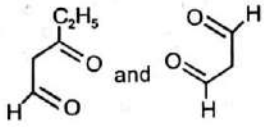
P 70 : (C) 7

Sol. (i) 2 moles of esters and 1 mole of nitrile group will consume 2 mole of RMgX each

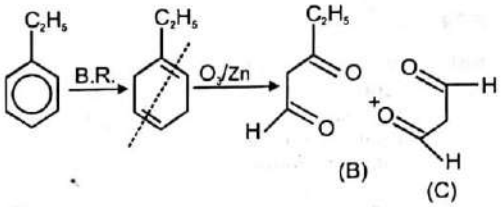
$\therefore$ No. of moles of RMgX = (2×2+2×1) = 6

(ii) 1 mole of ketone will consume 1 mole of RMgX

$\therefore$ (a) $\Rightarrow$ 6 + 1 = 7 moles

P 71 : (A) 

Sol.



P 72 : (D) 1 and 4

Sol. For a successful reaction, colliding system must aquire energy greater than activation energy and collision must occur in proper orientation.

P 73 : (C) At critical point in the Van der Waal's gas isotherm $\left(\frac{\partial P}{\partial V}\right)_{T_c} = 0$

Sol. (A) $\frac{T_c}{P_c} = \frac{\frac{8a}{27Rb}}{\frac{a}{27b^2}} = \frac{8b}{R}$

(B) $T_b = \frac{a}{Rb}$, $T_c = \frac{8a}{27Rb}$

$T_c = \frac{8}{27}T_b$, $T_c < T_b$.

P 74 : (D) 9

Sol. $q = \Delta U + (-W)$

$\Rightarrow \Delta U = q + w = 8 - 3 = 5\ J$

$\Rightarrow q_{ADB} = \Delta U + (-W)\ 5J + 4J = 9J$

P 75 : (B) Nitrogen at the centre of the tetrahedron, three of the vertices being occupied by three hydrogens.

Sol. In $\ddot{N}H_3$ central atom is nitrogen which is sp^3 hybridized. Hence, it will be the centre of tetrahedron with H-atoms at three vertices.

P 76 : (A) Self-reduction and carbon reduction

P 77 : (D) –1.468 V

Sol. $\Delta G_3^\circ = \Delta G_1^\circ + \Delta G_2^\circ$

$\Rightarrow -n_3FE_3^\circ = -n_1FE_1^\circ - n_2FE_2^\circ$

$\Rightarrow -5 \times E_3^\circ = -6\,(-1.45) - (1\times(+1.36))$

$\Rightarrow E_3^\circ = -1.468$ V

P 78 : (A) It exhibits geometrical isomerism

Sol. $[PtCl_2(H_2O)(NH_3)]$; Pt-oxidation state +2 ; Valence configuration = $5d^8$
Complex is diamagnetic and exhibits geometrical isomerism.

P 79 : (D) $\frac{M}{8}$

Sol. Potash alum is a double salt.
Total +ve charge = Total –ve charge
$K_2^{2+}SO_4^{2-}.Al_2^{+6}(SO_4)_3^{-2\times3}24H_2O$
Total +ve charge = 8
Total –ve charge = 8

$\therefore$ Ew = $\frac{M}{8}$

P 80 : (D) 4

P 81 : (C) $NaHCO_3$, Na_2CO_3

Sol. $NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$
(X) (Y)
$\xrightarrow{CO_2}$ $NaHCO_3$

P 82 : (C) $Cr(OH)_3$

Sol. It is a fact.

P 83 : (D) $Be(OH)_2$

Sol. The solubility of most salts decreases with increased atomic weight. But trend is reversed with the fluorides and hydroxides.

BeF_2, $Be(OH)_2$
$Mg(OH)_2$
$Ca(OH)_2$
$Sr(OH)_2$
$Ba(OH)_2$
BaF_2
K_{sp} ↓ decreases

P 84 : (B) 2-Hydroxypropane-1,2,3-tricarboxylic acid

P 85 : (A) $Me - O - \overset{\ominus}{C}H_2$

Sol. (A) In (B), (C) and (D), carbanion is stabilised by resonance, but in (A) it is not stabilised. Moreover, (+I) effect of (Me) group destabilises the carbanion in (A).

P 86 : (B)

P 87 : (D) It dissociates easily and acts as an oxidising agent in chemical reactions.

Sol. H_2O_2 is thermally unstable and it decomposes easily.

$H_2O_2(l) \longrightarrow H_2O(l) + \frac{1}{2}O_2(g)$

Its decomposition is catalysed by alkali metals present in traces in the glass of the vessel.

P 88 : (D) All correct

Sol. M.O. configuration up to $14e^-$

$\sigma1s^2 < \overset{*}{\sigma}1s^2 < \sigma2s^2 < \overset{*}{\sigma}2s^2 < \pi2py^2$
$= \pi2pz^2 < \sigma2px^2 < \overset{*}{\pi}2py^0 = \overset{*}{\pi}2pz^0$
Example :
For N_2, CN^-;
N_2, B. O. = 3.0 N_2^- = B. O. = 2.5
CN^-, B.O. = 3.0 N_2^{2-} b.o. = 2.0
M.O. configuration for more than 14 e^-
$\sigma1s^2 < \overset{*}{\sigma}1s^2 < \sigma2s^2 < \overset{*}{\sigma}2s^2 < \sigma2px^2 < \pi2py^2$
$= \pi2pz^2 < \overset{*}{\pi}2pz^1$
Example :

For O_2, NO :
O_2 B.O. = 2.0
O_2^+ B.O. = 2.5
O_2^{2+} B.O. = 3.0
NO^+, B.O. = 3.0
NO^-, B.O. = 2.0
NO^{2+}, B.O. = 2.5
B. O. of N – O bond in NO_2^+ is 2
$O = \overset{+}{N} = O$

P 89 : (C) The migrating group order is 3° alkyl >Phenyl > H > 2° alkyl > 1° alkyl > Me.

Sol. Migratory aptitude for Baeyer-Villige oxidation is
H > 3°alkyl > 2°alkyl > –phenyl > 1°alkyl > Me

P 90 : (A) Aqueous $NaHCO_3$

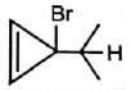
Sol. (A) Phenol is also called carbolic acid. Benzoic acid is more acidic than phenol. So the acid reacts with $NaHCO_3$ to give $CO_2(g)$, while phenol does not react.
(B) Only carbonyl compound gives 2,4-DNP test.
(C) Both acid and phenol react with aqueous NaOH to give their corresponding salts.
$PhCOOH + NaOH \rightarrow PhCOONa$
$PhOH + NaOH \rightarrow PhONa$
(D) Both acid and phenol react with NH_3.
$PhCOOH + NH_3 \rightarrow PhCONH_2$(benzamide) $+ H_2O$
$PhOH + NH_3 \rightarrow PhNH_2$(Aniline) $+ H_2O$

REPLICA 7

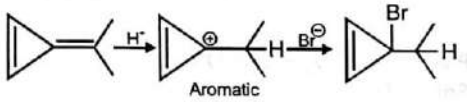
P 1 : (A) Syclic silicate having three O atoms shared by each tetrahedral unit.

Sol. Syclic silicate having two O atoms shared by each tetrahedral unit. So (A) is false.

P 2 : (B) Enantiomeric forms form stronger intramolecular H-bonds than meso form.

P 3 : (B) 

Sol. Involve aromatic carbocation.



P 4 : (A) $CaCO_3 \rightarrow CaO + CO_2$

Sol. (A) Oxidation state of Ca and C are +2 and +4, respectively, in both reactant and product; hence, not redox.

(B) $4e^- + O_2 \longrightarrow 2O^{2-}$ (Reduction)

$H_2 \longrightarrow 2H^{\oplus} + 2e^-$ (Oxidation)

Hence, redox

(C) $Na^- \longrightarrow Na^{\oplus} + e^-$ (Oxidation)

$2H^{\oplus} + 2e^- \longrightarrow H_2$ (Reduction)

Hence, redox

(D) $e^- + Mn^{3+} \longrightarrow Mn^{2+}$ (Reduction)

$2Cl^{\ominus} \longrightarrow Cl_2 + 2e^-$ (Oxidation)

Hence, redox

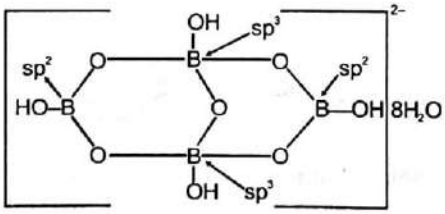
P 5 : (B) 8 : 1

P 6 : (D) Both (A) and (B) are correct

Sol. It is a fact.

P 7 : (D) I, II, III, IV

Sol.



the sp^2 boron can participate in $p\pi$-$p\pi$ back bonding with oxygen atoms and hence the six B-O bonds of two sp^2 boron-oxygen bonds are shorter.

P 8 : (D) $2HI + H_2SO_4 \rightarrow I_2 + SO_2 + 2H_2O$

Sol. H_2SO_4 oxidises iodide to iodine.

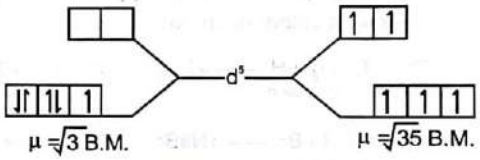
$SO_4^{2-} + 4H^+ + 2I^- \longrightarrow SO_2 + 2H_2O + I_2$

P 9 : (B) $Ln(OH)_3 + H_2.Ln_2O + H_2$

Sol. Lanthanoids react with dilute acids to liberate H_2 gas and they form oxides and hydroxides of the type M_2O_3 and $M(OH)_3$.

P 10 : (D) All the above are correct statements

Sol. There are two possible options.



P 11 : (A) 0.156 V

Sol. $Pb + SO_4^{2-} \rightleftharpoons PbSO_4 + 2e^-$; $E^\circ = 0.356V$

$2H^+ + 2e^- \rightleftharpoons H_2$; $E^\circ = 0$

Adding : $Pb + SO_4^{2-} + 2H^+ \rightleftharpoons PbSO_4 + H_2$

$E^\circ = 0.356$ V

Also, in anode chamber $[SO_4^{2-}]$

$= \sqrt{K_{sp}} = 10^{-4}$ M

$$E = E^\circ - \frac{0.059}{2}\log\frac{1}{10^{-4} \times (0.04)^2}$$

$= 0.356 - 0.20 = 0.156$ V

P 12 : (A) 1

Sol. Dilution does change the pH of a buffer solution.

P 13 : (B) 11.2 kJ

Sol. $\Delta U = C_V \Delta T$

$$\Rightarrow C_V = \frac{8000}{400} = 20 \text{ J} \quad \Rightarrow C_P = C_V + R$$

$= 20 + 8.3 = 28.3$

$\Rightarrow \Delta H = C_P \Delta T = 28.3 \times 400 = 11320$ J

$= 11.32$ kJ

P 14 : (B) $[Ag(CN)_2]^-$ $Ni(CO)_4$ $[Ag(S_2O)_2]^{3-}$

Sol. (I) $Ag_2S + CN^- \longrightarrow [Ag(CN)_2]^-$

(II) $Ni + 4CO \longrightarrow [Ni(CO)_4]$

(III) $AgBr + 2S_2O_3^{2-} \longrightarrow [Ag(S_2O_3)_2]^{3-}$

P 15 : (D) It neither confirms Cl^- nor Br^- unless it is passed through NaOH solution

Sol. When Cl^- salt is heated with $K_2Cr_2O_7$ + conc. H_2SO_4, red vapours of CrO_2Cl_2 are evolved which on passing into NaOH solution gives yellow solution of CrO_4^{2-} ions. Further it gives yellow ppt. of $PbCrO_4$ with $Pb(CH_3CO)_2$ and acetic acid.

$$4Cl^- + \underset{\text{reddish brown vapours}}{Cr_2O_7^-} + 6H^+ \longrightarrow 2CrO_2Cl_2 \uparrow + 3H_2O$$

$$\underset{\text{reddish brown vapours}}{CrO_2Cl_2} + 4OH^- \longrightarrow CrO_4^{--} + 2Cl^- + 2H_2O$$

While on heating a bromide salt with conc. H_2SO_4 and $K_2Cr_2O_7$, instead of CrO_2Br_2, reddish brown fumes of Br_2 are formed due to the reducting character of HBr formed during the reaction. Br_2 dissolves in NaOH and no colour is produced.

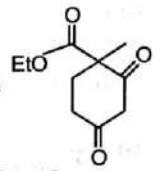
$$5Br^- + Cr_2O_7^- + H^+ \longrightarrow \underset{\text{Reddish brown vapours}}{2Br_2} \uparrow + Cr^{3+} + 7H_2O$$

$$2NaOH + Br_2 \longrightarrow \underset{\text{colourless}}{NaBr} + NaOBr + H_2O$$

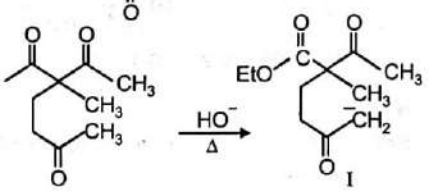
P 16 : (C) $C_6H_5\underset{\underset{CN}{|}}{C}HCH_2\overset{\overset{O}{\|}}{C}CH_3$

Sol. Conjugate addition of CN takes place.

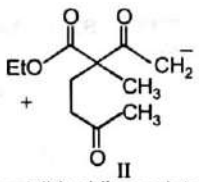
P 17 : (D)



Sol.



Enolate I can give (a) and (b) while enolate II gives (c)



P 18 : (D) Speed of new particle in an orbit will be same as speed of electron.

Sol.

$$r_n = \frac{n^2h^2}{4\pi^2mkZe^2}$$

$$v_n = \frac{2\pi KZe^2}{nh}$$

$$E_n = \frac{2\pi mK^2z^2e^4}{n^2h^2}$$

P 19 : (D) dissolves in water to form a mixture of nitrous and nitric acids

Sol. NO_2 exists as a dimer (N_2O_4). It is mixed anhydride of HNO_3 and HNO_2 and HNO_2

$$N_2O_4 + H_2O \longrightarrow \underset{\text{Nitric acid}}{HNO_3} + \underset{\text{Nitrous acid}}{HNO_2}$$

P 20 : (A) $H_2O_2 > O_3 > O_2$

Sol.

O = O	double bond
H – O – O – H	single bond
O – O → O	between double bond and single bond

$$\text{Bond length} \propto \frac{1}{\text{Number of bonds}}$$

P 21 : (D) $CF_4 < NF_3 < NH_3 < H_2O$

Sol. CF_4 has no dipole moment due to symmetric tetrahedron. NF_3 has less dipole moment then NH_3 due to vectorial addition of bond moments with lone pair moment.

P 22 : (D) small cation and a large anion

P 23 : (B) The second step is the slow step

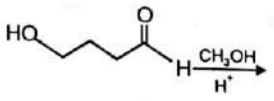
Sol. Assuming second step as rate determining (slowest) step : Rate = $k[Br_2][N_2O_2]$

also $[N_2O_2] = k[NO]^2$

$\Rightarrow$ Rate = $KK[NO]^2[Br_2]$

= $k'[NO]^2[Br_2]$

P 24 : (A)



Sol. Intramolecular reaction in the first step gives hemiacetal which reacts further with CH_3OH to give the desired acetal.

P 25 : (A) 725

Sol. $\Delta T = K_b \times m$

$2 = 0.76 \times m$

$$m = \frac{2}{0.76} = 2.63$$

Mole fraction of solute can be calculated as

$$m = \frac{x_B \times 1000}{(1-x_B)m_A}$$

$$2.63 = \frac{x_B \times 1000}{(1-x_B)\times 18}$$

$x_B = 0.045$

$$\frac{p_0 - p}{p_0} = x_B$$

$$\frac{760 - p}{760} = 0.045$$

or $p = 725.8$ mm

It is close to 725 mm

P 26 : (B) $X_5Y_4Z_8$

Sol. No. of X particles per unit cell = 4

No. of Y particles = 4

No. of Z particles = 8

Along one body diagonal 2X atoms from 2 corners, one Y particle and 2Z particles will be removed.

So, effective no. of X particles in a unit cell

$$= 4 - \frac{1}{8}\times 2 = \frac{15}{4}$$

Effective no. of Y particles in a unit cell

$= 4 - 1 = 3$

Effective no. of Z particles in a unit cell

$= 8 - 2 = 6$

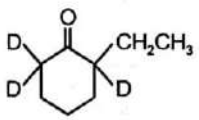
X : Y : Z

X : Y : Z

$\frac{15}{4}$: 3 : 6

5 : 4 : 8

P 27 : (B) Formation of micelles takes place below CMC.

P 28 : (C) 

Sol. All α–H will be replaced by deuterium via enol tautomer.

P 29 : (B) (i) $CH_3-\overset{O}{\overset{\|}{C}}-Cl$ (ii) $Br_2, FeBr_3$

(iii) Zn-Hg/HCl

Sol. Acetylation followed by meta bromination then reduction of keto group.

P 30 : (B) 3

Sol. 3 moles of phenyl hydrazine reacts with one mole of glucose to forms osazone.

P 31 : (C) 2/3

Sol. $PV^{5/3} = C$

$$\frac{T}{V}V^{5/3} = C$$

$$TV^{2/3} = C$$

$$x = \frac{2}{3}$$

P 32 : (B) 90 kW

Sol. $U = \frac{1}{2}CV^2 = \frac{1}{2} \times 40 \times 10^{-6} \times (3000)^2 = 180J$

The power delivered in 2 ms is

$$= \frac{180}{2\times 10^{-3}} = 90kW$$

P 33 : (C) 0.48 kg–m²

Sol. For torsional pendulum, time period is given by

$T = 2\pi\sqrt{\frac{I}{K}}$ where K is torisonal constant of string & it is same for both cases.

$$I_{rod} = \frac{mL^2}{12} = \frac{1\times 1.2^2}{12} = 0.12\ Kg\text{–}m^2$$

$$\frac{T_{rod}}{T_{plate}} = \left(\frac{I_{rod}}{I_{plate}}\right)^{1/2}$$

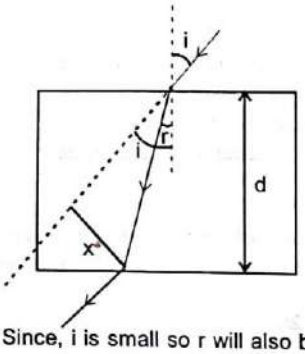
$$\Rightarrow \left(\frac{3}{6}\right)^2 = \frac{0.12}{I_{plate}} \Rightarrow I_{plate} = 0.48\ kg - m^2$$

P 34 : (D) $d\theta\left(\frac{\mu-1}{\mu}\right)$

Sol. Let the lateral displacement be x. Then

$$\frac{x}{\ell} = i - r$$

Also, $\frac{d}{\ell} = \cos r$



Since, i is small so r will also be very small.

$\therefore \quad \frac{d}{\ell} = 1$, i.e., $d \equiv \ell$

$\therefore \quad \frac{x}{d} = i - r$

or $\quad x = di\left[1-\frac{r}{i}\right]$

Also, $\quad \frac{\sin i}{\sin r} = \frac{i}{r} = \mu$

$\therefore \quad s = di\left[1-\frac{1}{\mu}\right]$

Put $\quad i = \theta$

$\therefore \quad x = d\theta\left(\frac{\mu-1}{\mu}\right)$

P 35 : (C) The full image will be formed, but it will be less bright.

Sol. Refracted rays intersect same as before central part is painted, but due to less aperture area intensity of image is less in comparision to before.

P 36 : (D) 84

Sol. $\lambda_1 = 5893$ Å, $n_1 = 62$ $n_2 = 4358$ Å

As field of view in case of both the wavelength is same, hence

$$n_1\beta_1 = n_2\beta_2$$

or $\quad n\left(\frac{D\lambda_1}{d}\right) = n\left(\frac{D\lambda_2}{d}\right)$

or $\quad n_2 = n_1\left(\frac{\lambda_1}{\lambda_2}\right) = 62 \times \frac{5893}{4358} = 84$

P 37 : (C) 90°

Sol. At Brewster's angle, angle between reflected and refracted rays is 90°.

P 38 : (B) greater than the velocity of sound

Sol. Doppler effect will not be applicable when the velocity of sound source is greater than the velocity of sound.

P 39 : (D) 15 ms^{-1}

Sol. The velocity of sound in air, $v_s = 300$ ms^{-1}

As the source is moving towards a stationary person with velocity v, the apparent frequency is given by :

$$n' = \frac{v_s}{v_s - v}n$$

or $\quad n'(v_s - v) = nv_s$

or $\quad 10{,}000(300 - v) = 9500 \times 300$

or $\quad v = \frac{1{,}50{,}000}{10{,}000} = 15$ ms^{-1}

P 40 : (C) 43

Sol. $L = 10 \log_{10}\left(\frac{I}{I_0}\right)$

$\therefore L = 10 \log_{10}\left(\frac{2\times10^{-8}}{10^{-12}}\right)$ $(\because I_0 = 10^{-12}$ W/v$^{-2})$

$L = 10 \log_{10}(2 \times 10^4)$

$L = 10[\log_{10} 2 + \log_{10}(10)^4]$

$L = 10\ (4.3)$

$L = 43$ decibel

P 41 : (A) 4.26 R

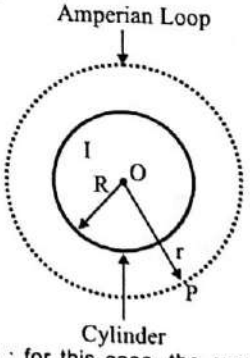
Sol. If rocket is fired with $0.9\,v_e$ then, $v_0 = 0.9\sqrt{2gR}$

For maximum height, $v = 0$

So, $\quad 0.9^2 \times 2gR = \frac{2gh}{1+\frac{h}{R}} \Rightarrow h = 4.26R.$

P 42 : (A) $\frac{\mu_0 I}{2\pi r}$

Sol. Figure shows the cross-sectional view of the cylinder.



$r > R$: for this case, the amperian loop is a circle of radius r concentric with the

cross-section figure. For this loop, L = 2πr and the current threading the loop is i = I. From Ampere's circuital law,

$$BL = \mu_0 i \Rightarrow B \times 2\pi r = \mu_0 I \Rightarrow B = \frac{\mu_0 I}{2\pi r}.$$

P 43 : (A) $\sqrt{\frac{2mc^2}{E}}$

Sol. For an electron with kinetic energy E :

$$\lambda = \frac{h}{\sqrt{2mE}}$$

and for a photon : $\lambda = \frac{hc}{E}$

$$\therefore \text{ Ratio} = \frac{hc}{E}.\frac{\sqrt{2mE}}{h} = \sqrt{\frac{2mc^2}{E}}$$

P 44 : (C) 30° or 150°

Sol. If the angle between $\vec{A}$ and $\vec{B}$ is θ, the cross product will have a magnitude

$$|\vec{A}\times\vec{B}| = AB\sin\theta$$

or $15 = 5 \times \sin\theta \Rightarrow \sin\theta = \frac{1}{2}$

$\Rightarrow$ $\theta = 30°$ or, 150°.

P 45 : (D) 5

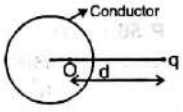
Sol. Let h_e and h_m be the maximum heights attained by the projectile when projected from earth and moon respectively. Now

$$h_e = \frac{u^2\sin^2\theta}{2g_e} \text{ which gives } \frac{h_m}{h_e} = \frac{g_e}{g_m} = 6.$$

P 46 : (B) $\frac{1}{4\pi\epsilon_0}\frac{q}{d^2}$ directed towards charge q

Sol. Net field at O will be zero as it is in the cavity of conductor for that charge will be distributed on the surface of conductor

So $\vec{E}_{\text{induced charge}} + \vec{E}_q = 0$



$$\vec{E}_{\text{induced charge}} = -\vec{E}_q$$

$$\vec{E}_{\text{induced charge}} = \frac{1}{4\pi\epsilon_0}\frac{q}{d^2} \text{ directed towards the charge q}$$

P 47 : (C) Y will decay at larger rate than X

Sol. $T_{1/2} = \frac{\ln 2}{\lambda}$

$$T_{1/2} = \frac{\ln 2}{\lambda_x} \text{ and Mean life } = \frac{1}{\lambda}$$

$$\frac{\ln z}{\lambda_x} = \frac{1}{\lambda_y}$$

$$\frac{dN}{dt} = \lambda N$$

$$\frac{R_1}{R_2} = \frac{\lambda_1}{\lambda_2} \Rightarrow \frac{\lambda_x}{\lambda_y} = \ln 2$$

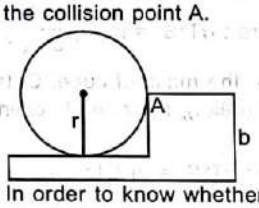
$$\therefore R_y > R_x$$

P 48 : (C) Only S-1 and S-3

Sol. Focal length of objective is large to increase sharpness of image.
Aperture of objective is large to increase intensity of image (because objects are at very large distance)

P 49 : (A) $\frac{2r\sqrt{gb}}{(2r-b)}$

Sol. After the collision, the hoop rotates around the collision point A.



In order to know whether the hoop can roll up the step or not we have to know the angular velocity immediately after the collision.

Angular momentum w.r.t. A before collision

$$L_1 = I_C\omega_1 + m(r-b)v$$

Where I_C = moment of inertia about the centre of hoop.

Angular momentum after collision w.r.t A

$$L_2 = I_C W_2 + mrv_2$$

$$L_2 = (I_C + mr^2)w_2, \text{ putting } I_C = mt$$

$$L_1 = L_2$$

$$\omega_2 = (2r - b)\, V(2r^2)$$

and kinetic energy after collision.

$$\frac{1}{2}(I_C + mr^2)\omega_2^2 \geq mgb$$

$$v \geq \frac{2r\sqrt{gb}}{(2r-b)}, \quad v_{min} = \frac{2r\sqrt{gb}}{(2r-b)}$$

P 50 : (B) $|w_{H.C}| > |w_{H.S}| > |w_{S.C}| > |w_{S.S}|$

Sol. Requirement of friction coefficient for rolling on inclined plane

$$\mu = \frac{\tan\theta}{1+\frac{mr^2}{I}}$$

So for (S.S.)(H.S.)(S.C.) and (H.C.) required μ is

$\frac{2}{7}\tan\theta$, $\frac{2}{5}\tan\theta$, $\frac{1}{3}\tan\theta$, $\frac{1}{2}\tan\theta$ respectively

as $\tan\theta = 1$, $\mu = 0.3$ so (S.S.) will roll and all others will slip.

$(w_{S.S.}) = 0$

In all other cases friction will be acting with same amount but slipping will be more in that object which has more moment of inertia.

$|w_{H.C.}| > |w_{H.S.}| > |w_{S.C.}|$

P 51 : (A) $\sqrt{\frac{\pi}{6}} : 1$

Sol. Rate of emission of energy = $\sigma T^4 S$

Let m_1 be the mass of the sphere, C its specific heat and $(d\theta/dt)_s$ is the rate of cooling.

For sphere : $\sigma T^4 S = m_1 C\left(\frac{d\theta}{dt}\right)_s$ ...(i)

Let m_2 be the mass of cube, C its specific heat and $(d\theta/dt)_c$ the rate of cooling.

For cube : $\sigma T^4 S = m_2 C\left(\frac{d\theta}{dt}\right)_c$ ...(ii)

From equations (i) and (ii),

$$\frac{(d\theta/dt)_s}{(d\theta/dt)_c} = \frac{m_2}{m_1} = \frac{R_s}{R_c}$$

or $$\frac{a^3\rho}{(4/3)\pi r^3\rho} = \frac{R_s}{R_c}$$

Where a is the side of cube and r is the radius of sphere, ρ is the density.

$$\therefore \quad \frac{R_s}{R_c} = \frac{3a^3}{4\pi r^3}$$

But since S is the same, so

$6a^2 = 4\pi r^2$

or $a^2 = (2/3)pr^2$

$$\therefore \frac{R_s}{R_c} = \frac{3(2\pi r^2/3)^{3/2}}{4\pi r^3} = \frac{2\pi\sqrt{2\pi}}{\sqrt{3}(4\pi)} = \sqrt{\frac{2\pi}{12}} = \sqrt{\frac{\pi}{6}}$$

P 52 : (B) r is decreasing if $\Delta\lambda$ is – ive

Sol. $\Delta\lambda = \lambda_{observed} - \lambda_{actual}$

$\lambda_{observed} < \lambda_{actual}$ ($\Delta\lambda$ = –ive) :
If distance between star and earth decreases.

$\lambda_{observed} > \lambda_{actual}$ ($\Delta\lambda$ = +ive) :
If distance between star and earth increases.

P 53 : (A) r is increasing with increasing rate

Sol. $\Delta\lambda$ is +ive $\Rightarrow$ r increases and

$\lambda_{observed}$ increases as star move away with increasing speed $\Rightarrow \frac{d}{dt}(\Delta\lambda)$ = +ive

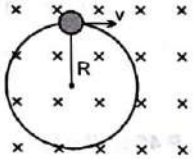
P 54 : (A) 1 Å

Sol. $\lambda' = \frac{C - V_s}{f}$ and $\lambda = \frac{C}{f}$

$$\Delta\lambda = \frac{-V_s}{f} = \frac{-10^5}{10^{15}} = -1\text{Å}$$

P 55 : (D) $B = \frac{\mu_0}{4\pi}\frac{e^3B^2}{m^2v}$

Sol. the radius of circle,



$$R = \frac{mv}{eB}$$

time period = $\frac{2\pi m}{eB}$

So effective current, $i = \frac{e^2B}{2\pi m}$

So, magnetic field at centre due to e $= \frac{\mu_0 i}{2R}$

$$Be = \frac{\mu_0 eB}{2mv}\frac{e^2B}{2\pi m} = \frac{\mu_0}{4\pi}\frac{e^3B^2}{m^2v}$$

and its direction is out of page

So, $B_{nett} = B - \frac{\mu_0}{4\pi}\frac{e^3B^2}{m^2v}$

P 56 : (D) $\ell_1 = \ell_2 < \ell_3$

Sol. If resistivity is uniform then position of null point does not depend on its value so $\ell_1 = \ell_2$.

If left half has resistivity ρ and right half has resistivity 2ρ then to get some ratio null point has to be shifted towards right $\ell_3 = \ell_2$.

P 57 : (A) $\frac{V^2.L}{2R^2}e^{-2}$

Sol. Transient decay current of R-L circuit is given as

$$I(t) = I_0e^{-t/\tau}$$

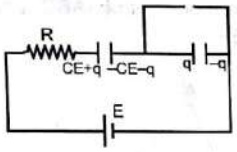
where $I_0 = \frac{V}{R}$ and $\tau = \frac{L}{R}$

Magnetic energy stored in L at $t = \tau$ is given as

$$W_B(\tau) = \frac{1}{2}LI^2(\tau) = \frac{V^2L}{2R^2}e^{-2}$$

P 58 : (A) zero

Sol. Initially C_1 has CE charge and C_2 has zero charge.



Let q charge flow after switch is open

So $\frac{CE+q}{C}+\frac{q}{C}=E \Rightarrow q = 0$

So no. charge flows, so no heat losses

P 59 : (A) 3°

Sol. $\delta_1 = \delta_2$

$(1.54 - 1)4 = (1.75 - 1)A$

$\frac{0.54\times4}{0.75} = A \Rightarrow A = 3°$

P 60 : (B) $1/n^5$

Sol. $B = \frac{\mu_0}{4\pi}\frac{qv\times r}{r^3}$

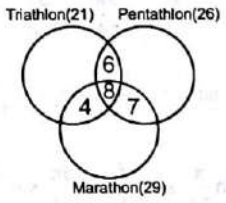
$$\propto \frac{V}{r^2} \quad ; \quad \propto \frac{(Z/n)}{(n^2/z)^2} \quad ; \quad \propto \frac{z^3}{n^5}$$

P 61 : (A) 10

Sol. Since there are 14 players who are in triathlon and pentathlon, and there are 8 who take part in all three games, there will be 6 who take part in only triathlon and pentathlon. Similarly,

Only triathlon and marathon = 12 – 8 = 4 &

Only pentathlon and Marathon = 15 – 8 = 7.



The figure above can be completed with values for each sport (only) plugged in.

P 62 : (A) 45°

Sol. A vector normal to the plane containing vectors $\hat{i}$ and $\hat{i}-\hat{j}$ is

$$\vec{p} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{vmatrix} = \hat{k}$$

A vector normal to the plane containing vectors $\hat{i}-\hat{j}, \hat{i}+\hat{k}$ is

$$\vec{q} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 0 \\ 1 & 0 & 1 \end{vmatrix} = -\hat{i}-\hat{j}+\hat{k}$$

Vector $\vec{a}$ is parallel to vector $\vec{p}\times\vec{q}$.

$$\vec{p}\times\vec{q} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 1 \\ -1 & -1 & 1 \end{vmatrix} = \hat{i}-\hat{j}$$

Therefore, a vector in direction of $\vec{a}$ is $\hat{i}-\hat{j}$.

Now, if θ is the angle between $\vec{a}$ and $\hat{i}-2\hat{j}+2\hat{k}$, then

$$\cos\theta = \pm\frac{1.1+(-1).(-2)}{\sqrt{1+1}\sqrt{1+4+4}} = \pm\frac{3}{\sqrt{2}.3}$$

$$= \pm\frac{1}{\sqrt{2}} \Rightarrow \theta = \frac{\pi}{4} \text{ or } \frac{3\pi}{4}.$$

P 63 : (D) $\frac{1}{64}$

Sol. $\sin\frac{\pi}{14}.\sin\frac{3\pi}{14}.\frac{\sin5\pi}{14}.\sin\frac{7\pi}{14}.\sin\frac{9\pi}{14}.\sin\frac{11\pi}{14}.$

$$\Rightarrow \sin\frac{13\pi}{14} = \sin\frac{\pi}{14}.\sin\frac{3\pi}{14}.\sin\frac{5\pi}{14}\times 1\times$$

$$\sin\left(\pi-\frac{5\pi}{14}\right).\sin\left(\pi-\frac{3\pi}{14}\right).\sin\left(\pi-\frac{\pi}{14}\right)$$

$$=\left[\sin\frac{\pi}{14}.\sin\frac{3\pi}{14}.\sin\frac{5\pi}{14}.\sin\frac{7\pi}{14}\right]^2$$

$$=\frac{1}{64}$$

P 64 : (B) $\dfrac{(a-d)(c-a)}{(b-c)(d-b)}$

Sol. $a\sin^2x + b\cos^2x = c$

$\Rightarrow a\tan^2x + B = c\sec^2x$

$\Rightarrow a\tan^2x + b = c(1+\tan^2x)$

Therefore, $\tan^2 x = \left(\dfrac{c-b}{a-c}\right)$.

Similarly, $\tan^2 y = \dfrac{d-a}{b-d}$

Therefore, $\dfrac{a^2}{b^2} = \dfrac{\tan^2 y}{\tan^2 x} = \dfrac{\left(\frac{d-a}{b-}\right)}{\left(\frac{c-}{a-c}\right)}$

$$= \frac{(d-a)(a-c)}{(b-d)(c-b)} = \frac{(a-d)(c-a)}{(b-c)(d-b)}$$

P 65 : (B) $e-\dfrac{e^2}{2}-\dfrac{3}{2}$.

Sol. $\displaystyle\int_0^1 f(x)g(x)dx = \int_0^1 e^x(x^2-e^x)dx$

$$= \int_0^1 x^2e^x dx - \int_0^1 e^{2x}dx \quad \left[\because f(x) = e^x \text{ satisfies } f'(x) = f(x) \text{ and } f(0) = 1\right]$$

$$= [x^2e^x]_0^1 - \int_0^1 (2x)e^x dx - \left[\frac{e^{2x}}{2}\right]_0^1$$

$$= e - 2\int_0^1 xe^x dx - \frac{1}{2}[e^2-1]$$

$$= e - 2\left[[xe^x]_0^1 - \int_0^1 e^x dx\right] - \frac{1}{2}e^2 + \frac{1}{2}$$

$$= e - 2[e-(e-1)] - \frac{1}{2}e^2 + \frac{1}{2}$$

$$= e - 2 - \frac{1}{2}e^2 + \frac{1}{2} = e - \frac{e^2}{2} - \frac{3}{2}$$

P 66 : (B) circle

Sol. Since, $a\cos\theta = a - x$

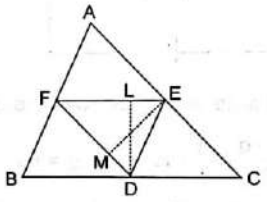
$a\sin\theta = y$

Squaring and adding, we have

$a^2 = (a-x)^2 + y^2 \Rightarrow (x-a)^2 + y^2 = a^2$

P 67 : (C) orthocenter of ΔABC is the circumcenter of ΔDEF

Sol.



(i) Area (ΔABC) = 4 area (ΔDEF)

(ii) Let DL and EM be altitudes of ΔDEF. Then DL and EM are perpendicular bisectors of the sides of ΔABC. Hence orthocenter of DEF is the circumcentre of ΔABC.

P 68 : (C) $0 < \lambda < 4$

Sol. $2s(2s-2a) = \lambda bc$

i.e., $4\dfrac{s(s-a)}{bc} = \lambda$ i.e. $\sin^2\dfrac{A}{2} = \dfrac{\lambda}{4}$

$\therefore \quad 0 < \dfrac{\lambda}{4} < 1$ i.e. $0 < \lambda < 4$

P 69 : (B) $-2 < x < 1$

Sol. If $y = |2x+1|$, then we have

$y^2 - y - 6 < 0$

$(y-3)(y+2) < 0$

$\Rightarrow \quad -2 < y < 3$

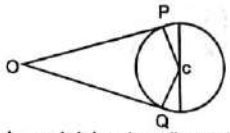
$\Rightarrow \quad |2x+1| < 3$

($\because |2x+1| > -2$ is always true)

$\Rightarrow \quad -2 < x < 1$

P 70 : (B) $x^2 + y^2 + gx + fy = 0$

Sol. Let C is the centre and $\angle OPC = \angle OQC = 90^\circ$ hence the circumcircle of ΔOPQ will pass through C also and OC will be a diameter of circumcircle and hence the equation of circumcircle is



$(x + g)(x) + (y + f)y = 0$

i.e. $x^2 + y^2 + gx + fy = 0$

P 71 : (C) (1, 2)

Sol. A(– 2, 0), B(1, 3) $P(t^2, 2t)$

$$\text{Area} = \frac{1}{2}\begin{vmatrix} -2 & 0 & 1 \\ 1 & 3 & 1 \\ t^2 & 2t & 1 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} -2 & 0 & 1 \\ 3 & 3 & 0 \\ t^2+2 & 2t & 0 \end{vmatrix}$$

$$= \frac{1}{2}|6t - 3t^2 - 6|$$

$$= \frac{1}{2}|3t^2 - 6t + 6|$$

$$= \frac{3}{2}|t^2 - 2t + 1 + 1|$$

$$= \frac{3}{2}|(t-1)^2 + 1|$$

$\therefore$ $t = 1$

$\therefore$ the point is (1, 2)

P 72 : (B) –2

Sol. $\frac{dy}{dx} = \frac{ax+3}{2y+1}$

i.e., $(2y + 1)\,dy = (ax + 3)\,dx$

$\therefore$ $y^2 + y = \frac{ax^2}{2} + 3x + c$

$\therefore$ $a = -2$

P 73 : (A) $y = \frac{x^3}{6} + c$

Sol. $xy = 2 \Rightarrow y = \frac{2}{x} \Rightarrow \frac{dy}{dx} = -\frac{2}{x^2}$

orthogonal trajectory is

$$-\frac{dx}{dy} = -\frac{2}{x^2} \Rightarrow x^2dx = 2dy$$

$$\Rightarrow \frac{x^3}{3} = 2y + c \Rightarrow y = \frac{x^3}{6} - c$$

P 74 : (A) f(x) is nowhere continuous

Sol. f(x) is continuous if $\sin x = 1 - 2\cos x$, which has infinite solutions

$$f\left(\frac{\pi}{6}\right) = f\left(\frac{-\pi}{6}\right) = 1 - 2\cos\frac{\pi}{6},$$

thus f is not injective.

P 75 : (B) 2

Sol. eccentricity of ellipse $\frac{x^2}{4} + \frac{y^2}{1} = 1$ is $e = \sqrt{\frac{3}{4}}$

eccentricity of the ellipse $\frac{x^2}{16} + \frac{y^2}{\lambda^2} = 1$ is

$\sqrt{\frac{16-\lambda^2}{16}}$

$\therefore$ $\frac{16-\lambda^2}{16} = \frac{3}{4}$ $\lambda^2 = 4$ $\lambda = 2$

or eccentricity of the ellipse $\frac{x^2}{16} + \frac{y^2}{\lambda^2} = 1$

is $\sqrt{\frac{\lambda^2-16}{\lambda^2}}$

$\therefore$ $\frac{\lambda^2-16}{\lambda^2} = \frac{3}{4}$ i.e. $4\lambda^2 - 64 = 3\lambda^2$

i.e. $\lambda^2 = 64$ $\lambda = 8$

P 76 : (B) 2

Sol. Since $\frac{e}{2}$ and $\frac{e'}{2}$ are eccentricities of a hyperbola and its conjugate

$\therefore$ $\frac{4}{e^2} + \frac{4}{e'^2} = 1$

i.e. $4 = \frac{e^2e'^2}{e'^2 + e'^2}$

line passing through the points (e, 0) and (0, e')

$e'x + ey - ee' = 0$

it is tangent to the circle $x^2 + y^2 = r^2$

$\therefore$ $\frac{ee'}{\sqrt{e^2 + e'^2}} = r$

$\therefore$ $r^2 = \frac{e^2e'^2}{e^2 + e'^2} = 4$

$\therefore$ $r = 2$

P 77 : (B) $\frac{^{15}C_3 \cdot 20}{4}$

Sol. Select say the vertex 1 of polygon. Now 2 & 20 cannot be taken. The vertices remaining are, 3, 4, 5,....., 17, 18, 19. 3 more are to be taken which are say o o o. Now we have 14 crosses left. x x x x..... x x. Then there are 15 gap and 3 can be taken in $^{15}C_3$ ways. The total ways = 20 x $^{15}C_3$, but the same quad. will appear 4 times. Hence the required answer is $\frac{^{15}C_3 \cdot 20}{4}$

P 78 : (B) $\left(\frac{-2}{5}, \frac{-13}{10}\right), \left(0, \frac{3}{2}\right)$ and $\left(\frac{-8}{5}, \frac{3}{10}\right)$

Sol. Let $L_1 \equiv 3x-4y+1 = 0$ and $L_2 \equiv 8x+6y+1 = 0$
In diagram A, B, C and D are four points which lie at a unit distance from the two lines. You can also observe that A, B, C and D lie on angle bisectors of L_1 and L_2.
Let (h, k) be the coordinates of a point of unit distance from each of the given lines.

$$\Rightarrow \frac{|3h-4k+1|}{\sqrt{3^2+4^2}} \text{ and } \frac{|8h+6k+1|}{\sqrt{8^2+6^2}}$$

$\Rightarrow 3h - 4k + 1 = \pm 5$ and $8h + 6k + 1 = \pm 10$

$\Rightarrow$ $3h - 4k - 4 = 0$ .....(i)
$3h - 4k + 6 = 0$ .....(ii)
$8h + 6k - 9 = 0$ .....(iii)
and $8h + 6k + 11 = 0$ .....(iv)

Solve (i) and (iii) to get : $(h, k) \equiv \left(\frac{6}{5}, \frac{-1}{10}\right)$

Solve (i) and (iv) to get : $(h, k) \equiv \left(\frac{-2}{5}, \frac{-13}{10}\right)$

Solve (ii) and (iii) to get : $(h, k) \equiv \left(0, \frac{3}{2}\right)$

Solve (ii) and (iv) to get : $(h, k) \equiv \left(\frac{-8}{5}, \frac{3}{10}\right)$

Hence the required four points are $\left(\frac{6}{5}, \frac{-1}{10}\right)$, $\left(\frac{-2}{5}, \frac{-13}{10}\right), \left(0, \frac{3}{2}\right)$ and $\left(\frac{-8}{5}, \frac{3}{10}\right)$

P 79 : (A) $\frac{3}{8}$

Sol. The event 'exactly one die shows an even number and sum of the other two is even' is the same as the event 'exactly one die shows an even number'.
Hence the required probability

$$= {}^3C_1 \left(\frac{1}{2}\right)\left(\frac{1}{2}\right)^2 = \frac{3}{8}$$

P 80 : (A) 1/72

Sol. Prob. that four balls are placed in their own boxes

$$= {}^7C_4 \times \frac{1}{7} \times \frac{1}{6} \times \frac{1}{5} \times \frac{1}{4} = \frac{7!}{4!3!} \times \frac{3!}{7!}$$

$$= \frac{1}{4!} = \frac{1}{24}$$

Let E be the event take none of the remaining 3 balls to go into their correct boxes

$$\therefore \quad f(E) = 1 - \left(1 - \frac{1}{2!} + \frac{1}{3!}\right) = \frac{1}{2!} - \frac{1}{3!}$$

$$= \frac{1}{2} - \frac{1}{6} = \frac{1}{3}$$

$$\therefore \quad \text{the reqd. point} = \frac{1}{24} \times \frac{1}{3} = \frac{1}{72}$$

P 81 : (C) $[2, \infty)$

Sol. We have $\log_{0.04}(x-1) \geq \log_{0.02}(x-1)$

$$\Rightarrow \log_{(0.2)^2}(x-1) \geq \log_{0.02}(x-1)$$

$$\Rightarrow \frac{1}{2}\log_{0.2}(x-1) \geq \log_{0.02}(x-1)$$

$\Rightarrow \log_{0.2}(x-1) \geq \log_{0.02}(x-1)$
$\Rightarrow \log_{0.2}(x-1) \leq 0$
$\Rightarrow (x-1) \geq (0.2)^0 \Rightarrow x \geq 2 \Rightarrow x \in [2, \infty)$

P 82 : (A) $\frac{1}{2}\left(\sqrt{5}-1\right)$

Sol. We have $a = ar + ar^2$

$$\Rightarrow r^2 + r - 1 = 0 \Rightarrow r = \frac{-1 \pm \sqrt{1+4}}{2}$$

For terms to be positive, we have

$$\Rightarrow r = \frac{\sqrt{5}-1}{2}$$

P 83 : (D) None of these

Sol. $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 = 10$

$x_1 \geq 3,\ x_2, x_3, \ldots, x_7 \leq 7$

$\therefore$ the corresponding multinomial is

$(x^3 + x^4 + \ldots + x^9)(1 + x + x^2, \ldots, x^7)^6$

Coefficient x^{10} in

$x^3 (1 + x + x^2 + \ldots + x^6)(1 + x + x^2 + \ldots + x^7)^6$

$$= \text{coefficient of } x^7 \text{ in } \left(\frac{1-x^7}{1-x}\right)\left(\frac{1-x^8}{1-x}\right)^6$$

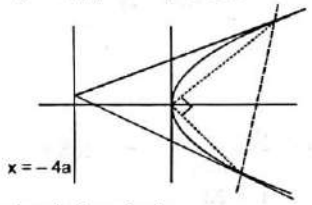
$= $ coefficient of x^7 in $(1 - x^7)(1 - x)^{-7}$

$= 1 \times {}^{13}C_7 - 1 \quad = {}^{13}C_6 - 1$

$\therefore$ The number of solutions is ${}^{13}C_6 - 2$

P 84 : (A) 90°

Sol. $P(-4a, \beta) \qquad \beta \in$ real



chord of contact

$y.\beta. = 2a(x - 4a)$

$2ax - \beta y - 8a^2 = 0$

Homogenizing

$$y^2 = 4ax\frac{(2ax - \beta y)}{8a^2}$$

$2ay^2 = 2ax^2 - \beta xy$

$2ax^2 - \beta xy - 2ay^2 = 0$

clearly coeff. of x^2 + coeff. of $y^2 = 0$ for all values of β

P 85 : (A) $\{-1, 1\}$

Sol. Equation of focal chord

$y = m(x - 4)$

$mx - y - 4m = 0$

this is the tangent to $(x - 6)^2 + y^2 = 2$

$$\therefore \left|\frac{6m - 4m}{\sqrt{m^2 + 1}}\right| = \sqrt{2}$$

i.e. $4m^2 = 2(m^2 + 1) \qquad m = \pm 1$

P 86 : (A) $\frac{n\pi}{4}$ or $n\pi \pm \frac{\pi}{3}$

Sol. $(\sin 6\theta + \sin 2\theta) + \sin 4\theta = 0$

$\Rightarrow 2\sin 4\theta \cos 2\theta + \sin 4\theta = 0$

$\Rightarrow \sin 4\theta(2\cos 2\theta + 1) = 0$

$\Rightarrow \sin 4\theta = 0$ or $2\cos 2\theta + 1 = 0$

$$\Rightarrow \sin 4\theta = \sin 0 \text{ or } \cos 2\theta = \cos\frac{2\pi}{3}$$

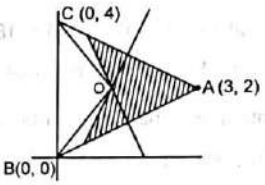
$$\Rightarrow 4\theta = n\pi \text{ or } 2\theta = 2n\pi \pm \frac{2\pi}{3}$$

$$\Rightarrow \theta = \frac{n\pi}{4} \text{ or } \theta = n\pi \pm \frac{\pi}{3}.$$

P 87 : (C) 3

Sol. In the figure O is circumcentre and shaded region shows location of P.

Distance of P from side BC is maximum if P is at A.



P 88 : (C) $\frac{13}{\sqrt{75}}$

Sol. The equation of plane passing through $(-1, -2, -1)$

and normal $\perp$ to L_1 to L_2 i.e. parallel to $-i - 7j + 5k$ is

$a(x + 1) + b(y + 2) + c(z + 1) = 0$

or, $-(x + 1) - 7(y + 2) + 5(z + 1) = 0$

i.e. $x + 7y - 5z + 10 = 0$

Distance of (1, 1, 1) from this plane is

$$\left|\frac{1 + 7 - 5 + 10}{5\sqrt{3}}\right| = \frac{13}{\sqrt{75}}.$$

P 89 : (C) $a > 1$

Sol. $x^2 + (a - b)x + (1 - a - b) = 0$

$\because D > 0$

$\Rightarrow (a - b)^2 - 4 \times 1 \times (1 - a - b) > 0$

$a^2 + b^2 - 2ab - 4 + 4a + 4b > 0$

$\because b^2 + 2b(2 - a) + (a^2 + 4a - 4) > 0$

$\because$ Now $4(2 - a)^2 - 4 \times 1 \times (a^2 + 4a - 4) < 0$

$4 + a^2 - 4a - a^2 - 4a + 4 < 0$

$8a - 8 > 0$

$a > 1$

P 90 : (A) $x = -1/4$

REPLICA 8

P 1 : (C) 7 : 5 : 2

Sol. $w = nR\,\Delta T$

$$\Delta U = \frac{5}{2}nR\Delta T$$

$$\Delta Q = \frac{7}{2}nR7\Delta$$

$$\Delta Q : \Delta V : W = 7 : 5 : 2$$

P 2 : (A) 0.3s

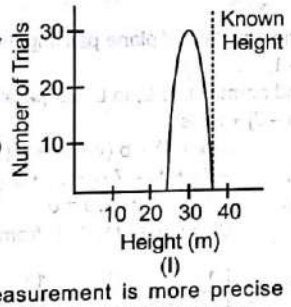
Sol. We see that $m \times k = 0.2 \times 90 = 18$ kg Nm^{-1}

$\Rightarrow \sqrt{km} = 4.243$ kgs^{-1} and $b = 0.04$kg s^{-1}.

As b is much less than $\sqrt{km}$, hence the time period T is given by

$$T = 2\sqrt{\frac{m}{k}} = 2\sqrt{\frac{0.2}{90}} = 0.3s$$

P 3 : (A)



(I)

Sol. Measurement is more precise if readings are close to actual reading.

P 4 : (C) mgh

Sol. Constant speed $\Rightarrow$ No change in KE

$\Rightarrow$ work doen by friction = work done by mg.

P 5 : (D) $\left(\frac{K_2}{K_1}\right)^{1/2}$

Sol. $V_{max} \Rightarrow WA$

$W_A A_A = W_B A_B$

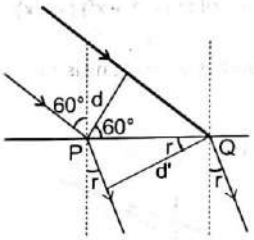
$$\frac{A_A}{A_B} \Rightarrow \frac{W_B}{W_A} \Rightarrow \sqrt{\frac{K_2}{K_1}}$$

P 6 : (C) 3.26 cm

Sol. Let d' be the diameter of refracted beam. Then

$d = PQ \cos 60°$

and $d' = PQ \cos r$



i.e., $\frac{d'}{d} = \frac{\cos r}{\cos 60°} = 2\cos r$

or $d' = 2d \cos r$

$$\sin r = \frac{\sin i}{\mu} = \frac{\sqrt{3}/2}{3/2} = \frac{1}{\sqrt{3}}$$

$$\cos r = \sqrt{1 - \sin^2 r} = \sqrt{2/3}$$

$$d' = (2)(2)\sqrt{\frac{2}{3}}$$

$$= 4\sqrt{\frac{2}{3}} = 3.26 \text{ cm}$$

P 7 : (C) Power is called a diopter and does not depend on the system of unit used to measure f.

Sol. Power of lens is $D = \frac{n_s}{f}$

Where n_s is medium of surrounding generally n_s is taken as 1 (for air)

S.I. unit of power is dioptres, f is in meter

P 8 : (D) (9/4) fold

Sol. We know that a phase change of π occurs when the reflection takes place at the boundary of denser medium. This is equivalent to a path difference of $\lambda/2$

Total phase difference = $\pi - \pi = 0$, i.e., two waves superimpose in phase. Hence resultant amplitude

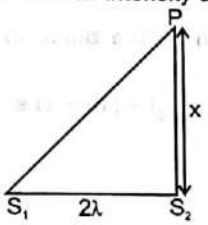
$$= \sqrt{I} + \sqrt{I/4} = \frac{3}{2}\sqrt{I}$$

So, resultant intensity

$$= \left(\frac{3}{2}\sqrt{I}\right)^2 = I\frac{9}{4} = \frac{9}{4}\text{fold}$$

P 9 : (A) $\frac{7\lambda}{12}$

Sol. Path difference at S_2 is 2λ. Therefore for minimum intensity at P



$$S_1P - S_2P = \frac{3\lambda}{2} \neq \frac{\lambda}{2} \ldots(i)$$

$$\text{or } \sqrt{4\lambda^2 + x^2} - x = \frac{3\lambda}{2}$$

Solving this equation, we get ;

$$x = \frac{7\lambda}{12}.$$

P 10 : (C) $H_0 \cos^2(\pi x/\beta)$

Sol. $D = x\frac{d}{D}$

$\therefore$ Phase difference $= \phi = \frac{2\pi}{\lambda}\Delta$

Let a = amplitude at the screen due to each slit

$\therefore\ I_0 = KA^2 = K(4a)^2\cos^2(\phi/2)$

$= I_0\cos^2(\pi\Delta/\lambda)$

$$= I_0\cos^2\left(\frac{\pi}{\lambda}\frac{xd}{D}\right) = I_0\cos^2(\pi x/\beta)$$

P 11 : (B) 6

Sol. When the train is approaching.

$$n_1 = \frac{v}{v - v_s} \times n = \frac{320}{320 - 4} \times 243$$

$$= \frac{80}{79} \times 243$$

When the train is receding,

$$n_2 = \frac{v}{v + v_s} \times n = \frac{320}{324} \times 243 = \frac{80}{81} \times 243$$

Beat frequency is ,

$$n = n_1 - n_2 = 80 \times 243\left(\frac{1}{79} - \frac{1}{81}\right) = 6\text{Hz}$$

P 12 : (D) straight line

Sol. Since the velocity of charged particle (aquired due to electric field) is always parallel to the magnetic field; the magnetic field exerts no force on the particle. Hence the charged particle moves along a straight line.

P 13 : (A) Four times

Sol. $L = (\mu_0 n^2)$(volume)

Now, n is doubled and length is the same.
So L becomes four times.

P 14 : (D) 0°

Sol. $\phi = \vec{B}.\vec{A} = BA\cos\omega t$ and $e = BA\omega\sin\omega t$

for maximum emf : $\sin\omega t = 1$

$\Rightarrow \cos\omega t = 0$

$\Rightarrow \omega t = 90°$, this is the angle between area vector and $\vec{B}$.

$\therefore$ angle between $\vec{B}$ and plane is 0° at that time.

P 15 : (D) longitudinal waves

Sol. Polarizer absorbs electric field perpendicular to axis of rotation therefore longitudinal wave connot be polarised.

P 16 : (D) 5.67 eV

Sol. We use $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}} \Rightarrow E = \frac{h}{2m\lambda^2}$

Thus we have $\Delta E = \frac{h^2}{2m}\left(\frac{1}{\lambda_1^2} - \frac{1}{\lambda_2^2}\right)$

Using $\lambda_1 = 0.5 \times 10^{-9}$ m

$\lambda_2 = 2 \times 10^{-9}$ m

$\Delta E = 5.67$ eV

P 17 : (A) $15\hat{i} + 20\hat{j}$

Sol. Let $C = 3\lambda\hat{i} + 4\lambda\hat{j}$

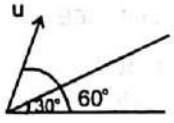
Now, $(3\lambda)^2 + (4\lambda)^2 = 7^2 + 24^2$

$\Rightarrow \quad \lambda = 5$

$$\vec{C} = 2\hat{i} + 3\hat{j}$$

P 18 : (C) 2 sec.

Sol.



$u = 10\sqrt{3}$ m/s

Time of flight on the inclined plane

$$T = \frac{2u\sin\alpha}{g\cos\beta}$$

given $\alpha = 30°$ & $\beta = 30°$ & $u = 10\sqrt{3}$ m/s

$$T = \frac{2\times10\sqrt{3}\sin30°}{10\cos30°}$$

so $T = 2$ sec .

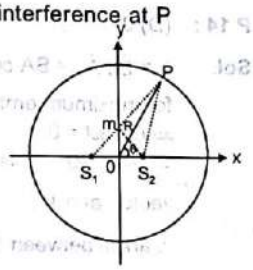
P 19 : (C) 60

Sol. For constructive interference at P

$S_1P - S_2P = \lambda$

$\Rightarrow S_1M = \lambda$

$2\lambda\cos\theta = \lambda$



$$\cos\theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{3}$$

P 20 : (A) 0.029 m/s

Sol. $(P_{sys})_i = (P_1 + P_2)$

As no force acts on it

$$O = \frac{h}{\lambda} - mv$$

$$v = \frac{h}{\lambda m}$$

$$= \frac{6.63\times10^{-34}}{590\times10^{-9}\times23\times1.67\times10^{-27}}$$

$$= \frac{663}{590\times23\times1.67} = 0.29 \text{ m/s}$$

P 21 : (B) 330 ± 0.44 m/s

Sol. $V = 2f(\ell_2 - \ell_1)$

$= 2(220)(110-35)\times10^{-2} = 330$ m/s

$$\frac{\Delta V}{V} = \frac{\Delta(\ell_2-\ell_1)}{\ell_2-\ell_1} = \frac{\Delta\ell_2+\Delta\ell_1}{\ell_2-\ell_1}$$

$$\Delta V = \frac{2\times10^{-1}}{75}\times330 = 0.88 \text{ m/s}$$

$V = (330 \pm 0.88)$ m/s

Note :- According to IIT-JEE

$$\frac{\Delta V}{V} = \frac{\Delta(\ell_2-\ell_1)}{\ell_2-\ell_1} = \frac{1\times10^{-1}}{75}$$

$\Delta V = 0.44$ m/s

P 22 : (D) $(10 + \pi)$ s

Sol. Time taken by block to reach point A,

$$t_1 = \frac{S}{v} = \frac{10}{2} = 5s$$

Time taken by block to reach from A to the position of maximum compression.

$$t_2 = \frac{T}{4} = \frac{\pi}{2}\sqrt{\frac{m}{k}} = \frac{T}{2}s$$

$\therefore$ Time taken by the block to reach P again.

$$t = 2(t_1 + t_2) = (10 + \pi) \text{ s}$$

P 23 : (D) $\frac{1}{4}CV_0^2$

Sol. Let the left side capacitor be connected to a voltage V_0. At an arbitrary instant of time t the charge of the right side capacitor is q and the current in the circuit is I.

In the circuit at time t we have,

$$\frac{q}{C} - \left(\frac{CV_0 - q}{C}\right) + \frac{dq}{dt}R = 0$$

or $$\frac{dq}{CV_0 - 2q} = \frac{dt}{RC}$$

integrating within the limits

$$\int_0^q \frac{dq}{CV_0 - 2q} = \frac{1}{RC}\int_0^t dt$$

$$\therefore \quad q = \frac{CV_0}{2}\left(1 - e^{-2t/RC}\right) \quad ...(i)$$

Differentiating eq. (i) we get,

$$\frac{dq}{dt} = I = \left(\frac{V_0}{R}\right)e^{-2t/RT}$$

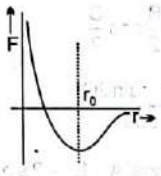
Clearly, we have

$$q = \frac{CV_0}{2} \text{ and } I = 0 \text{ for } t \to \infty$$

$$H = \int I^2Rdt = \int_0^\infty \left(\frac{V_0}{R}\right)^2 \left(e^{-2t/RT}\right)^2 Rdt$$

$$= \frac{1}{4}CV_0^2$$

P 24 : (A)



Sol. $U(r) = \frac{a}{r^{12}} - \frac{b}{r^6}$

$f(r) = -\frac{dU}{dr} = \frac{12a}{r^{13}} - \frac{6b}{r^7}$

$f(r) = 0 \Rightarrow r = \left(\frac{a}{2b}\right)^{1/6}$ unique point

P 25 : (B) $\frac{C}{\ell}\log\left(\frac{T_2}{T_1}\right)$

Sol. Consider an element of rod at a distance x and thickness dx. Then

$Q = KA\frac{dT}{dx} = \frac{CA}{T}\frac{dT}{dx}$

$\therefore$ q(Heat flow density) $= \frac{Q}{A} = \frac{C}{T}\frac{dt}{dx}$

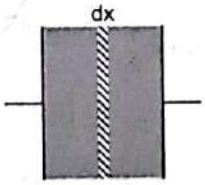
$q\int_0^1 dx = C\int\frac{T_2}{T_1}\frac{dT}{T}$

or $q\ell = C\log\left(\frac{T_2}{T_1}\right)$

or $q = \frac{C}{\ell}\log\left(\frac{T_2}{T_1}\right)$

P 26 : (B) $\frac{\epsilon_0 A\alpha}{(1-e^{-\alpha d})}$

Sol. $dc = \frac{k\varepsilon_0 A}{d} = \frac{e^{\alpha x}\varepsilon_0 A}{dx}$



all capacitor are in series

$\frac{1}{c_{eq}} = \frac{1}{c_1} + \frac{1}{c_2} ..... + \frac{1}{c}$

$\int_0^d \frac{dx}{e^{\alpha x}\varepsilon_0 A} = [e^{\alpha d} - 1]$

$c = \frac{\alpha\varepsilon_0 A}{1-e^{-\alpha d}}$

P 27 : (D) zero in all the four cases

Sol. All processes are cyclic and gas returns to its initial state. So internal energy will not change as it only depends on temperature.

P 28 : (B) 10 rad/sec

Sol. The body of mass 10 kg will not leave contact from the platform if

$mg = m\omega^2 a$

or $\omega = \sqrt{\frac{g}{a}} = \sqrt{\frac{10}{10\times10^{-2}}} = 10$ rad/sec.

P 29 : (A) $\sqrt{2}\,\alpha$

Sol. Frequency $f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$

initially spring cosntant = k

So, $\alpha = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$

finally after cutting, spring constant of each part will be 2k

So $f = \frac{1}{2\pi}\sqrt{\frac{2k}{m}}$

$f = \sqrt{2}\alpha$

P 30 : (C) field crossed over by concentric bright and dark bands

P 31 : (D) 1/4

Sol. $\lim_{n\to\infty}\frac{1.n + 2.(n-1) + 3.(n-2) + .. + n\{n-(n-1)\}}{1^2 + 2^2 + ...n^2}$

$= \lim_{n\to\infty}\frac{\sum_{K=1}^{n} K[n-(K-1)]}{\sum n^2}$

$= \frac{(n+1)\sum_{K=1}^{n} K - \sum_{K=1}^{n} K^2}{\sum n^2}$

$$= \lim_{n\to\infty} \frac{\frac{n(n+1)^2}{2} - \frac{n(n+1)(2n+1)}{6}}{\frac{n(n+1)(2n+1)}{6}} = \frac{\frac{1}{2}-\frac{2}{6}}{\frac{2}{6}} = \frac{1}{2}$$

P 32 : (B) 6

Sol. q = Area of parallelogram with $\overline{OA}$ and $\overline{OC}$ as adjacent sides

$= |\overline{OA} \times \overline{OC}| = |\vec{a} \times \vec{b}|$

p = Area of quadrilateral OABC

$$= \frac{1}{2}|\overline{OA} \times \overline{OB}| + \frac{1}{2}|\overline{OB} \times \overline{OC}|$$

$$= \frac{1}{2}\left[|\vec{a} \times (10\vec{a} + 2\vec{b})| + |\vec{a} \times (10\vec{a} + 2\vec{b}) \times \vec{b}|\right]$$

$$= \frac{1}{2}|(12\vec{a} + \vec{b})| = 6|\vec{a} \times \vec{b}| \Rightarrow k = 6$$

P 33 : (D) $\frac{\pi^2}{32}$

Sol. Let $\ell = \int_0^{\pi} \frac{x^3 \cos^4 x \sin^2 x}{\pi^2 - 3\pi x + 3x^2} dx$

$$= \int_0^{\pi} \frac{(\pi - x)^3 \cos^4 (\pi - x) \sin^2 (\pi - x)}{\pi^2 - 3\pi(\pi - x) + 3(\pi - x)^2} dx$$

$$= \int_0^{\pi} \frac{(\pi^3 - 3\pi^2 x + 3\pi x^2 - x^3)\cos^4 x \sin^2 x}{\pi^2 - 3\pi x + 3x^2} dx$$

$$= \int_0^{\pi} \frac{(\pi^3 - 3\pi^2 x + 3\pi x^2)\cos^4 x \sin^2 x}{(\pi^2 - 3\pi x + 3x^2)} dx - \ell$$

$$\Rightarrow \quad 2\ell = \pi \int_0^{\pi} \frac{(\pi^2 - 3\pi x + 3x^2)\cos^4 x \sin^2 x}{(\pi^2 - 3\pi x + 3x^2)} dx$$

$$\therefore \quad \ell = \pi \int_0^{\pi} \cos^4 x \sin^2 x dx = 2\pi \cdot \frac{3}{6}\cdot\frac{1}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}$$

$$\therefore \quad \ell = \frac{\pi^2}{32}$$

P 34 : (B) $8\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$

Sol. $\frac{\sin 2A + \sin 2B + \sin 2C}{\cos A + \cos B + \cos C - 1}$

$$= \frac{2\sin(A+B).\cos(A-B) + 2\sin C \cos C}{2\cos\frac{A+B}{2}\cos\frac{A-B}{2} + 1 - 2\sin^2\frac{C}{2} - 1}$$

$$= \frac{2\sin C \cos(A-B) + 2\cos C \sin C}{2\sin\frac{C}{2}\cos\frac{A-B}{2} - 2\sin^2\frac{C}{2}}$$

$$= \frac{2\sin C[\cos(A-B) + \cos(A+B)]}{2\sin\frac{C}{2}\left[\cos\frac{(A-B)}{2} - \cos\frac{(A+B)}{2}\right]}$$

$$= \frac{4\sin A \sin B \sin B}{4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}}$$

$$= \frac{4 \times 2\sin\frac{A}{2} \times \cos\frac{A}{2} \times 2\sin\frac{B}{2}\cos\frac{B}{2} \times 2\sin\frac{C}{2}\cos\frac{C}{2}}{4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}}$$

$$= 8\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$$

P 35 : (D) none of these

Sol. Put θ = 0° and 90° then

$$\frac{A\cos\alpha + B\sin\beta}{A'\sin\alpha + B'\cos\beta} = \frac{-A'\sin\alpha + B\cos\beta}{A'\cos\alpha - B'\sin\beta}$$

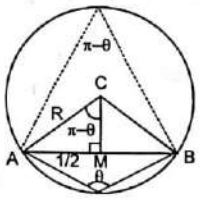
$AA'(\cos^2\alpha + \sin^2\alpha) - BB'(\sin^2\beta + \cos^2\beta) =$
$(AA' - AB')(\sin\alpha\cos\beta - \cos\alpha\sin\beta)$
Or $(AA' - BB') = (A'B - AB')\sin(\alpha - \beta)$

P 36 : (A) $\frac{1}{2}$

Sol. In triangle AMC

$$\sin(\pi - \theta) = \frac{1/2}{R}$$

$$\therefore R\sin\theta = \frac{1}{2}$$



P 37 : (C) 1

Sol. Let $\Delta = \begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$

$\Rightarrow \Delta = \sin^3 x + 2\cos^3 x - 3 \sin x \cos^2 x = 0$

Since, $\frac{\pi}{4} \le x \le \frac{\pi}{4} \Rightarrow \cos x \neq 0$ in this range

Thus, dividing the whole expression by $\cos^3 x$, we get,

$\Rightarrow \quad \tan^3 x + 2 - 3\tan x = 0$

$\Rightarrow \quad \tan^3 x - 3 \tan x + 2 = 0$

$\Rightarrow \quad (\tan x - 1)^2 (\tan x + 2) = 0$

Either $\tan x = 1$ or $\tan x = -2$

Now, $-\frac{\pi}{4} \le x \le \frac{\pi}{4}$

$\Rightarrow \quad -1 \le \tan x \le 1$

Thus, $\tan x = -2$ is not defined and hence the only solution is $\tan x = 1$

P 38 : (A) HP

Sol. Clearly $x = \frac{1}{1-a}, y = \frac{1}{1-b}, z = \frac{1}{1-c}$

Since a, b, c are in AP

$\Rightarrow \quad 1 - a, 1 - b, 1 - c$ are also in AP

$\Rightarrow \quad \frac{1}{1-a}, \frac{1}{1-b}, \frac{1}{1-c}$ are in HP

Therefore, x, y, z are in HP

P 39 : (B) –64

Sol. Adding each of the four equations in the expression we get:

$16(a + d) + 20(b + c) = - 32.$

Also, by adding the second and the third equations we get $a + b + c + d = 0$, which means that $(a + d) = - (b + c)$.

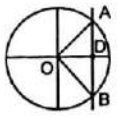
Then from : $16(a + d) + 20(b + c) = - 32.$ we have : $- 16(b + c) + 20(b + c) = -32 \Rightarrow 4(b + c) = - 32.$ Hence, $(b + c) = - 8$ and $(a + d) = 8$. Hence, the multiplication of $(a = d) (b + c) = - 64.$

P 40 : (B) a circle

Sol. Let D is the mid point of chord AB hence the distance $OD = \frac{r}{\sqrt{2}}$

Equation of AB is $x = \frac{r}{\sqrt{2}}$

coordinate of $A\left(\frac{r}{\sqrt{2}}, \frac{r}{\sqrt{2}}\right), B\left(\frac{r}{\sqrt{2}}, \frac{-r}{\sqrt{2}}\right)$



Let coordinates of moving point P be $(r \cos\theta, r \sin\theta)$

$\therefore$ centroid of ΔPAB is

$$\left(\frac{r\cos\theta + \sqrt{2}\, r}{3}, \frac{r\sin\theta}{3}\right)$$

locus of P is $(3x - \sqrt{2}\, r)^2 + (3y)^2 = r^2$

which is a circle

P 41 : (A) $\left(2t_4, \frac{2}{t_4}\right)$

Sol. Hyperbola $xy = 4$ [compare with $xy = c^2, c = 2$]

Equation of normal $xt^3 - yt - 2t^4 - 2 = 0$

passes through (6, 6) $t^4 - 3t^4 + 3t - 1 = 0$

The above equation gives four values of t which corresponds to the points $P(t_1)$, $Q(t_2)$, $R(t_3)$ & $S(t_4)$

$t_1 t_2 t_3 t_4 = - 1 \qquad .....(1)$

orthocentre of $\Delta PQR = \left(-\frac{c}{t_1 t_2 t_3}, -c\, t_1 t_2 t_3\right)$

$= \left(-\frac{2}{t_1 t_2 t_3}, -2\, t_1 t_2 t_3\right) = \left(2t_4, \frac{2}{t_4}\right)$

P 42 : (A) $2y + 3 = 0$

Sol. Assymptotes are

$x - y = 0 \qquad .....(i)$

$x + y + 3 = 0 \qquad .....(ii)$

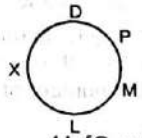
Transverse axis is the bisector of that angle between (i) & (ii) in which the point (2, 3) does not lie

$\therefore$ the required equation is $\frac{x-y}{\sqrt{2}} = \frac{x+y+3}{\sqrt{2}}$

i.e. $2y + 3 = 0$

P 43 : (B) 240

Sol. D P M L A A = D M P L X (say)



$4 ! . {}^5C_2 = 240$

P 44 : (B) 44

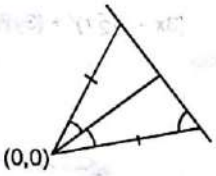
Sol. ${}^8C_5$ – [only top remains empty + bottom empty]

${}^8C_5 - [{}^6C_5 + {}^6C_5] = 56 - 12 = 44$

P 45 : (A) $\frac{36}{13}$

Sol. Distance of (0, 0) from the line $7x + 3y - 6 = 0$

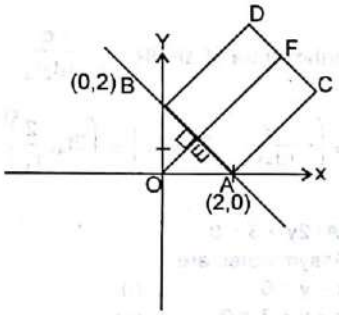
$$\frac{6}{\sqrt{4+9}} = \frac{6}{\sqrt{13}}$$



$\therefore$ area of the Δ is $\left(\frac{6}{\sqrt{13}}\right)^2 = \frac{36}{13}$

P 46 : (C) $3\sqrt{2}$

Sol.



$\Rightarrow AB = \sqrt{4+4} = 2\sqrt{2}$

$\because OE = \sqrt{2}$

$\Rightarrow OF = \sqrt{2} + 2\sqrt{2} = 3\sqrt{2}$

P 47 : (C) $\frac{53}{54}$

Sol. Player A can win if A throws (1, 6) or (6, 1) and B throws ((1, 1,), (2, 2), (3, 3), (4, 4), (5, 5) or (6, 6)). Thus the number of ways is 12.
similarly the number of ways in which B can win is 12.
total number of ways in which either A wins or B wins = 24
Thus the number of ways in which none of the two wins = $6^4 - 24$.

$\therefore$ the required probability = $\frac{6^4 - 24}{6^4} = \frac{53}{54}$.

P 48 : (C) 2

Sol. If $a(b - c)x + b(c - a)xy + c(a - b)y^2 = 0$ is a perfect square, then

$b^2 (c - a)^2 y^2 - 4ac(b - c)(a - b)y^2 = 0$

$\Rightarrow \quad b^2 (c - a)^2 - 4ac(b - c)(a - b) = 0$

$\Rightarrow \quad b^2 (c + a)^2 = 4ac(ab - ac + bc)$

$\Rightarrow \quad \{b(c + a)\}^2 - 4abc(a + c) + 4a^2c^2 = 0$

$\Rightarrow \quad \{b(c + a) - 2ac\}^2 = 0 \Rightarrow b = \frac{2ac}{a+c}$

Now, $\frac{\log(a+c) + \log(a - 2b + c)}{\log(a - c)}$

$$= \frac{\log(a+c) + \log\left(a - \frac{4ac}{a+c} + c\right)}{\log(a-c)} = \frac{\log(a-c)^2}{\log(a-c)}$$

$= 2$

P 49 : (B) Centroid of triangle PQR is on y-axis

Sol. P(2, 4) $\therefore$ $t = 1$

$\therefore$ parameter of Q is $t = -1 - 2 = -3$

$\therefore$ coordinates of Q are (18, –12)

parameter of R is $t = 3 - \frac{2}{-3} = \frac{11}{3}$

$\therefore$ coordinates of R are $\left(\frac{242}{9}, \frac{44}{3}\right)$

$\therefore$ centroid of PQR is

$$\left(\frac{2 + 18 + \frac{242}{9}}{3}, \frac{4 - 12 + \frac{44}{3}}{3}\right)$$

$= \left(\frac{422}{27}, \frac{20}{9}\right)$. Point of intersection of tangent at P and Q is (–6, –4)

Equation of normal at (8, 8) is

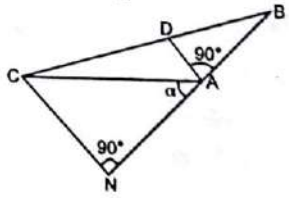
$y = -2x + 24$

$-12 = -2 \times 18 + 24$

$\therefore$ the normal passes through Q.

P 50 : (C) $\tan A + 2 \tan B = 0$

Sol. We have BD = DC and $\angle DAB = 90°$. Draw CN $\perp$ to BA produced. Then in ΔBCN, we have $DA = \frac{1}{2}$ CN and AB = AN.



Let $\angle CAN = \alpha$

$\tan A = \tan(\pi - \alpha)$

$= -\tan\alpha$

$= -\frac{CN}{NA} = -2\frac{AD}{AB} = -2\tan B$

$\Rightarrow \quad \tan A + 2\tan B = 0.$

P 51 : (C) an ellipse with eccentricity $\frac{\sqrt{3}}{2}$

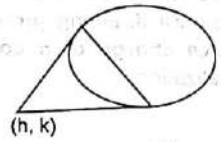
Sol. Let the point of intersection of the tangents be (h, k)

Equation of chord of contact

$hx + 2ky - 4 = 0$

It is tangent to the circle $x^2 + y^2 = 1$

$\therefore \quad \frac{4}{\sqrt{h^2+4k^2}} = 1 \quad \therefore 16 = h^2 + 4k^2$

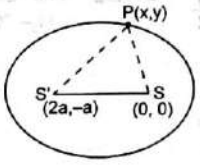


$\therefore$ locus is $x^2 + 4y^2 = 16$ i.e. $\frac{x^2}{16} + \frac{y^2}{4} = 1$

its eccentricity $= \sqrt{\frac{4-1}{4}} = \frac{\sqrt{3}}{2}$

P 52 : (A) $-2 < a < 2$

Sol. If given equation represents an ellipse



$\Rightarrow SS' < PS + PS' \qquad \Rightarrow \sqrt{5a^2} < 2\sqrt{5}$

$\Rightarrow |a| < 2$

P 53 : (B) $(2, -1)$

Sol. Let image of (0, 0) in $4x - 2y - 5 = 0$ be (x, y) then

$\frac{x-0}{4} = \frac{y-0}{-2} = -2\,\frac{(-5)}{20}$

$\frac{x}{4} = \frac{y}{-2} = \frac{1}{2} \quad \Rightarrow x = 2 \;\&\; y = -1$

P 54 : (C) 5/16

Sol. Required probability :

$= \left({}^3C_0\right)^2\left(\frac{1}{8}\right)^2 + \left({}^3C_1\frac{1}{8}\right)^2 + \left({}^3C_2\frac{1}{8}\right)^2 + \left({}^3C_3\frac{1}{8}\right)^2$

$= \frac{1+9+9+1}{8\times 8} = \frac{20}{8\times 8} = \frac{5}{16}$

P 55 : (C) 4

Sol. $y^2 = k\left(x - \frac{8}{k}\right)$

Equation of directrix is $x - \frac{8}{k} = -\frac{k}{4}$

$\Rightarrow \quad \frac{8}{k} - \frac{k}{4} = 1 \quad \Rightarrow \quad k = 4$

P 56 : (B) 30π

Sol. Here, $\cos x\left(\frac{1}{4}\cos^2 x - \frac{3}{4}\sin^2 x\right) = \frac{1}{4}$ or

$\frac{\cos x}{4}\left(4\cos^2 x - 3\right) = \frac{1}{4}$ or $\cos 3x = 1$

$\Rightarrow 3x = 2n\pi$

$\Rightarrow x = \frac{2n\pi}{3}$, where n = 0,1,2,3,4,5,6,7,8,9;

$\therefore$ The required sum $\frac{2\pi}{3}\sum_{n=0}^{9} n = 30\pi$

P 57 : (B) 2n

Sol. $\because \sum_{i=1}^{2n}\sin^{-1}x_i = n\pi = \sum_{i=1}^{2n}\left(\frac{\pi}{2}\right)$

Which is possible only when

$\sin^{-1}x_i = \frac{\pi}{2}\,\forall i$

Therefore, $x_i = 1\,\forall i$

Then $\sum_{i=1}^{2n} x_i = \sum_{i=1}^{2n} 1 = 2n$

P 58 : (B) $\frac{-\hat{i} - 7\hat{j} + 5\hat{k}}{5\sqrt{3}}$

Sol. Direction ratio of L_1 are 3, 1, 2 and of L_2 are 1, 2, 3

Vector perpendicular to L_1 and L_2 is

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & 2 \\ 1 & 2 & 3 \end{vmatrix}$$

$= -\hat{i} - 7\hat{j} + 5\hat{k}$ unit vector is $\frac{-\hat{i} - 7\hat{j} + 5\hat{k}}{5\sqrt{3}}$

P 59 : (A) $p^3 - q(3p - 1) + q^2 = 0$

Sol. $x^2 + px + q = 0 \Rightarrow \alpha, \alpha^2$

$\alpha + \alpha^2 = -p$

$\alpha^3 = q$

$\because (\alpha + \alpha^2)^3 = \alpha^3 + \alpha^6 + 3\alpha^3(\alpha + \alpha^2)$

$-p^3 = q + q^2 + 3q(-p)$.

P 60 : (B) two straight lines and a circle, when a = b, and c is of sign opposite to that of a

Sol. $(ax^2 + by^2 + c)(x^2 - 5xy + 6y^2) = 0$

$\Rightarrow x = 3y$ or $x = 2y$ or $ax^2 + by^2 + c = 0$

If a = b and c is of opposite sign, then it will represent a circle.

P 61 : (B) The vapour will contain higher percentage of toluene

Sol. A : Benzene B : Toluene

$p = p_A + p_B$

$p = p_A^0 x_A + p_A^0 x_B$

$= 75 \times \frac{1}{2} + 22 \times \frac{1}{2}$

$= 37.5 + 11 = 48.5$

Mole fraction of benzene in vapour,

$y_A = \frac{p_A}{p} = \frac{37.5}{48} = 0.78$

Similarly, mole fraction of toluene in vapour, $y_B = 0.22$

$\therefore$ The vapour will contain higher percentage of benzene.

P 62 : (A) The density of the crystals having Schottky defect is larger than that of the perfect crystal

P 63 : (A) 0.15 A

Sol. $k = \frac{\Lambda C}{1000} = \frac{195 \times 0.05}{1000} = 9.75 \times 10^{-3}\ S\,cm^{-1}$

Also, $G = \frac{1}{R} = k\left(\frac{A}{\ell}\right)$

$= 9.75 \times 10^{-3}\ S\ cm^{-1} \times \frac{1.5 cm^2}{0.50 cm}$

$= 29.25 \times 10^{-3}\ ohm^{-1}$

Also, $V = IR \Rightarrow I = \frac{V}{R} = 5 \times 29.25 \times 10^{-3}$

$= 0.146 \approx 0.15$ A

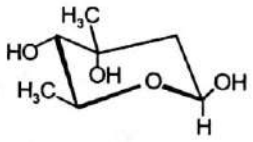
P 64 : (B)

Sol. MnO_2 selectively oxidises allylic or benzylic hydroxyl group to (C = O) group, whereas Jones reagent oxidises 1° and 2° ROH to aldehydes and ketones, respectively.

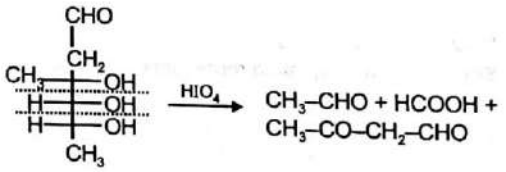
P 65 : (C) The group has ability to delocalize the positive charge of the arenium ion

Sol. Ortho-para directing group stabilizes the positive charge of σ complex through delocalization.

P 66 : (A)

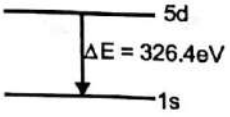


Sol. It's open chain structure is



P 67 : (D) B^{+4}

Sol. z = ?



$\Delta E = 326.4 = 13.6\ z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] eV$

$\Rightarrow 326.4 = 13.6 \times z^2 \left[\frac{1}{(1)^2} - \frac{1}{(5)^2}\right]$

$z = 5$

P 68 : (A) O_2 acts as an oxidising agent

Sol. (A) $K \longrightarrow K^{\oplus} + e^-$ (oxidation, acts as reducing agent)
$e^- + O_2 \longrightarrow O_2^{1-}$ (reduction, acts as oxidising agent)

P 69 : (C) Density of the gas

Sol. The quantity that does not change for a sample of a gas in sealed tube. It is cooled from 120°C to 90°C at constant volume is density of the gas because during the process, total mass of gas remains constant.

P 70 : (B) 2

Sol. $4S^3 = 4 \times 10^{-12} \Rightarrow S = 10^{-4}$ M
Now let molarity of $CaCl_2 = x \Rightarrow [Cl^-] = 2x$
If S' be solubility in presence of $CaCl_2$ then

$$S' = \frac{S}{4 \times 10^8} = 2.5 \times 10^{-13}$$

$\Rightarrow K_{sp} = 4 \times 10^{-12} = 2.5 \times 10^{-13}(2x)^2$
$\Rightarrow x = 2M$

P 71 : (B) 5.568 m^2g^{-1}

Sol. No. of molecules per gram of N_2 in monolayer

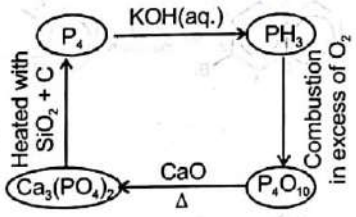
$$= \frac{6 \times 10^{23}}{22{,}400} \times 1.30 = 3.48 \times 10^{-19}$$

Cross-sectional area of a molecule
$= 1.6 \times 10^{19}$ m^2
$\therefore$ Area covered by molecules per gram
$= 3.48 \times 10^{19} \times 1.6 \times 10^{-19} = 5.568$ m^2
$\therefore$ Surface area = 5.568 m^2g^{-1}

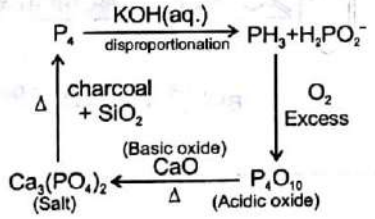
P 72 : (A) The main raw materials required in Solvay process are NaCl and NH_4Cl

Sol. Raw materials is Solvay proces are $CaCO_3$ and NaCl.

P 73 : (A)



Sol.



P 74 : (D) dipole-induced dipole interaction

Sol. Xenon is trapped in the cavities formed when water is frozen under high pressure of gas and the nature of bonding between xenon and water molecule is dipole-induced dipole interaction.

P 75 : (C) For every process, in an isolated system that has no macroscopic kinetic or potential energy, $\Delta E = 0$

Sol. In isolated system q = 0 also there is no macroscopic potential or kinetic energy, W = 0, hence $\Delta E = 0$.

P 76 : (A) Half-life changes only on changing temperature of the reaction.

Sol. Half-life also changes on changing the condition of catalyst.

P 77 : (A) I > II > III

Sol. CFSE↑, λ↓

P 78 : (D) all ture

Sol. $FeCl_3$ (Yellow) + Reducing agent $\longrightarrow$ Fe^{+2} (Green)

P 79 : (C) 17

Sol. % ionic character $= \frac{\mu_{exp}}{\mu_{calc}} \times 100$

$\mu_{H-Cl} = 1.03$ D $= 1.03 \times 10^{-18}$ esu. cm
$d_{H-Cl} = 1.275$ Å $= 1.275 \times 10^{-8}$ cm

$$\% \text{ I.C.} = \frac{1.03 \times 10^{-18}}{(4.8 \times 10^{10}\text{ esu}) \times 1.275 \times 10^{-8}} \times 100$$

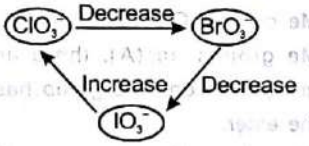
$\equiv 17\%$

P 80 : (A) two sigma and two pi bonds

Sol. $Mg_2C_3 + H_2O \longrightarrow CH_3C \equiv CH$

Two σ and two π bonds.

P 81 : (B)



Sol. Bond angle
$ClO_3^- > BrO_3^- > IO_3^-$

P 82 : (D) mixture of $CaCl_2$ and $Ca(OCl_2)$

Sol. $Ca(OH)_2 + 2Cl_2 \rightarrow Ca(OCl_2) + CaCl_2 + 2H_2O$

P 83 : (A) Ag^+ polarises Br^- and I^-

Sol. Due to polarisation, covalent character increases so colour intensifies.

P 84 : (B) $C_6H_6 = C_6D_6 = C_6T_6$

Sol. Sulphonation is a reversible process and shows 1° kinetic isotope effect, whereas other SE reactions (i.e., nitration, halogenation, etc.) do not show 1° kinetic isotope effect.

P 85 : (A) III > IV > I > II

P 86 : (B) Strong bases are good leaving groups.

Sol. Weak bases are good leaving groups.

Acidic order: $H_2O > ROH > CH \equiv CH > NH_3$

Basic order: $\overset{\ominus}{O}H < RO^{\ominus} < CH{\equiv}\overset{\ominus}{C}{}^{\ominus} < \overset{\ominus}{N}H_2$

Weaker the base, better is the leaving group.

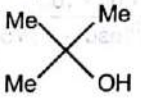
Leaving group order:

$$\overset{\ominus}{O}H > RO^{\ominus} > CH{\equiv}\overset{\ominus}{C}{}^{\ominus} > \overset{\ominus}{N}H_2$$

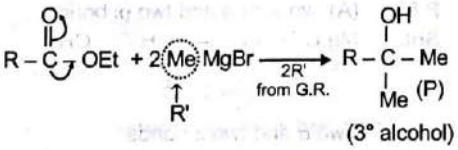
P 87 : (B) $Me-CH=CH_2$

Sol. When alkene is symmetrical, the same product is obtained by (i) catalytical hydration, (ii) mercuration demercuration, and (iii) hydroboration oxidation reactions.

P 88 : (A)



Sol.



The product (P) should contain at least two Me groups. Only (A) contains at least two Me groups. In (A), there are three Me groups, so one Me group has come from the ester.

Therefore, the structure of the ester is $\left(\begin{matrix} O \\ \| \\ Me-C-OEt \end{matrix}\right)$ (ethyl acetate or ethyl ethanoate)

P 89 : (D) $PhCH_2OH \xrightarrow{H^+/KMnO_4/\Delta}$

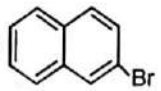
Sol. A. Lindlar's catalyst converts $\left(\begin{matrix} O \\ \| \\ Ph-C-Cl \end{matrix}\right)$ group to (PhCHO).

B. Lithium aluminum t-butoxide also converts (PhCOCl group to (PhCHO)).

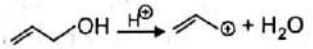
C. DIBAL-H converts PhC ≡ N to PhCHO.

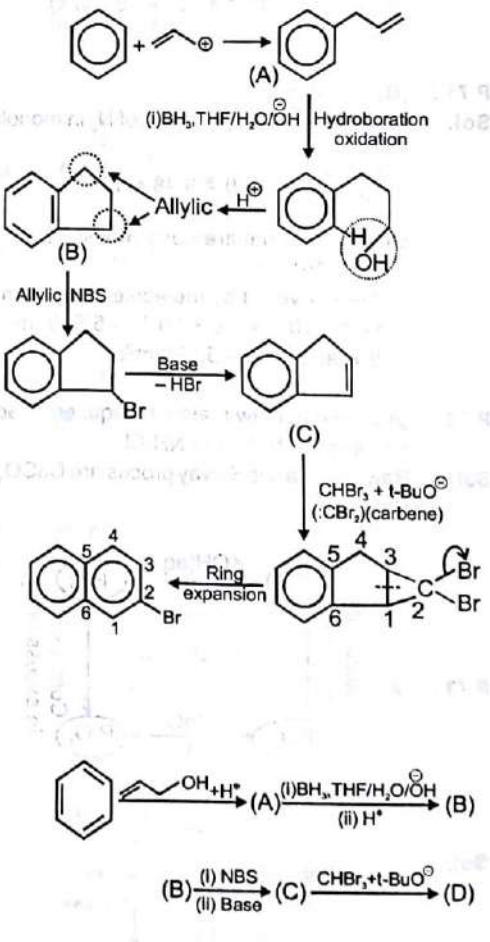
D. $H^+/KMnO_4/\Delta$ converts 1° alcohol to carboxylic acid.

P 90 : (A)



Sol.





REPLICA 9

P 1 : (C) $1/n = 1/n_1 + 1/n_2 + 1/n_3$

Sol. $n_1 = \frac{1}{2\ell_1}v$

$n_2 = \frac{1}{2\ell_2}v$

$n_3 = \frac{1}{2\ell_3}v$

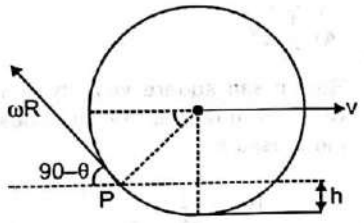
$n = \frac{1}{2\ell}v$

$\ell = \ell_1 + \ell_2 + \ell_3$

$\frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3}$

P 2 : (B) $R - \frac{v}{\omega}$

Sol. $\omega R\cos(90-\theta) = v$



$\Rightarrow \quad \sin\theta = \frac{V}{\omega R}$

$h = R - R\sin\theta = R - R\frac{V}{\omega R} = R - \frac{V}{\omega}$

P 3 : (D) 50

Sol. By Snell's law, $\mu \sin i$ = constant

$\therefore \mu \sin 30° = (\mu - m\Delta\mu)\sin 90°$

$\mu \sin 30° = \mu - m\Delta\mu$

$m\Delta\mu = \mu - \mu \sin 30°$

$m = \frac{\mu(1-\sin 30°)}{\Delta\mu}$

Here, $\mu = 1.5$, $\Delta\mu = 0.015$

$\therefore \quad m = \frac{1.5\left(1-\frac{1}{2}\right)}{0.015} = \frac{0.75}{0.015} = 50$

P 4 : (C) the sharpness of fringes will decrease as we move away from central fringe

Sol. Due to the two different wavelength there will be two interference patterns which will superimpose over each other due to which sharpness decreases as we move away from central bright fringe.

P 5 : (A) 4290 Å

Sol. Light of wavelength λ is stored reflected if

$$2\mu t = (2n + 1)\frac{\lambda}{2}$$

If λ = 4000 Å, n = 3.25

If λ = 7000 Å, n = 1.66

Hence, within 4000Å and 7000Å the integer can take values 2 and 3 from eqn.(i)

λ = 6000 Å and 4290 Å

Hence, option (a) is : λ = 4290 Å

P 6 : (B) 60

Sol. $V = \frac{C_1V_1 + C_2V_2}{C_1 + C_2} = 34$ volt

As $V_1 > V_2$

$\Rightarrow$ $Q = C_1(V_1 - V) = 10(40 - 34) = 60\ \mu C$

P 7 : (B) $\bar{v}$ must be perpendicular to $\bar{E}$

Sol. $q\bar{E} = -\left(\bar{v} \times \bar{B}\right)$

$\bar{E} = -\left(\bar{v} \times \bar{B}\right)$

$\Rightarrow \bar{E} \perp \bar{V}$ and $\bar{E} \perp \bar{B}$.

P 8 : (C) 200 P_m

Sol. Initial wavelength of electron is

$\lambda = \frac{h}{P_i} \quad \Rightarrow \quad d\lambda = \frac{h}{P_i^2}dP_i$

$\Rightarrow \frac{d\lambda}{h/P_i} = \frac{dP_i}{P_i} \Rightarrow \frac{d\lambda}{\lambda} = \frac{dP_i}{P_i}$

$\Rightarrow \frac{0.5}{100} = \frac{P_m}{P_i} \Rightarrow P_i = 200\ P_m$

P 9 : (D) $(31\hat{i}+40\hat{j}+120\hat{k})$m

Sol. $\vec{v}=\frac{3\hat{i}+4\hat{j}+12\hat{k}}{13}\times 65$

$\vec{r}=\hat{i}+\vec{v}\times 2$

$=31\hat{i}+40\hat{j}+120\hat{k}$

P 10 : (C) 100 ms^{-1}

Sol. $v_1 = 50 - gT$

$v_2 = -50 - gT \Rightarrow v_r = v_1 - v_2 = 100$ m/sec

P 11 : (D) The speed of propagation as well as the wavelength

Sol. We know that velocity of transverse wave on a string is given as $v=\sqrt{\frac{T}{\mu}}$

It is given that $\mu_2 > \mu_1$ so the speed of wave will change and as frequency is constant so wavelength will also change.

P 12 : (D) kT where k is Boltzmann's constant and T is the absolute temperature.

Sol. $\varepsilon_0 E^2$ and $\frac{B^2}{\mu_0}$ have dimensions same as energy per volume.

y has dimensions of pressure

$=\frac{F}{A}=\frac{\text{energy}}{\text{volume}}$

KT has dimensions of energy/molecule so it would be different from others.

P 13 : (D) the particle has a constant acceleration along +ve y-axis.

Sol. Given x = kt ...(i) and y = kt(1–pt) ....(ii)

by these two eq. if we eliminate t then

$y = x - px^2$ ∴ The trajectory is a parabola.

$v=\frac{dx}{dt}=k$ ∴ velocity of the particle along x -axis is constant.

$a_y = -kp < 0$

∵ k and p are both negative $a_y < 0$ so a_y is along –ve y-axis

P 14 : (D) 8.02×10^{-13} A-m^2, 8×10^5 A/m

Sol. Maximum dipole moment when all dipoles are arranged in same directions

$$M=\frac{(1\times10^{-6})^3\left(\frac{7.9}{10^{-6}}\right)}{55}\times 6.023\times10^{23}\times9.27\times10^{-24}$$

$M = 8.02 \times 10^{-13}$ A-m^2

$$\text{Magnetisation} = \frac{M}{\text{volume}}=\frac{8.02\times10^{-13}}{10^{-18}}$$

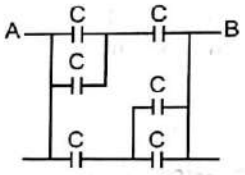
$\simeq 8\times10^5$ A/m

P 15 : (C) 4C/3

Sol. The network is equivalent to therefore equivalent capacitance

= [2 C series C]/[C series 2C]

$=\left[\frac{2C\times C}{2C+C}\right]=\frac{4C}{3}$



P 16 : (A) $\frac{1}{3}\rho\bar{c}^2$

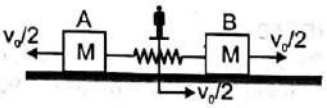
Sol. Root mean square velocity is velocity wrt COM so movement of box does not effect the pressure.

$$P=\rho\frac{RT}{M}=\frac{1\rho}{3}\left(\frac{3RT}{M}\right)=\frac{1}{3}\rho\bar{C}^2$$

P 17 : (D) both blocks come to rest simultaneously

Sol. In COM frame they will perform SHM with

$\omega=\sqrt{\frac{2k}{m}}$



So time period $T=2\pi\sqrt{\frac{m}{2k}}$

Amplitude of one particle $A=\frac{V_0}{2\omega}$

So total compression is $2A=\frac{V_0}{\omega}=V_0\sqrt{\frac{M}{2k}}$

Block B will dome at rest after T/2 time.

P 18 : (B) 16°C

Sol. If we apply Newton's law of cooling Avg method

$$\frac{80-64}{5} = C[72 - T_s]$$

$$\frac{64-52}{5} = C[58 - T_s]$$

$$\frac{16}{12} = \frac{72-T_s}{58-T_s}$$

$$\Rightarrow T_s = 16°C$$

P 19 : (B) 495 m

Sol. Let x = width of valley
and y = distance of man from one cliff
then (x – y) = distance of man from second cliff
Hence, $2y = 1 \times 330$
and $2(x - y) = 2 \times 330 + 330$

$$\therefore \quad x = \frac{990}{2} = 495m$$

P 20 : (A) $I\omega_1\omega_2$

Sol. If angular momentum is changing its direction, then rate of change of angular momentum :

$$\frac{d\vec{L}}{dt} = \vec{L} \times \vec{\omega}$$ ($\vec{\omega}$ is angular velocity of axis)

$$\text{So, } \left|\frac{d\vec{L}}{dt}\right| = I\omega_1\omega_2 \sin 90°$$

$$\Rightarrow \left|\frac{d\vec{L}}{dt}\right| = I\omega_1\omega_2$$

P 21 : (A) 1 sec

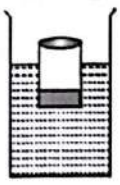
Sol. Volume of liquid displaced = Ax
Volume of liquid raised = 3Ah

$$Ax = 3Ah \Rightarrow h = \frac{x}{3}$$

F_{net} force in upward direction = $(Ah')d_Lg$

$$A\frac{4x}{3}d_Lg = m\omega^2x$$

$$A\frac{4x}{3}d_Lg = Ad_m H \times \omega^2$$



$$\omega = \sqrt{\frac{4d_Lg}{3(Hd_m)}}$$

$$T = 2\pi\sqrt{\frac{3(Hd_m)}{4g\,d_L}}$$

$$T = 2\pi\sqrt{\frac{3(1m)}{4g} \times \frac{1}{3}}$$

$$T = 1 \text{ sec}$$

P 22 : (C) $\frac{1}{8}$m

Sol. The wall acts like a rigid boundary and reflects this wave and sends it back towards the open end. At the end an antinode is formed and a node is formed at the wall. The distance between antinode and node is $\frac{\lambda}{4}$. Therefore, if v be the frequency of node emitted, then

$$\lambda = \frac{c}{v} \Rightarrow \lambda = \frac{300}{600} = \frac{1}{2}m$$

Maximum amplitude is obtained at distance

$$= \frac{\lambda}{4} = \frac{1}{2} \times \frac{1}{4} = \frac{1}{8}m$$

P 23 : (B) 100 m/s²

Sol. $y = x^2$

$$\Rightarrow \frac{dy}{dt} = 2x\frac{dx}{dt}$$

$$\Rightarrow \frac{d^2y}{dt^2} = 2\left(\frac{dx}{dt}\right)^2 + 2x\frac{d^2x}{dt^2}$$

at x = 0, particle will have only acceleration in y direction

$$\text{So, } \frac{d^2x}{dt^2} = 0$$

$$\Rightarrow a = a_y = \frac{d^2y}{dt^2} = 2\left(\frac{dx}{dt}\right)^2 = 100 \text{ m/s}^2$$

P 24 : (A) 3.86 atm

Sol. Number of oxygen moles required

$$n = \frac{(1)(30)}{R(300)}$$

For cylinder v = 5.2 lt, T = 200
Let pressure = P atm

$$n = \frac{(P)(5.2)}{R(200)}$$

$$P = \frac{30}{5.2} \times \frac{200}{300} = 3.86 \text{ atm}$$

P 25 : (D) $\frac{1}{4}$

Sol. We use

$$\frac{h}{mv} = \lambda$$

$$\Rightarrow \quad \frac{E_k}{E_{ph}} = \frac{\frac{1}{2}mv^2}{\frac{hc}{\lambda}} = \frac{m^2v^2\lambda}{2hcm} = \frac{v}{2c} = \frac{1}{4}$$

P 26 : (A) 3.27eV

Sol. The light contains two different frequencies. The one with larger frequency will cause photoelectrons with largest kinetic energy. This larger frequency is

$$\nu = \frac{\omega}{2\pi} = \frac{8\times10^{15}s^{-1}}{2\pi}$$

The maximum kinetic energy of the photoelectrons is

$K_{max} = h\nu - W$

$$= (4.14 \times 10^{-15}\ \text{eV-s}) \times \left(\frac{8\times10^{15}}{2\pi}s^{-1}\right) - 2.0\text{eV}$$

$$= 5.27\ \text{eV} - 2.0\ \text{eV} = 3.27\ \text{eV}$$

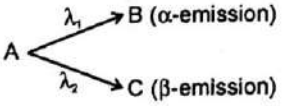
P 27 : (D) On increasing the power of cathode, λ_k decreases

Sol. As value of Z increases the difference of energies of energy levels also increase so wavelength corresponding to each characteristic line decreases hence option (A) is correct. As applied voltage across the tube increases, cut off wavelength decreases hence option (B) is correct. If power of cathode is increased which is done by increase in current or increase in number of electrons per unit time incident on the target, then due to this more X-ray photons are emitted per unit time from the target which increases the intensity of X-rays emitted so option (C) is correct

$\Rightarrow$ D is false

P 28 : (A) 648 years

Sol.



So net decay constant $\lambda = \lambda_1 + \lambda_2$

$$\frac{1}{t_{1/2}} = \left(\frac{1}{t_{1/2}}\right)_\alpha + \left(\frac{1}{t_{1/2}}\right)_\beta \qquad \left\{\lambda = \frac{1}{t_{1/2}}\ell n2\right\}$$

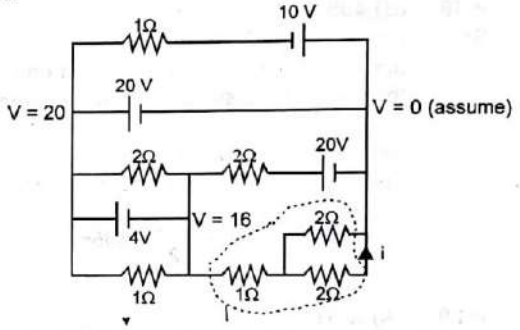
$$\frac{1}{t_{1/2}} = \frac{1}{1620} + \frac{1}{405} \Rightarrow t_{1/2} = 324 \text{ years}$$

So for $\frac{3^{th}}{4}$ decay it will bake two half life.

So time = 648 years

P 29 : (B) 4A

Sol.

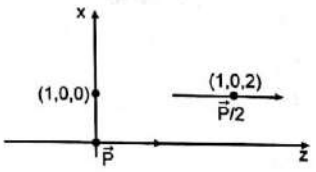


potential difference across this configuration = 16V, $R_{eff} = 2\Omega$

$\therefore$ current of 8A will flow in 1Ω resistance & hence i = 4 A .

P 30 : (B) $\frac{-7p}{32\pi\epsilon_0}\hat{k}$

Sol.



The given point is at axis of $\frac{\vec{P}}{2}$ dipole and at equatorial line of $\vec{P}$ dipole so total field at given point is.

$$\vec{E} = -\frac{k\vec{P}}{(1)^3} + \frac{2k(\vec{P}/2)}{(2)^3} = k\vec{P}\left(-1+\frac{1}{8}\right) = \frac{-7\vec{P}}{32\pi\varepsilon_0}$$

P 31 : (C) 5

Sol. Initial pH of water = 7

Final $[H^+] = \frac{1}{100} = 10^{-2}$, pH = 2

Change in pH = 7 – 2 = 5

P 32 : (D) log A : where A is pre-exponential factor

Sol. $\log k = \log A = -\frac{E_a}{23RT}$

$\Rightarrow \lim_{T\to\infty} \log k = \log A$

P 33 : (B) $(CH_3)_2\overset{OMe}{\underset{|}{C}}C\equiv CCH_3$

Sol. $CH_3C\equiv CH \xrightarrow{CH_3MgBr} CH_4 + CH_3-C\equiv C^{\ominus}$

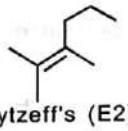
$\xrightarrow{CH_3-\overset{O}{\overset{||}{C}}-CH_3} CH_3-C\equiv C-\overset{O^{\ominus}}{\overset{|}{C}}-CH_3$

$\xrightarrow{I-CH_3} CH_3-C\equiv C-\overset{OCH_3}{\overset{|}{C}}-CH_3$ (CH_3 on C)

P 34 : (D) Enthalpy of adsorption ($\Delta H_{adsorption}$) is low and positive.

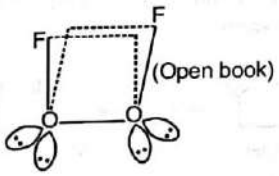
P 35 : (A) $AgNO_3$, in ethanol

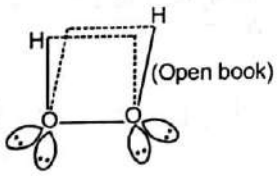
Sol. The bromide will give a light yellow precipitate of AgBr.

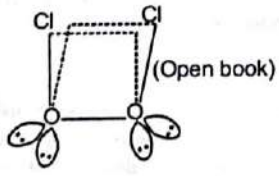
P 36 : (B) 

Sol. Saytzeff's (E2) elimination will give the desired product.

P 37 : (D) H_2C_2

Sol. (A) 

(B) 

(C) 

(D) H—C ≡ C —H (sp) Linear

P 38 : (C) $+\frac{18}{7}$, +2, +3

Sol. $(Fe_4[FeCN)_6]_3 \rightarrow 4Fe^{3+} + 3[3Fe(CN)_6]^{+}$

P 39 : (A) Gypsum contains a lower percentage of calcium than Plaster of Paris.

Sol. Gypsum $\Rightarrow CaSO_4.2H_2O$

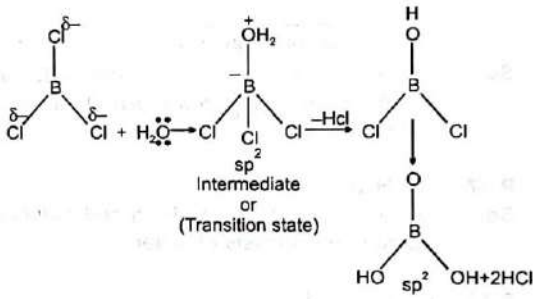
P.O.P $\Rightarrow CaSO_4.1/2\ H_2O$

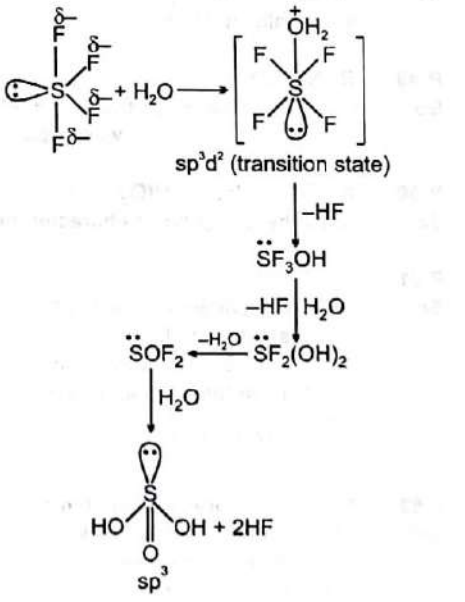
P 40 : (C) Adehydes are less sterically hindered.

Sol. In aldehydes, one of the group at carbonyl carbon is hydrogen that posses least steric hindrance where as in ketones, both the groups at carbonyl carbon are alkyl groups, former is more reactive towards nucleophilic addition reaction.

P 41 : (C) sp^3, sp^3d^2

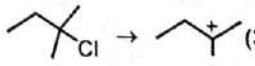
Sol.





P 42 : (D) different with 1, 0 and 2 lone pair of electrons respectively.

P 43 : (D) 2-chloro-2-methyl butane will react more rapidly because the carbocation formed will have 3 electron donating substituents.

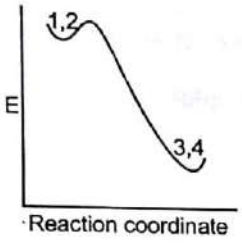
Sol.  (3° carbocation)

P 44 : (B) Chlorophyll

P 45 : (D) all of these

Sol. It is a strong oxidising agent.

P 46 : (A)



Sol. A more reactive (less stable) acid derivative is being converted into more stable acid derivative.

P 47 : (C) $H_2O(\ell)$; $H_2O(\ell)$

Sol. Water acts both as oxidising and reducing agent in electrolysis of water.

P 48 : (D) I, III and IV

Sol. *Trans*-$[Co(en)_2Cl_2]^+$ is optically inactive.

P 49 : (B) Na_2SO_4

Sol. $Ba(NO_3)_2 + Na_2SO_4 \rightarrow BaSO_4\downarrow + 2NaNO_3$
white ppt.

P 50 : (A) $TiO_2 < ZrO_2 < HfO_2$

Sol. Down the group basic character increases.

P 51 : (A) 11.11%

Sol. Two Cu^+ replaced by one Cu^{2+}
Total loss of Cu = 0.2 mole
So, mole of Cu^{2+} present in $Cu_{1.8}S$ = 0.2
% of Cu^{2+} in total copper content

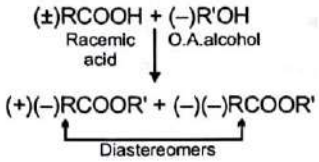
$$= \frac{0.2}{1.8} \times 100 = 11.11$$

P 52 : (C) low pressure and high temperature

Sol. When real gases behave ideally, it is critical temperature which is greater than normal and pressure becomes lower.

P 53 : (C) Diastereomeric mixture

Sol.



P 54 : (B) 8.0 kg

Sol. The reaction involved is
$2KOH + SO_2 \longrightarrow K_2SO_3 + H_2O$
Moles of KOH reacting = 250
$\Rightarrow$ Moles of SO_2 reacted = 125

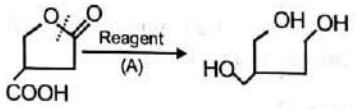
$$\text{Mass of } SO_2 = \frac{125 \times 64}{1000} = 8.0 \text{ kg}$$

P 55 : (D) 4

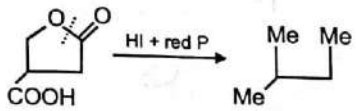
Sol. $IE = 13.6\ z^2$
$\Rightarrow 870.4 = 13.6\ z^2$
$\Rightarrow z^2 = 64 \Rightarrow z = 8$
$\therefore$ electronic configuration would be
$1s^2\ 2s^2\ sp^4$
$\therefore$ Total electrons in $\ell = 1$ will be 4.

P 56 : (A) LAH

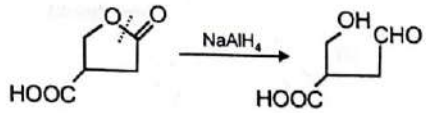
Sol. (A) LAH reduces both ester and acid to alcohols.



(B) HI + red P reduces both ester and acid to alkane.



(C) $NaAlH_4$ reduces only ester to aldehyde and alcohol.



(D) Zn-Hg/HCl only reduce carbonyl compounds.

P 57 : (D) 3 > 2 > 1

Sol. 1 is least basic as there is no lone pair on nitrogen. 2 is weaker base than 3 due to more resonance delocalization of lone pair in 2.

P 58 : (C) all are diamagnetic

Sol. No unpaired electron present after hybridization.

P 59 : (D) +8.96 kJ mol^{-1} ; towards reactant

Sol. $\Delta G = \Delta G^\circ + RT \ln Q$

$$= 4.73 \times 10^3 + 298 \times 8.3 \ln \frac{(2.1)^2}{0.8}$$

$= 8.959 \times 10^3$ J = 8.96 KJ

Reaction will proceed to left.

P 60 : (D) It forms dichloromethane with PCl_5.

Sol. Statements (A), (B) and (C) are correct, but statement (D) is incorrect, although

$HCOOH + PCl_5$ gives $H-\overset{\overset{O}{\|}}{C}-Cl$ (Formyl chloride).

P 61 : (D) $(-\infty, -2/3) \cup (1, \infty)$

Sol. $\log_2 (4x^2 - x - 1) - \log_2 (x^2 + 1) > 0$

$$\Rightarrow \frac{4x^2 - x - 1}{x^2 + 1} > 1 \Rightarrow 3x^2 - x - 2 > 0$$

$\Rightarrow (3x + 2)(x - 1) > 0$

$\Rightarrow x < -2/3$ or $x > 1$

$\Rightarrow x \in (-\infty, -2/3) \cup (1, \infty)$

P 62 : (A) directrix

Sol. Let coordinate of P, Q, R Q are as follows P_1 $(at_1^2, 2at_1)$, $P_2(at_2^2, 2at_2)$

$$Q_1\left(\frac{a}{t_1^2}, -\frac{2a}{t_1}\right), Q_2\left(\frac{a}{t_2^2}, -\frac{2a}{t_2}\right)$$

equation of $P_1 P_2$ is

$y(t_1 + t_2) = 2(x + a t_1 t_2)$

equation of $Q_1 Q_2$ is

$y(t_1 + t_2) + 2(x t_1 t_2 + a) = 0$

$\Rightarrow 0 = 2(1 + t_1 t_2)(x + a) \Rightarrow x + a = 0$

$\Rightarrow$ which is directrix

P 63 : (C) $a^2 + b^2 = c^2 + d^2$

Sol. $a \sec\alpha = d + c \tan\alpha$ ...(1)

And $b \sec\alpha = c - d \tan\alpha$ ...(2)

Squaring and adding equations (1) and (2)

$(a^2 + b^2) \sec^2\alpha = d^2 + c^2 \tan^2\alpha$

$2dc \tan\alpha + c^2 + d^2 \tan^2\alpha - 2dc \tan\alpha$

$(a^2 + b^2) \sec^2\alpha = c^2(\tan^2\alpha + 1) + d^2(1 + \tan^2\alpha)$

$= (c^2 + d^2) \sec^2\alpha$

Hence, $a^2 + b^2 = c^2 + d^2$

P 64 : (A) a = –40

Sol. Three points $A(\vec{a}), B(\vec{b}), C(\vec{c})$ are collinear if $\overline{AB} \parallel \overline{AC}$, $\overline{AB} = -20\hat{i} - 11\hat{j}$;

$\overline{AC} = (a - 60)\hat{i} - 55\hat{j}$

$$\overline{AB} \parallel \overline{AC} \Rightarrow \frac{a - 60}{-20} = \frac{-55}{-11} \text{ or } a = -40$$

P 65 : (D) none of these

Sol. Given limit is $\lim\limits_{x \to 0} \frac{|\sin x|}{x}$

$$\lim_{x \to 0^+} \frac{|\sin x|}{x} = 1 \text{ and } \lim_{x \to 0^-} \frac{-\sin x}{x} = -1$$

P 66 : (B) 15 sq. unit

Sol. Required area $= \int_{-2}^{3} [|x - 3|]\,dx$

$$\int_{-2}^{-1} |[x-3]|dx + \int_{-1}^{0} |[x-3]|dx + \int_{0}^{1} |[x-3]|dx + \int_{1}^{2} |[x-3]|dx + \int_{2}^{3} |[x-3]|dx$$

$$= \int_{-2}^{-1} 5.dx + \int_{-1}^{0} 4.dx + \int_{0}^{-1} 3.dx + \int_{2}^{1} 2.dx + \int_{2}^{3} 1.dx$$

$= 5(1) + 4(1) + 3(1) + 2(1) + 1(1)$

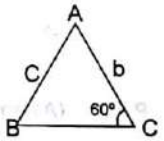
= 15 sq. unit

P 67 : (C) 105°

Sol. Let $a = \sqrt{3} + 1$ $b = \sqrt{3} - 1$

$c^2 = a^2 + b^2 - 2ab \cos 60^\circ = 6$

$c = \sqrt{6}$



$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$= \frac{3 + 1 - 2\sqrt{3} + 6 - (3 + 1 + 2\sqrt{3})}{2(\sqrt{3} - 1)\sqrt{6}}$$

$$= \frac{1 - \sqrt{3}}{2\sqrt{3}} = \cos 105^\circ$$

P 68 : (D) (–3, 3)

Sol. If α, β are roots of $x^2 + ax + 1 = 0$, then $|\alpha - \beta| < \sqrt{5}$

$$\Rightarrow \left|\sqrt{(\alpha + \beta)^2 - 4\alpha\beta}\right| < \sqrt{5}$$

$\Rightarrow \quad \left|\sqrt{a^2-4}\right| < \sqrt{5}$

$\Rightarrow \quad \sqrt{a^2-4} < \sqrt{5}$

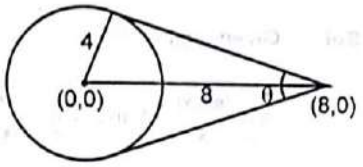
$\Rightarrow \quad a^2 - 4 < 5$

$\Rightarrow \quad a^2 < 9$

Hence, $a \in (-3, 3)$

P 69 : (C) $\frac{\pi}{3}$

Sol.



$\therefore \quad \sin\frac{\theta}{2} = \frac{4}{8} = \frac{1}{2}$

$\therefore \quad \frac{\theta}{2} = \frac{\pi}{6}$

$\therefore \quad \theta = \frac{\pi}{3}$

P 70 : (A) g = f

Sol. If tangents from origin to the circle $x^2 + y^2 + 2gx + 2fy + f^2 = 0$ are perpendicular implies origin lies on the director circle of the given circle

$\Rightarrow \quad \sqrt{2}\sqrt{g^2+f^2-f^2}$

$= \sqrt{(0+g)^2+(0+f)^2} = \sqrt{g^2+f^2}$

$2g^2 = g^2 + f^2$

$\Rightarrow \quad g^2 = f^2 \Rightarrow \quad g = \pm f$

P 71 : (A) an ellipse of eccentricity $\sqrt{3}/2$

Sol. $2px + y\sqrt{1-p^2} = 1$

i.e. $y = -\frac{2p}{\sqrt{1-p^2}}x + \frac{1}{\sqrt{1-p^2}}$ .....(i)

Let $m = -\frac{2p}{\sqrt{1-p^2}}$. Then $1 - p^2 = \frac{4}{4+m^2}$.

$\therefore$ equation of the line (i) becomes

$y = mx + \frac{\sqrt{4+m^2}}{2}$

i.e. $y = mx + \sqrt{\frac{m^2}{4}+1}$

$\therefore$ the curve is an ellipse for which $b^2 = \frac{1}{4}$

and $a^2 = 1$ for ellipse $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$

$\therefore \quad e^2 = \frac{3}{4} \quad \therefore \quad e = \frac{\sqrt{3}}{2}$

P 72 : (C) 20

Sol. Consider A's to be different

$\Rightarrow$ Number of ways $(6-1)! = 120$ then A_1, A_2 & A_3 can be arranged in 3! ways

$\Rightarrow \frac{120}{6} = 20$

P 73 : (B) 21

Sol. coefficient of x^8 in $(x + x^2 + ... + x^6)^6$

$= \text{coefficient of } x^2 \text{ in } \left(\frac{1-x^6}{1-x}\right)^6$

$= \text{coefficient of } x^2 \text{ in } (1-x)^{-6} = {}^7C_2 = 21$

P 74 : (A) x – 5y + 9 = 0

Sol. Two perpendicular lines passing through (1,2) and equally inclined to 3x - 2y = 4 have their slopes as

$\frac{3/2+1}{1-3/2} = -5$ and $\frac{1}{5}$

so possible lines are x – 5y + 9 = 0 and 5x + y – 7 = 0

P 75 : (C) $|\alpha + \beta + 2|$

Sol. $\text{ar}(\Delta PBC) = \frac{1}{2}\left|\begin{vmatrix}\alpha & \beta & 1\\ -3 & 5 & 1\\ 4 & -2 & 1\end{vmatrix}\right|$

$\Rightarrow \text{ar}(\Delta PBC) = \frac{1}{2}|7\alpha + 7\beta - 14|$

Also, $\text{ar}(\Delta PBC) = \frac{1}{2}\left|\begin{vmatrix}6 & -3 & 1\\ -3 & 5 & 1\\ 4 & -2 & 1\end{vmatrix}\right|$

$\frac{\text{ar}(\Delta PBC)}{\text{ar}(\Delta ABC)} = |\alpha + \beta - 2|$

P 76 : (A) 1383

Sol. The value of the expression would be dependent on the individual values of each of the terms in the expression. $[315^{1/3}]$ would give us a value of 6 and so would all the terms upto $[342^{1/3}]$. (as $6^4 = 216$ and $7^3=343$) Hence, the value of the expression from $[315^{1/3}] + [316^{1/3}] + ...[342^{1/3}] = 28 \times 6 = 168$

Similarly, the value of the expression from $[343^{1/3}] + [344^{1/3}] + ... + [511^{1/3}] = 169 \times 7 = 1183$.

Similarly, the value of the expression from $[512^{1/3}] + [513^{1/3}] + ... + [515^{1/3}] = 4\times8 = 32$.

thus the answer = 168 + 1183 + 32 = 1383.

P 77 : (B) $(-11, -17)$

Sol. $|PA - PB|$ will be maximum if P, A & B will be collinear . Hence $2a - b + 5 = 0$ and the

$$\text{determinant} \begin{vmatrix} a & b & 1 \\ 4 & -2 & 1 \\ 2 & -4 & 1 \end{vmatrix} = 0$$

P 78 : (B) $\frac{2}{5}$

Sol. Two adjacent rows can be selected out of 5 rows in 4 ways. Total ways of selecting 2 rows is ${}^5C_2$ hence $P = \frac{4}{{}^5C_2} = \frac{2}{5}$

P 79 : (B) $\sqrt{\frac{q}{p}}$

Sol. Given, $\operatorname{cosec}\theta = \frac{p+q}{p-q} \Rightarrow \frac{1}{\sin\theta} = \frac{p+q}{p-q}$,

Apply componendo and dividendo, we get

$$\frac{1+\sin\theta}{1-\sin\theta} = \frac{p+q+p-q}{p+q-p-q}$$

$$\Rightarrow \left[\frac{\cos\frac{\theta}{2}+\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}-\sin\frac{\theta}{2}}\right]^2 = \frac{p}{q}$$

$$\Rightarrow \left[\frac{1+\tan\theta/2}{1-\tan\theta/2}\right]^2 = \frac{p}{q} \Rightarrow \tan^2\left(\frac{\pi}{4}+\frac{\theta}{2}\right) = \frac{p}{q}$$

$$\Rightarrow \cot^2\left(\frac{\pi}{4}+\frac{\theta}{2}\right) = \frac{p}{q}$$

P 80 : (D) none of these

Sol. Here, $\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+....\text{to }\infty = \frac{\pi^2}{8}$

Let $\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+....\text{to }\infty = x$. Then

$$x = \frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+....\text{to }\infty$$

$$= \left(\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+....\text{to }\infty\right) + \left(\frac{1}{2^2}+\frac{1}{4^2}+\frac{1}{6^2}+....\text{to }\infty\right)$$

$$= \frac{\pi^2}{8} + \frac{1}{2^2}\left(\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+....\text{to }\infty\right)$$

$$= \frac{\pi^2}{8} + \frac{1}{4}\left(\frac{\pi^2}{8}\right) = \frac{5\pi^2}{32}$$

Note : This is the Basel problem which asks for the precise summation of the reciprocals of the squares of the natural numbers, till infinity.

Euler found the exact sum to be $\frac{\pi^2}{6}$.

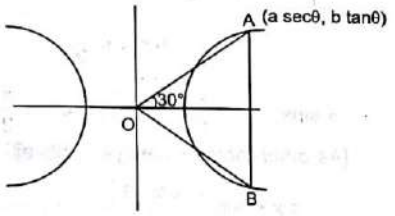
P 81 : (D) 16

Sol. $xy = 16 \quad \therefore \quad c = 4$

distance between foci = $4c = 16$

P 82 : (A) $e > \frac{2}{\sqrt{3}}$

Sol.



$$\frac{b}{a}\sin\theta = \tan 30°$$

$$\operatorname{cosec}\theta = \frac{\sqrt{3}\,b}{a} > 1 \quad [\because \theta \neq \frac{\pi}{2}]$$

$$3(e^2 - 1) > 1$$

$$e^2 - 1 > \frac{1}{3} \Rightarrow e^2 > \frac{4}{3} \Rightarrow e > \frac{2}{\sqrt{3}}$$

P 83 : (D) 13/16

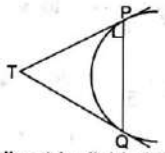
Sol. 1 – P (Determinant has negative value)

$$1 - \frac{3}{16} = \frac{13}{16}\left(\begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix}; \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix}; \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}\right)$$

P 84 : (C) 1 : 1

Sol. Let P (t), then Q $\left(-t-\frac{2}{t}\right)$

$\therefore$ T $\left(a\left(-2-t^2\right), -\frac{2a}{t}\right)$ and P $(at^2, 2at)$



Let the directrix divide it in the ratio k : 1

then $-a = \frac{kat^2 + a(-2-t^2)}{k+1}$

$-ka - a = kat^2 - 2a - at^2$

$a = a\,(kt^2 - t^2 + k)$

$t^2 + 1 = k(t^2 + 1)$ i.e. $k = 1$

P 85 : (C) $0 < x < \sin^{-1}\left(\frac{\sqrt{5}-1}{2}\right)$

Sol. The given inequality is equivalent to

$\cos x > \tan x > 0 \; ; \; 0 < x < \frac{\pi}{2}$

$\Rightarrow \cos^2 x > \sin x > 0 \; ; \; 0 < x < \frac{\pi}{2}$

$\Rightarrow \sin^2 x + \sin x - 1 < 0 \; ; \; 0 < x < \frac{\pi}{2}$

$\Rightarrow \left[\sin x - \left(\frac{-1+\sqrt{5}}{2}\right)\right]\left[\sin x - \left(\frac{-1-\sqrt{5}}{2}\right)\right] < 0$

$\; ; \; 0 < x < \frac{\pi}{2}$

$\Rightarrow \sin x - \left(\frac{-1+\sqrt{5}}{2}\right) < 0 \; ; \; 0 < x < \frac{\pi}{2}$

[As other factor is always positive]

$\Rightarrow 0 < x < \sin^{-1}\left(\frac{\sqrt{5}-1}{2}\right)$.

P 86 : (B) 2abc

Sol. Let $\sin^{-1} a = A$, $\sin^{-1} b = B$, $\sin^{-1} c = C$

Therefore, $\sin A = a$, $\sin B = b$, $\sin C = c$

and $A + B + C = \pi$, then

$\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$

Now $a\sqrt{(1-a^2)} + b\sqrt{(1-b^2)} + c\sqrt{(1-c^2)}$

$= \sin A \cos A + \sin B \cos B + \sin C \cos C$

$= \frac{1}{2}[\sin 2A + \sin 2B + \sin 2C]$

$= 2 \sin A \sin B \sin C$

$= 2abc$

P 87 : (A) $2^{2n-1} + \frac{(2n-1)!}{n!(n-1)!}$

Sol. ${}^{2n}C_0 + {}^{2n}C_1 + \ldots\ldots {}^{2n}C_n$

$= {}^{2n}C_0 + {}^{2n}C_1 + \ldots + {}^{2n}C_{2n} - ({}^{2n}C_{n+1} + \ldots + {}^{2n}C_{2n})$

$= 2^{2n} - ({}^{2n}C_{n-1} + {}^{2n}C_{n-2} + \ldots\ldots + {}^{2n}C_0)$

$= 2^{2n} - ({}^{2n}C_0 + {}^{2n}C_1 + \ldots + {}^{2n}C_{n-1} + {}^{2n}C_n - {}^{2n}C_n)$

$\therefore 2({}^{2n}C_0 + {}^{2n}C_1 + \ldots\ldots {}^{2n}C_n) = 2^{2n} + {}^{2n}C_n$

$\therefore {}^{2n}C_0 + {}^{2n}C_1 + \ldots\ldots + {}^{2n}C_n$

$= 2^{2n-1} + \frac{2n\{(2n-1)!\}}{2n(n!)\{(n-1)!\}}$

$= 2^{2n-1} + \frac{(2n-1)!}{n!(n-1)!}$

P 88 : (A) $\beta = 0$, $\delta = 0$

Sol. equation is $\begin{vmatrix} -\lambda & -2 & 1 \\ 2 & -\lambda & -3 \\ -1 & 3 & -\lambda \end{vmatrix} = 0$

$\lambda^3 + 14\lambda = 0$ using Caley-Hamilton theorem

$A^3 + 14A = 0$

Hence, $\beta = 0$ $\gamma = 14$ $\delta = 0$

P 89 : (C) $\phi(x) = x + c$, $c \in R$

Sol. $\int \tan^4 x\, dx = \int \tan^2 x\,(\sec^2 x - 1)\, dx$

$= \int \tan^2 x \; d(\tan x) - \int (\sec^2 x - 1)\, dx$

$= \frac{1}{3} \tan^3 x - \tan x + x + c$

P 90 : (B) $2y = 8x - 7$

Sol. $f'(1) = \lim_{h\to 0} \frac{f(1+h) - f(1)}{h} = \frac{\frac{\sinh}{h^2 + 2h} - \frac{1}{2}}{h}$

$= \frac{2\sinh - h^2 - 2h}{2h^2(h+2)} = \frac{2(\sinh - h) - h^2}{4h^2}$

$\because \lim_{h\to 0} \frac{\sinh - h}{h^2} = 0 \Rightarrow h = -\frac{1}{4}$

Thus tangent has slope $-\frac{1}{4}$ and therefore slope of normal is 4.

Hence equation of normal is

$\left(y - \frac{1}{2}\right) = 4\,(x - 1) \Rightarrow y = 4x - \frac{7}{2}$

REPLICA 10

P 1 : (A) $R/(\gamma - 1) + R/(1 - x)$

Sol. $PV^x = C$

$$C = \frac{R}{1-x} + \frac{R}{\gamma - 1}$$

P 2 : (A) $v_{max} \propto m^{1/9}$

Sol. The size of the cube is $L \propto m^{1/3}$, so $A \propto m^{2/3}$ since the density is constant.
Constant speed required,
$m \propto P = Fe \propto Av^3 \propto m^{2/3}v^3$
Therefore, $m \propto v^9$

P 3 : (C) $\lambda = b^2/3d$

Sol. Path difference = $(S_2P - S_1P)$
From figure, $(S_2P)^2 - (S_1P)^2 = b^2$
or $(S_2P - S_1P)(S_2P + S_1P) = b^2$

$$\text{or} \quad (S_2P - S_1P) = \frac{b^2}{2d}$$

For dark fringes, $\frac{b^2}{2d} = (2n + 1)\ \frac{\lambda}{2d}$

For n = 0, $\frac{b^2}{2d} = \frac{\lambda}{2}$ or $\lambda = \frac{b^2}{d}$

For n = 1, $\frac{b^2}{2d} = \frac{3\lambda}{2}$ or $\lambda = \frac{b^2}{3d}$

P 4 : (C) $\sqrt{3} \times 10^8$

Sol. From Brewster's law, $m = \tan i_p$

$$\text{or} \quad \frac{c}{v} = \tan 60°$$

$$\therefore v = \frac{3\times10^8}{\sqrt{3}}$$

$$= \sqrt{3} \times 10^8 \text{ m/s}$$

P 5 : (A) 6.93s

Sol. The time, $T_{1/2}$, for the amplitude to drop to half of its initial value is given by

$$T_{1/2} = \frac{\ln(1/2)}{b/2m} = \frac{0.693\times2\times0.2}{40\times10^{-3}} = 6.93\text{s}$$

P 6 : (A) 2.66×10^4 km

Sol. Here time period of satellite with respect to observer on equator is 24 Hrs and satellite is moving from west to east, so angular velocity of satellite with respect to earth's axis of rotation (considered as fixed) is,

$\omega = \frac{2\pi}{T_S} + \frac{2\pi}{T_E}$ where T_S & T_E are time period of satellite & earth respectively

$$\Rightarrow \quad \omega = \frac{\pi}{6}(\text{hr}^{-1}) = 1.45\times10^{-4}\text{rad/s}$$

$$\Rightarrow \quad r^{3/2} = \frac{\sqrt{GM}}{\omega}$$

Substituting the values we get,
$\Rightarrow \quad r = 2.66 \times 10^7$ m $= 2.66 \times 10^4$ km

P 7 : (D) $\frac{1}{2}\varepsilon_0\left(\frac{V^2}{d^2}\right)$

Sol. $u_{vol} = \frac{1}{2}\varepsilon_0 E^2$

$$\Delta V = E.d \quad \Rightarrow \quad E = \frac{V}{d}$$

$$u_{vol} = \frac{1}{2}\varepsilon_0\frac{v^2}{d^2}$$

P 8 : (B) 0.2 H

Sol. Average emf $= L\frac{\Delta i}{\Delta t}$

$$8 = L\frac{(4-2)}{0.05} \ ; \text{ Solving, } L = 0.2\text{H}$$

P 9 : (B) $\lambda_e > \lambda_p = \lambda_n > \lambda_\alpha$

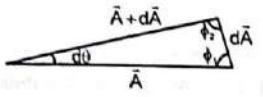
Sol. De Broglie wavelength $\lambda = \frac{h}{mv}$

$$\Rightarrow \quad \lambda = \frac{h}{\sqrt{2mE}} \quad \Rightarrow \quad \lambda \propto \frac{1}{\sqrt{m}}.$$

we know $m_\alpha > m_p = m_n > m_e$
$\Rightarrow \quad \lambda_e > \lambda_p = \lambda_n > \lambda_\alpha$

P 10 : (A) $\vec{B} \cdot \vec{A} = 0$

Sol. Let $\vec{A}$ at any instant of time 't' changes to $\vec{A} + d\vec{A}$ at instant t + dt as shown in diagram.



$\because$ magnitude of vector does not change

$$|\vec{A}| = |\vec{A} + d\vec{A}|$$

Hence $\phi_1 = \phi_2 = \pi/2$ [as $d\theta \to 0$]

or $d\vec{A} \perp \vec{A}$ or $\frac{d\vec{A}}{dt} \perp \vec{A}$

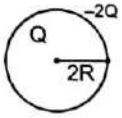
P 11 : (C) $-5\ ms^{-1}$

Sol. Acceleration between 8 and 10 s (or at t = 9 s):

$$a = \frac{v_2 - v_1}{t_2 - t_1} = \frac{5-15}{10-8} = -5ms^{-2}$$

P 12 : (D) zero

Sol. Initial electrostatic energy (u_i)



$$= \frac{k(2Q)^2}{2R} - \frac{2kQ^2}{R} = 0$$

Final electrostatic energy (u_f)

$$= \frac{k(2Q)^2}{2(2R)} + \frac{k.Q(-2Q)}{2R} = 0$$

Change = $u_f - u_i = 0$

P 13 : (D) $\frac{r_1 r_2^2}{R^3}$

Sol. Electric field at r_1 is given as

$$E_1 = \frac{KQr_1}{R^3}$$ {where Q is total charge}

Electric field at r_2 is given as

$$E_2 = \frac{KQ}{r_2^2}, \text{ So, } \frac{E_1}{E_2} = \frac{r_1 r_2^2}{R^3}$$

P 14 : (C) $2.1 \times 10^{-8}\ (\hat{k})$

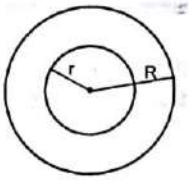
Sol. $$B = \frac{E}{C} = \frac{6.3}{3\times10^8} = 2.1\times10^{-8}T$$

Also direction of wave propagation is along $\vec{E}\times\vec{B}$

So $\vec{B} = (2.1\times10^{-8}T)\hat{k}$

P 15 : (C) $\frac{\rho_0 R}{3\varepsilon_0}$

Sol. Charge enclosed by a sphere of



$$q = \int_0^r \rho dV$$

$$= \int_0^r \rho(4\pi r^2 dr)$$

$$= 4\pi\rho_0 \int_0^r \left(r^2 - \frac{r^3}{R}\right) dr$$

$$= 4\pi\rho_0 \left[\frac{r^3}{3} - \frac{r^4}{4R}\right]$$

Total charge on the sphere,
Put r = R

$$Q = 4\pi\rho_0 \left[\frac{R^3}{3} - \frac{R^3}{4R}\right] = \frac{\pi\rho_0 R^3}{3}$$

Inside the sphere

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2} = \frac{\rho_0}{\varepsilon_0}\left[\frac{r}{3} - \frac{r^2}{4R}\right] = \frac{\pi\rho_0 R^3}{3}$$

Outside the sphere

$$E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2} = \frac{\rho_0}{\varepsilon_0}\frac{R^3}{3r^2}$$

For maximum field,

$$\frac{dE}{dr} = 0$$

$$\Rightarrow \quad \frac{\rho_0}{\varepsilon_0}\left[\frac{1}{3} - \frac{r}{2R}\right] = 0$$

$$\therefore \quad F_{max} = \frac{\rho_0}{\varepsilon_0}\left[\frac{2}{9}R - \frac{1}{9}R\right] = \frac{\rho_0 R}{9\varepsilon_0}$$

P 16 : (A) $\frac{t_1 t_2}{t_1 + t_2}$

Sol. Let R_1 and R_2 be the resistances of the 2 windings respectively.
Since in every case the kettle is connected to the same electric mains, hence we get

$$H_1 = \frac{V^2}{R_1}t_1,\ H_2 = \frac{V^2}{R_2}t_2$$

But $H_1 = H_2$

Hence, $\frac{V^2}{R_1}t_1 = \frac{V^2}{R_2}t_2$

or $\frac{t_1}{t_2}=\frac{R_1}{R_2}$

When the windings are connected in parallel

$$R_{eq}=\frac{R_1R_2}{R_1+R_2}$$

or $\frac{R_{eq}}{R_2}=\frac{R_1}{R_1+R_2}$

$$\frac{R_2}{R_{eq}}=\frac{R_2}{R_1}+1=\frac{t_2}{t_1}+1$$

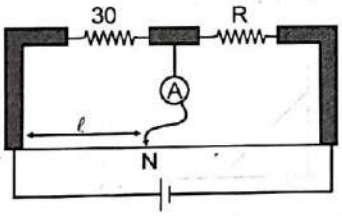
Let H' be the heat produced in this case. Then

$$H'=\frac{V^2}{R_{eq}}t'=\frac{V^2}{R_2}t_2$$

$$t'=\frac{R_{eq}t_2}{R_2}=\frac{t_1t_2}{t_1+t_2}$$

P 17 : (D) 45 Ω

Sol. As experimenter had no idea about end correction so he has used the equation



$\frac{30}{R}=\frac{\ell}{100-\ell}$ ...(i)

but in practicality, the equation must be

$\frac{30}{R}=\frac{\ell+2}{(100-\ell)+3}$ ...(ii)

As he got the same result so

$\frac{\ell}{100-\ell}=\frac{\ell+2}{(100-\ell)+3}$

$\Rightarrow \ell = 40$ cm

So, by equation (i)

R = 45 Ω

P 18 : (B) 90 cal/gm

Sol. According to principle of calorimetry

$ML_F + Ms\Delta T = (ms\Delta T)_{water} + (ms\Delta T)_{flask}$

$\Rightarrow \quad 50L_F + 50\times1\times(40-0) = 200\times1\times(70-40)+W(70-40)$

or $50L_F+2000 = (200 + W)30$

or $5L_F = 400 + 3W$ ...(i)

Now, the system contains (200 + 50) gm of water at 40°C, so when further 80 gm of ice is added,

$80L_F + 80\times1\times(10-0) = 250\times1\times(40-10)+W(40-10)$

or $80L_F+800=(250+W)30$

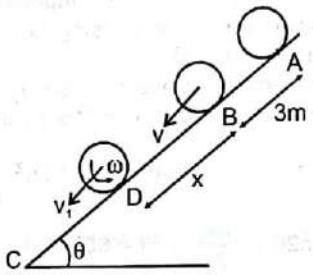
or $8L_F = 670 + 3W$ ...(ii)

Solving equations (i) and (ii)

$L_F = 90$ cal/gm and $W = \frac{50}{3}$ gm.

P 19 : (C) 7 m

Sol.



The solid cylinder will finally roll as for rolling, friction required is $\frac{\tan\theta}{3}=0.25$

Let it starts rolling at point D.

So v = 6 [By $v^2 = u^2 + 2as$]

$v_1 = 6 + [g\sin\theta - \mu g\cos\theta]t$

$v_1 = \omega R = \alpha tR$

$\tau_{net} = \mu mg\cos\theta R = \frac{mR^2}{2}\alpha \Rightarrow \alpha = \frac{2\mu g\cos\theta}{R}$

$2\mu g\cos\theta t = 6 + g\sin\theta t - \mu g\cos\theta t$

$\Rightarrow t = 1$ sec

$S = 7$ m [by $s = ut + \frac{1}{2}at^2$]

P 20 : (A) The amount the wire is stretched by the hanging mass is 7.84×10^{-4} m

Sol. Stress = y (strain)

$\frac{mg}{A}=y\frac{\Delta\ell}{\ell}$

$\frac{4(9.8)}{5\times10^{-3}\times(10^{-4})}=2\times10^{11}\left(\frac{\Delta\ell}{\ell}\right)$

$0.8 \times 9.8 \times 10^7 = 2 \times 10^{11} \times \frac{\Delta\ell}{2}$

$784 \times 10^{-4} = \Delta\ell$

P 21 : (B) 485 Hz

Sol. Velocity of the car = $\frac{36000\text{m}}{3600\text{s}}$ = 10 m/s

$v_{emitted}$ = 500 Hz and v_{sound} = 330 m/s

$v_{obs} = v_{emitted} \frac{(v_{sound} - v_{obs})}{v_{sound}}$

Observed frequency $v_{obs} = 500\frac{(330-10)}{330}$

or v_{obs} = 485 Hz

P 22 : (A) 30°

Sol. intensity of unpolarised light (I_0) = 32 W/m²
Intensity of emergent light (I_3) = 3 W/m²
As we know intensity of unpolarised light becomes half after passing first polarizer
i.e., I_1 = 16 W/m² .....(i)
After passing second polariser $I_2=16\cos^2\theta$
After third polariser, $I_3 = (16\cos^2\theta)\sin^2\theta$

$\Rightarrow 3 = 16\cos^2\theta \sin^2\theta \Rightarrow (\sin2\theta)^2 = \frac{3}{4}$

$\Rightarrow \sin2\theta = \frac{\sqrt{3}}{2} \Rightarrow 2\theta = 60° \Rightarrow \theta = 30°$

P 23 : (C) 19

Sol. $\alpha = \frac{\beta}{1-\beta} \quad \Rightarrow \quad \alpha = \frac{0.95}{0.05} = 19$

$\beta = \frac{\alpha}{1-\alpha} = \frac{0.95}{1-0.95} = 19$

P 24 : (C) 20 μC

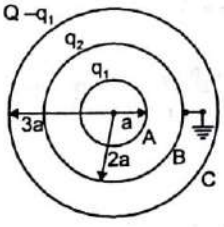
Sol. $\frac{1}{4\pi\epsilon_0}\cdot\frac{\left(\frac{Q}{2}\right)\cdot\left(\frac{Q}{2}\right)}{(0.3\text{m})^2} = (100\text{ N/m})(0.1\text{ m})$

$Q^2 = \frac{4\times100\times(0.1)\times(0.3)^2}{9\times10^9}$

$Q = 20\times10^{-6}$ C = 20 μC

P 25 : (B) $-\frac{8Q}{11}$

Sol. From given conditions,
$V_A = V_C$ and $V_B = 0$



$\Rightarrow V_B = \frac{K(Q-q_1)}{3a} + \frac{Kq_2}{2a} + \frac{3q_1}{2a} = 0$

$\Rightarrow 2Q + q_1 + 3q_2 = 0$ .... (1)

Using $V_A = V_C$

$\frac{K(Q-q_2)}{3a} + \frac{Kq_2}{3a} + \frac{Kq_1}{3a}$

$= \frac{Kq_1}{a} + \frac{K(Q-q_1)}{3a} + \frac{Kq_2}{2a}$

$\Rightarrow q_1 = -\frac{q_2}{4}$ .... (2)

Using it in (1), $q_2 = -\frac{8}{11}Q$

P 26 : (A) 1 A

Sol. When switch K_1 is opended and K_2 is closed it becomes L–C circuit so applying energy conservation :

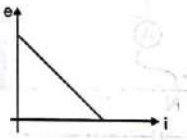
$\frac{Q_0^2}{2C} = \frac{1}{2}Li^2$; $Q_0 = C_{eq}V = \frac{C_1C_2}{C_1+C_2}\cdot V$

$= (20\times10^{-6})$

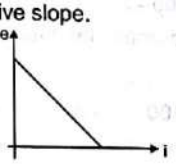
$\frac{(20\times10^{-6})^2}{2\times2\times10^{-6}} = \frac{1}{2}(0.2\times10^{-3})\,i^2$

$\Rightarrow i = 1$ A

P 27 : (A)



Sol. The potential difference across the inductor is e = E– iR.
Hence the plot of e versus i is a straight line with negative slope.



P 28 : (B) 69 dB

Sol. Let intensity due to a single person = I
then 10 log I/I_0 = 60
Also, Intensity due to 8 persons = 8I.
∴ final decibal level

$= 10\log\left(\frac{8I}{I_0}\right) = 10\left(\log\frac{I}{I_0} + \log8\right)$

$= 60 + 10\log8 = 60 + 30\log2$
$= 60 + 30\,(.3010) \simeq 69.$

P 29 : (A) The value of aV_0 is 4N.

Sol. Area under the curve is equal to number of molecules of the gas sample. Hence

$$N = \frac{1}{2} . a . V_0 \Rightarrow aV_0 = 2N$$

$$V_{avg} = \frac{1}{N}\int_0^{\infty} v\, N(V)dV = \frac{1}{N}\int_0^{V_0} C.\left(\frac{a}{V_0}.V\right)dV$$

$$= \frac{2}{3} V_0 \Rightarrow \frac{V_{avg}}{V_0} = \frac{2}{3}$$

$$v_{rms}^2 = \frac{1}{N}\int_0^{\infty} V^2 N(V)dV$$

$$= \frac{1}{N}\int_0^{V_0} V^2\left(\frac{a}{V_0}.V\right) dV = \frac{V_0^2}{2} \Rightarrow \frac{V_{rms}}{V_0} = \frac{1}{\sqrt{2}}$$

Area under the curve from 0.5 V_0 to V_0 is $\frac{3}{4}$ of total area.

P 30 : (B) $RT_0 (18\ \ell n\, 3 - 8)$

Sol. Work done in process AB

$$\Delta W_{AB} = \frac{2R\left[\frac{T_0}{3} - 3T_0\right]}{\left(\frac{1}{3} - 1\right)}$$

$\therefore$ $W = -8RT_0$

for process B $\rightarrow$ C, $W_{BC} = 0$

and C $\rightarrow$ A (isothermal process)

$$\Delta W_{CA} = 2.R\,(3T_0)\ \ell n \frac{V_e}{V_i} \qquad \left(\frac{V_e}{V_i} = 27\right)$$

$\Delta W_{CA} = 18\ RT_0\ \ell n\, 3$

Now total work done

$\Delta W = \Delta W_{AB} + \Delta W_{BC} + \Delta W_{CA}$

$= -8RT_0 + 0 + 18RT_0\ \ell n\, 3$

$\Delta W = RT_0 (18\ \ell n\, 3 - 8)$

P 31 : (A) a square

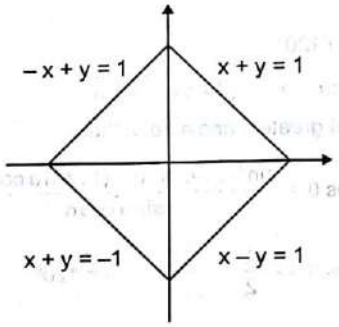
Sol. Let the two perpendicular lines be the coordinates of the point such that the sum of distances from two perpendicular lines is unity.

Hence locus of (h, k) is $|x| + |y| = 1$

Case I : $x \geq 0$ and $y \geq 0$ {1st Quadrant}

$x + y = 1$

Case II : The equation $|x| + |y| = 1$ represents a square as shown in the following figure.



P 32 : (A) $\frac{(n+1)^2}{4n}$

Sol. $\tan(\theta - \phi) = \frac{\tan\theta - \tan\phi}{1 + \tan\theta\tan\phi} = \frac{(n-1)\tan\phi}{1 + n\tan^2\phi}$

$$= \frac{(n-1)}{(\cot\phi + n\tan\phi)}$$

For minimum of $\cot\phi + n\tan\phi$,

$\tan^2\phi = \frac{1}{n}$

maximum value of $\tan^2(\theta - \phi)$ is

$$\frac{(n-1)^2}{(2n\tan\phi)^2} = \frac{(n-1)^2}{4n^2\tan^2\phi} = \frac{(n-1)^2}{4n}$$

P 33 : (D) ±1

Sol. Taking first two members, then

$2\cos(\alpha + \theta)\sin(\gamma + \alpha)$

$= 2\cos(\beta+\theta)\sin(\beta + \gamma)\sin(2\alpha + \theta + \gamma) - \sin(\theta - \gamma)$

$= \sin(2\beta + \theta + \gamma) - \sin(\theta - \gamma)\sin(2\alpha + \theta + \gamma)$

$= \sin(2\beta + \theta + \gamma)$

$2\alpha + \theta + \gamma = n\pi + (-1)^n (2\beta + \theta + \gamma)$

For n = 0, $2\alpha + \theta + \gamma = 2\beta + \theta + \gamma$

$\alpha = \beta$

Similarly, taking last two members we get

$\beta = \gamma$

Hence, $\alpha = \beta = \gamma$

Also take n = 1, –1,

Then we get $\alpha + \beta + \gamma + \theta = \frac{\pi}{2}, -\frac{\pi}{2}$

$$K = \frac{\cos(\alpha + \theta)}{\sin(\beta + \gamma)} = \frac{\cos\left(\frac{\pi}{2} - (\beta + \gamma)\right)}{\sin(\beta + \gamma)}$$

$$= \frac{\sin(\beta + \gamma)}{\sin(\beta + \gamma)} = 1$$

$$\text{And } k = \frac{\cos\left(-\frac{\pi}{2} - (\beta + \gamma)\right)}{\sin(\beta + \gamma)} = -1$$

Hence, $k = \pm 1$

P 34 : (C) 120°

Sol. $\sin\alpha, \cos\alpha, \sqrt{1+\sin\alpha\cos\alpha}$

Let greatest angle be θ, then

$$\cos\theta = \frac{\sin^2\alpha+\cos^2\alpha-(1+\sin\alpha\cos\alpha)}{2\sin\alpha\cos\alpha}$$

$$\cos\theta = -\frac{1}{2} \quad \Rightarrow \quad \theta = 120°$$

P 35 : (C) 446

Sol. $$T_n = \frac{1^3+2^3+3^3+....+n^3}{1+3+5+....n \text{ terms}} = \frac{\sum n^3}{\frac{n}{2}[2.1+(n-1)2]}$$

$$\Rightarrow T_n = \frac{1}{4}.\frac{n^2(n+1)^2}{n^2} = \frac{1}{4}(n^2+2n+1)$$

Now, $S_n = \frac{1}{4}(\Sigma n^2 + 2\Sigma n + 1)$

$$= \frac{1}{4}\left[\frac{n(n+1)(2n+1)}{6} + 2.\frac{n(n+1)}{2} + n\right]$$

$$= \frac{n}{24}[2n^2 + 3n + 1 + 6n + 6 + 6]$$

$$= \frac{n}{24}[2n^2 + 9n + 13]$$

Putting n = 16, we get,

$$S_n = \frac{16}{24}[2(256) + 144 + 13] = \frac{2}{3}(669) = 446$$

P 36 : (D) $\left(x-\frac{23}{3}\right)^2 + \left(y-\frac{23}{3}\right)^2 = 80$

Sol. The point $\left(\frac{23}{3}, \frac{23}{3}\right)$ lie on the y = x and length of perpendicular from this point to the line x + 2y – 3 = 0 is equal to $\left|\frac{\frac{23}{3}-3}{\sqrt{1+4}}\right|$

$$= \frac{20}{\sqrt{5}} = 4\sqrt{5}$$

hence the equation of circle is

$$\left(x-\frac{23}{3}\right)^2 + \left(y-\frac{23}{3}\right)^2 = 80$$

P 37 : (B) a hyperbola with eccentricity $\sqrt{2}$

Sol. Let P be the point $\left(ct, \frac{c}{t}\right)$, then

coordinates of A are (ct, 0) and

coordinates of B are $\left(0, \frac{c}{t}\right)$.

$\therefore$ coordinates of mid point of AB are $\left(\frac{ct}{2}, \frac{c}{2t}\right)$.

$\therefore$ the locus of the mid point is $xy = \frac{c^2}{4}$ which is a rectangular hyperbola.

$\therefore$ eccentricity = $\sqrt{2}$

P 38 : (D) Parabola

Sol. Let equation of circle be

$(x-h)^2 + (y-K)^2 = r^2$ .....(1)

$\because$ (1) passes through (a, 0)

$\Rightarrow$ $(a-h)^2 + K^2 = r^2$....(2)

$\because$ (1) touches x + y = 0

$\therefore$ $\frac{h+K}{\sqrt{2}} = r$ .....(3)

from (2) and (3)

$$(a-h)^2 + K^2 = \frac{h^2+K^2+2hK}{2}$$

$h^2 + K^2 - 2hK - 4ah + 2a^2 = 0$

$\therefore$ locus of centre (h, K) of (1) is

$x^2 + y^2 - 2xy - 4ax + 2a^2 = 0$ ....(4)

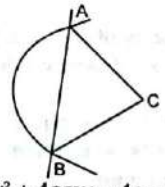
$\because$ $h^2 = ab$

$\Rightarrow$ (4) represents a parabola

P 39 : (A) $\left(\frac{4am^2}{\ell^2}, \frac{4am}{\ell}\right)$

Sol. y – coordinates of the points of intersection of the line $\ell x + my = 1$ and the parabola $y^2 = 4ax$ are given by

$$y^2 - 4a\,\frac{(1-my)}{\ell} = 0$$



i.e. $\ell y^2 + 4amy - 4a = 0$

$\therefore \quad y_1 + y_2 = \frac{-4am}{\ell}$

y_1, y_2 are y-coordinates of A and B respectively
since $y_1 + y_2 + y_3 = 0$
(y_3 is the y-coordinate of D)

$\therefore y_3 = \frac{4am}{\ell}$

P 40 : (B) 3360

Sol. (i) Total no. of rectangles including squares $= {}^{15}C_2 \times {}^{9}C_2$

$= \frac{15\times14}{2} \times \frac{9\times8}{2} = 105 \times 36 = 3780$

(ii) Total no. of squares {squares of sides 1, 2, 3, etc.} =
$4\times8 + 13\times7 + 12\times6 + 11\times5 + 10\times4 + 9\times3 + 8\times2 + 7\times1$
$= 112 + 91 + 72 + 55 + 40 + 27 + 16 + 7 = 420$

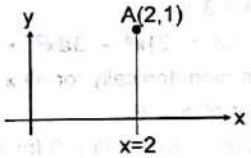
(iii) Total no. of rectangles excluding squares
$= 3780 - 420 = 3360$

P 41 : (A) 134055

Sol. Middle place 1, 4 & 9
Two terminal positions 1, 2,....., 9
Hence total numbers = 9. 9. 3 = 243
For the middle place 1, 4 & 9 will come 81 times
$\Rightarrow$ sum = 81 × 10 (1 + 4 + 9) — A
For units place each digit from 1 to 9 will appear 27 times
$\Rightarrow$ sum = 27 × 10 (1 + 2 +..... + 9) — B
For hundredth's place, similarly sum
= 27 × 10 (1 + 2 +..... + 9) — C
A + B + C gives the required sum

P 42 : (A) (0, 0)

Sol.



Altitude from A is $\perp$ to the x-axis
$\therefore$ side BC will be || to x-axis.
$\therefore$ slope of BC = 0
slope of internal angle bisector through C = $1-\sqrt{2}$
Let the slope of AC = m

$\therefore \frac{m-(1-\sqrt{2})}{1+m(1-\sqrt{2})} = \frac{1-\sqrt{2}-0}{1-0}$

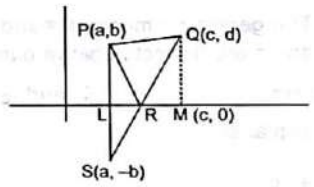
$\Rightarrow m = -1$
$\Rightarrow$ equation of AC is given by
$y - 1 = -1(x - 2)$
$y = -x + 3$
$\therefore$ point of intersection of AC & internal angle bisector from C will be point C $\equiv$ (3, 0)
Now BC || x – axis
as C lies on x – axis B will also lie on x – axis

Ar (ΔABC) $= \frac{3}{2} = \frac{1}{2}$ base × height

$\Rightarrow 3 =$ (base) × 1
$\Rightarrow$ base = 3.
$\therefore$ points an x–axis at distance of 3 units from (3, 0) are (0, 0) and (6, 0)
But for (6,0), $y = (1-\sqrt{2})(x - 3)$ will not be internal angular bisector
$\therefore$ point B $\equiv$ (0, 0)

P 43 : (C) internally in the ratio b : d

Sol.



Let S be the point (a, – b)
Equation of the line SQ

$y + b = \frac{b+d}{c-a}(x - a)$

It meets x-axis.

$\therefore \quad x = \frac{bc+ad}{b+d}$

it divides LM in the ratio b : d internally.

P 44 : (A) $\frac{5}{9}$

Sol. $P(\text{Even}) = \frac{2}{3}$; $P(\text{Odd}) = \frac{1}{3}$

The event 'sum is even' is equivalent to the event 'either both show even numbers or both show odd numbers'

$\therefore$ required probability $= \frac{2}{3}\times\frac{2}{3} + \frac{1}{3}\times\frac{1}{3} = \frac{5}{9}$

P 45 : (A) $\frac{4}{13}(3\sqrt{3}-1)$

Sol. line $y = \sqrt{3}x$ and curve
$x^3 + y^3 + 3xy + 5x^2 + 3y^2 + 4x + 5y - 1 = 0$
Solving them,

$\Rightarrow x^3 + 3\sqrt{3}x^3 + 3\sqrt{3}x^2 + 5x^2 + 9x^2 + 4x + 5\sqrt{3}x - 1 = 0$

Let roots x_1, x_2, x_3
then $x_1x_2x_3 =$

Co-ordinates of A, B, C are $(x_1, \sqrt{3}x_1)$, $(x_2, \sqrt{3}x_2)$ and $(x_3, \sqrt{3}x_3)$ respectively.
then OA. OB. OC $= 8x_1x_2x_3$

$= \frac{8}{3\sqrt{3}+1} = \frac{8(3\sqrt{3}-1)}{26} = \frac{4}{13}(3\sqrt{3}-1)$

P 46 : (B) the zero function

Sol. Required area $= \int_c^d y\,dx = \int_c^d f(x)\,dx$
= Independent of d, it is possible only when $f(x) = 0$

P 47 : (B) The geometric mean of a and b

Sol. a,b and c are distinct negative numbers and vectors $a\hat{i}+a\hat{j}+c\hat{k}$, $\hat{i}+\hat{k}$ and $c\hat{i}+c\hat{j}+b\hat{k}$ are coplanar

$$\begin{vmatrix} a & a & c \\ 1 & 0 & 1 \\ c & c & b \end{vmatrix} = 0$$

or $ac + c^2 - ab - ac = 0$
or $c^2 = ab$
Hence, a, b, c are in GP.
Hence c is the geometric mean of a and b.

P 48 : (D) infinite

Sol. We have, $4(\log_2 x)^2 + 1 = 2\log_2 y$
and $\log_2 x^2 \geq \log_2 y$
$\Rightarrow 4(\log_2 x)^2 + 1 \leq 2\log_2 x^2$
$\Rightarrow 4(\log_2 x)^2 - 4\log_2 x + 1 \leq 0$
$\Rightarrow (2(\log_2 x) - 1)^2 \leq 0$
$\Rightarrow 2\log_2 x - 1 \leq 0$

$\Rightarrow \log_2 x \leq \frac{1}{2} \Rightarrow x \in [0, \sqrt{2}]$

Now, $\log_2 x^2 \geq \log_2 y$
$\Rightarrow x^2 \geq y$ and $y > 0 \Rightarrow 2^3 y > 0 \Rightarrow 0 < y \leq 2$

Hence, the ordered pairs are $(\sqrt{2}, y)$, where $0 < y \leq 2$

P 49 : (C) ω^2

Sol. $p^{1/3} = (-p)^{1/3}(-1)^{1/3}$
$= -(-p)^{1/3} - (-p)^{1/3}\omega - (-p)^{1/3}\omega^2$
$= -q - q\omega - q\omega^2$ where $q = (-p)^{1/3}$
Let $\alpha = -q$, $\beta = -q\omega$, $\gamma = -q\omega^2$

Therefore, $\frac{x\alpha + y\beta + z\gamma}{x\beta + y\gamma + z\alpha} = \frac{-q(x + y\omega + z\omega^2)}{-q(x\omega + y\omega^2 + z)}$

$= \frac{x\omega^3 + y\omega^4 + z\omega^2}{x\omega + y\omega^2 + z} = \omega^2$

P 50. (D) $y^2 = a(a - 2x)$

Sol. Let the point of intersection of the tangent be (h, k)
$\therefore$ Equation of chord of contact is
$hx + ky = a^2$
It is a tangent to the circle $x^2 + y^2 - 2ax = 0$

$\therefore \left| \frac{ha - a^2}{\sqrt{h^2 + k^2}} \right| = a$

$\therefore (h - a)^2 = h^2 + k^2$
i.e., $-2ah + a^2 = k^2$
$\therefore$ the locus $y^2 = a(a - 2x)$

P 51 : (B) $\sqrt{\frac{1}{3}}$

Sol. For conjugate diameters

$m_1m_2 = -\frac{b^2}{a^2} \Rightarrow 1 \times \frac{-2}{3} = \frac{-b^2}{a^2}$

$\Rightarrow \frac{2}{3} = \frac{a^2(1-e^2)}{a^2}$

$\Rightarrow 1 - e^2 = \frac{2}{3} \Rightarrow e = \frac{1}{\sqrt{3}}$

P 52 : (D) $-\infty < a \leq 3$

Sol. If $f(x) = (a + 2)x^3 - 3ax^2 + 9ax - 1$ decreases monotonically for all $x \in R$, then
$f'(x) \leq 0$ for all $x \in R$
$\Rightarrow 3(a + 2)x^2 - 6ax + 9a \leq 0$ for all $x \in R$
$\Rightarrow (a + 2)x^2 - 2ax + 3a \leq 0$ for all $x \in R$
$\Rightarrow a + 2 < 0$ and discriminant ≤ 0
$\Rightarrow a < -2$ and $-8a^2 - 24a \leq 0$
$\Rightarrow a < -2$ and $a(a + 3) \geq 0$
$\Rightarrow a < -2$ and $a \leq 3$ or $a \geq 0$
$\Rightarrow a \leq 3 \Rightarrow -\infty < a \leq -3$

P 53 : (D) Cannot be determined

Sol. Let the common root be 'm'. Since we know that $Q_1(10) = 0$, it means that 10 would be one of the roots of Q_1. By the same logic it is given to us that 8 is one of the roots of Q_2. using this information we have :

$Q_1 = c_1 \times (x - 10) \times (x - m)$ and

$Q_2 = c_2 \times (x - 8) \times (x - m)$.

Note : c_1 and c_2 are constant (each not equal to 0).

The only information we have beyond this is that the product $Q_1(4) \times Q_2(5) = 36$. However, it is evident that by replacing $x = 4$ and $x = 5$ in the expressions for Q_1 and Q_2 respectively, we would not get any conclusive value for m since the value of m would depend on the values of c_1 and c_2. You can see this happening here :

$c_1 \times -6 \times (4 - m) \times c_2 \times -3 \times (5 - m) = 36$

$c_1 \times c_2 \times (20 - 9m + m^2) = 2$.

In this equation it can be clearly seen that the value of the common root 'm' would be dependent on the values of c_1 and c_2 and hence we cannot determine the anser to the question.

P 54 : (A) $\frac{1}{2}$

Sol. $\sum_{r=1}^{5}\cos(2r-1)\frac{\pi}{11}$

$$\cos\frac{\pi}{11}+\cos\frac{3\pi}{11}+\cos\frac{5\pi}{11}+\cos\frac{7\pi}{11}+\cos\frac{9\pi}{11}$$

$$=\frac{2\sin\frac{\pi}{11}}{2\sin\frac{\pi}{11}}\left(\cos\frac{\pi}{11}+\cos\frac{3\pi}{11}+\cos\frac{5\pi}{11}+\cos\frac{7\pi}{11}+\cos\frac{9\pi}{11}\right)$$

$$=\frac{\left(\begin{array}{c}\sin\frac{2\pi}{11}+\sin\frac{4\pi}{11}-\sin\frac{2\pi}{11}+\sin\frac{6\pi}{11}-\sin\frac{4\pi}{11}\\ +\sin\frac{8\pi}{11}-\sin\frac{6\pi}{11}+\sin\frac{10\pi}{11}-\sin\frac{8\pi}{11}\end{array}\right)}{2\sin\frac{\pi}{11}}$$

$$=\frac{\sin\frac{10\pi}{11}}{2\sin\frac{\pi}{11}}=\frac{\sin\left(\pi-\frac{\pi}{11}\right)}{2\sin\frac{\pi}{11}}=\frac{1}{2}$$

P 55 : (A) –1/12

Sol. $\sin^{-1}6x = -\pi/2 - \sin^{-1}\ 6\sqrt{3x}$

$\sin(\sin^{-1}6x) = \sin(-\pi/2 - \sin^{-1}6\sqrt{3x})$

Now, let $= \sin^{-1}6\sqrt{3x} = \theta$, $-\pi/2 \geq \theta \geq \pi/2$

Therefore,

$\sin\theta = 6\sqrt{3x} = \cos\theta = \sqrt{(1-108x^2)}$

Therefore, $\theta = \sin^{-1}6\sqrt{3x} = \cos^{-1}\sqrt{(1-108x^2)}$

$6x = -\sqrt{(1-108x^2)}$, RHS is negative,

Therefore, x is negative

Squaring, $36x^2 = 1 - 108x^2$

$\Rightarrow x = \pm 1/12$, $x = -1/12$

P 56 : (D) 3

Sol. Put $\theta = 0 \Rightarrow x + iy = 1 \Rightarrow x = 1$ & $y = 0$

$\Rightarrow 4x - x^2 - y^2 = 3$. Alternatively

$$\cos\theta + i\sin\theta + 2 = \frac{3(x-iy)}{x^2+y^2} \Rightarrow \cos\theta + 2$$

$$= \frac{3x}{x^2+y^2} \ \&\ \sin\theta = -\frac{3y}{x^2+y^2}$$

$\Rightarrow \cos^2\theta + \sin^2\theta$

$$= \left(\frac{3x}{x^2+y^2}-2\right)^2 + \frac{9y^2}{\left(x^2+y^2\right)^2}$$

$= 1$ or $(3x - 2x^2 - 2y^2)^2 + 9y^2 = (x^2 + y^2)^2$ or $9x^2 + 4(x^2 + y^2)^2 - 12x(x^2 + y^2) + 9y^2 = (x^2 + y^2)^2$

or $3 + x^2 + y^2 - 4x = 0$

$\Rightarrow 4x - x^2 - y^2 = 3$

P 57 : (B) 4th & 5th

Sol. $\Rightarrow \left|\frac{T_{r+1}}{T_r}\right| > 1$

$$\frac{{}^{9}C_r}{{}^{9}C_{r-1}}\left(\frac{2x}{a}\right) > 1$$

$$\frac{10-r}{r}\times\frac{2}{3} > 1$$

$5r < 20$

$r < 4$

$\Rightarrow T_4$ & T_5 will be the greatest terms .

P 58 : (C) $\sqrt{2}\ \tan^{-1}\left(\sqrt{\frac{f(x)g(x)-1}{2}}\right) + c$

Sol. $\int \frac{f(x)g'(x)+g(x)f'(x)}{(f(x)g(x)+1)\sqrt{f(x)g(x)-1}}dx =$

$= \int \frac{dt}{(t+1)\sqrt{t-1}}$ where $t = f(x) . g(x)$

$= 2\int \frac{du}{u^2+2}$ where $\sqrt{t-1} = u$

$= \sqrt{2}\ \tan^{-1}\frac{u}{\sqrt{2}} + c = \sqrt{2}\ \tan^{-1}\sqrt{\frac{f(x)\,g(x)-1}{2}} + c$

P 59 : (C) 'h' is decreasing whenever 'f' is decreasing

Sol. Here $h(x) = f(x) - [\{f(x)\}^2 + \{f(x)\}^3]$

$\Rightarrow h'(x) = f'(x) - 2f(x)f'(x) + 3\{f(x)\}^2 f'(x)$

$= f'(x)[1 - 2f(x) + 3\{f(x)\}^2]$

$= f'(x)(3y^2 - 2y + 1)$ where $y = f(x)$

The discriminant of $3y^2 - 2y + 1 = 4 - 12$

$= -8 < 0$

and so its sign is the same as both co-efficient of y^2 i.e. > 0

$\therefore h'(x) = f'(x)$ (a positive quantity)

$\Rightarrow$ sign of $h'(x)$ is the same as that of $f'(x)$

$\Rightarrow$ either $h'(x) > 0$ & $f'(x) > 0$

or $h'(x) < 0$ & $f'(x) < 0$. Hence $h(x)$ & $f(x)$ increase and decrease together.

P 60 : (B) -25

Sol. Since $\vec{u}+\vec{v}+\vec{w} = 0$, we have

$|\vec{u}+\vec{v}+\vec{w} = 0|^2 = 0$

Or $|\vec{u}|^2 + |\vec{v}|^2 + |\vec{w}|^2 + 2(\vec{u}.\vec{v}+\vec{v}.\vec{w}+\vec{w}.\vec{u}) = 0$

Or $9 + 16 + 25 + 2\ (\vec{u}.\vec{v}+\vec{v}.\vec{w}+\vec{w}.\vec{u}) = 0$

Or $\vec{u}.\vec{v} + \vec{v}.\vec{w} + \vec{w}.\vec{u} = -25$

P 61 : (D) Particle size of the adsorbent will not affect the amount of adsorption

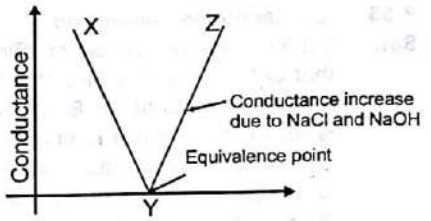
P 62 : (C) Na^+ has the higher equivalence conductance than H_3O^+

Sol. When NaOH solution is added to HCl solution then conductance of solution depends on Na^+ and H_3O^+ ion.

$NaOH + HCl \longrightarrow NaCl + H_2O$

$H_2O + HCl \longrightarrow H_3O^+ + Cl^-$

Therefore, at first, conductance decreases due to decrease in H^+ ions.



After the equivalent point, NaCl and NaOH solution increases the conductance with increase in OH^- ion. Thus, (a), (b) and (d) statements are correct.

P 63 : (D) 1.75 g/cc

Sol. Given, $a - 2r = 60.3$ and for bcc. $4r = \sqrt{3}a$

$\Rightarrow a - \frac{\sqrt{3}}{2}a = 60.3 \Rightarrow a = 450$ pm

Density $(\rho) = \frac{2\times 48}{6.023\times10^{23}\times(4.5)^3\times10^{-24}}$

$= 1.75$ g/cc.

P 64 : (C) 7.5×10^{-18}

Sol.

	$2KCN + AgNO_3 \longrightarrow$	$Ag(CN)_2^{\ominus}$	+ KCN_3	+ $K^{\oplus}$
	0.1 0.03	0	0	0
	(0.1–0.06) 0	0.03	0.03	0.03

$[Ag(CN)_2^{\ominus}] = 0.03$ M

Now use

	$Ag(CN)_2 \rightleftharpoons$	$Ag^{\oplus}$	+ $2CN$
	0.03	0	0.04
	(0.03 – a)	a	0.04 + a

Since K_c is too small and dissociation of $Ag(CN)_2^-$ is very less and thus,

$$K_c = \frac{[Ag^{\oplus}]\times[CN]^{\ominus}]^2}{[Ag(CN)_2^-]} = \frac{a\times(0.04)^2}{0.03}$$

$\therefore a = 7.5 \times 10^{-18}$

P 65 : (D) $\Delta H°$ opposes the reaction, but $\Delta S°$ favours it.

Sol. $\Delta H° > 0$ but $\Delta S° > 0$. $\Delta H°$ opposing spontaneity but $\Delta S°$ favouring spontaneity.

P 66 : (A) $(SiO_4)^{4-}$

Sol. Quartz, mica, asbestos have common unit $(SiO_4)^{4-}$.

P 67 : (A) Ratio of concentration of solute in g/L to $\% \left(\frac{\omega}{v}\right)$ is 10.

Sol. (I) $\dfrac{\left[\dfrac{w_{solute}}{V_{Solution}}\right]}{\left[\dfrac{n_{solute}}{V_{solution}}\right]} = 25 \Rightarrow M_{solute} = 25$

(II) $\dfrac{\left[\dfrac{w_{solute}}{V_{Solution}}\right]\times 100}{\left[\dfrac{w_{solute}}{V_{solution}}\right]\times 100} = 0.8 \Rightarrow d_{solute} = \dfrac{1}{0.8}$ g/ml

Option (A)

$$\frac{\left[\dfrac{w_{solute}}{V_{Solution}}\right]}{\dfrac{w_{solute}}{V_{solution}\text{ in me}}\times 100} = 10$$

Option (B)

$d_{solution} = \dfrac{1}{0.8}$ g/ml

Option (C)

If molarity = 2M $\Rightarrow W_{solution} = \dfrac{1000}{0.8}$ g

$\Rightarrow W_{solution} = 50$ g
$\Rightarrow W_{solvent} = 1.2$ kg

$\therefore$ Molality(m) $= \dfrac{2\text{mol}}{1.2\text{kg}} = \dfrac{5}{3}$ m

Option (B)

$$\frac{\left[\dfrac{w_{solute}}{w_{Solution}}\times 100\right]}{\dfrac{w_{solute}}{V_{solution}}\times 10^6} = \frac{1}{10^4}$$

P 68 : (C) $H_3Te^+ < H_3Se^+ < H_3S^+ < H_3O^+$

Sol. (A) $NH_3 < PH_3 < AsH_3 < SbH_3$
(As E.N. value of C.A. ↑ bond angle ↓)
(B) $OF_2 < OH_2 < OCl_2$
(C) $H_3Te^+ < H_3Se^+ < H_3S^+ < H_3O^+$
(D) $BF_3 = BCl_3 = BBr_3 = BI_3$

P 69 : (C)

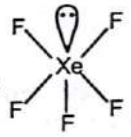
Sol. It gives phenol.

P 70 : (B) Bent-T-shape, square pyramidal

Sol. $[XeF_3^+]$:

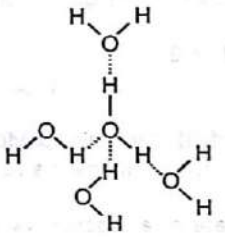
Bent - T- shape

(sp^3d-hybridisation)

$[XeF_5^+]$: 

Square pyramidal (sp^3d^2-hybridisation)

P 71 : (D) Each O atom has four H-Bonds around it

Sol. 

P 72 : (C) 0.0024

Sol. 100 mL of 0.024 M = 0.0024 mol of the complex
complex is $[Co(en)_2Cl_2]Cl$ and thus, one Cl^- is formed per mol of the complex which gives 1 mole of AgCl.
Hence, AgCl formed = 0.0024 mol

P 73 : (C) $CH_3CH_2C{\equiv}CH$

Sol. Terminal alkyne forms precipitate of silver acetylide with Tollen's reagent.

P 74 : (C) both (A) and (B)

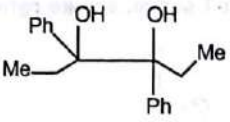
P 75 : (B) 1.18 Å

Sol. $d_{C-X} = (r_C + r_X) - 0.09$ (Δ E.N.)

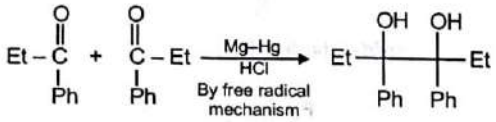
$$= \left(\frac{1.54}{2} + \frac{1.00}{2}\right) - (0.09 \times 1)$$

$= (0.77 + 0.5) - (0.09)$
$= 1.18$ Å

P 76 : (D)



Sol. It is an exmple of bimolecular reduction (formation of pinacol).



P 77 : (B) Volumetric method using KI and titration of the liberated iodine using hypo solution

Sol. $2KI + H_2O + O_3 \longrightarrow 2KOH + I_2 + O_2$

P 78 : (B) IV<I<III<II

Sol. In aqueous solution, both inductive effect and extent of hydration of conjugate acid determine the basic strength and the overall order is :

$NH_3 < 3° < 1° < 2°$

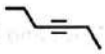
P 79 : (C) II + III

Sol. In both II and III, the major product would be 1-butene, due to Hofmann's rule.

P 80 : (B) Plot of rate of a reaction vs time is a straight line passing from origin but with –ve slope

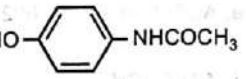
Sol. Rate of reaction decreases exponentially with time.

P 81 : (C)



Sol. Deprotonation followed by chain extension

P 82 : (B)



P 83 : (D) SbH_3

Sol. On moving top to bottom, size of central atom increases. Bond length of X–H bond increases and bond dissociation energy decreases. Hence reducing nature increases.

NH_3
PH_3 | Bond length increases
AsH_3 | Bond dissociation energy decreases
SbH_3 ↓ Reducing character increases

Hence, SbH_3 acts as strongest reducing agent among these.

P 84 : (C) The reaction occurs readily with primary alkyl halides.

Sol. In S_N2 reaction, reactivity order is : 1° > 2° > 3°.

P 85 : (C) Wolff-Kishner reduction

Sol. Wolff-Kishner reduction convert $\left(\diagup C = O\right)$ group to $(-CH_2-)$, and simultaneously dehydrohalogenates to alkene.

P 86 : (A) Ca

Sol. Lower the value of E° stronger the reducing agent.

P 87 : (D) The given wave function shows angular dependence too.

P 88 : (C) $[Co(NH_3)_5Cl]Cl_2$

Sol. Molality of solution = 0.1

Depression in freezing point = 0.558°C = ΔT_f

$K_f(H_2O) = 1.86$ kg/mol/K

As, $\Delta T_f = K_f \times i \times m$ (i = van't Hoff factor)

$0.558 = 1.86 \times 0.1 \times i$

$$i = \frac{0.558}{0.186} = 3$$

or, $i \approx 3$ indicates that complex ionize to form 3 ions since coordination number is 6 hence X = 5.

i.e., $\underset{\text{1 cation}}{COCl_3.5NH_3} \rightarrow \underset{\text{2 anions}}{[Co(NH_3)_5Cl]^{2+}} + 2Cl^-$

So, the complex is $[Co(NH_3)_5Cl]Cl_2$

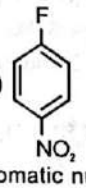
P 89 : (A) It will remain same

Sol. $[Ni(H_2O)_6]^{2+} + 6NH_2 \rightleftharpoons [Ni(NH_3)_6]^{2+} + 6H_2O$

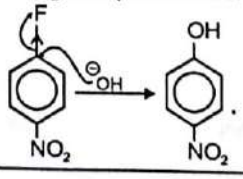
sp^3d^2 Hyb. sp^3d^2 Hyb.

$\mu_{eff} = 2.8$ BM $\mu_{eff} = 2.8$ BM

P 90 : (A)



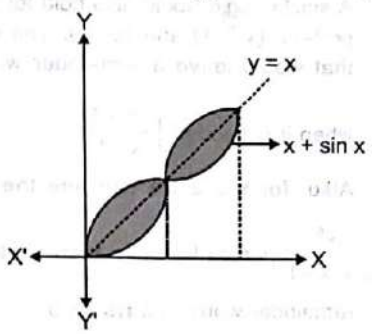
Sol. Aromatic nucleophilic substitution reactions are favoured by strong EWG, –I effect of F is highest (more EN of F).



REPLICA 11

P 1 : (C) 8 sq. units

Sol. Required area $= 4\left[\int_0^{\pi}(x+\sin x)dx - \int_0^{\pi} x dx\right]$



$= 4\left[\frac{x^2}{2} - \cos x\right]_0^{\pi} - \left[\frac{x^2}{2}\right]_0^{\pi}$

$= 4\left\{\left[\frac{\pi^2}{2} - \cos\pi\right] - (0 - \cos 0) - \left[\left(\frac{\pi^2}{2} - 0\right)\right]\right\}$

$= 4 \times 2 = 8$ sq. units

P 2 : (B) $\theta \neq (2n+1)\frac{\pi}{2}$

Sol. $\begin{vmatrix} 1 & 1 & 1 \\ 0 & \sec^2\theta & 1 \\ \text{cosec}^2\theta & 0 & 1 \end{vmatrix} = 0$

Applying $C_2 \to C_2 - C_1$ and $C_3 \to C_3 - C_1$

$\Rightarrow \begin{vmatrix} 1 & 0 & 0 \\ 0 & \sec^2\theta & 1 \\ \text{cosec}^2\theta & -\text{cosec}^2\theta & 1-\text{cosec}^2\theta \end{vmatrix} = 0$

$\Rightarrow \sec^2\theta(1-\text{cosec}^2\theta) + \text{cosec}^2\theta = 0$

$\Rightarrow \frac{1}{\cos^2\theta}\left(1 - \frac{1}{\sin^2\theta}\right) + \frac{1}{\sin^2\theta} = 0$

$\Rightarrow 4\left(\frac{\sin^2\theta - 1 + \cos^2\theta}{4\sin^2\theta\cos^2\theta}\right) = 0$

$\Rightarrow 4\frac{\sin^2\theta + \cos^2\theta - 1}{\sin^2 2\theta} = 0$

which is always true provided, $\sin^2 2\theta \neq 0$

$\Rightarrow 2\theta \neq n\pi \pm 0 \Rightarrow \theta \neq \frac{n\pi}{2}$.

Therefore best fit option is option(B)

P 3 : (A) $\leq \frac{\alpha+\beta+\gamma+\delta}{16}$

Sol. $\log_{abcd} x = \frac{1}{\log_x abcd}$

$= \frac{1}{\log_x a + \log_x b + \log_x c + \log_x d}$

$= \frac{1}{\alpha^{-1} + \beta^{-1} + \gamma^{-1} + \delta^{-1}}$

Also, AM $\geq$ GM

$\frac{\alpha+\beta+\gamma+\delta}{4} \leq \frac{4}{\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}+\frac{1}{\delta}}$

or $\frac{1}{\alpha^{-1}+\beta^{-1}+\gamma^{-1}+\delta^{-1}} \leq \frac{\alpha+\beta+\gamma+\delta}{16}$

P 4 : (B) $1+(p-r)^2$

Sol. From the given equations we have $\Sigma \tan\alpha = p$, $\Sigma \tan\alpha \tan\beta = 0$, and $\tan\alpha\tan\beta\tan\gamma = r$

So that $(1 + \tan^2\alpha)(1 + \tan^2\beta)(1 + \tan^2\gamma)$

$= 1 + \Sigma\tan^2\alpha + \Sigma\tan^2\alpha\tan^2\beta + \tan^2\alpha\tan^2\beta$

$= 1 + p^2 - 2pr + r^2 = 1+(p-r)^2$

P 5 : (A) 1

Sol. $\sin^2\gamma = \frac{\sin^2\alpha}{\tan\alpha}\left[\tan\alpha - \left(\frac{\tan\alpha - \tan\beta}{1+\tan\alpha\tan\beta}\right)\right]$

$= \frac{\sin^2\alpha\tan\beta(1+\tan^2\alpha)}{\tan\alpha(1+\tan\alpha\tan\beta)}$

$\Rightarrow \sin^2\gamma\,(1 + \tan\alpha\tan\beta) = \frac{\tan\beta\sin^2\alpha}{\tan\alpha\cos^2\alpha}$

$= \tan\alpha\tan\beta$

$\Rightarrow \tan^2\gamma = \tan\alpha\tan\beta$.

P 6 : (A) $A > \frac{\pi}{2}$

Sol. $a^2\cos^2A = b^2 + c^2$

$\Rightarrow a^2\cos^2A - a^2 = b^2 + c^2 - a^2 = 2bc\cos A$

$\therefore A > \frac{\pi}{2}$

P 7 : (C) $\arg(z-1) = \alpha$

Sol. Let $z - 1 = r(\cos\theta + i\sin\theta) = re^{i\theta}$

Given expression $= re^{i\theta}.e^{-i\alpha} + \frac{1}{re^{i\theta}}.e^{i\alpha}$

$= re^{i(\theta-\alpha)} + \frac{1}{r}e^{-i(\theta-\alpha)}$

Since, imaginary part of given expression is zero, we have $r\sin(\theta-\alpha) - \frac{1}{r}\sin(\theta-\alpha) = 0$

$r^2 - 1 = 0 \quad \Rightarrow r^2 = 1$

$\Rightarrow \quad r = 1 \quad \Rightarrow |z-1| = 1$

$\sin(\theta-\alpha) = 0 \quad \Rightarrow \theta - \alpha = 0$

$\Rightarrow \quad \theta = \alpha \quad \Rightarrow \arg(z-1) = \alpha$

P 8 : (C) (2, 2)

Sol. The length of the chord of contact is smallest when the point in nearest to the circle. The point on the line $x + y = 4$ which is nearest to (0, 0) (centre of the circle) is the foot of perpendicular drawn from (0, 0) to the line $x + y = 4$

i.e. $\frac{x-0}{1} = \frac{y-0}{1} = -\frac{-4}{2} = 2 \quad x = 2, y = 2$

P 9 : (C) 5

Sol. The value of f(x) as given is $f(x) = x^4 + x^3 + x^2 + x + 1 = 1 + x + x^2 + x^3 + x^4 + x^5$. This can be visualised as a geometric progression with 5 terms with the first term 1 and common ratio x. The sum of the GP

$= f(x) = \frac{x^5-1}{x-1}$.

the value of $f(x^5) = x^{20} + x^{15} + x^{10} + x^5 + 1$ and this can be rewritten as :

$F(x^5) = (x^{20} - 1) + (x^{15} - 1) + (x^{10} - 1) + (x^5 - 1) + 5$.

When this expression is divided by $f(x) = \frac{x^5-1}{x-1}$ we get each of the first four terms of the expression would be divisible by it, i.e., $(x^{20}-1)$ would be divisible by $f(x) = \frac{x^5-1}{x-1}$ and would leave no remainder (because $x^{20}-1$ can be rewritten in the form $(x^5-1)\times(x^{15}+x^{10}+x^5+1)$ and when you divide this expression by $\left(\frac{x^5-1}{x-1}\right)$ we get the remainder as 0.)

A similar logic would also hold for the terms $(x^{15}-1)$, $(x^{10}-1)$ and (x^5-1). The only term that would leave a remainder would be 5 when it is divided $\left(\frac{x^5-1}{x-1}\right)$

Also, for $x \geq 2$ we can see the value of $\left(\frac{x^5-1}{x-1}\right)$ would be more than 5. Hence, the remainder would always be 5.

P 10 : (A) $x + 2y + 1 - 4\sqrt{5} = 0$

Sol. Slope of the line = 2

$\therefore$ slope of the tangents $= -\frac{1}{2}$

equation of the circle is $(x-3)^2 + (y+2)^2 = 4^2$

$\therefore$ equations of the tangents are

$y + 2 = -\frac{1}{2}(x-3) \pm 4\sqrt{1+\frac{1}{4}}$

i.e. $2y + 4 = -x + 3 \pm 4\sqrt{5}$

i.e. $x + 2y + 1 \pm 4\sqrt{5} = 0$

i.e. $x + 2y + 1 + 4\sqrt{5} = 0$ and

$x + 2y + 1 - 4\sqrt{5} = 0$

P 11 : (C) $p^2(4p-3)$

Sol. Maximum no of triangles

$= {}^{3p}C_3 - 3({}^pC_3)$

$= \frac{3p(3p-1)(3p-2) - 3p(p-1)(p-1)}{6}$

$= \frac{1}{2}[9p^2 - 9p + 2 - p^2 + 3p - 2]$

$= p[4p^2 - 3p]$

$= p^2[4p - 3]$

P 12 : (C) 124

Sol. Case 1: Mr. 'B' is present ⇒ 'A' is excluded & 'C' included
Hence number of ways = $^4C_2 . {}^4C_1 = 24$
Case 2: Mr. 'B' is absent ⇒ no constraint
Hence number of ways = $^5C_3 . {}^5C_2 = 100$
Total = 124

P 13 : (D) none

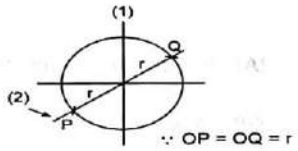
Sol. imposing the given condition $\sin^2\theta - \cos^2\theta = 1$ for which $\theta = \pi/2$ but for this value of θ vertices are not possible.

P 14 : (C) $\frac{12}{4+\sqrt{3}}$

Sol. $x^2 + y^2 + xy - 3 = 0$ ... (1)

$y = \frac{x}{\sqrt{3}}$ ... (2)

$\because$ centre of ellipse is (0, 0) and (2) also passes through (0, 0)
$\therefore$ chord will be bisect at (0, 0).



$\because$ equation PQ is $\frac{x-0}{\cos\frac{\pi}{6}} = \frac{y-0}{\sin\frac{\pi}{6}} = \pm r.$

$\therefore \left(\frac{r\sqrt{3}}{2}, \frac{r}{2}\right)$ lies on ... (1)

$\Rightarrow \frac{3r^2}{4} + \frac{r^2}{4} + \frac{r^2\sqrt{3}}{4} - 3 = 0$

$\Rightarrow r^2 = \frac{3\times 4}{(4+\sqrt{3})}$ $\Rightarrow r^2 = \frac{12}{4+\sqrt{3}}$

$\therefore$ $|OP||OQ| = \frac{12}{4+\sqrt{3}}$

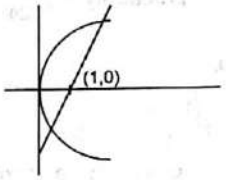
P 15 : (D) 5/21

Sol. $P(A/A\cup B) = \frac{P(A)(A\cup B)}{P(A\cup B)} = \frac{P(A)}{P(A)+P(B)}$

$= \frac{0.10}{0.10+0.32}$

P 16 : (C) $\sqrt{10}$

Sol. Parametric co-ordinate of line is
$x = 1 + r\cos\theta$
$y = r\sin\theta$



solving the line with parabola,
then, $r^2\sin^2\theta = 1 + r\cos\theta$
$\Rightarrow r^2\sin^2\theta - r\cos\theta - 1 = 0$

where $\theta = \frac{\pi}{4}$

$r_1 + r_2 = \sqrt{2}$ & $r_1 r_2 = -2$

$|PA| + |PB| = r_1 - r_2 = \sqrt{(r_1+r_2)^2 - 4r_1r_2} = \sqrt{10}$

P 17 : (B) $f(x) \le 1$

Sol. Given $f'(x) \le \frac{1}{2}$

$\Rightarrow \frac{d}{dx}(f(x) - \frac{x}{2}) \le 0$

$\Rightarrow$ $h(x) = f(x) - \frac{x}{2}$ is decreasing in nature

Now h(0) = 0 and h(x) is decreasing

$\Rightarrow f(x) - \frac{x}{2} \le 0$

$\Rightarrow f(x) \le \frac{x}{2}$

$\Rightarrow f(x) \le 1$ since $x \in \{0,2\}$

P 18 : (A) $\frac{25}{256}$

Sol. Sum can be 23 by following ways :

(i) 11 + 11 + 01 + 00 + 00

number of ways = $\frac{5!}{2!2!} = 30$

(ii) 11 + 10 + 01 + 01 + 00

Number of ways = $\frac{5!}{2!} = 60$

(iii) 10 + 10 + 01 + 01 + 01

Number of ways = $\frac{5!}{2!3!} = 10$

total number of favourable ways = 100
total number of ways = $(4)^5 = 1024$

probability = $\frac{100}{1024} = \frac{25}{256}$

P 19 : (C) a_0

Sol. $a_1 = \sqrt{\frac{(1+a_0)}{2}}$

Put $a_0 = \cos\theta \Rightarrow a_1 = \cos(\theta/2)$

$a_2 = \sqrt{\frac{(1+a_1)}{2}} = \cos(\theta/2^2)$

$a_1a_2a_3\ldots$ to $\infty = \lim_{n\to\infty} a_1a_2a_3\ldots a_n$

$$\lim_{n\to\infty}\prod_{r=1}^{n} a_r \times \lim_{n\to\infty}\prod_{r=1}^{n}\left\{\frac{2\sin\left(\frac{\theta}{2^r}\right)\cos\left(\frac{\theta}{2^r}\right)}{2\sin\left(\frac{\theta}{2^r}\right)}\right\}$$

$$\lim_{n\to\infty}\prod_{r=1}^{n}\left\{\frac{\sin\left(\frac{\theta}{2^{r-1}}\right)}{2\sin\left(\frac{\theta}{2^r}\right)}\right\}$$

$$\lim_{n\to\infty}\left\{\frac{\sin\theta}{2\sin\left(\frac{\theta}{2}\right)}\cdot\frac{\sin\left(\frac{\theta}{2}\right)}{2\sin\left(\frac{\theta}{2^2}\right)}\cdot\frac{\sin\left(\frac{\theta}{2^2}\right)}{2\sin\left(\frac{\theta}{2^3}\right)}\cdots\cdots\frac{\sin\left(\frac{\theta}{2^{n-1}}\right)}{2\sin\left(\frac{\theta}{2^n}\right)}\right\}$$

$$\lim_{n\to\infty}\left\{\frac{\sin\theta}{\frac{2^n\sin\left(\frac{\theta}{2^n}\right)}{\left(\frac{\theta}{2^n}\right)}\cdot\frac{\theta}{2^n}}\right\}$$

$$= \frac{\sin\theta}{\theta(1)} = \frac{\sin\theta}{\theta}$$

$$= \cos\left(\frac{\sqrt{1-a_0^2}}{a_1a_2a_3\ldots\infty}\right) = \cos\left(\frac{\sin\theta}{\frac{\sin\theta}{0}}\right) = \cos\theta = a_0$$

P 20 : (D) $(1/2)\{\sqrt{2n+1}-1\}$

Sol. The required sum to n terms is

$$\frac{1}{\sqrt{1}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{5}} + \frac{1}{\sqrt{5}+\sqrt{7}} + \ldots + \frac{1}{\sqrt{2n-1}+\sqrt{2n+1}}$$

$$= \frac{1}{2}[(\sqrt{3}-\sqrt{1})+(\sqrt{5}-\sqrt{3})+(\sqrt{7}-\sqrt{5})] + (\sqrt{2n+1}-\sqrt{2n-1})$$

$$= \frac{1}{2}(\sqrt{2n+1}-1)$$

P 21 : (B) $\left(\frac{4}{5}, -\frac{1}{5}\right)$

Sol. $x - y - 5 = 0$ ... (1)

$\frac{x^2}{4} + \frac{y^2}{1} = 1$ ... (2)

$\because$ any point P on (1) is (h, h – 5)
$\therefore$ equation of chord of contact is:
$\Rightarrow$ $xh + 4hy - 20y = 4$
$\Rightarrow$ $h(x + 4y) - 4(5y + 1) = 0$ ... (3)
$\because$ (3) always passes through point of intersection of $5y + 1 = 0$ and $x + 4y = 0$

$\Rightarrow$ required fixed point $\left(\frac{4}{5}, \frac{-1}{5}\right)$

P 22 : (A) an ellipse of eccentricity $\sqrt{3}/2$

Sol. $2px + y\sqrt{1-p^2} = 1$

i.e. $y = -\frac{2p}{\sqrt{1-p^2}}x + \frac{1}{\sqrt{1-p^2}}$ .....(i)

Let $m = -\frac{2p}{\sqrt{1-p^2}}$. Then $1 - p^2 = \frac{4}{4+m^2}$.

$\therefore$ equation of the line (i) becomes

$y = mx + \frac{\sqrt{4+m^2}}{2}$

i.e. $y = mx + \sqrt{\frac{m^2}{4}+1}$

$\therefore$ the curve is an ellipse for which

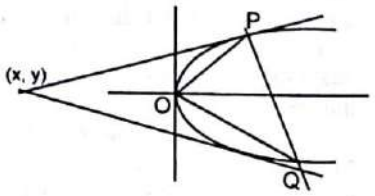
$b^2 = \frac{1}{4}$ and $a^2 = 1$

$\therefore$ $e^2 = \frac{3}{4}$ $\therefore$ $e = \frac{\sqrt{3}}{2}$

P 23 : (A) $x + 4a = 0$

Sol. Let $y = mx + c$ is the varibale chord which is subtending right angle at the vertex of parabola hence equation of pair of straight lines OP and

OQ can be given by making a homogeneous second degree equation with the help of parabola and chord as follows



$$y^2 - 4ax\left(\frac{y-mx}{c}\right) = 0$$

since these lines are making 90° with each other

hence $1 + \frac{4am}{c} = 0$

$\Rightarrow$ $c = -4am$

$\Rightarrow$ equation of chord is

$y = m(x - 4a)$

which is always passing through (4a, 0)

Let point of intersection of tangents at the extremities is (x_1, y_1)

hence equation of chord of contact is

$y y_1 = 2a(x + x_1)$

which is passing through (4a, 0)

$\Rightarrow$ $x_1 + 4a = 0$

$\Rightarrow$ $x + 4a = 0$ is the required locus.

P 24 : (B) $y = \tan^{-1}\frac{1}{2}, x = -5\sqrt{5}$

where $2n\pi + < y < 2n\pi + \frac{3\pi}{2}$

Sol. Clearly, $x \neq 0$ dividing the equations, we get

$$\frac{\cos^3 y + 3\cos y \sin^2 y}{\sin^3 y + 3\cos^2 y \sin y} = \frac{14}{13}$$

by componendo and dividendo, we get

$$\frac{(\cos y + \sin y)^3}{(\cos y - \sin y)^3} = \frac{14+13}{14-13}$$

$$\text{or } \left(\frac{\cos y + \sin y}{\cos y - \sin y}\right)^3 = 27 = (3)^3$$

$$\text{or } = \frac{\cos y + \sin y}{\cos y - \sin y} = \frac{3}{1}$$

dividing numerator and denominator by cosy, we get

$$\frac{1+\tan y}{1-\tan y} = \frac{3}{1} \text{ or } \frac{2\tan y}{2} = \frac{2}{4}.$$

$$\sin y = \frac{1}{\sqrt{5}}, \cos y = \frac{2}{\sqrt{5}}$$

(when y is in 1st quadrant)

and $\sin y = -\frac{1}{\sqrt{5}}$ and $\cos y = -\frac{2}{\sqrt{5}}$

(when y is in 3rd quadrant)

When y is in first quadrant.

$$x\left[\frac{8}{5\sqrt{5}} + 3\frac{2}{\sqrt{5}}\cdot\frac{1}{5}\right] = 14$$

$\Rightarrow x = -5\sqrt{5}$

When y is in 3rd quadrant

$$x\left[\frac{-8}{5\sqrt{5}} + 3\left(\frac{-2}{\sqrt{5}}\right)\cdot\frac{1}{5}\right] = 14$$

$\Rightarrow x = -5\sqrt{5}$

Hence $y = \tan^{-1}\frac{1}{2}, x = 5\sqrt{5}$

where $2n < y < 2n\pi + \frac{\pi}{2}$

and $y = \tan^{-1}\frac{1}{2}, x = -5\sqrt{5}$

where $2n\,\pi + < y < 2n\pi + \frac{3\pi}{2}$

P 25 : (A) $\sqrt{\left(\frac{x^2+1}{x^2+2}\right)}$

Sol. $\sin[\cot^{-1}\{\cos(\tan^{-1}x)\}]$

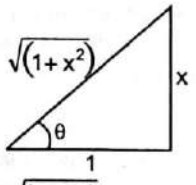
$= \sin\{\cot^{-1}(\cos\theta)\}$

Where $\theta = \tan^{-1}x$ or $x = \tan\theta$

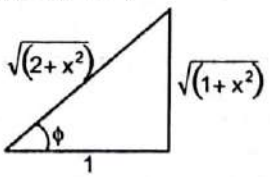
$$= \sin\left\{\cot^{-1}\left(\frac{1}{\sqrt{(1+x^2)}}\right)\right\}$$

$$= \sin\left(\tan^{-1}\sqrt{(1+x^2)}\right)$$

$= \sin\phi$, where $\phi = \tan\phi\sqrt{(1+x^2)}$



or $\tan\phi = \sqrt{(1+x^2)}$



$$= \sqrt{\left(\frac{1+x^2}{2+x^2}\right)}$$

P 26 : (A) 47

Sol. 1640 can be prime factorised as $2^3 \times 5^1 \times 41^1$. this number would have a total of 16 factors.

Similarly, 1728 can be prime factorised as $2^6 \times 3^3$. Hence, it would have 28 factors. While $448 = 2^6 \times 7^1$ would have 14 factors Thus there are a total of (16 + 28 + 14) = 58 factors amongst x, y and z.

However some of these factors must be common between x,y,z. Hence in order to find the number of natural numbers that would divide at least one amongst x, y, z we will need to account for double and triple counted numbers amongst these 58 numbers (by reducing the count by 2 for each triple counted number and by reducing the count by 1 for each double counted number).

The number of cases of triple counting would be for all the common factors of (x, y, z). This number can be estimated by finding the HCF of x, y and z and counting the number of factors of the HCF. The HCF of 1640, 1728 and 448 is 8 and hence the factors of 8, i.e., 1,2,4 and 8 itself must have got counted in each of the 3 counts done above. Thus each of these 4 numbers should get subtracted twice to remove the triple count. This leaves us with 58–4×2=50 numbers.

We still need to eliminate numbers that have been counted twice, i.e., numbers, which belong to the factors of any two of these numbers (while counting this we need to ensure that we do not count the triple counted numbers again). This can be visualised in the following way;

Number of common factors that are common to only 1640 and 1728 and not to 448 (x and y but not z)

$1640 = 2^3 \times 5^1 \times 4^1$

$1728 = 2^6 \times 3^3$

It can be seen from these 2 standard forms of the numbers that the highest common factors of these 2 numbers is 8. Hence there is no new number to be subtracted for double counting in this case

The case of 1640 and 448 is similar because $1640 = 2^3 \times 5^1 \times 41^1$ while $448 = 2^6 \times 7^1$ and HCF = 2^3 and hence they will not give any more numbers as common factors apart from 1,2,4 and 8.

Thus there is no need of adjustment for the pair 1640 and 448.

Finally when we look for 1728 and 448, we realise that the HCF = 2^6 = 64 and hence the common factor between 448 and 1728 are 1,2,4,8,16,32,64. But we are looking for factors which are common for 1728 and 448 but not common to 1640 to estimate the double counting error for this case.

Hence we can eliminate the number 1,2,4,8 from this list and conclude that there are only 3 numbers 16,32 and 64 that divide both 1728 and 448 but do not divide 1640.

If we subtract these numbers once each, from the 50 numbers we will end up with 50–3×1=47.

The complete answer can be visualised as 16+28+14–4×2–3×1=47.

P 27 : (A) atleast one root

Sol. Consider $f(x) = \sin x - e^{-x}$. Let $f(x) = 0$ has real roots say α, β

$\therefore$ Rolle's Theorem is applicable to $f(x)$ in $[\alpha, \beta]$

$\therefore$ $f'(x) = 0$ has atleast one root in (α, β)

i.e. $\cos x + e^{-x} = 0$ has atleast one root in (α, β)

$\therefore$ $e^x \cos x = -1$ has atleast one root in (α, β)

P 28 : (B) $4x + 3y + 5z = 50$

Sol. Let $(3\lambda + 1, \lambda + 2, 2\lambda + 3)$ lie on line (I) and line (II)

Hence $\dfrac{3\lambda + 1 - 3}{1} = \dfrac{\lambda + 2 - 1}{2} \Rightarrow \lambda = 1$

Hence point of intersection P of the two lines can be given as (4, 3, 5)

Equation of plane perpendicular to OP where O is (0, 0, 0) and passing through P is

$4x + 3y + 5z = 50$

P 29 : (C) 0

Sol. $\because \sqrt{2+\sqrt{2+2\cos 4x}} = \sqrt{2-2\cos 2x}$

$\because \pi < 2x < \dfrac{3\pi}{2} = 2 \sin x$

$\therefore \sqrt{2+\sqrt{2+2\cos 4x}} - 2 \sin x = 0$

$\therefore \int e^x .0 \, dx = 0$

P 30 : (B) $3\left(\dfrac{f''(x)}{f(x)} - \dfrac{g''(x)}{g(x)}\right)$

Sol. $f(x)\, g(x) = 1$

$f'(x)\, g(x) + f(x)\, g'(x) = 0$

$f''(x)\,g(x) + 2\,f'(x)g'(x) + f(x)\,g''(x) = 0$

$f'''(x)\,g(x) + 3f''(x)\,g'(x) + 3f'(x)\,g''(x) + f(x)\,g'''(x) = 0$

$f'''(x)\,g(x) + f(x)\,g'''(x) = -3\,(f''(x)\,g'(x) + f'(x)\,g''(x))$

$$\frac{f'''(x)}{f'(x)} \cdot f'(x)\,g(x) + f(x)\,g'(x)\,\frac{g'''(x)}{g'(x)}$$

$$= -3\left(\frac{f''(x)}{f(x)}f(x)g'(x) + \frac{g''(x)}{g(x)}.g(x)f'(x)\right)$$

$\because f'(x)g(x) = -f(x)g'(x)$

$$\frac{f'''(x)}{f'(x)} - \frac{g'''(x)}{g'(x)} = 3\left(\frac{f''(x)}{f(x)} - \frac{g''(x)}{g(x)}\right)$$

P 31 : (A) –0.5 D

Sol. This is the case of myopic person. The focal length of his spectacles lenses should be such that if an object is placed at a distance of 200 cm, the image should be formed at infinity.

Thus, $u = +200$ cm, $v = \infty$, $f = ?$

So, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ or $\frac{1}{-\infty} - \frac{1}{200} = \frac{1}{f}$

$\therefore$ $f = 200$ cm (concave lens)

$$P = \frac{200}{f} = \frac{100}{-200} = -0.5\ D$$

P 32 : (A) 4308 Å, 5091 Å, 6222 Å

Sol. The film's bright when the path difference

$$2\mu t\cos r = (2n-1)\frac{\lambda}{2}, \quad n = 1, 2, 3,....$$

$$\therefore \lambda = \frac{4\mu t\cos r}{(2n-1)}$$

$$= \frac{4\times1.4\times10000\times10^{-10}\times\cos0^\circ}{(2n-1)} = \frac{56000}{(2n-1)}\text{Å}$$

$\therefore \lambda$ = 56000Å, 18666Å, 8000Å, 6222Å 5091Å, 4308Å, 3733Å

The wavelengths which are not within specified range are to be rejected.

P 33 : (C) 2 cm, 4 cm

Sol. Fringe width $\beta \propto \lambda$. Therefore, λ and hence β will decrease 1.5 times when immersed in the liquid. The distance between central maxima and 10th maxima is 3 cm in vacuum. When immersed in liquid it will reduce to 2 cm. Position of central maxima will not change while 10th maxima will be obtained at y = 4 cm.

P 34 : (D) 0.02

Sol. For the first minimum, $a\sin\theta = \lambda$

where, a = width of slit

$$a = \frac{\lambda}{\sin\theta} = \frac{6980\times10^{-10}}{\sin2^\circ} = 2\times10^{-5}\ m = 0.02m$$

P 35 : (C) 5

Sol. $v_a = 5$m/s, n = 165 Hz, v = 335 m/s

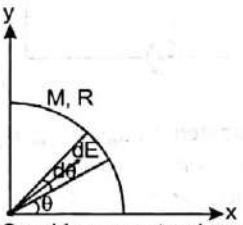
$$n' = \frac{v+v_a}{v-v_s}\times n = \frac{335+5}{335-5}\times620 = 700\ Hz$$

Therefore, number of beats, heard per second

$= n' - n = 170 - 165 = 5$

P 36 : (B) $\frac{2\sqrt{2}GM}{\pi R^2}$

Sol.



Consider a quarter ring of mass M and radius R.

Linear mass density, $\lambda = \frac{M\times2}{\pi R}$

Mass of the element, $dm = \lambda R d\theta$

Gravitational field intensity due to element at center is given by

$$dE = \frac{Gdm}{R^2} = \frac{G\times2M\times Rd\theta}{\pi R\times R^2}$$

$$d\vec{E} = \frac{2GM}{\pi R^2}\left[\cos\theta d\theta\hat{i} + \sin\theta d\theta\hat{j}\right]$$

$$\vec{E} = \frac{2GM}{\pi R^2}\left[\int_0^{\frac{\pi}{2}}\cos\theta d\theta\hat{i} + \int_0^{\frac{\pi}{2}}\sin\theta d\theta\hat{j}\right] = \frac{2GM}{\pi R^2}(\hat{i}+\hat{j})$$

P 37 : (D) 150 volt

Sol. Common potential,

$$V = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$$

Here, $C_1 = 10$ pF, $V_1 = 200$ V

$C_2 = 20$ pF, $V_2 = 100$ V

$$V = \frac{10\times200 + 20\times100}{10+20} = \frac{4000}{30}V = 133.3\ V$$

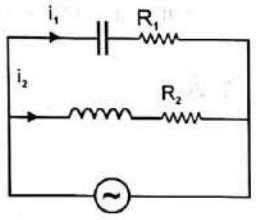
P 38 : (C) 0, 6×10^{-2} J

Sol. Since the magnetic field is uniform, the force acting on the solenoid is zero.
torque $\tau = MB \sin\theta = 1.5 \times (8 \times 10^{-2}) \times \sin 30° = 6 \times 10^{-2}$ J

P 39 : (C) 1514 watt

Sol. $i_{1_{rms}} = \dfrac{E_{rms}}{\sqrt{X_C^2 + R_1^2}} = \dfrac{130}{13} = 10$ A

$i_{2_{rms}} = \dfrac{E_{rms}}{\sqrt{X_L^2 + R_2^2}} = 13$ A



Power dissipated = $i_{1rms}^2 R_1 + i_{2rms}^2 R_2$ =
$10^2 \times 5 + 13^2 \times 6$
= power delivered by battery
= 500 + 169 × 6
= 1514 watt

P 40 : (C) $\sqrt{17}$ km

Sol. $\vec{S} = \vec{S}_1 + \vec{S}_2 + \vec{S}_3$

$5\hat{e} = 2\hat{e} + (4\hat{e} - 4\hat{N}) + \vec{S}_3$

$\vec{S}_3 = 4\hat{N} - \hat{e} \Rightarrow |\vec{S}_3| = \sqrt{17}$ km

P 41 : (D) $60 \sin (0.5\times10^3 x + 1.5\times10^{11} t)$

Sol. E = BC
$E = 60 \sin (0.5\times10^3 x + 1.5\times10^{11} t)$

P 42 : (B) 10

Sol. The resistance and current in each bulb is given as

$$R_1 = \frac{V^2}{P} = 200\Omega$$

$$I_1 = \frac{P}{V} = \frac{1}{2}A$$

If m lamps are connected in parallel, resistance of external circuit is given as

$$R = \frac{R_1}{m} = 200/m$$

For the lamps to burn normally, the current from the battery should be :

$$I = mI_1 = \frac{m}{2}A$$

According to Ohm's law

$$E = I(R + r)$$

$$\Rightarrow \quad \frac{m}{2}\left(\frac{200}{m} + 10\right) = 150$$

$$\Rightarrow \quad m = 10$$

P 43 : (D) $\pi\sqrt{\dfrac{m}{3(C^2d)^{1/3}}}$

Sol. $U = Cx^4 - 4dx$

$$\Rightarrow F = -\frac{dU}{dx} = -4Cx^3 + 4d$$

So, at equilibrium F = 0

$$\Rightarrow x = \left(\frac{d}{c}\right)^{1/3}$$, this is stable equilibrium point.

So for small amplitude change, in F will give restoring force

$$\frac{dF}{dx} = -12Cx^2$$

So, at $x = \left(\dfrac{d}{c}\right)^{1/3}$

$$\Rightarrow dF = -12C\left(\frac{d}{c}\right)^{1/3} dx \Rightarrow \text{SHM}$$

So, $\omega = \sqrt{\dfrac{12C^{2/3}d^{1/3}}{m}}$

$$\Rightarrow T = 2\pi\sqrt{\frac{m}{12(C^2d)^{1/3}}} = \pi\sqrt{\frac{m}{3(C^2d)^{1/3}}}$$

P 44 : (C) 40 g more than R

Sol. The decrease in weight of the solid is 40 g; same is added to that of water and beaker. If we draw the FBD of water and beaker, an additional force of buoyancy on water by ball in downward will be there which is equal to 40 gram so additional weight of 40 gram is to be kept on another pan.

P 45 : (B) 1 kg

Sol. Heat required by 10 kg water to change its temperature from 20°C to 80°C in one hour is,

$Q_1 = (ms\Delta T)_{water}$
$= (10\times10^3)\times1\times(80-20)=600\times10^3$ cal.

If m gm is the mass of steam condensed per hour, the heat released by steam to convert into water at 90°C is :

$Q_2 = (ms\Delta T)_{steam} + mL_{steam} + (ms\Delta T)_{water}$
$= m[1\times(150-100)+540+1\times(100-90)]$
$= 600$ mcal

According to the problem,
$Q_1 = Q_2$
$600 \times 10^3 = 600$ m or $m = 10^3$ gm = 1kg.

P 46 : (A) 8

Sol. For $\rho = \rho_0 r$

$$g = \frac{\rho_0 r^2}{4}.(4\pi G) = \pi G\rho_0 r^2$$

$$\text{Now } \frac{dp}{dr} = -\rho g = -\pi G\rho_0{}^2 r^3$$

$$\text{Now } \frac{\left(\frac{dP}{dr}\right)_{r=\frac{R}{2}}}{\left(\frac{dP}{dr}\right)_{r=\frac{R}{4}}} = \frac{\left(\frac{R}{2}\right)^3}{\left(\frac{R}{4}\right)^3} = 8$$

P 47 : (C) $\left(\frac{2\pi^2 n^2 r^2}{5s}\right)$

Sol. [Δ K.E.] 50% = ms ΔT

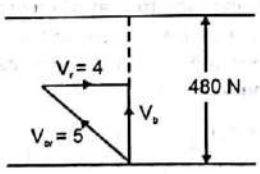
$$\frac{1}{2}\times\left[\frac{2}{5}mr^2\right]\omega^2\left[\frac{1}{2}\right] = ms\,\Delta T$$

$$\left[\frac{2\pi^2 n^2 r^2}{5s}\right] = \Delta T$$

P 48 : (B) 160 s

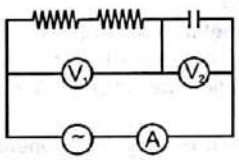
Sol. $V_b = \sqrt{5^2-4^2} = 3$ m/s

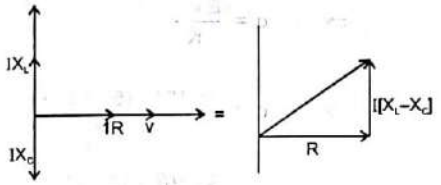
$$t = \frac{480}{3} = 160 \text{ s}$$



P 49 : (C) Reading of V_1 is 40 volt

Sol.





$$\text{Impedance} = Z = \sqrt{R^2+(X_L-X_C)^2} = 5\Omega$$

$$I_{RMS} = \frac{V_{RMS}}{Z} = 20\sqrt{2}$$

$V_L = 140\sqrt{2}$, $V_2 - V_C = 60\sqrt{2}$, $V_R = 60\sqrt{2}$

$$\cos\theta = \frac{R}{Z} = \frac{3}{5} \Rightarrow V_1 = \sqrt{v_2^2+v_R^2} = 40\sqrt{29}$$

P 50 : (A) $\frac{1}{5}$ m/s

Sol. According to mirror formula

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f} \quad \text{or} \quad v = \frac{fu}{u-f}$$

When an object is at a distance of 9 m from the concave mirror

$u = -9$ m, $f = -1$ m

$$\therefore \quad v = \frac{(-1)(-9)}{-9+1} = -\frac{4}{3}\text{m}$$

As the object moves at a constant speed of 5 ms^{-1}, after is the position of image for

$u' = -9$ m + 5 m = – 4 m

$$\therefore \quad v' = \frac{(-1)(-4)}{-4+1} = -\frac{4}{3}\text{m}$$

The shift in the position of image in 1 s is:

$$= v - v' = -\frac{9}{8}+\frac{4}{3} = \frac{-27+32}{24} = \frac{5}{24}$$

$$= \frac{1}{4.8} = \frac{1}{5}$$

∴ Average speed of the image = $\frac{1}{5}$ ms^{-1}

P 51 : (B) 8.3×10^{-5}C

Sol. Given, diameter of sphere (d) = 1m
Radius of sphere (r) = 0.5 m
Breakdown field for air (E) = 3×10^6 V/m

We know that $E = \frac{Kq}{r^2}$ (for sphere)

$$\Rightarrow \quad q = \frac{E \times r^2}{K}$$

$$\Rightarrow \quad q = \frac{3\times10^6 \times (5\times10^{-1})^2}{9\times10^9}$$

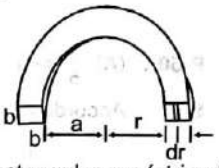
$$\Rightarrow \quad q = \frac{3\times10^6 \times 25\times10^{-2}}{9\times10^9}$$

$$\Rightarrow \quad q = \frac{25}{3}\times10^{-5} = 8.3 \times 10^{-5}\ C$$

P 52 : (B) $L = \frac{\mu_0 N^2 b}{2\pi}\log_e\left(1+\frac{b}{a}\right)$

Sol. Half the toroid is shown in the given figure If current I flows through it, magnetic field B inside it is given as

$$B = \frac{\mu_0 NI}{2\pi r}$$



the flux through the rectangular arc (strip of width dr) is

$$d\phi = b(bdr) = \frac{\mu_0 NIb}{2\pi}\frac{dr}{r}$$

∴ Total flux

$$\phi = \frac{\mu_0 NIb}{2\pi}[\log_e r]_a^{a+b}$$

$$= \frac{\mu_0 NIb}{2\pi}\log_e\left(\frac{a+b}{a}\right)$$

Total flux linked with N-turns

$$= \frac{\mu_0 N^2 Ib}{2\pi}\log_e\left(\frac{a+b}{a}\right)$$

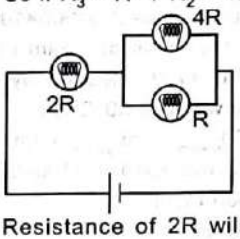
By definition $\phi = LI$

$$L = \frac{\mu_0 N^2 b}{2\pi}\log_e\left(1+\frac{b}{a}\right)$$

P 53 : (D) $B_1 > B_3 > B_2$

Sol. Resistance of a bulb is $R = \frac{V^2}{P}$

So if $R_3 = R \Rightarrow R_2 = 4R$ and $R_1 = 2R$



Resistance of 2R will have more power than the total power of R and 4R as it will have more potential drop across R and 4R, bulb with R resistance have more power so they are having same drop but R- has smaller resistance.

P 54 : (C) $E_1 < E_2 < E_3$

Sol. $\Delta E \propto Z^2$
Thus it is maximum for Li^{++} and minimum for H.

P 55 : (A) $h\left(1+\frac{m}{M}\right)$

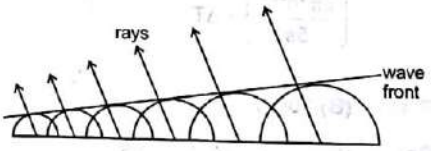
Sol. $\text{time} = \frac{h}{u}$

$$\text{speed (upward) of man} = \frac{mu}{M}$$

$$\text{Required height} = h+\frac{m}{M}h$$

P 56 : (A) It will tilt towards A

Sol. By huygen principle, resultant wave front is as shown



So rays are bent towards end A

P 57 : (C) 1.6 kHz

Sol. The reflected appears to propagate in a direction opposite to that of direct sound of engine. Thus, the source and the observer can be presumed to approach, each other, with same velocity.

$$v' = \frac{v(v+v_0)}{(v-v_s)} = v\left(\frac{v+v_s}{v-v_s}\right) \qquad [\because v_a = v_s]$$

$$\text{or } v' = 1.2\left(\frac{350+50}{350-50}\right) = \frac{1.2\times400}{300} = 1.6 \text{ kHz}$$

P 58 : (B) $\frac{I_0}{2}$

Sol. Intensity after passing through 1st polaroid $= I_0/2$

Intensity after passing through 2nd polaroid

$$= \frac{I_0}{2}\cos^2\theta$$

Intensity after passing through n^{th} polaroid

$$I_{final} = \frac{I_0}{2}\cos^{2(n-1)}\theta = \frac{I_0}{2}\cos^{2n-2}\left(\frac{\pi}{2n}\right)$$

$$I_{final} = \lim_{n\to\infty}\frac{I_0}{2}\cos^{2n-2}\left(\frac{\pi}{2n}\right) = \frac{I_0}{2}$$

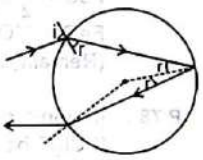
P 59 : (B) $\pi + 2i - 4r$

Sol. Total deviation

$\delta = \delta_1 + \delta_2 + \delta_3$

$= i - r + \pi - 2r + i - r$

$= \pi + 2i - 4r$

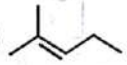


P 60 : (D) Magnetic field does interact with stationary charges i.e., charges at rest.

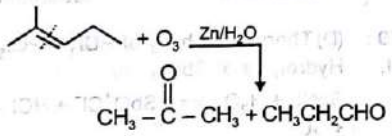
Sol. Magnetic force is interaction force between moving charge particles.

So magnetic field connot apply force on stationary charge.

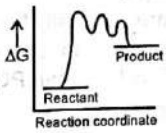
P 61 : (C)



Sol.



P 62 : (D)

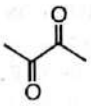


Sol. It has $\Delta G > 0$ with two intermediates.

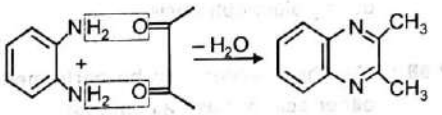
P 63 : (C) Electrons flow in the external circuit from cathode to anode

Sol. In electrochemical cells, electrons flow in the external circuit from anode to cathode because anode has the –ve charge whereas cathode is +ve charged electrode.

P 64 : (A)



Sol.



P 65 : (D) 1.2×10^5 mol/L

Sol. Henry law for CO_2

$$p_{CO_2} = K_{CO_2}C_{CO_2}$$

here $K_{CO_2} = 32$ L/atm/mol

In the unopened bottle, $p_{CO_2} = 5.0$ atm and

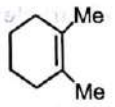
$$C_{CO_2} = \frac{p_{CO_2}}{K_{CO_2}} = \frac{5.0\,atm}{32\,L\,atm/mol} = 0.16\ mol/L$$

In the opened bottle, the CO_2 in the soda eventually reaches equilibrium with the atmospheric pressure CO_2, so

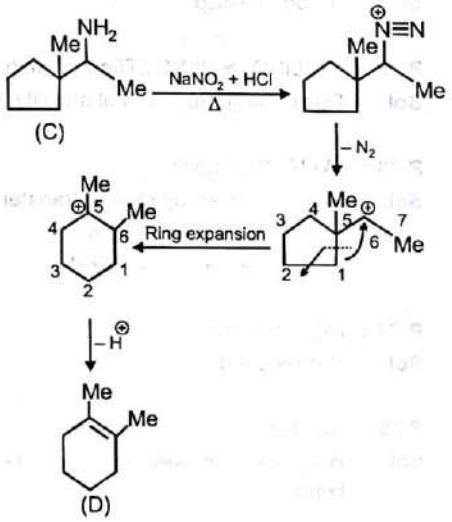
$$p_{CO_2} = 4 \times 10^{-4}\ atm$$

$$C_{CO_2} = \frac{p_{CO_2}}{(K_{CO_2})} = \frac{5\times10^{-4}\,atm}{32\,L\,atm/mol} = 1.2\times10^{5}\ mol/L$$

P 66 : (C)



Sol.



P 67 : (C) A colloid of $Fe(OH)_3$ is prepared by adding a little excess (required to completely precipitate Fe^{3+} ions as $Fe(OH)_3$)) of NaOH in $FeCl_3$ solution the particles of this sol will move towards anode during electrophoresis.

P 68 : (A) The reaction can be performed using either acid or base as catalyst.

Sol. Acetal formation is not catalysed by base.

P 69 : (A) 34%

Sol. $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

$K_p = P_{CO^2} = 1.16$ atm

$$n(CO_2) = \frac{PV}{RT} = \frac{1.16\text{atm} \times 10.0\text{L}}{(0.0821\text{L atm mol}^{-1}\text{K}^{-1})(1073\text{K})}$$

$= 0.132$ mol

Moles of $CaCO_3$ initially present = 20/100 = 0.2 mol

% of dissociation of $CaCO_3 = \frac{0.132}{0.2} \times 100 =$ 66%

% of $CaCO_3$ left = 100 – 66 = 34%

P 70 : (A) be scattered

Sol. If a silver foil is bombarded with α-particles, majority of the α-particles would be scattered.

P 71 : (A) bent T shape

Sol. 3σ bond +2ℓ.p.

P 72 : (D) $LiNO_3 > RbNO_3$:Thermal stability order

Sol. Down the group thermal stability increases.

P 73 : (A) $[Fe(H_2O)_5NO]^{2+}$

Sol. Ring is formed by charge transfer

$NO \longrightarrow NO^+ + e^-$

$Fe^{2+} + e^- \longrightarrow Fe^{2+}$

P 74 : (A) Cis–platin

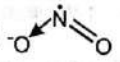
Sol. Not required

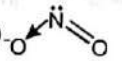
P 75 : (C) H_2O_2

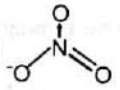
Sol. H_2O_2 has – OH polar and – O – O – nonpolar bond.

P 76 : (C) 1 2 4 3

Sol. (A) $O{=}\dot{N}{=}O$ $\theta = 180°$

(B)  $\theta = 132°$

(C)  $\theta = 115°$

(D)  $\theta = 120°$

P 77 : (D) copper has less affinity for oxygen than sulphur at high temperature

Sol. $2CuFeS_2 + O_2 \longrightarrow Cu_2S + 2FeS + SO_2$

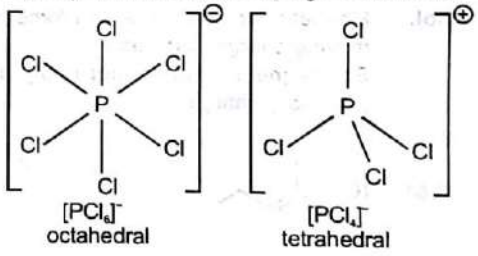
$FeS + \frac{3}{2}O_2 \longrightarrow FeO + SO_2\uparrow$

$FeO + SiO_2 \longrightarrow FeSiO_3$

(Remain Cu_2S is called matte)

P 78 : (D) ionic solid with $[PCl_4]^+$ tetharedral and $[PCl_6]^-$ octahedral

Sol. In solid state PCl_5 exists as an ionic solid with $[PCl_4]^+$ tetrahedral and $[PCl_6]^-$ octahedral.



P 79 : (D) Thermal stability of $PCl_3 < PCl_5$

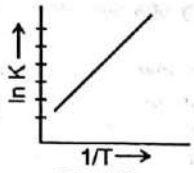
Sol. Hydrolysis of $SbCl_3$

$SbCl_3 + H_2O \rightleftharpoons SbO^+Cl^- + HCl \longrightarrow SbO^+ + 3Cl^-$

NF_3 does not undergo hydrolysis, due to high N–F bond strength.

Pentahalides are thermally less stable than the corresponding trihalides e.g., thermal stability of PCl_5 is less than PCl_5.

$PCl_5 \xrightarrow{\Delta} PCl_3 + Cl_2$

P 80 : (B) 

Sol. $\ln K = \frac{\Delta S^\circ}{R} - \frac{\Delta H^\circ}{RT}$; $\Delta S^\circ > 0$, $\Delta H^\circ < 0$

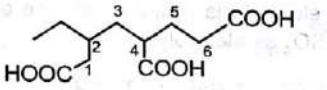
⇒ Intercept is positive and slope is also positive.

P 81 : (A) The reaction will occur and the temperature of the solution will increase

Sol. The reaction will occur spontaneously releasing energy ($\Delta U < 0$), hence temperature of solution will increase.

P 82 : (B) 2-ethyl-1,4,6-hexanetricarboxylic acid

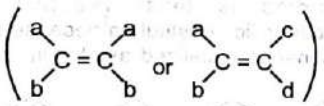
Sol.



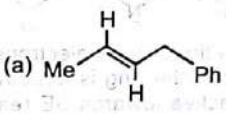
It has more than two carbon containing principal functional groups hence the longest chain containing maximum number of such functional groups become the parent chain.

P 83 : (A) 1-Phenyl-2-butene

Sol. (A) compounds of the type



show G.I. so (a) shows G.I



P 84 : (D) C_6D_6

Sol. Sulphonation followed by desulphonation with D_2SO_4/D_2O will replace all H by D.

P 85 : (A) 64.36 %

Sol. Vol. of all atoms in 1.68 g of Ar

$$= \frac{1.68}{40} \times 6 \times 10^{23} \times \frac{4}{3} \times \frac{22}{7} \times (1.50 \times 10^{-8})^3$$

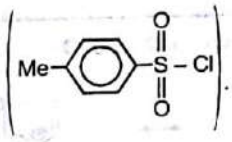
$= 0.3564$

Vol. of solid Ar = 1 cm^3

% of empty space = $(1 - 0.3564) \times 100$

$= 64.36$

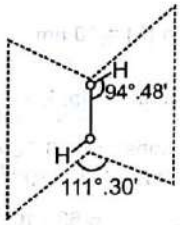
P 86 : (C) The (OH) group is converted into a good leaving group by reacting alcohols with TsCl (p-toluene sulphonyl chloride)



Sol. $EtO^{\ominus}$ is a strong base and so a poor leaving group.

P 87 : (C) Two hydroxyl groups in H_2O_2 lie in the same plane

Sol.

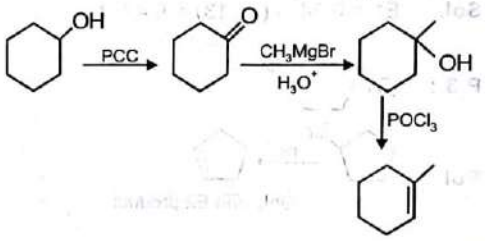


However, all the four atoms do not lie in the same plane

H_2O_2 molecule is not polar.

P 88 : (A) (i) PCC, (ii) MeMgBr (iii) $POCl_3$

Sol.



P 89 : (B) 10

Sol. At 298 K

$$A \rightleftharpoons B$$

gives, $\Delta_r H° = -54.08 kJ\ mol^{-1}$

$\Delta_r S° = 10.0\ J/mol$

from Gibb's free energy change,

$\Delta G° = \Delta_r H° - T\Delta_r S°$

$\Delta G° = -54.08 - 298 \times 10 = -51.1\ kJ$

$\Delta G° = -51100\ J$

For calculation of $\log_{10}K$, we use relationship of $\Delta G°$ and K.

$\Delta G° = -2.303\ RT\log_{10}K$

$$\text{or } \log_{10}K = \frac{-51100}{-2.303 \times 8.314 \times 298} = 9.95 = 10$$

P 90 : (A) Williamson ether synthesis

Sol. In Williamson ether synthesis, carbon-oxygen bond formation takes place.

REPLICA 12

P 1 : (A) 2.4×10^{-21} J

Sol. Given, wavelength (λ) = 10 nm

Form λ and KE relationship, $\lambda = \frac{h}{\sqrt{2KEm}}$

where, h = plank constant = 6.62×10^{-34} Js
Mass of electron (m) = 9.1×10^{-31} kg

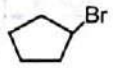
Thus, $10 \times 10^{-9} = \frac{6.62 \times 10^{-34}\,Js}{\sqrt{2 \times KE \times 9.1 \times 10^{-31} kg}}$

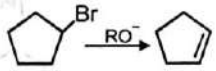
$$\sqrt{2KE} = \frac{6.62 \times 10^{-34}}{\sqrt{9.1 \times 10^{-31} \times 10^{-9}}}$$

$KE = 2.4 \times 10^{-21}$ J

P 2 : (C) 0.47 V

Sol. $E° = 0.34 - (-0.13) = 0.47$ V

P 3 : (B) 

Sol. 

Only one E2 product

P 4 : (C) 139 L

Sol. For step II : $p^{\frac{1-\gamma}{\gamma}}\ T$ = constant

$$p_2^{\frac{1-\gamma}{\gamma}} T_2 = p_3^{\frac{1-\gamma}{\gamma}} T_3$$

$$p_3\left(\frac{T_2}{T_3}\right)^{\frac{\gamma}{1-\gamma}} p_2 = \left(\frac{600}{300}\right)^{-\frac{7}{2}} \times 2$$

$$= \frac{1}{2^{7/2}} \times 2 = \frac{1}{4\sqrt{2}}\,atm$$

$$V_3 = \frac{RT_3}{p_3} = \frac{0.082 \times 300}{\frac{1}{4\sqrt{2}}} = 139L$$

P 5 : (C) Both (A) and (B)

Sol. Concentration : The ore is concentrated first by gravity method then by froth floatation process.

Roasting : Conc. ore is then roasted at 900° – 1000°C to convert ZnS and $ZnSO_4$ to ZnO.

Reduction of coke : The oxide ZnO is reduced with coke at about 1500° to obtain metallic zinc.

Electrolytic refining : Pure zinc is obtained by electrolysis zinc in presence of acidified $ZnSO_4$ as electrolyte.

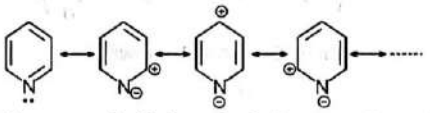
P 6 : (C) Volume was increased

P 7 : (D) H; HH; HH

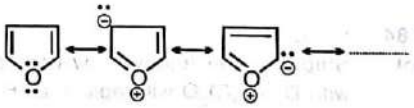
Sol. More substituted alkene is more stable (thermodynamically).

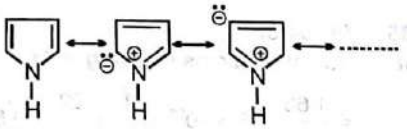
P 8 : (A)

Sol. Pyridine is least reactive towards electrophilic substitution because pyridine is resonance stabilised as shown.

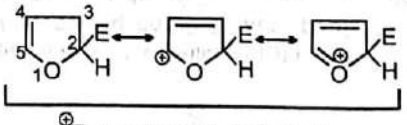


Because of withdrawal of electrons from the ring by N atom, the ring is deactivated, So, it is least reactive towards SE reaction.

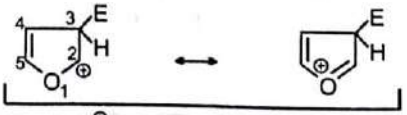




SE reaction in pyrrole, thelophene, and furan takes place at C-2 or C-5 positions because there are three resonating structures when $E^{\oplus}$ attacks at C-2 or C-5 and two resonating structures when $E^{\oplus}$ attacks at C-3.



$^{\oplus}E$ attacks at C - 2 or C - 5



$^{\oplus}E$ attacks at C-3

P 9 : (C) 0.66

Sol. Vapour pressure of two isomeric liquid X and Y are 200 torr and 100 torr respectively. So, total pressure for two liquid X, Y

$p_T = p_x + p_y$

$= \frac{200}{2} + \frac{100}{2} = 100 + 50 = 150$ torr.

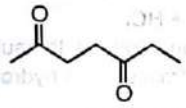
From partial partial pressure equation,
For X liquid (p_T) $y_1 = 100$ torr. ...(i)
For Y liquid (p_T) $y_2 = 50$ torr. ...(ii)

From eq. (i) ÷ (ii), $\frac{y_1}{y_2} = \frac{100/150}{50/150} = \frac{2/3}{1/3}$

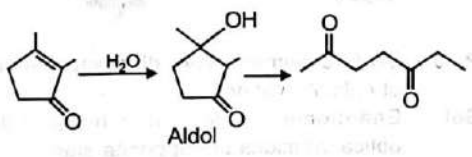
So, mole fraction of X = $\frac{2}{3} = 0.66$

P 10 : (B) heating metallic iron in a stream of dry chlorine gas

Sol. It is a fact.

P 11 : (C) 

Sol.



P 12 : (A) 4

Sol. $t_{1/2} \propto \frac{1}{[A]_0^{n-1}}$

$\Rightarrow \frac{(t_{1/2})_2}{(t_{1/2})_1} = 8 = \left(\frac{[A]_{01}}{[A]_{02}}\right)^{n-1} = 2^{n-1}$

P 13 : (D) presence of many hydroxyl groups on the polymer backbone.

Sol. Cellulose is a polyhydroxy compound, contain several free hydroxy group on its backbone capable of forming intermolecular H-bonds with water.

P 14 : (C) 130

Sol. Given, Conductivity (κ) = 1.29×10^{-2} S cm^{-1}
Resistance (R) = 28.44 Ω
From equation of molar conductivity,

$$\lambda_m = \frac{\kappa \times 1000}{\text{Molarity / Concentration (mol L}^{-1})}$$

$$\text{or } \lambda_m = \frac{\text{Conductance (c)} \times \text{cell constant(G*)} \times 1000}{\text{Molarity / Concentration (m)}}$$

$$\text{or } \lambda_m = \frac{\left[\frac{1}{R} \times \frac{l}{a}\right] \times 1000}{m} \quad ...(i)$$

For calculation of cell constant

$$k = \frac{1}{R} \times \frac{l}{a}; 1.29 \times 10^{-2}\ \text{Scm}^{-1} = \frac{1}{28.44\Omega} \times \frac{l}{a}$$

$$\text{or } \frac{l}{a} = 1.29 \times 10^{-2} \times 28.44 \quad ...(ii)$$

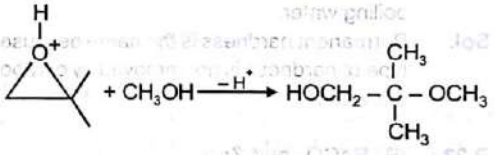
Now, put the value from Eq. (ii) in Eq. (i)

$$\lambda_m = \frac{1}{28.44} \times \frac{1.29 \times 10^{-2} \times 28.44 \times 1000}{0.5}$$

$= 130\ \text{S cm}^2\ \text{mol}^{-1}$

P 15 : (B) CH_3OH, H_2SO_4

Sol.



P 16 : (C) $2(\sqrt{2})^3 : (\sqrt{3})^3$

Sol. $a_{fcc} = 2\sqrt{2}\,r$

$$a_{bcc} = \frac{4r}{\sqrt{3}}$$

$$\text{density} = \frac{Z \times M}{N_A \times a^3}$$

P 17 : (A) the colloidal particles migrate in an applied electric field

Sol. When electric potential is applied across two Pt-electrodes, dipping in a colloidal solution the particles move towards positive or negative ends or electrodes.

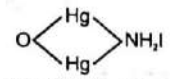
P 18 : (A) heating of NH_4NO_3 or NH_4NO_2

Sol. $NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$

$NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$

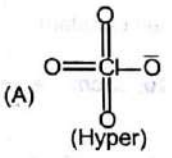
P 19 : (D) all of the above are correct statement
Sol. Desilverisation process.

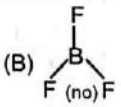
P 20 : (A) K_2HgI_4
(Nessler's reagent)
(X)

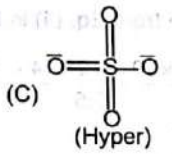


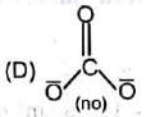
(Iodide of Million's base)
(Y)

P 21 : (B) 1, 3
Sol. Hypervalent ⇒ violating octet rule









⇒ 1, 3 are Hypervalent.

P 22 : (D) Permanent hardness can be removed by boiling water.
Sol. Permanent hardness is the name because this type of hardness is not removed by only boiling the water.

P 23 : (B) $BaCO_3$ and ZnS

Sol. $S^{2-} + BO_3^{3-}, PO_4^{3-}, C_2O_4^{2-}, CO_3^{2-}$

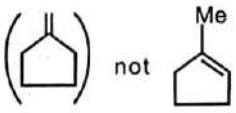
Most of cation
water insoluble
$Ag^+, Hg^{+2}, Pb^{+2}, Cu^+$
Most of anion
water insoluble

$BaCO_3$ and ZnS insoluble in water but soluble in dilute HCl solution

$$BaCO_3 + 2HCl \longrightarrow BaCl_2 + H_2CO_3$$
$$ZnS + HCl \longrightarrow ZnCl_2 + H_2S$$

P 24 : (D)All of the above.

P 25 : (C) $\rightarrow O^{\ominus}$ at high temperature
Sol. Reagent (c) is a bulky base, it will give

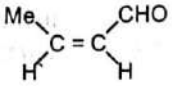


not Saytzeff product

obtained by reagent a, b, d (less-bulky base).

P 26 : (A) $[CCl_6]^{-2}$
Sol. C atom does not have vacant d-orbitals.

P 27 : (B)



Sol. Lindlar's catalyst reduces (C ≡ C) to (C = C) in syn (*cis*) addition and simultaneoulsy reduce (–COCl) group to (–CHO) group.

P 28 : (C) SO_3
Sol. SO_3 acts as an electrophile in electrophile substitution reaction.

P 29 : (B) $NaHSO_4 + HCl$
Sol. When aqueous sodium thiosulphate reacts with Cl_2 then from sodium hydrogen sulphate.

$$\underset{\text{Sodium thio sulphate}}{Na_2S_2O_3} + 4Cl_2 + 5H_2O \rightarrow 2\underset{\text{Sodium hydrogen sulphate}}{NaHSO_4} + 8HCl$$

P 30 : (A) Enantiomers have different magnitude of optical rotations.
Sol. Enantiomers have same magnitude of optical rotations but opposite sign.

P 31 : (A) 30 m/s

Sol. $n' = n\left(\frac{v - v_a}{v - v_s}\right)$

or $605 = 550\left(\frac{335 - 0}{3300 - v_s}\right)$

$\therefore$ $v_s = 30$ m/s

P 32 : (C) The value of F is zero
Sol. As the wheel is going with constant velocity without slipping so there is no friction force acting on wheel or on plank.
We don't need any force to stop the plank.

P 33 : (A) 2×10^{-8}
Sol. Magnetic field,

$$B = \frac{E}{c} = \frac{6}{3 \times 10^8} = 2 \times 10^{-8} \text{ Tesla}$$

P 34 : (D) More than 2L

Sol. The image of A is at a distance of

$$\sqrt{(2L)^3+\left(\frac{L}{2}\right)^2} \text{ from P}$$

P 35 : (C) of the same size as the object and would be erect

Sol. Equivalent focal length

$$\frac{1}{f_{eq}}=\frac{1}{f_1}+\frac{1}{f_2}=\frac{1}{20}+\frac{1}{-20}=0 \Rightarrow f_{eq}\to\infty$$

$\Rightarrow$ Combination is acting as a parallel slab. So magnification is +1.

P 36 : (B) $\frac{2\pi}{\lambda}(n_1L_1-n_2L_2)$

Sol. Optical path for 1st ray = n_1L_1

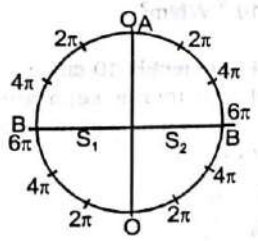
Optical path for 2nd ray n_2L_2

$\therefore$ Path diff. = $(n_1L_1 = n_2L_2)$

Now, Phase diff. = $\frac{2\pi}{\lambda}(n_1L_1-n_2L_2)$

P 37 : (B) 12

Sol.



$(S_1A - S_2B) = 0 \Rightarrow \Delta X = 0$

$\frac{2\pi}{\lambda}(0) \Rightarrow \Delta\theta = 0$

$(S_1B - S_2B) = 3\lambda \Rightarrow 6\pi$

12^{th} bright fringe

P 38 : (D) 350 Hz

Sol. Here, speed of train, $v_T = 10\ ms^{-1}$

Speed of sound, $v = 340\ ms^{-1}$

Frequency of whistle, $v = 340\ ms^{-1}$

The frequency of whistle heard by the stationary observer on the platform is

$$v' = v\left(\frac{v}{v-v_T}\right) = 340\left(\frac{340}{340-10}\right)$$

$$= 340\left(\frac{340}{340-10}\right) = 350 \text{ Hz}$$

P 39 : (A) 3.46 s

Sol. For calculating the time, for its mechanical energy to drop to half its initial value we make use of equation for energy of damped oscillation. From this equation we have,

$$\frac{E(t)}{E(0)}=e^{\frac{bt}{m}} \qquad \Rightarrow \frac{1}{2}=e^{\frac{bt}{m}}$$

$$\Rightarrow \ln 2=\frac{bt}{m} \qquad \Rightarrow t=\frac{0.2\times 0.693}{0.04}$$

$\Rightarrow t = 3.46$ s

P 40 : (D) the voltage across the capacitor increases

Sol. Charging battery is removed

$\therefore$ Q = constant

Distance between the plates is increased.

$\therefore$ D decreases $\left(QC=\frac{\varepsilon_0 A}{d}\right)$

Now, $V=\frac{Q}{C}$, Q is constant and C is decreasing

$\therefore$ V increases

$U=\frac{1}{2}\frac{Q^2}{C}$, again Q is constant and C is decreasing

$\therefore$ U increases

P 41 : (D) 50 Hz

Sol. Frequency of the current remains same, only magnitude of current changes in a tranformer.

P 42 : (B) Maximum

Sol. Displacement current will be equal to the current in the circuit which will be maximum at t = 0 as capacitor is uncharged and offers no resistance.

P 43 : (A) 28

Sol. The maximum number of electrons that can be accommodated in orbits with n = 3 is

$2 \times 1^2 + 2 \times 2^2 + 2 \times 3^2 = 28$

P 44 : (B) 12 inch at an angle 45° from initial line.

P 45 : (A) 0.0059 m/s

Sol. $\frac{1}{V}+\frac{1}{-6}=\frac{1}{0.2} \Rightarrow V=\frac{6}{29}$

$$V_I=-\frac{V^2}{u^2}V_0=-\left(\frac{6/29}{6}\right)^2(5)$$

$V_I \simeq 0.0059$ m/s

P 46 : (A) 71.4°C

Sol. Given, speed of heavy metal block = 60 km/h, Coefficient of friction between the block and surface = 0.6
Specific heat of block material = 0.1 Cal/g-°C
We know that, distance = velocity × time

$$\text{Distance} = \frac{60}{60}\times 10 = 10 \text{ km} = 10000 \text{ m}$$

Work done by friction, W = μ mg.s
$W = 0.6 \times m \times 10 \times 1 \times 10^4 = 6m \times 10^4$ J
According to question, 50% of work done converts into heat so.

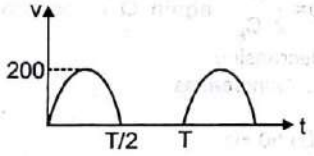
$$\frac{1}{2}W = ms\Delta T$$

$$\Rightarrow \frac{1}{2}\times (6m \times 10^4) = m\times 0.1\times 4.2\times \Delta T\times 10^3$$

$$\Delta T = \frac{60}{2\times 0.1\times 4.2} = \frac{30}{0.42} = 71.4°C$$

P 47 : (B) $\frac{100}{\sqrt{2}}$ V

Sol. Voltage-time graph across Resistance is



$$\text{So} \quad v_{rms}^2 = \frac{\int_0^{T/2} v^2 dt}{\int_0^T dt} = \frac{v_0}{2\sqrt{2}} = \frac{100}{\sqrt{2}} \text{ volt}$$

P 48 : (B) 10 cm

Sol. For the rod A,

$$a_A = \frac{\Delta L}{L\times \Delta T} = \frac{0.050}{25\times 100} = 2\times 10^{-5} \text{ per °C}$$

For rod B,

$$a_B = \frac{\Delta L'}{L'\times \Delta T'} = \frac{0.040}{40\times 100} = 1\times 10^{-5} \text{ per °C}$$

Let the composite rod C is made of protions L_A of A and L_B of rod B. then
$L_A + L_B = 50$ ...(i)
After expansions, let the lengths of the two rods be L'_A and L'_B respectively. Then

$L'_A + L'_B = 50.03$ ...(ii)

But $L'_A = L_A (1 + 2 \times 10^{-5} \times 50)$

and $L'_B = L_B (1 + 10^{-5} \times 50)$

$\therefore$ $L_A(1+2\times 10^{-5}\times 50)+L_B(1+10^{-5}\times 50) = 50.03$ ....(iii)

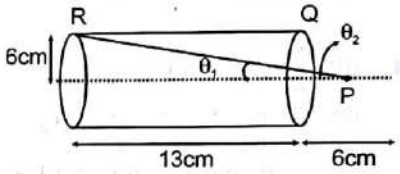
From eq. (i) we have
$L_A = 50 - L_B$
Substituting this value in eq. (iii), we get
$(50 - L_B)(1.001) + L_B(1.0005) = 50.03$
Solving we get, $L_B = 40$ cm
$L_B = 40$ cm and $L_A = 10$ cm.

P 49 : (B) 9.2×10^{-4} wb/m²

Sol. Magnetic field at P is given as

$$B_1 = \frac{1}{2}\mu_0 nI(\cos\theta_1 - \cos\theta_2)$$

The field is in the direction along the symmetry axis.



$B = B_1 + B_2 + 2B_1$
$= 2\times 3.14\times 10^{-7}\times 3000$

$$\times 1\left[\frac{16}{\sqrt{16^2+6^2}} - \frac{3}{\sqrt{3^2+6^2}}\right]$$

$= 0.92 \times 10^{-3}$ Wb/m²

P 50 : (B) Height of object is 10 cm

Sol. As object and Image separation is not changing, this is question of displacement method $m_1m_2 = 1$

$$\frac{h_{i1}}{h_0}\times\frac{h_{i2}}{h_0} = 1$$

$$\frac{20}{h_0}\times\frac{5}{h_0} = 1 \quad \Rightarrow \quad h_0 = 10 \text{ cm}$$

P 51 : (B) 2 cm

Sol. $P_B - P_C = \rho gh$

$$P_A - P_B = \frac{\rho\omega^2 x^2}{2} \quad \{x = 5 \text{ cm}\}$$

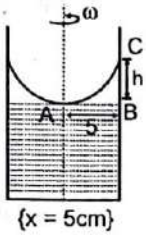
But $P_A = P_C$ as open to atmosphere

$$gh = \frac{\omega^2 x^2}{2}$$

$$20h = (4\pi)^2 (5 \times 10^{-2})^2$$

$$\Rightarrow h = \frac{16\pi^2\times 25\times 10^{-4}}{20} \text{ m}$$

$h = 2$ cm



{x = 5cm}

P 52 : (A) $t_2 : t_1$

Sol. The rate of heat loss is $H = \frac{kA\Delta T}{\ell}$

$\Rightarrow H \propto k$

So time to melt will be

$t = \frac{Q}{H} = \frac{mL}{H}$

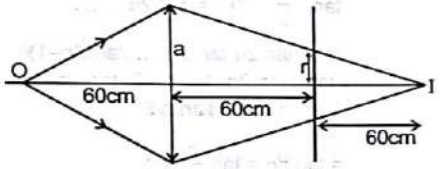
$t \propto \frac{mL}{k}$

$\Rightarrow \frac{t_1}{t_2} = \frac{k_2}{k_1}$

$\Rightarrow k_1 : k_2 = t_2 : t_1$

P 53 : (A) 30 cm

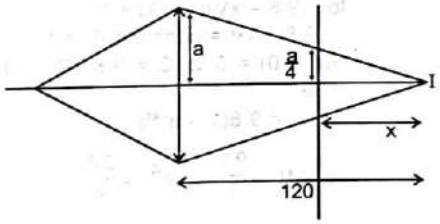
Sol.



for u = –60 cm, v = 120cm

$\frac{r}{a} = \frac{60}{120} \Rightarrow r = \frac{a}{2}$

Intensity of patch increased to 4I $\Rightarrow$ radius of patch becomes $\frac{r}{2}\left(\text{Area}\frac{1^{th}}{4}\text{ times}\right) = \frac{a}{4}$



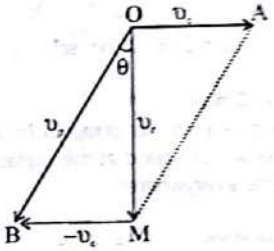
$\frac{x}{a/4} = \frac{120}{a} \Rightarrow x = 30$ cm

$\Rightarrow$ Screen is shifted by 30 cm.

P 54 : (A) 30°

Sol. In the figure υ_c represents the velocity of the car and υ_r that of the parcel. M is the position of the man. From parallelogram law, the direction of the resultant velocity υ_r must be along the direction along which the man is standing. It follows from the figure that angle θ is given by

$\sin\theta = \frac{\upsilon_c}{\upsilon_p} = \frac{10}{10\sqrt{2}} = \frac{1}{\sqrt{2}}$ or $\theta = 45°$



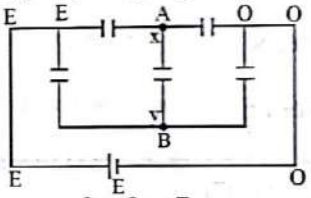
P 55 : (C) E/9

Sol. By applying conservation of charge.

$2C[x - E] + C[x - 0] + 3C(x - y) = 0$

$6x - 3y = 2E$ .....(1)

$C[y - E] + 3C[y - x] + 2C(y - 0) = 0$



$6y - 3x = E$ .....(2)

$y = \frac{4E}{9}, \quad x = \frac{5E}{9}$

$x - y = \frac{E}{9} = \Delta V_{AB}$

P 56 : (B) 50%

Sol. Speed of electron is inversely proportional to n so for excitation from n = 1 to n = 2 the velocity will reduce to half.

P 57 : (B) 0.8×10^{15} Hz

Sol. Threshold frequency, $\nu = \frac{\phi}{h}$

$= \frac{3.3 \times 1.6 \times 10^{-19}}{6.63 \times 10^{-34}}$

$= 7.96 \times 10^{14}$ Hz

Hence (B) is most appropriate.

P 58 : (D) 256

Sol. Wein's displacement law is :

$\lambda_m . T = b$

i.e. $T \propto \frac{1}{\lambda_m}$

Here, λ_m becomes half.
$\therefore$ Temperature doubles.
Also $e = \sigma T^4$

$$\Rightarrow \frac{e_1}{e_2} = \left(\frac{T_1}{T_2}\right)^4 \Rightarrow e_2 = \left(\frac{T_1}{T_2}\right)^4 . e_1 = (2)^4 . 16$$

$= 16.16 = 256$ J m^{-2} s^{-1}

P 59 : (A) 12 mW

Sol. Imagine a cylinder of radius 7m and length 10m. Intensity of sound at the surface of cylinder is same everywhere.

Therefore $I = \frac{P}{2\pi rL}$

(As sound is propagating radially out only, sound energy does not flow out through the ends)
$\therefore$ $I = 50$ W/m^2
Energy intercepted by the detector
$= I \times A = 12$ mW

P 60 : (C) 660 N

Sol. $T = 2\pi \sqrt{\frac{l}{g}}$

when the temperature is raised length changes to $\ell (1 + \alpha \Delta T)$
when the lift acceleration upwards
$g_{eff} = g + a$
new period of pendulum,

$$T' = 2\pi\sqrt{\frac{l(1+\alpha\Delta T)}{g+a}}$$

$$T' = T \Rightarrow g\frac{(1+\alpha\Delta T)}{(g+a)} = 1$$

or $a = g \alpha \Delta T = 10 \times 20 \times 10^{-4} \times 5.0$
$= 1$ m/sec^2
$w_{eff} = m (g + a)$
$= 60 (10 + 1) = 660$ N

P 61 : (D) $2x + 5 = 0$

Sol. Since the normals are perpendicular
$\therefore$ the tangent will also be perpendicular to each other
$\therefore$ they will intersect on the directrix
Equation of parabola is
$y^2 - 4y + 4 = 2x + 4$
$(y - 2)^2 = 2(x + 2)$
$\therefore$ equation of the directrix is

$x + 2 = -\frac{1}{2}$ i.e. $2x + 5 = 0$

P 62 : (B) $(2, -1/2)$

Sol. $\frac{dy}{dx} = \frac{a}{x} + 2bx + 1$

$\left.\frac{dy}{dx}\right|_{x=-1} = 0$ and $\left.\frac{dy}{dx}\right|_{x=2} = 0$

$\Rightarrow a + 2b - 1 = 0, a + 8b + 2 = 0$
$\Rightarrow a = 2, b = -1/2$

P 63 : (C) 1

Sol. $4n\alpha = \pi$
$\tan(2n - 1)\alpha = \tan(2n\alpha - \alpha)$

$= \tan\left(\frac{\pi}{2} - \alpha\right) = \cot\alpha$

$\tan(2n - 2)\alpha = \tan(2n\alpha - \alpha) =$

$\tan\left(\frac{\pi}{2} - 2\alpha\right) = \cot 2\alpha$ .....

$\tan\alpha \tan 2\alpha \tan 3\alpha$ .....$\tan(2n-1)\alpha$
$(\tan\alpha \tan(2n-1)\alpha(\tan 2\alpha \tan(2n-1)\alpha)$....$\tan n\alpha$
$(1, 1, 1,$ .....$1)\tan n\alpha$

$= \tan n\alpha = \tan\frac{\pi}{4} = 1$

P 64 : (D) $\frac{9.8}{k}$ m/s

Sol. Let V(t) be the velocity of the object at time t

Given $\frac{dV}{dt} = 9.8 - kV \Rightarrow \frac{dV}{9.8 - kV} = dt$

Integrating, we get
$\log(9.8 - kV) = -kt + C$
$\Rightarrow 9.8 - kV =$ constant C. e^{-kt}
But $V(0) = 0 \Rightarrow C = 9.8$. Thus $9.8 - kV = 9.8e^{-kt}$
$\Rightarrow kV = 9.8(1 - e^{-kt})$

$\Rightarrow V(t) = \frac{9.8}{k}(1 - e^{kt}) < \frac{9.8}{k}$

for all t. Hence. V(t) cannot exceed $\frac{9.8}{k}$

P 65 : (A) $\left(\frac{2}{a}\right)^{1/3}$

Sol. $\cos^2\theta = \frac{a^2 - 1}{3}$

$$\tan^2\frac{\theta}{2} = \frac{1-\cos\theta}{1+\cos\theta} = \frac{\sqrt{3} - \sqrt{a^2-1}}{\sqrt{3} + \sqrt{a^2-1}}$$

Therefore, $\tan^{2/3}\alpha \frac{\sqrt{3}-\sqrt{a^2-1}}{\sqrt{3}+\sqrt{a^2-1}}$ or

$$\frac{\sin^{2/3}\alpha}{\sqrt{3}-\sqrt{a^2-1}} = \frac{\cos^{2/3}\alpha}{\sqrt{3}+\sqrt{a^2-1}}$$

$$\text{or } \frac{\sin^2\alpha}{\left(\sqrt{3}-\sqrt{a^2-1}\right)^3} = \frac{\cos^2\alpha}{\left(\sqrt{3}+\sqrt{a^2-1}\right)^3}$$

$$\text{or } \frac{\sin^2\alpha}{\left(\sqrt{3}-\sqrt{a^2-1}\right)^3} = \frac{\cos^2\alpha}{\left(\sqrt{3}+\sqrt{a^2-1}\right)^3}$$

$$= \frac{1}{\left(\sqrt{3}-\sqrt{a^2-1}\right)^3+\left(\sqrt{3}+\sqrt{a^2-1}\right)^3}$$

$\sin^{2/3}\alpha + \cos^{2/3}\alpha =$

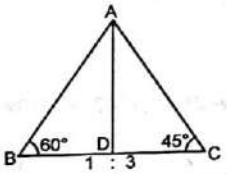
$$= \frac{\left(\sqrt{3}-\sqrt{a^2-1}\right)+\left(\sqrt{3}+\sqrt{a^2-1}\right)}{\left[\left(\sqrt{3}-\sqrt{a^2-1}\right)^3+\left(\sqrt{3}+\sqrt{a^2-1}\right)^3\right]^{1/3}}$$

$$= \frac{2\sqrt{3}}{\left[2(\sqrt{3})^3+2.3\sqrt{3}(a^2-1)\right]^{1/3}}$$

$$= \frac{2\sqrt{3}}{\left(6\sqrt{3}a^2\right)^{1/3}} = \left(\frac{2}{a}\right)^{2/3}$$

P 66 : (A) $\frac{1}{\sqrt{6}}$

Sol.



$$\frac{a/4}{\sin\angle BAD} = \frac{AD}{\sin 60} \text{ in } \Delta ABD$$

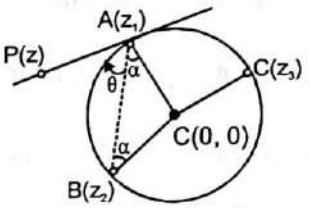
$$\text{and } \frac{3a/4}{\sin\angle CAD} = \frac{AD}{\sin 45^\circ} \text{ in } \Delta ADC$$

$$\therefore \frac{a}{4\sin\angle BAD}\cdot\frac{\sqrt{3}}{2} = \frac{3a}{2\sin\angle CAD}\cdot\frac{1}{2!}$$

$$\therefore \frac{\sin\angle BAD}{\sin\angle CAD} = \frac{\sqrt{3}}{8}\times\frac{4\sqrt{2}}{3} = \frac{1}{\sqrt{6}}$$

P 67 : (B) $\frac{1}{2}\arg\left(\frac{z_1}{z_2}\right)$

Sol.



Let $\angle BAP = \theta$

Since we know that the radius meets the point of contact of tangent to the circle at right angles, such that

$$\theta + \alpha = 90^\circ \Rightarrow \alpha = 90^\circ - \theta$$

Again in isosceles triangle ABC, we have

$$\angle CAB = \angle CBA = 90^\circ - \theta$$

such that $\angle BCA = 2\theta$

Given that $\theta = \arg\left(\frac{z-z_1}{z_2-z_1}\right)$. (1)

Now, in ΔABC, we have

$$\arg\left(\frac{z_1-0}{z_2-0}\right) = 2\theta \Rightarrow \theta = \frac{1}{2}\arg\left(\frac{z_1}{z_2}\right)$$

Substituting respective value in Eq. (1), we get,

$$\arg\left(\frac{z-z_1}{z_2-z_1}\right) = \frac{1}{2}\arg\left(\frac{z_1}{z_2}\right)$$

P 68 : (D) none of these

Sol. $S_p = 0 \therefore \frac{p}{2}[2a + (p-1)d] = 0 \Rightarrow d = \frac{-2a}{p-1}$

Sum of next q terms = Sum of an AP, whose first term will be $T_{p+1} = a + pd$

$$\Rightarrow S = \frac{q}{2}[2(a+pd) + (q-1)d]$$

$$= \frac{q}{2}[2a + (p-1)d + (p+q)d]$$

$$= \frac{q}{2}\left[0 - (p+q)\frac{2a}{p-1}\right]$$

$$= -a\frac{(p+q)q}{p-1}, \text{ by (1)}$$

P 69 : (B) form an equilateral triangle

Sol. Let the given position vectors be of points A, B and C, respectively, then

$$|\overline{AB}| = \sqrt{(\beta-\alpha)^2+(\gamma-\beta)^2+(\alpha-\gamma)^2}$$

$$|\overline{BC}| = \sqrt{(\gamma-\beta)^2+(\alpha-\gamma)^2+(\alpha-\beta)^2}$$

$$|\overline{CA}| = \sqrt{(\alpha-\gamma)^2+(\beta-\alpha)^2+(\gamma-\beta)^2}$$

$\because \quad |\overline{AB}| = |\overline{BC}| = |\overline{CA}|$

P 70 : (D) $y^2 = a(a - 2x)$

Sol. Let the point of intersection of the tangent be (h, k)

$\therefore$ Equation of chord of contact is

$hx + ky = a^2$

It is a tangent to the circle $x^2 + y^2 - 2ax = 0$

$$\therefore \quad \left|\frac{ha-a^2}{\sqrt{h^2+k^2}}\right| = a$$

$\therefore \quad (h - a)^2 = h^2 + k^2$

i.e., $\quad -2ah + a^2 = k^2$

$\therefore$ the locus $y^2 = a(a - 2x)$

P 71 : (D) $\left(\frac{1}{2}, -\sqrt{2}\right)$

Sol. Let the circle be $x^2 + y^2 + 2gx + 2fy = 0$

since it passes through (1, 0)

$\therefore 1 + 2g = 0$

$\therefore$ equation of the circle is $x^2 + y^2 - x + 2fy = 0$

its centre is $\left(\frac{1}{2}, -f\right)$ and radius = $\sqrt{\frac{1}{4}+f^2}$

radius of the given circle = 3

$\therefore 3 - \sqrt{\frac{1}{4}+f^2}$ = distance between the centres = $\sqrt{\frac{1}{4}+f^2}$

$$\therefore \sqrt{\frac{1}{4}+f^2} = \frac{3}{2} \quad \Rightarrow \frac{1}{4} + f^2 = \frac{9}{4} \quad f = \sqrt{2}$$

$\therefore$ centre is $\left[\frac{1}{2}, \sqrt{2}\right)$ or $\left(\frac{1}{2}, -\sqrt{2}\right)$

P 72 : (C) 3

Sol. Coordinates of point G is G $\left(\frac{b}{3}, \frac{c}{3}\right)$

$$\text{Let } E = \frac{(AB^2)+(BC^2)+(CA^2)}{(GA)^2+(GB)^2+(GC)^2}$$

$$\Rightarrow E = \frac{4a^2+(a+b^2)+c^2+(a-b)^2+c^2}{\left(\frac{b}{3}-a\right)^2+\left(\frac{c}{3}\right)^2+\left(\frac{b}{3}+a\right)^2+\left(\frac{c}{3}\right)^2+\left(\frac{2b}{3}\right)^2+\left(\frac{2c}{3}\right)^2}$$

$$\Rightarrow E = \frac{4a^2+2c^2+2a^2+2b^2}{\frac{2b^2}{9}+2a^2+\frac{6c^2}{9}+\frac{4b^2}{9}}$$

$$\Rightarrow E = \frac{6a^2+2b^2+2c^2}{\frac{1}{9}(6b^2+18a^2+6c^2)}$$

$$\Rightarrow E = \frac{2(3a^2+b^2+c^2)}{\frac{1}{9}6(3a^2+b^2+c^2)} = 3$$

P 73 : (B) $P \in (-\infty, -1]$

Sol. Go through trial and error for a fast approach. At p = 2, both roots are negative. so, option (c) is rejected.

At p = 0, roots are imaginary., so option (d) is rejected.

At p = –1, we have both roots positive and equal

At p > -1 roots are imaginary, thus option (b) is correct.

P 74 : (B) (0, 2)

Sol. $(5x-1)^2 + (5y-2)^2 = (\lambda^2-2\lambda+1)(3x+4y-1)^2$

$$\left(x-\frac{1}{5}\right)^2 + \left(y-\frac{2}{5}\right)^2 = (\lambda-1)^2\left(\frac{3x+4y-1}{5}\right)^2$$

$$\therefore \sqrt{\left(x-\frac{1}{5}\right)^2+\left(y-\frac{2}{5}\right)^2}$$

$$= |\lambda-1|\left|\frac{3x+4y-1}{5}\right|$$

$|\lambda - 1| < 1$

$0 < \lambda < 2$

P 75 : (B) $\sin^{-1} 1$

Sol. Let the tangents at P & Q meet at (h, k), then

equation of PQ is : $\frac{hx}{a} + \frac{ky}{b} = a + b$

i.e. $y = \frac{-bh}{ak} x + (a+b)\frac{b}{x}$

it is tangent to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\therefore (a+b)^2 \frac{b^2}{k^2} = a^2 \frac{b^2h^2}{a^2k^2} + b^2$

i.e. $(a + b)^2 = h^2 + k^2$

$\therefore$ locus of (h,k) is $x^2 + y^2 = (a + b)^2$ which is

director circle of $\frac{x^2}{a} + \frac{y^2}{b} = a + b$

Hence the angle is $\frac{\pi}{2} = \sin^{-1} 1$.

P 76 : (D) 16

Sol. The required no. of ways
= Total no. of ways of selecting any three vertices
– (no. of ways in which two consecutive and one part vertex is chosen)
– (no. of ways in which three consective vertices are chosen)
$= {}^nC_3 - n \,.\, n - {}^4C_1 - n \quad (n = 8)$
$= 56 - 8 \,.\, 4. - 8$
$= 56 - 32 - 8 = 16$

P 77 : (D) 10

Sol. (i) 3 balls in one urn + 1 ball in each of 3 urns
(ii) 2 balls in each of two urns + 1 ball in each of 8 urns
(i) B B B ; W W W; W B B W W B = 4
(ii) B B B B; B B B W; B B W W; W W W W; W W W B;
W B W B = 6 $\Rightarrow$ 10

P 78 : (A) 192

Sol. $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2 = (2\Delta)^2 = 4\left(\frac{\sqrt{3}}{4} \times 4^2\right)^2 = 192$

P 79 : (A) $\left(\frac{2}{5}, -\frac{1}{25}\right)$

Sol. $x = \underset{c \to 1}{\text{Limit}} \frac{5(c-1)(c+1)}{5(c-1)(3c+2)} = \frac{2}{5}$;

$y = \underset{c \to 1}{\text{Limit}} \frac{1-c}{5(c-1)(3c+2)} = -\frac{1}{5}$

P 80 : (A) $\frac{1}{4}$

Sol. $P(B/A \cup B^C) = \frac{P(B \cap (A \cup B^C))}{P(A \cup B^C)} = \frac{P(A \cap B)}{P(A \cup B^C)}$

Given $P(A \cap B^C) = P(A) - P(A \cap B) = 0.5$
$\Rightarrow \quad P(A \cap B) = 0.2$
$P(A \cup B^C) = P(A) + P(B^C) - P(A \cap B^C)$
$= 0.7 + 0.6 - 0.5 = 0.8$

Hence $P(B/A \cup B^C) = \frac{0.2}{0.8} = \frac{1}{4}$

P 81 : (A) T T T

Sol. S_1 : $P_1 = \frac{1}{6}$

$P_2 = \frac{5}{6^3} \quad \therefore \quad P_1 > P_2$

$\therefore$ S_1 is true.

S_2 : B throws 10, 11, or 12. The case are
6 + 4, 5 + 5, 4 + 6
6 + 5, 5 + 6, 6 + 6

$\therefore$ prob. $= \frac{6}{36} = \frac{1}{6}$

$\therefore$ S_2 is true

S_3 : $P(A_1 \cup A_2) = 1 - P(A_1 \cup A_2)'$
$= 1 - P(A_1' \cap A_2')$

also $P(A_1 \cup A_2) = 1 - P(A_1')\ P(A_2')$

$\therefore$ $P(A_1' \cap A_2') = P(A_1')\ P(A_2')$

$\therefore$ A_1' and A_2' are independent

$\therefore$ A_1 and A_2 are independent

P 82 : (C) –1

Sol. We must have $x^2 + 6x + 8 > 0$,
$2x^2 + 2x + 3 > 0$
and $x^2 + 6x + 8 \neq 2x^2 + 2x + 3 \neq 1$
$\therefore (x + 4)(x + 2) > 0$ and $(x + 1/2)^2 + 5/4 > 0$

$\Rightarrow x \in (-\infty, -4) \cup (-2, \infty)$ and $x \neq -3 \pm \sqrt{2}$ ..(i)
and $x^2 - 2x > 0$ i.e., $x \in (-\infty, 0) \cup (2, \infty)$
The given equation can be written as

$\log_{(2x^2+2x+3)} (x^2 - 2x) = 1$

$\Rightarrow \quad x^2 - 2x = 2x^2 + 2x + 3$
$\Rightarrow \quad x^2 + 4x + 3 = 0 \Rightarrow x = -1, -3$
$x = -3$ does not satisfy Equation (i) so $x = -1$

P 83 : (A) $\frac{\pi}{3}$

Sol. $\tan A + \tan B + \tan C = \tan A \tan B \tan C$

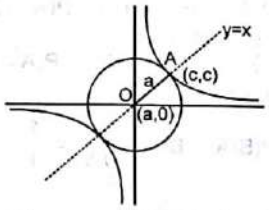
$\Rightarrow \quad 3 = \sqrt{3} \tan C$

$\Rightarrow \quad \tan C = \sqrt{3}$

$\Rightarrow \quad C = \frac{\pi}{3}$

P 84 : (B) $c^2 = \frac{a^2}{2}$

Sol. Hyperbola $xy = c^2$ & $x^2 + y^2 = a^2$ have only two common points if they touch each other as



$OA = a = \sqrt{c^2 + c^2} \quad \Rightarrow \quad 2c^2 = a^2$

$\Rightarrow \quad c^2 = \frac{a^2}{2}$

which is the required relation.

P 85 : (A) $x + 3 = 0$

Sol. Let equation of tangent to the parabola $y^2 = 4(x + 1)$ is

$y = m(x + 1) + \frac{1}{m}$

and tangent to the parabola $y^2 = 8(x + 2)$ is

$y = m'(x + 2) + \frac{2}{m'}$

since $mm' = -1 \qquad \therefore m' = \frac{-1}{m}$

then the tangents are $y = mx + m + \frac{1}{m}$ ....(1)

and $y = -\frac{1}{m}x - \frac{2}{m} - 2m$ ....(2)

by subtracting (2) from (1)

$0 = \left(m + \frac{1}{m}\right)x + \left(m + \frac{1}{m}\right) + 2\left(m + \frac{1}{m}\right)$

$\Rightarrow \quad x + 3 = 0$ is the locus.

P 86 : (B) $\frac{2}{11}$

Sol. Let $3P(A) = 2P(B) = P(C) = p$ which gives $p(A) = \frac{p}{3}$, $P(B) = \frac{p}{2}$ and $P(C) = p$

Now since A, B, C are mutually exclusive and exhaustive events, we have

$P(A) + P(B) + P(C) = 1$

$\Rightarrow \frac{p}{3} + \frac{p}{2} + p = 1 \Rightarrow p = \frac{6}{11}$

Hence $P(A) = \frac{p}{3} = \frac{2}{11}$

P 87 : (A) measure of angle B is $\pi/3$

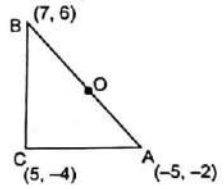
Sol. A (–5, –2) ; B (7, 6) ; C (5, –4)

$m_{AB} = \frac{8}{12} = \frac{2}{3} \qquad ; m_{BC} = \frac{10}{2} = 5$

$m_{CA} = \frac{-2}{10} = \frac{-1}{5} \qquad ; m_{BC} \times m_{CA} = -1$

$\therefore$ ΔABC is right angled at C

$\tan B = \left|\frac{5 - \frac{2}{3}}{1 + 5 \cdot \frac{2}{3}}\right|$



$\tan B = 1 \qquad \Rightarrow \qquad B = \frac{\pi}{4}$

$\therefore$ ΔABC is right isosceles

circumcenter is mid point of AB whose co-ordinates are (1, 2)

and orthocenter is C (5, –4)

P 88 : (D) $2\sin\theta$

Sol. $\sqrt{2 + \sqrt{2(1 + \cos 4\theta)}} = \sqrt{2 + 2|\cos 2\theta|}$

$= \sqrt{2(1 - \cos 2\theta)}$

$= 2|\sin\theta| = 2\sin\theta$ as $\frac{\pi}{2} < \theta < \frac{3\pi}{4}$

P 89 : (C) $x \in \left(2 - \sqrt{(9 - 2\pi)}, 2 + \sqrt{(9 - 2\pi)}\right)$

Sol. $\sin^{-1}(\sin 5) > x^2 - 4x$

$\Rightarrow \quad \sin^{-1}\{\sin(5 - 2\pi)\} > x^2 - 4x$

$\Rightarrow \quad 5 - 2\pi > x^2 - 4x$

$\Rightarrow \quad 9 - 2\pi > (x - 2)^2$

$\Rightarrow \quad (x - 2)^2 < 9 - 2\pi$

$\Rightarrow \quad -\sqrt{(9 - 2\pi)} < x - 2 < \sqrt{(9 - 2\pi)}$

$\Rightarrow \quad 2 - \sqrt{(9 - 2\pi)} < x < 2 + \sqrt{(9 - 2\pi)}$

Therefore, $x \in \left(2 - \sqrt{(9 - 2\pi)}, 2 + \sqrt{(9 - 2\pi)}\right)$

P 90 : (B) $\langle 1, 1, \sqrt{2} \rangle$

Sol. Let, A (1, 0, 0) and B(0, 1, 0)

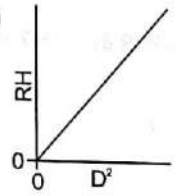
direction ratio of AB are $\langle -1, 1, 0 \rangle$

Normal is perpendicular to AB and makes an angle $\frac{\pi}{4}$ with the direction $\langle 1, 1, 0 \rangle$

$\therefore$ direction ratios of the normal are $\langle 1, 1, \sqrt{2} \rangle$.

REPLICA 13

P 1 : (D)



Sol. Let v be the speed with which the bead leaves the wire. Then by conservation of energy,

$$\frac{1}{2}v^2 = 2gR$$

Let t be the time, the bead is a projectile. Then, D = vt

and $H = \frac{1}{2}gt^2 = \frac{1}{2}g\left(\frac{D}{v}\right)^2$

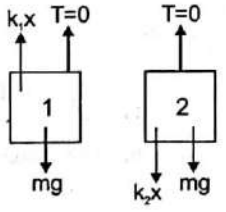
Combine, and $H = \frac{1}{8}\frac{D^2}{R}$

so that the graph of HR against D^2 is a straight line

P 2 : (A) Either $\left(a_1 = a_2 = \frac{(k_1+k_2)x}{2m}\right)$ or

$$\left(a_1 = \frac{k_1x}{m} - g \text{ and } a_2 = \frac{k_2x}{m} + g\right)$$

Sol. If x is greater than $\frac{mg}{k_1}$, then acceleration of block 1 will be in upward direction which causes tension in the string to be 0.



$-mg + k_1x = ma$

$a = \frac{k_1x}{m} - g$ upward for block 1

$K_2x + mg - 0 = ma$

$a = \frac{K_2x}{m} + g$ downward

If x is less than $\frac{mg}{k_1}$, then string will remain taut

$mg - k_1x - T = ma$

$T - k_2x - mg = ma$

$-(k_1 + k_2)x = 2\,ma$

$a = \frac{(k_1+k_2)x}{2m}$ upward

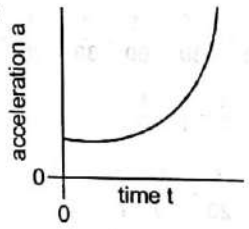
P 3 : (B) 9 ΔP

Sol. $\Delta P = P_1 - P_2 = \frac{1}{2}\rho\left(v_2^2 - v_1^2\right)$

$$= \frac{\rho}{2}\left(\left(\frac{\theta}{A_2}\right)^2 - \left(\frac{\theta}{A_1}\right)^2\right)$$

$$\Delta P' = \frac{1}{2}\rho\left(\left(\frac{3\theta}{A_2}\right)^2 - \left(\frac{3\theta}{A_1}\right)^2\right) = 9\Delta P$$

P 4 : (C)



Sol. The acceleration tangent to the incline of the helix is given by $a_t = g\sin\theta$
where θ is the angle of the incline of the helix. As it accelerates, it moves faster, according to $v = a_t t$
Since this is motion in a circle, then there is a radial acceleration given by

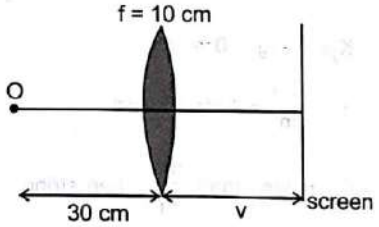
$$a_r = \frac{v^2}{r} = \frac{a_t^2}{r}t^2$$

The magnitude of the aceeleration is then

$$a = \sqrt{a_r^2 + a_t^2} = a_t\sqrt{a_t^2\frac{t^4}{r^2} + 1}$$

P 5 : (D) 20

Sol. According to lens formula, $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$



Here, u = –30 cm f = 10 cm

$$\frac{1}{v}-\frac{1}{-30}=\frac{1}{10}$$

$$\frac{1}{v}=\frac{1}{10}-\frac{1}{30}=\frac{1}{15}$$

or v = 15 cm

Now a concave lens of focal length 'f' is placed in contact with convex lens. In order to obtain a sharp image, again the screen has to be moved by 45 cm.

New image distance,

u' = v + 45 cm

15 cm + 45 cm = 60 cm

Let F be focal length of the combination.

Again using lens formula

$$\frac{1}{F}=\frac{1}{v'}-\frac{1}{u}$$

$$\frac{1}{F}=\frac{1}{60}-\frac{1}{-30}=\frac{1}{60}+\frac{1}{30}=\frac{1}{20} \text{ or } F = 20 \text{ cm}$$

As $\frac{1}{F}=\frac{1}{f}+\frac{1}{f'}$

$$\frac{1}{20}=\frac{1}{10}+\frac{1}{f'}$$

or $\frac{1}{f'}=\frac{1}{20}+\frac{1}{10}=-\frac{1}{20}$

or f' = –20 cm, |f'| - 20 cm

P 6 : (D) 14 cm

Sol. Given that : d_1 = 16 cm and d_2 = 9 cm

Now, $2d = \sqrt{d_1 d_2} = \sqrt{16\times 9} = 12$ cm

P 7 : (C) 9 : 1

Sol. $\frac{I_{max}}{I_{min}}=\frac{(a_1+a_2)^2}{(a_1-a_2)^2}=\frac{(2+1)^2}{(2-1)^2}=\frac{9}{1}$

P 8 : (A) Doppler's law

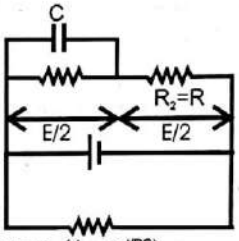
Sol. Whenever there is a relative motion between a sounding body and a listener, there occurs an apparent change in the pitch of the sounding body. This law is called Doppler's law.

P 9 : (B) 18.7 kPa

Sol. $\Delta P = \rho gh = 1060(9.8) = 18.7$ kPa.

P 10 : (C) $\frac{EC}{2}(1-e^{-2t/RC})$

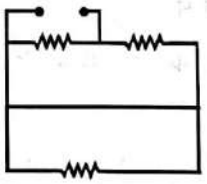
Sol.



$q = q_0 (1 - e^{-t/RC})$

$q_0 = \frac{CE}{2}$

And equivalent time constant can be calculated for discharging circuit.



$R_{eq} = \frac{R}{2} \Rightarrow \tau = \frac{RC}{2}$

$q = \frac{CE}{2}\left[1-e^{-2t/RC}\right]$

P 11 : (C) remain same

Sol. $T = \frac{2\pi m}{qB}$; no effect of speed on T.

P 12 : (B) 0.4 A

Sol. $I_s = P_s/V_s$

$\Rightarrow \quad I_s = \frac{4.4\times 10^3}{11\times 10^3} = 0.4A$

P 13 : (A) when electric field is changing in the circuit

Sol. Displacement current is defined as $I_d = \epsilon_0 \frac{dE}{dt}$.

So, it will be caused when there is a change in electric field.

P 14 : (C) $\frac{3r^2\theta I}{2}$

Sol. We know magnetic dipole moment
$\mu = IA$ {where A is area of loop}
the area of loop ABCD =
Area [OBC] – area [OAD]

$$A = \frac{(2r)^2\theta}{2} - \frac{r^2\theta}{2} = \frac{3r^2\theta}{2}$$

$$\text{So, } \mu = \frac{3r^2\theta I}{2}.$$

P 15 : (D) 4.8×10^{10}

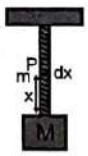
Sol. $n_e \cdot n_H = n_i^2$
$n_e = 5 \times 10^{22} - 0.05 \times 10^{22}$
$= 4.95 \times 10^{22}$
$[4.95 \times 10^{22}]\ [n_H] = 2.25 \times 10^{32}$
$n_H = 4.8 \times 10^{10}$ atoms/m^3

P 16 : (C) Magnetic field lines always form closed loops.

Sol. (A) False : An electric field requires a difference in potential. Therefore a region can exist in which the potential is constant but non-zero and for which the electric field is zero.
(B) False : A moving charged particle experiences a force in a magnetic field only if there is a component of the field perpendicular to the particle's motion. Therefore a particle moving parallel to a magnetic field experiences no force.
(C) True : Isolated magnetic north and south poles have not been observed (unlike isolated +ve and –ve charges). hence there is nothing upon which a magnetic field line can terminate and hence they must always form closed loops.
(D) False : Since the speed of light in the bubble is greater than in the surrounding water, the convex shape of the bubble produces a diverging lens.

P 17 : (A) 10.38 cm

Sol. Tension in the rope at a point at distance x from bottom



$$T = \left(M + \frac{m}{\ell}x\right)g$$

$$\therefore v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{\ell g\left(M + \frac{m}{\ell}x\right)}{m}}$$

Now, frequency of wave on string,

$$f = \frac{v_0}{\lambda_0} = \frac{1}{\lambda_0}\sqrt{\frac{\ell g M}{m}}$$

$\therefore$ Wavelength at point P

$$\lambda = \frac{v}{f} = \lambda_0\sqrt{\frac{M + \frac{m}{\ell}x}{M}}$$

$$= 6\sqrt{\frac{2 + \frac{6}{12}x}{2}}$$

$$= 6\sqrt{1 + 0.25x}\ \text{cm}$$

$$\text{At } x = \frac{1}{3} = 4\ \text{cm}$$

$$\ell = 6\sqrt{2} = 8.46\ \text{cm}$$

$$\text{At } x = \frac{2\ell}{3} = 8\ \text{cm}$$

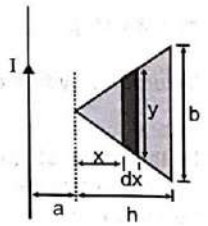
$$\lambda = 6\sqrt{3} = 10.38\ \text{cm}$$

P 18 : (A) rotate but not move toward or away from B

Sol. A will move perpendicular to B.

P 19 : (C) $4\times10^{-9}[1+\ell n^2]$

Sol. Consider an elementary section of coil as shown in figure.



Area of this section.

$$dA = y\,dx$$

$$\text{but } \frac{y}{b} = \frac{x}{h} \Rightarrow y = \frac{b}{h}x$$

$$dA = \frac{b}{h}x\,dx$$

Magnetic flux linked with this section,

$$d\phi = BdA = \frac{\mu_0 I}{2\pi(a+x)}\cdot\frac{b}{h}x\,dx$$

$$\Rightarrow \quad d\phi = \frac{\mu_0 Ib}{2\pi h}\cdot\frac{x}{a+x}dx$$

$\therefore$ Total magnetic flux linked with the coil

$$\phi = \frac{\mu_0 Ib}{2\pi h}\int_0^h \frac{x}{a+x}dx$$

$$= \frac{\mu_{0h}Ib}{2\pi h}[x - a\ell n|a+x|]_0^h$$

$$= \frac{2\times10^{-7}\times20\times10^{-2}}{10\times10^{-2}}$$

$$\left[10\times10^{-2} + 10\times10^{-2}\ell n\left|\frac{2}{1}\right|\right]$$

$= 4n \times 10^{-9}[1 + \ell n2]H$

P 20 : (B) $U_{C_3} > U_{C_4} = U_{C_1} > U_{C_2}$

Sol. In steady state C_2 is shorted potential difference across C_3 is maximum and equal to EMF.
Potential drop across C_1 and C_4 will same and will be equal to half of emf.

P 21 : (A) 1.5 g/cc

Sol. Volume of glass in hydrometer = $2\pi r\ell \times 0.1$

$$= 2 \times \frac{22}{7} \times 0.6 \times 20 \times 0.1 \text{ cc}$$

Mass of glass = $\frac{44}{7} \times 0.6 \times 20 \times 0.1 \times 2.6$ g

Total mass of hydrometer

$$= \left[\frac{44}{7} \times -0.6 \times 20 \times 0.1 \times 2.6\right] + 13.54$$

m = 33.15 g

Volume of hydrometer = $\frac{22}{7} \times 0.6 \times 0.6 \times 20$

= 22.63cc
In the state of floatation, when it is fully immersed (for measured density of liquid to be minimum)
Upthrust = weight of hydrometer
Vdg = mg

or $d = \frac{m}{V} = \frac{33.15}{22.63}$ = 1.5 gm/cc.

P 22 : (A) 6.1 m/s

Sol. 75% of spring energy will give K.E. of dart

$$\Rightarrow \frac{3}{4}\left(\frac{1}{2}kx^2\right) = \frac{1}{2}mv^2$$

$$\Rightarrow \frac{3}{4}\left(\frac{1}{2}100(0.1)^2\right) = \frac{1}{2}(20\times10^{-3})v^2$$

$$\Rightarrow \frac{3}{4} = 20\times10^{-3}v^2$$

$$\Rightarrow v^2 = \frac{3}{8\times10^{-2}} \text{ m/s} \Rightarrow v = 6.1 \text{ m/s}$$

P 23 : (D) 225 m/s

Sol. Impulse = $P_f - P_i$
$8,00,000 = mv_f - [10000]100$
m = remaining mass = 8000
v_f = 225 m/s

P 24 : (A) 1/3 sec.

Sol. First block remove from equilibrium line 1st time at time $t = \frac{T}{12} = \frac{1}{2}\left[2\pi\sqrt{\frac{4}{\pi^2}}\right] = \frac{1}{3}$ sec.

{because amplitude is 10cm}
2nd block comes in front of equilibrium line 1st time at time $t = \frac{T}{6} = \frac{1}{6}(4) = \frac{2}{3}$ sec.

Clearly detector detects light first time at $t = \frac{1}{3}$ sec.

P 25 : (C) $2\left(\frac{3}{4\pi}\right)^{1/3}$

Sol. Rate of cooling $\frac{dT}{dt} = \frac{\sigma Ae}{ms}(T^4 - T_0^4)$
Let radius of sphere is R and side of cube 'a' then $m_1 = m_2$

$$\Rightarrow \rho.\frac{4}{3}\pi R^3 = \rho.a^3 \Rightarrow R = \left(\frac{3}{4\pi}\right)^{\frac{1}{3}}.a$$

Now ratio of area $\frac{A_2}{A_1} = \frac{6a^2}{4\pi R^2}$

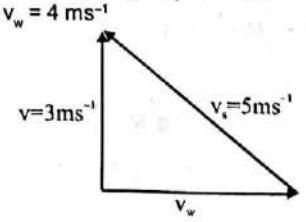
$$= \frac{6}{4\pi}\left(\frac{4\pi}{3}\right)^{2/3} = 2\left(\frac{3}{4\pi}\right)^{1/3}$$

Now given, $\left(\frac{dT}{dt}\right)_1 = \frac{dT}{dt_2} \Rightarrow e_1A_1 = e_2A_2$

$$\frac{e_1}{e_2} = \frac{A_2}{A_1} = 2\left(\frac{3}{4\pi}\right)^{\frac{1}{3}}$$

P 26 : (B) 3 ms^{-1}

Sol. In order to cross the river in the shortest time, the resultant velocity v of the swimmer must be perpendicular to the velocity v_w of water, as shown in the figure. It follows from the figure :
$v_s^2 = v^2 + v_w^2$ or $v_w^2 = v_s^2 - v^2 = 25 - 9 = 16$
or $v_w = 4$ ms^{-1}



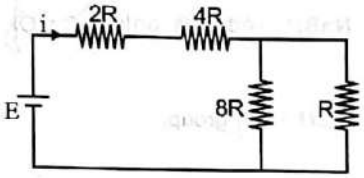
P 27 : (A) $P_{50} > P_{100} > P_{200} > P_{25}$

Sol. Resistance of a bulb, $R = \frac{V_{rating}^2}{P_{rating}}$

So, resistance of 200W, 100W, 50W and 25W can be taken as R, 2R, 4R and 8R respectively.

Total resistance $\frac{62R}{9}$.

So, $i = \frac{9E}{62R}$



So, Power in 100 W bulb $= \left(\frac{9E}{62R}\right)^2 2R = P_{100}$

Power in 50W bulb $= \left(\frac{9E}{62R}\right)^2 4R = P_{50}$

Power in 24W bulb $= \left(\frac{E}{62R}\right)^2 8R = P_{25}$

Power in 200W bulb $= \left(\frac{8E}{62R}\right)^2 R = P_{200}$

$\Rightarrow P_{50} > P_{100} > P_{200} > P_{25}$

P 28 : (C) 2.8×10^{-11} J

Sol. Peak value of E is $E_0 = 100\ NC^{-1}$

Volume of cylinder is $V = \pi r^2 l$. Average energy density is

$$u = \frac{1}{2}\varepsilon_0 E_0^2$$

$\therefore$ Energy contained in the cylinder is

$$U = uV = \pi r^2 l \times -\varepsilon_0 E_0^{\ 2}$$

$$\frac{1}{2} \times 8.85 \times 10^{-12} \times (100)^2 = 2.8 \times 10^{-11}\ J$$

P 29 : (B) 3.315×10^{-19} J

Sol. We use $\phi = \frac{12431}{6000} eV$

$\Rightarrow \quad \phi = 2.07 eV$

$\Rightarrow \quad \phi = 2.07 \times 1.6 \times 10^{-19}$

$\Rightarrow \quad \phi = 3.315 \times 10^{-19}$ J

P 30 : (C) 7 W

Sol. The energy of incident photons is given by

$h\nu = eV_s + \phi_0 = 2 + 5 = 7$ eV

(V_s is stopping potential and ϕ_0 is work function)

$\Rightarrow$ Saturation current $= 10^{-5}$ A

$$= \frac{\eta P}{h\nu} e = \frac{10^{-5} P}{7 \times e} e$$

Thus $P = 7W$

P 31 : (C) BF_3

Sol. Only in BF_3 $p\pi$-$p\pi$ back bonding is possible due to which B–F bond length is shortest. In other compounds, p-orbital at boron is not vacant hence $p\pi$-$p\pi$ back bonding is not possible.

P 32 : (A) $9\ \frac{L}{mol}$

Sol.

	$2A$	+ B	$\rightleftharpoons$ $2C$
$t = 0$	6M	3M	2M
$t = t_{eq}$	2M	1M	6M

$$K_C = \frac{[C]^2}{[A]^2[B]} = \frac{(6)^2}{(2)^2 \times 1} = \frac{36}{4} = 9\frac{L}{mol}$$

P 33 : (D) 450

Sol. $W_{glucose} = 180$

We know that, $\pi = CRT$

Since, R × T = Constant

$$\pi_{gaseous} = \pi_{He}$$

$$\frac{2}{180} \times \frac{1000}{100} = \frac{5}{M} \times \frac{1000}{100}$$

$$M = \frac{180 \times 5}{2} = 450g$$

P 34 : (C) –0.0591 V

Sol. $E_{Red} = E^\circ - \frac{0.0591}{n} \log \frac{C_2}{C_1}$

$$= 0.347 - \frac{0.0591}{2} \log 1 = 0.347$$

$$E_{oxi} = -0.347 - \frac{0.0591}{2} \log \frac{[0.01]}{1}$$

$$= -0.347 - \frac{0.0591}{2} \log 10^{-2}$$

$$= -0.347 + \frac{2 \times 0.0591}{2} - 0.3879$$

$$E_{cell} = E_{oxi} + E_{Red} = -0.2879 + 0.347$$

$$= -0.0591$$

P 35 : (A) 4

Sol. From relationship between density and number of atoms in hexagonal close packing structure.

$$d = \frac{Z \times m}{a^3 \times N_a} \quad [a^3 = a \times a \times c]$$

where, d = (density) = 0.92 g/cm^3
m(mass) of water = 18g
a(length) = 4.53 Å = 4.53 × 10^{-8} cm
c(length) = 7.41 Å = 7.41 × 10^{-8} cm

$$\text{So, } Z = \frac{d \times a \times a \times c \times N_A}{m}$$

$$Z = \frac{0.92 \times (4.53 \times 10^{-8}) \times 7.41 \times 10^{-8} \times 6.023 \times 10^{23}}{18}$$

$Z = 4.6 \simeq 4$

P 36 : (C) $HOIO_2$

Sol. $I_2 + 10HNO_3 \longrightarrow 2HIO_3 + 10NO_2 + 4H_2O$

P 37 : (A) Visible

Sol. Given, ionisation energy of certain element = 412 kJ/mol

Ionisation energy for first excited state = 126 kJ/mol.

Difference in ionisation energy

$\Delta E = 412 - 126$

= 286 kJ/mole

Energy for per atom = $\Delta E / N_A$

$$= \frac{286 \text{ kJ/mol}}{6.023 \times 10^{23}}$$

= 47.7 × 10^{-20} J/ mol

From photonic equation

$\Delta E = h\nu$

or $\Delta E = \frac{hc}{\lambda} \quad \left[\because \nu = \frac{C}{\lambda}\right]$

or $\lambda = \frac{ch}{\Delta E}$

$$\lambda = \frac{3 \times 10^8 \text{ m/s} \times 6.62 \times 10^{-34} \text{ Js}}{4.47 \times 10^{-20} \text{ J/A}}$$

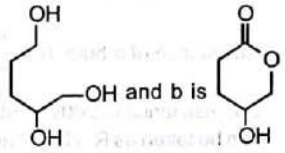
λ = 4155.5 Å

Above wavelength lies at visible region.

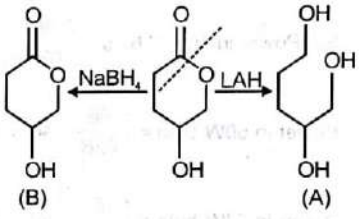
P 38 : (A) Mixture of NH_4Cl and $NaNO_3$ oh heating gives N_2 gas

P 39 : (B) $MgCO_3 < CaCO_3 < SrCO_3 < BaCO_3$

Sol. Thermal stability increases down the group.

P 40 : (C) a is  and b is

Sol. LAH reduces both ester (here cyclic ester) and $\left(\text{C}=\text{O}\right)$ group to alcohols, whereas $NaBH_4$ reduces only $\left(\text{C}=\text{O}\right)$ group to $\left(\text{CH}-\text{OH}\right)$ group.



(B) (A)

P 41 : (B) is always true because entropy is a state function

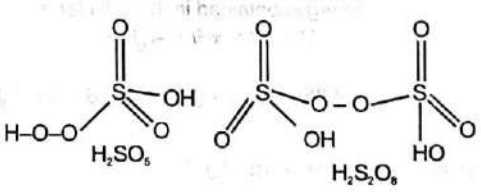
Sol. State functions do not depend on path followed.

P 42 : (C) 84

Sol. $K[PtCl_3(n^2\text{–}C_2H_4)]$ E.A.N. of Pt = 78 – 2 + 3 × 2 + 2 = 84.

P 43 : (A) H_2SO_5 and $H_2S_2O_8$

Sol. Peroxoacids of sulphur must contain one –O–O– bond as shown below :



P 44 : (D) iron, manganese and carbon

Sol. Spiegeleisen is alloy of Fe + Mn + C

P 45 : (A) EDTA in the form of calcium dihydrogen salt

Sol. EDTA (in the form of dihydrogen salt) forms soluble compound with Pb^{2+} and is thus drained out.

P 46 : (C) will be maximum for n = 3

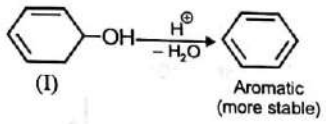
Sol. n cannot be 4 because to have n = 4 the valence shell should contain nine electrons to form AXL_4. Thus, n can have a maximum value 3 as in XeF_2.

P 47 : (B) $CaCO_3$

Sol. $CaCO_3$ on heating gives CaO(basic) and CO_2(acidic).

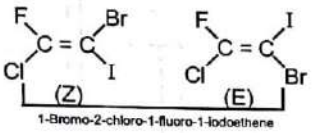
P 48 : (A) (I)

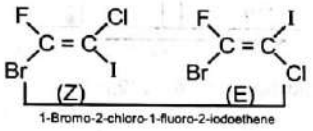
Sol.

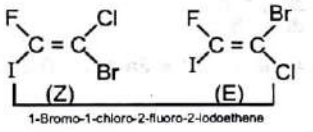


P 49 : (D) 6

Sol.







P 50 : (A) 5

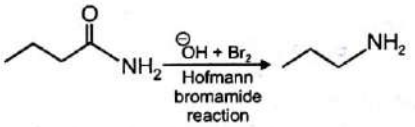
Sol. $A_n \underset{1-x}{\longrightarrow} \underset{nx}{nA}$

At "t" : $1 - x = nx$

$$\Rightarrow x = \frac{1}{n+1} = \frac{1}{6} \qquad \Rightarrow n = 5$$

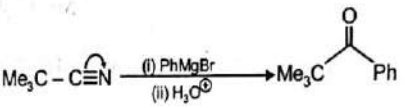
P 51 : (C)

Sol.



P 52 : (A) $Me_3C–\overset{O}{\overset{\|}{C}}–Ph$

Sol.



P 53 : (C) (I) = (II) > (IV) > (III)

Sol. Rate is faster when the substituent activates the ring (+ I or/and + R, o/p) and the rate is slower when the substituent deactivates the ring (–I, R; m). Halogen deactivates the ring (–I, +R, –I > +R) but the orientation is o/p.

Rate of $C_6H_6 = C_6D_6$, since no kinetic isotope effect is observed when H is replaced by D.

Hence the order is as given in (c).

(I) = (II) > (IV) > (III).

P 54 : (D)

Sol. (A) ⇒

(B) ⇒

(C) ⇒ Alkaline hydrolysis of B;

P 55 : (A) 0.30 mol L^{-1}

Sol. End point is obtained after 15 mL 0.2 M (3m mol) H_2SO_4 is added. Therefore 6 m mol KOH were present.

$$\Rightarrow [KOH] = \frac{6}{20} = 0.3 \text{ M}$$

P 56 : (A) –277 kJ/mol

Sol. $\Delta H_{reaction} = \left[2\Delta H_f^0(CO_2) + 3\Delta H_f^0(H_2O)\right] - \left[\Delta H_f^0(C_2H_5OH)\right]$

$-1368 = [2 \times (-393.5) + 3(-286)]$

$$-\left[\Delta H_f^0(C_2H_5OH)\right]$$

$$-1368 = [-787 + (-858)] - \left[\Delta H_f^0(C_2H_5OH)\right]$$

$$\left[\Delta H_f^0(C_2H_5OH)\right] = -277 \text{ kJ/mol}$$

P 57 : (A) 1° amine

Sol. 1° amine or 1° aromatic amine can be prepared by gabriel synthesis.

P 58 : (A) Sucrose

P 59 : (B) Only II

Sol. Soap forms insoluble precipitate when used with hard water because hard water contains calcium and magnesium ion.

P 60 : (D) all of these

P 61 : (C) $y^2 = a(x - 3a)$

Sol. Any normal to the parabola $y^2 = 4ax$ is

$y = -tx + 2at + at^3$

Let it pass through the point (h, k)

then $k = -th + 2at + at^3$

i.e. $at^3 + (2a - h)t - k = 0$

Since two normals are perpendicular to each other

$\therefore t_1 t_2 = -1$

$t_1 t_2 t_3 = \frac{k}{a}$ $\quad\therefore\quad$ $t_3 = -\frac{k}{a}$

Since t_3 is root of the equation

$$\therefore a\left(-\frac{k}{a}\right)^3 + (2a - h)\left(-\frac{k}{a}\right) - k = 0$$

$$\text{i.e. } -\frac{k^2}{a^2} + \frac{(h-2a)}{a} - 1 = 0$$

i.e. $-k^2 + ha - 2a^2 - a^2 = 0$

$\therefore$ the locus is

$y^2 = ax - 3a^2 = a(x - 3a)$

P 62 : (C) 0

Sol. $\left(\frac{4}{9}, \frac{10}{9}\right)$

$$\frac{4}{9}x^2 + \frac{10}{9}x + \frac{4}{9} - y = 0$$

$$\therefore \quad \left(\frac{4}{9}\right)^3 + \frac{10}{9} \times \frac{4}{9} + \frac{4}{9} - \frac{10}{9} < 0$$

$\therefore$ the point lies in the interior of the parabola, hence no tangent is possible

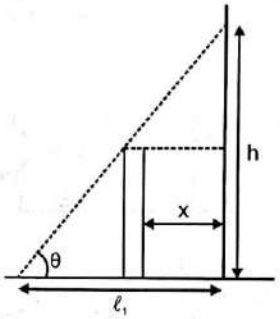
P 63 : (B) $\frac{-2\ell_2}{5}$ m/sec.

Sol. Clearly $\frac{h}{\ell_1} = \frac{\ell_2}{\ell_1 - x}$

$\Rightarrow h\ell_1 - hx = \ell_1\ell_2$ (Since h is decreasing put a –ve sign)

$$\Rightarrow \ell_1\frac{dh}{dt} - h\frac{dx}{dt} - x\frac{dh}{dt} = 0$$

$$\Rightarrow (\ell_1 - x)\frac{dh}{dt} = \frac{h\ell_1}{10}$$



at mid point, $x_1 = \ell_1/2, h = \ell_2$

$$\Rightarrow \frac{dh}{dt} = \frac{2\ell_2}{5}$$

$f'(x) = 2e^x + ae^{-x} + 2a + 1 \geq 0$ clearly $a \geq 0$

P 64 : (A) $\left(1 - \frac{\pi}{4} + \sqrt{2}\right)$

Sol. Given, $\int_{\pi/4}^{\beta} f(x)dx = \beta\sin\beta + \frac{\pi}{4}\cos\beta + \sqrt{2}\beta$

On differentiating w.r.t. β on both sides we get

$$f(\beta) = \sin\beta + \beta\cos\beta - \frac{\pi}{4}\sin\beta + \sqrt{2}$$

Put $\beta = \frac{\pi}{2}$

Then $f\left(\frac{\pi}{2}\right) = \sin\frac{\pi}{2} + \frac{\pi}{2}\cos\frac{\pi}{2} - \frac{\pi}{4}\sin\frac{\pi}{2} + \sqrt{2}$

$= 1 - \frac{\pi}{4} + \sqrt{2}$

P 65 : (C) $\left(\frac{5}{2},\frac{\sqrt{3}}{2}\right)$

Sol. Let P(x', y') be the coordinates of the point obtained by rotating the axes through an angle of 60°.
Therefore, the transformation matrix can be written as :

$$\begin{bmatrix}x'\\y'\end{bmatrix}=\begin{bmatrix}\cos\theta & \sin\theta\\-\sin\theta & \cos\theta\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix}$$

$$\Rightarrow \begin{bmatrix}2\\-\sqrt{3}\end{bmatrix}=\begin{bmatrix}\frac{1}{2} & \frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2} & \frac{1}{2}\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix}$$

$$\Rightarrow \begin{bmatrix}2\\-\sqrt{3}\end{bmatrix}=\begin{bmatrix}\frac{x}{2}+\frac{\sqrt{3}y}{2}\\-\frac{\sqrt{3}y}{2}+\frac{y}{2}\end{bmatrix}$$

$\Rightarrow x + \sqrt{3}\,y = 4$ and $\sqrt{3}\,x - y = 2\sqrt{3}$
Solving the above equations we have

$$(x, y) \equiv \left(\frac{5}{2},\frac{\sqrt{3}}{2}\right).$$

P 66 : (C) 33

Sol. In order to think about this question you need to think about whether the number can be divided by the initial numbers below the square root like 2, 3, 4, 5.... and so on. Let us say if we think of a number that is not divisible by 2, in such a case if we take the number to be divisible by 3, 5 and 7, then the largest factors of x that we will get would be $\frac{x}{3},\frac{x}{5}$ and $\frac{x}{7}$.

Even in this situation, the percentage value of these factors as a percentage of x would only be for $\frac{x}{3}$ = 33.33%, $\frac{x}{5}$ = 20% and $\frac{x}{7}$ = 14.28 %
Hence, if we try to think of a situation where $a = \frac{x}{3}$, $b = \frac{x}{5}$ and $c = \frac{x}{7}$ the value of a + b + c would give us only (33.33% + 20% + 14.28%) = 67.61% of x, which is not equal to 100%

Since the problem states that a + b + c = x, the value of a + b + c should have added up to 100% of x.
The only situation, for this to occur would be if

$a = \frac{x}{2}$ = 50 % of x , $b = \frac{x}{3}$ = 33.33% of x

and $c = \frac{x}{6}$ = 16.66% of x.

This means that 2, 3 and 6 should divide x necessarily. In other words x should be a multiple of 6. The multiples of 6 below 200 are 6, 12, 18....198, a total of 33 numbers.
hence the correct answer is c.

P 67 : (B) $\frac{\sin\theta}{\sin\phi}$

Sol. $x\sin\phi = \tan\theta - x\cos\phi.\tan\theta$

$$x = \frac{\tan\theta}{\sin\phi+\cos\phi\tan\theta} = \frac{\sin\theta}{\sin\theta\cos\phi+\cos\theta\sin\phi}$$

Similarly, $y = \frac{\sin\phi}{\sin\theta\cos\phi+\cos\theta\sin\phi}$

Therefore, $\frac{x}{y} = \frac{\sin\theta}{\sin\phi}$

P 68 : (B) $\sqrt{3}\,x^2 - (1-a)x + a\sqrt{3} = 0$

Sol. $\tan x \tan y = a$

And $\tan(x + y) \tan\left(\frac{\pi}{6}\right)$

$$\frac{\tan x+\tan y}{1-\tan x\tan y} = \frac{1}{\sqrt{3}}$$

$\tan x \tan y = (1 - a)$
Equation whose roots are tanx and tany is

$$x^2 - \frac{(1-a)}{\sqrt{3}}x + a = 0$$

$$\sqrt{3}\,x^2 - (1-a)x + a\sqrt{3} = 0$$

P 69 : (A) 0

Sol. $\vec{A}.(\vec{B}+\vec{C})\times(\vec{A}+\vec{B}+\vec{C})$

$$= \vec{A}.[\vec{B}\times\vec{A}+\vec{B}\times\vec{B}+\vec{B}\times\vec{C}+\vec{C}\times\vec{A}+\vec{C}\times\vec{B}+\vec{C}\times\vec{C}]$$

$$= \vec{A}.\vec{B}\times\vec{A}+\vec{A}.\vec{B}\times\vec{C}+\vec{A}.\vec{C}\times\vec{A}+\vec{A}.\vec{C}\times\vec{B}$$

(using $\vec{a}\times\vec{a} = 0$)

$$= 0+[\vec{A}\vec{B}\vec{C}]+0+[\vec{A}\vec{C}\vec{B}]$$

$$= [\vec{A}\vec{B}\vec{C}]-[\vec{A}\vec{C}\vec{B}] = 0$$

P 70 : (D) c^2

Sol. $a^2\left(\sin^2\frac{C}{2}+\cos^2\frac{C}{2}\right) + b^2\left(\sin^2\frac{C}{2}+\cos^2\frac{C}{2}\right)$

$- 2ab\left(\cos^2\frac{C}{2}-\sin^2\frac{C}{2}\right)$

$= a^2 + b^2 - 2ab\cos C = c^2$

P 71 : (C) rectangle

Sol. We have, $z_1 - z_4 = z_2 - z_3$ or $\frac{z_1+z_3}{2}=\frac{z_2+z_4}{2}$

i.e., the diagonals bisect each other. Therefore, it is a parallelogram.

Also, amp $\frac{z_4-z_1}{z_2-z_1}=\frac{\pi}{2}$

$\Rightarrow$ angle at z_1 is a right angle, therefore, it is a rectangle.

P 72 : (D) $\frac{n-1}{\sqrt{a_1}+\sqrt{a_n}}$

Sol. Let $a_1, a_2, a_3, \ldots\ldots a_n$ are in AP
Let the common difference is d. Also let

$$E = \frac{1}{\sqrt{a_1}+\sqrt{a_2}}+\frac{1}{\sqrt{a_2}+\sqrt{a_3}}+..+\frac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}}$$

Rationalizing, we get :

$$E = \frac{\sqrt{a_1}-\sqrt{a_2}}{a_1-a_2}+\frac{\sqrt{a_2}-\sqrt{a_3}}{a_2-a_3}+..+\frac{\sqrt{a_{n-1}}-\sqrt{a_n}}{a_{n-1}-a_n}$$

$$\Rightarrow E= \frac{\sqrt{a_1}-\sqrt{a_2}}{-d}+\frac{\sqrt{a_2}-\sqrt{a_3}}{-d}+...+\frac{\sqrt{a_{n-1}}-\sqrt{a_n}}{-d}$$

$$\Rightarrow E= -\frac{1}{d}\left\{\sqrt{a_1}-\sqrt{a_2}+\sqrt{a_2}-\sqrt{a_3}+...+\sqrt{a_{n-1}}-\sqrt{a_n}\right\}$$

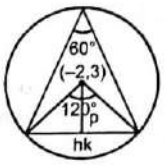
$$\Rightarrow E = -\frac{1}{d}\left(\sqrt{a_1}-\sqrt{a_n}\right)=-\frac{1}{d}\left\{\frac{a_1-a_n}{\sqrt{a_1}+\sqrt{a_n}}\right\}$$

$$\Rightarrow E = -\frac{1}{d}\left\{\frac{a_1-\{a_1+(n-1)d\}}{\sqrt{a_1}+\sqrt{a_n}}\right\}$$

Therefore, $E = \frac{n-1}{\sqrt{a_1}+\sqrt{a_n}}$

P 73 : (B) $(x + 2)^2 + (y-3)^2 = 6.25$

Sol. Radius = 5 $\Rightarrow$ p = 5 cos 60° = 2.5



$\Rightarrow$ locus is $(h + 2)^2 + (k - 3)^2 = 6.25$

P 74 : (B) 9

Sol. $\frac{x^2}{16}+\frac{y^2}{b^2}=1$ ... (1) $a = 4$

$\because$ Focus of (1) is (ae, o)

$\because$ $\frac{x^2}{4}-\frac{y^2}{3}=1$ ... (2)

$\because$ $e_1^2 = 1+\frac{3}{4}=\frac{7}{4} \Rightarrow e_1=\frac{\sqrt{7}}{2}$

$\therefore$ focus of (2) is $(\sqrt{7}, 0)$

$\therefore$ $ae = \sqrt{7}$

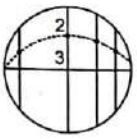
$\Rightarrow$ $4e = \sqrt{7} \Rightarrow 16e^2 = 7$

$\Rightarrow$ $16\left(1-\frac{b^2}{16}\right)=7 \Rightarrow 16 - b^2 = 7$

$\Rightarrow$ $b^2 = 9$

P 75 : (B) $\frac{4}{5}$

Sol. By property



$\frac{b}{a}=\frac{3}{5}$

Also $b^2 = a^2(1 - e^2)$ for an ellipse

$\Rightarrow$ $\frac{9}{25}=1-e^2$ or $e=\frac{4}{5}$

P 76 : (B) 7

Sol. 2. m !. n ! = 1209600 $\Rightarrow$ m ! n ! = 604800 = (7!) (5 !) $\Rightarrow$ m = 7

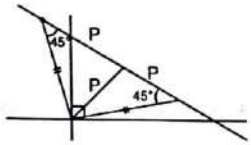
P 77 : (A) 6(7! – 4!)

Sol. $\frac{9!}{2!3!}$ – number of ways when balls of the same colour are together

$= \frac{9!}{2!3!} - 3!\,4! = 6(7! - 4!)$

P 78 : (A) independent of α

Sol.



For the isosceles (right angled) triangle formed; p is the length of the altitude to the hypotenuse. Hence the area of the triangle = $\frac{1}{2} 2p.\ p = p^2$

P 79 : (C) $P\left(\frac{A}{B}\right) > \frac{1}{2}$

Sol. $P(A \cup B) \geq P(A), P(A \cap B) \leq P(A)$
$\Rightarrow P(A \cup B) = P(A \cap B) = P(A)$
$\Rightarrow A = B$

$\therefore P(A/B) = \frac{P(A \cap B)}{P(B)} = 1$

But P(A) may not be equal to 1

P 80 : (D) None of these

Sol. Since $P(C) = 1 \ \therefore C = S$ (Sample space)

$\therefore$ $P(A \cap B \cap C) = P(A \cap B) = P(A)\ P(B)$
$= P(A)\ P(B)\ P(C)$

$\therefore$ A, B, C are independent

$P(A \cap C) = P(A) = P(A)\ .\ P(C)$

$P(B \cap C) = P(B) = P(B)\ P(C)$

$\therefore$ A, B and C are pairwise independent.

P 81 : (D) $(a - b)^2$

Sol. $u^2 = (a^2 \cos^2\theta + b^2 \sin^2\theta) + (a^2 \sin^2\theta + b^2 \sin^2\theta)$

$$+ 2\sqrt{a^2\cos^2\theta + b^2\sin^2\theta}\ \sqrt{a^2\sin^2\theta + b^2\cos^2\theta}$$

$$= a^2 + b^2$$

$$+ 2\sqrt{\sin^2\theta\cos^2\theta\left(a^4 + b^4\right) + a^2b^2\left(\sin^4\theta + \cos^4\theta\right)}$$

$$= (a^2 + b^2) + 2$$

$$\sqrt{a^2b^2\left(1 - 2\sin^2\theta\cos^2\theta\right) + \left(a^4 + b^4\right)\sin^2\theta\cos^2\theta}$$

$$= (a^2 + b^2) + 2\sqrt{a^2b^2 + \left(a^2 - b^2\right)^2 \sin^2\theta\cos^2\theta}$$

$$= (a^2 + b^2) + 2\sqrt{a^2b^2 + \frac{\left(a^2 - b^2\right)^2}{4}\sin^2 2\theta}$$

$$\text{Max } u^2 = (a^2 + b^2) + 2\sqrt{a^2b^2 + \frac{\left(a^2 - b^2\right)^2}{4}}$$

Min. $u^2 = (a^2 + b^2) + 2ab$

Therefore, Difference =

$$2\sqrt{a^2b^2 + \frac{\left(a^2 - b^2\right)^2}{4}} - 2ab$$

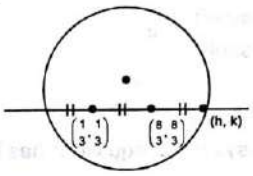
$$= \sqrt{4a^2b^2 + a^4 + b^4 - 2a^2b^2} - 2ab$$

$$= \sqrt{\left(a^2 + b^2\right)^2} - 2ab$$

$$= a^2 + b^2 - 2ab = (a - b)^2$$

P 82 : (C) 20

Sol. $\left(\frac{8}{3}, \frac{8}{3}\right)$ becomes mid point of $\left(\frac{1}{3}, \frac{1}{3}\right)$ and (h, k)



$$\Rightarrow \quad \frac{16}{3} = h + \frac{1}{3}\ ;\ \frac{16}{3} = k + \frac{1}{3}$$

$\Rightarrow$ $h = 5\ ;\ k = 5$

Hence (5, 5) must lie on circle

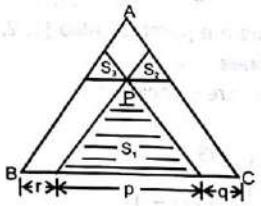
$x^2 + y^2 - 4x - 2y - c = 0$

$\Rightarrow$ $25 + 25 - 20 - 10 - c = 0$

$\Rightarrow$ $c = 20$

P 83 : (C) $\left(\sqrt{s_1} + \sqrt{s_2} + \sqrt{s_3}\right)^2$

Sol. $\frac{s_1}{s} = \frac{p^2}{a^2} \quad \Rightarrow \quad \sqrt{\frac{s_1}{s}} = \frac{p}{a}$



similarly $\sqrt{\frac{s_2}{s}} = \frac{q}{a}$ and $\sqrt{\frac{s_3}{s}} = \frac{r}{a}$

$$\Rightarrow \quad \frac{\sqrt{s_1} + \sqrt{s_2} + \sqrt{s_3}}{\sqrt{s}} = \frac{p + q + r}{a} = 1$$

$\Rightarrow s = \left(\sqrt{s_1} + \sqrt{s_2} + \sqrt{s_3}\right)^2$]

P 84 : (D) 4

Sol. Given $= \frac{\sqrt{3}}{\sin 20^\circ} - \frac{1}{\cos 20^\circ}$

$$= \frac{\sqrt{3}\cos 20^\circ - \sin 20^\circ}{\sin 20^\circ \cos 20^\circ}$$

$$= \frac{4.\left(\frac{\sqrt{3}}{2}\cos 20^\circ - \frac{1}{2}\sin 20^\circ\right)}{2\sin 20^\circ \cos 20^\circ}$$

$$= 4.\frac{\sin 60^\circ \cos 20^\circ - \cos 60^\circ \sin 20^\circ}{\sin 40^\circ}$$

$$= 4\frac{\sin 40^\circ}{\sin 40^\circ} = 4$$

P 85 : (D) –5

Sol. The system of equations has infinite solutions,

if $\Delta = \begin{vmatrix} 3 & -1 & 4 \\ 1 & 2 & -3 \\ 6 & 5 & \lambda \end{vmatrix} = 0$

$\Rightarrow 3(2\lambda + 15) - 1(-\lambda - 20) + 6(3 - 8) = 0$
$\Rightarrow 6\lambda + 45 + \lambda + 20 - 30 = 0$
$\Rightarrow 7\lambda = -35$ Therefore, $\lambda = -5$

P 86 : (A) coincident

Sol. Direction ratios of the lines are <1, 2, 3 > and <– 2, – 4, – 6 >

Thus $-\frac{2}{1} = -\frac{4}{2} = -\frac{6}{3}$

$\therefore$ the lines are parallel. Also (1, 2, 3) lies on both the lines
$\therefore$ the lines are coincident

P 87 : (A) $A = \tan^{-1} 2/3$

Sol. Let $\sin^{-1}\sqrt{(13/17)} = \theta$

$\Rightarrow \quad \sin\theta = \sqrt{(13/17)}$

$\cot\theta = 2/\sqrt{3}$

By question $\sin A = \cot\theta = 2/\sqrt{13}$

$\sin(\tan^{-1} 2/3) = 2/\sqrt{13}$

$\therefore \quad A = \tan^{-1} 2/3$

P 88 : (B) 2

Sol. $\frac{1}{r^2} + \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2}$

$$= \frac{s^2 + (s-a)^2 + (s-b)^2 + (s-c)^2}{\Delta^2}$$

$$= \frac{4s^2 - 2s(a+b+c) + a^2 + b^2 + c^2}{\Delta^2}$$

$$= \frac{a^2 + b^2 + c^2}{\Delta^2}$$

$\therefore n = 2$

P 89 : (D) $-\sqrt{2} < \lambda < \frac{5}{3\sqrt{2}}$

Sol. for $x = 1 + \frac{\lambda}{\sqrt{2}}$

$x + 2y = 1$ gives

$$2y = 1 - \left(1 + \frac{\lambda}{\sqrt{2}}\right) = -\frac{\lambda}{\sqrt{2}}$$

$$y = -\frac{\lambda}{2\sqrt{2}}$$

for $x = 1 + \frac{\lambda}{\sqrt{2}}$, $2x + 4y = 15$ gives

$$y = \frac{15 - 2 - \sqrt{2}\lambda}{4} = \frac{13 - \sqrt{2}\lambda}{4}$$

$$\therefore \quad -\frac{\lambda}{\sqrt{2}} < 2 + \frac{\lambda}{\sqrt{2}} < \frac{13 - \sqrt{2}\lambda}{4}$$

$$\Rightarrow \quad -\sqrt{2} < \lambda < \frac{5}{3\sqrt{2}}$$

P 90 : (D) 0

Sol. Given $\sin\theta = 3\sin(\theta + 2\alpha)$
$\Rightarrow \sin(\theta + \alpha - \alpha) = 3\sin(\theta + \alpha + \alpha)$
$\Rightarrow \sin(\theta + \alpha)\cos\alpha - \cos(\theta + \alpha)\sin\alpha$
$= 3\sin(\theta + \alpha)\cos\alpha + 3\cos(\theta + \alpha)\sin\alpha$
$\Rightarrow -2\sin(\theta + \alpha)\cos\alpha = 4\cos(\theta + \alpha)\sin\alpha$

$$\Rightarrow -\frac{\sin(\theta+\alpha)}{\cos(\theta+\alpha)} = \frac{2\sin\alpha}{\cos\alpha}$$

$\Rightarrow \tan(\theta + \alpha) + 2\tan\alpha = 0$

REPLICA 14

P 1 : (B) $1 - 2a^2 - 2b^2$

Sol. $\sin(\theta + \alpha) = a$, $\sin(\theta + \beta) = b$

$\theta + \alpha = \sin^{-1}a$, $\theta + \beta = \sin^{-1}b$

$\alpha - \beta = \sin^{-1}a - \sin^{-1}b$

$= \frac{\pi}{2} - \cos^{-1}a - \frac{\pi}{2} + \cos^{-1}b$

$= \cos^{-1}b - \cos^{-1}a$

$= \cos^{-1}\left\{ab + \sqrt{1-b^2}\sqrt{1-a^2}\right\}$

$= \cos^2(\alpha - \beta) = ab + \sqrt{1 - a^2 - b^2 + a^2b^2}$

$= \cos^2(\alpha - \beta) = +a^2b^2 + 1 - a^2 - b^2 + a^2b^2$

$2ab\sqrt{1 - a^2 - b^2 + a^2b^2}$

$= \cos 2(\alpha - \beta) - 4ab\cos(\alpha - \beta)$

$= 4a^2b^2 + 2 - 2a^2 - 2b^2 +$

$4ab\sqrt{(1 - a^2 - b^2 + a^2b^2)} - 1$

$= 4a^2b^2 - 4ab\sqrt{(1 - a^2 - b^2)}$

$= 4a^2b^2 - 4ab\sqrt{1 - a^2 - b^2}$

$= 1 - 2a^2 - 2b^2$

P 2 : (A) $\cos^{-1}\left(\frac{19}{35}\right)$

Sol. Angle between two plane faces is equal to the angle between the normals n_1 and n_2 to the planes.

n_1, the normal to the face OAB is given by

$$\overrightarrow{OA} \times \overrightarrow{OB} = \begin{vmatrix} i & j & k \\ 1 & 2 & 1 \\ 2 & 1 & 3 \end{vmatrix} = 5i - j - 3k \qquad (1)$$

n_2, the normal to the face ABC, is given by

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} i & j & k \\ 1 & -1 & 2 \\ -2 & -1 & 1 \end{vmatrix} = i - 5j - 3k \qquad (2)$$

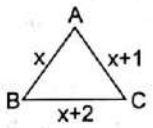
If θ be the angle between n_1 and n_2, then $\cos\theta$

$= \frac{n_1 . n_2}{|n_1||n_2|} = \frac{5.1 + 5 + 9}{\sqrt{35}\sqrt{35}}$

$\cos\theta = \frac{19}{35} \Rightarrow \theta = \cos^{-1}\left(\frac{19}{35}\right)$.

P 3 : (B) 4, 5, 6

Sol. $\because$ $x \in N$, $\therefore$ $A = 2C$



$\because$ $\frac{x+2}{\sin A} = \frac{x+1}{\sin B} = \frac{x}{\sin C}$

$\Rightarrow$ $\frac{x+2}{2\sin C\cos C} = \frac{x}{\sin C}$

$\Rightarrow$ $\frac{x+2}{2x} = \cos C$

$\Rightarrow$ $\frac{x+2}{2x} = \frac{(x+2)^2 + (x+1)^2 - x^2}{2(x+2)(x+1)}$

$\Rightarrow$ $(x + 2)^2 = x[2x + 1]$

$x^2 - 3x - 4 = 0$

$(x - 4)(x + 1) = 0$

$\Rightarrow$ $x = 4$

$\therefore$ required sides are 4, 5, 6

P 4 : (D) 4

Sol. Given that

$\tan\theta - \cot\theta = a$ (1)

and $\sin\theta + \cos\theta = b$ (2)

Now, $(b^2 - 1)^2 (a^2 + 4) = \{(\sin\theta + \cos\theta)^2 - 1\}^2 \{\tan\theta - \cot\theta)^2 + 4\}$

$= [1 + \sin 2\theta - 1]^2[\tan^2\theta + \cot^2\theta - 2 + 4]$

$= \sin^2 2\theta(\text{cosec}^2\theta + \sec^2\theta)$

$= 4\sin^2\theta\cos^2\theta\left[\frac{1}{\sin^2\theta} + \frac{1}{\cos^2\theta}\right] = 4$

P 5 : (A) $\tan A = 1$, $\tan B = 2$, $\tan C = 3$

Sol. $\because$ In any ΔABC

$\tan A + \tan B + \tan C = \tan A \tan B \tan C$.

$\Rightarrow$ only (A) option is possible

P 6 : (B) $ax^2 + 2hxy + by^2 + g\alpha + f\beta + c = 0$

Sol. Let the coordinates of point P(x, y) in new system is P(p, q) such that

$P = x - \alpha$ $\quad q = y - \beta$

$x = p + \alpha$ $\quad y = q + \beta$

Now,

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

$\Rightarrow a(p + \alpha)^2 + 2h(p + \alpha)(q + \beta) + b(q + \beta)^2 + 2g(p + \alpha) + 2f(q + \beta) + c = 0$

$\Rightarrow ap^2 + 2a\alpha p + a\alpha^2 + 2h(pq + p\beta + q\alpha + \alpha\beta) + bq^2 + 2b\beta q + b\beta^2 + 2gp + 2\alpha g + 2fq + 2\beta f + c = 0$

$\Rightarrow ap^2 + 2hpq + bq^2 + 2(\alpha a + h\beta + g)p + 2(b\beta + h\alpha + f)q + a\alpha^2 + 2h\alpha\beta + b\beta^2 + 2\alpha g + 2\beta f + c = 0$

According to given condition :

$a\alpha + h\beta + g = 0$ and $b\beta + h\alpha + f = 0$ ..(1)

$\Rightarrow ap^2 + 2hpq + bq^2 + a(\alpha a + h\beta + g) + \beta(b\beta + h\alpha + f) + \alpha g + \beta f + c = 0$

$\Rightarrow ap^2 + 2hpq + bq^2 + \alpha(0) + \beta(0) + g\alpha + f\beta + c = 0$

using (1)

Replace $p \to x$ and $q \to y$ we have

$ax^2 + 2hxy + by^2 + g\alpha + f\beta + c = 0$

P 7 : (A) 10

Sol. You can easily see that for the first term sum of the series is $\sqrt{2} - 1$, for 2 terms we have the sum as $\sqrt{3} - 1$ and so on. For the given series of 120 terms the sum would be $\sqrt{121} - 1 = 10$.

P 8 : (D) $-i$

Sol. Using $\sum_{k=1}^{n}\left[\sin\left(\frac{2k\pi}{n+1}\right) \mp i\cos\left(\frac{2k\pi}{n+1}\right)\right] = \pm i$

$\therefore \sum_{k=1}^{10}\left[\sin\frac{2k\pi}{11} \mp i\cos\frac{2k\pi}{11}\right] = -i$

P 9 : (A) $\frac{l^2 - a^2}{2S - (1+a)}$

Sol. Let n be the number of terms and d be the common difference of the AP, then

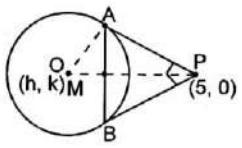
$l = a + (n - 1)d$ and $S = \frac{n}{2}(a + l)$

$\Rightarrow \frac{l - a}{n - 1} = d$, where $n = \frac{2S}{a + l}$

Therefore, $d = \frac{(l - a)}{\frac{2S}{l + a} - 1} = \frac{l^2 - a^2}{2S - (l + a)}$

P 10 : (B) $x^2 + y^2 - 5x + 9/2 = 0$

Sol.



Since triangle ABP is right-angled hence

$AM = PM = \sqrt{(x-5)^2 + k^2}$

'O' is the centre, OA in radius.

hence $AM = \sqrt{OA^2 - OM^2}$

$\Rightarrow (x - 5)^2 + k^2 = 16 - (h^2 + k^2)$

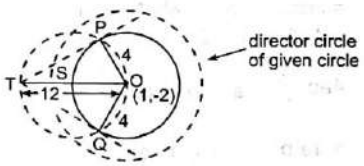
$\Rightarrow x^2 + y^2 - 5x + 9/2 = 0$ is the locus.

P 11 : (D) $12 - 4\sqrt{2}$

Sol. $(x - 1)^2 + (y + 2)^2 = 16$

$(x - 1)^2 + (y - 2)^2 = 32$

$\Rightarrow OS = 4\sqrt{2}$



$\therefore$ reqd. distance

$TS = OT - SO$

$TS = 12 - 4\sqrt{2}$

P 12 : (C) $a^2\ell^2 + b^2m^2 = 2n^2$

Sol. Let θ be eccentric angle of one of the end points. its coordinates are $(a\cos\theta, b\sin\theta)$

$\therefore$ the other end is $(-a\sin\theta, b\cos\theta)$

since both lie on $\ell x + my + n = 0$

$\therefore \ell a\cos\theta + mb\sin\theta = -n$

and $-\ell a\sin\theta + mb\cos\theta = -n$

Squaring and adding

$\ell^2a^2 + m^2b^2 = 2n^2$

P 13 : (C) 210

Sol. Of the total 7! cases in which the persons can be seated ABCD can occur in any order in 4! ways. Hence in $1/24^{th}$ of these cases, order will be ABCD. Hence $\frac{7!}{24} = 210$

P 14 : (A) N is divisible by 4 distinct primes

Sol. $770 = 2 \times 5 \times 7 \times 11$

$x_1 = 2^{\alpha_1}\ 5^{\beta_1}\ 7^{\gamma_1}\ 11^{\delta_1}$

$x_2 = 2^{\alpha_2}\ 5^{\beta_2}\ 7^{\gamma_2}\ 11^{\delta_2}$

$x_3 = 2^{\alpha_3}\ 5^{\beta_3}\ 7^{\gamma_3}\ 11^{\delta_3}$

$x_4 = 2^{\alpha_4}\ 5^{\beta_4}\ 7^{\gamma_4}\ 11^{\delta_4}$

Thus $2^{\alpha_1+\alpha_2+\alpha_3+\alpha_4}$ occurs at a factor of 770 where $\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 = 1$ it has 4 solutions.

Similarly $5^{\beta_1+\beta_2+\beta_3+\beta_4}$ has 4 solutions,

$7^{\gamma_1+\gamma_2+\gamma_3+\gamma_4}$ has 4 solutions

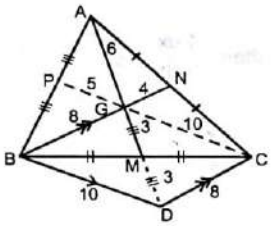
and $11^{\delta_1+\delta_2+\delta_3+\delta_4}$ has 4 solutions.

Thus the total number of solutions is 256.

P 15 : (C) 72 sq cm

Sol. Produce the median AM to D such that GM = MD . Join D to B and C . Now GBDC is a parallelogram . Note that the sides of the Δ GDC are 6, 8, 10 $\Rightarrow$ $\angle GDC = 90^\circ$

$$\left.\begin{array}{l}\text{Area of } \Delta ADC = \dfrac{12 \cdot 8}{2} = 48 \\ \text{Area of } \Delta MDC = \dfrac{3 \cdot 8}{2} = 12\end{array}\right]$$



$\Rightarrow$ Area of Δ AMC = 36

$\Rightarrow$ Area of Δ ABC = 72 cm^2

P 16 : (A) $\dfrac{1}{2}$

Sol. Since the tosses are independent

$\therefore$ P (head on 5th toss) = $\dfrac{1}{2}$

P 17 : (A) $\dfrac{2\sqrt{2}}{3}$

Sol. $\dfrac{2}{3} = \dfrac{\pi a^2 - \pi ab}{\pi a^2} = 1 - \dfrac{b}{a} = 1 - \sqrt{1-e^2} \Rightarrow e^2 = \dfrac{8}{9}$

$\Rightarrow e = \dfrac{2\sqrt{2}}{3}$

P 18 : (D) $3\omega^2$

Sol. We have $x^3 - 3x^2 + 3x + 7 = 0$

$\Rightarrow (x-1)^3 = (-2)^3 \Rightarrow \left(\dfrac{x-1}{-2}\right)^3 = (1)^3$

$\Rightarrow x - 1 = -2, -2\omega, -2\omega^2$

Since roots are α, β, γ

$a = -1 = -2,\ \beta - 1 = -2\omega,\ \gamma - 1 = -2\omega^2$

$$\frac{\alpha-1}{\beta-1} + \frac{\beta-1}{\gamma-1} + \frac{\gamma-1}{\beta-1} = \frac{1}{\omega} + \frac{1}{\omega} + \omega^2 = 3\omega^2$$

P 19 : (A) $(x^2 + y^2)^2 = 16x^2 - 9y^2$

Sol. $x^2 + y^2 = 16$ .....(1)

$9x^2 - 16y^2 = 144$

$$\Rightarrow \quad \frac{x^2}{16} - \frac{y^2}{9} = 1 \quad \text{.....(2)}$$

$\Rightarrow$ $a^2 = 16;\ b^2 = 9$

Let midpoint of any chord of (1) be (x_1, y_1)

$\therefore$ equation of this chord is $T = S_1$

$\Rightarrow$ $xx_1 + yy_1 = x_1^2 + y_1^2$

$$\Rightarrow \quad y = -x\left(\frac{x_1}{y_1}\right) + \left(\frac{x_1^2 + y_1^2}{y_1}\right) \text{.....(3)}$$

Now if (3) is tangent to (2)

$\Rightarrow$ $c^2 = a^2 m^2 - b^2$

$$\Rightarrow \quad \left(\frac{x_1^2 + y_1^2}{y_1}\right)^2 = 16 \times \frac{x_1^2}{y_1^2} - 9$$

$\Rightarrow$ $\therefore$ locus of (x_1, y_1) is $(x^2 + y^2)^2 = 16x^2 - 9y^2$

P 20 : (A) 50

Sol. Let the equation of the line be

$y - 2t = 2(x - t^2)$

$y = 2x + 2t - 2t^2$, then

$y^2 = 2(y - 2t + 2t^2)$

$y^2 - 2y + 4t - 4t^2 = 0$

$(y - 2t)(y + 2t - 2) = 0$

$\therefore$ $y = 2 - 2t$ is the other end.

$\therefore$ The ends of the chord are

$(t^2, 2t)$ and $((1-t)^2, 2(1-t))$

the point which divides this chord in the ratio 1 : 2 internally is

$$\left(\frac{2t^2 + (1-t)^2}{3}, \frac{4t + 2 - 2t}{3}\right)$$

its locus is given by eliminating 't' between

$$x = \frac{3t^2 - 2t + 1}{3},\ y = \frac{2t + 2}{3}$$

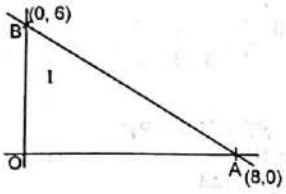
$$\left(y - \frac{8}{9}\right)^2 = \frac{4}{9}\left(x - \frac{2}{9}\right)$$

$$a = \frac{2}{9}\ ;\ b = \frac{8}{9}\ ;\ \ell = \frac{4}{9}$$

$$\therefore\ 81(a + b\ell) = 81\left(\frac{2}{9} + \frac{32}{81}\right) = 50$$

P 21 : (B) (2, 2)

Sol. OA = 8, OB = 6, AB = 10



$\therefore$ Incentre

$$I \equiv \left(\frac{8\times0+6\times8+10\times0}{8+6+10}, \frac{8\times6+6\times0+10\times0}{8+6+10}\right)$$

$I \equiv (2, 2)$

P 22 : (D) $2\sqrt{2}$

Sol. Foot of perpendicular from focus on tangent at vertex (h, k) is

$$\frac{h-2}{3} = \frac{k-3}{4} = \frac{-(6+12+7)}{25} = -1$$

$\therefore$ $h = -1$ and $k = -1$.

Hence equation of tangent at vertex is $y = x$

equation of axis is $x + y = 5$

so vertex is $\left(\frac{5}{2}, \frac{5}{2}\right)$

Length of latus rectum = 4(distance between vertex and focus) $= 4 \times \frac{1}{\sqrt{2}} = 2\sqrt{2}$

P 23 : (C) 0.68

Sol. Let C be the event that the person will have his teeth cleaned and F and E be the event of getting cavity filled or tooth extracted, respectively. We are given

$P(C) = 0.48$, $P(F) = 0.25$, $P(E) = 0.20$

$P(C \cap F) = 0.09$

$P(C \cap E) = 0.12$, $P(E \cap F) = 0.07$

and $P(C \cap F \cap E) = 0.03$

Now, $P(C \cup F \cup E) = P(C)+P(F)+P(E)$

$-P(C \cap F)-(C \cap E)-P(F \cap E)+P(C \cap F \cap E)$

$= 0.84+0.25+0.20-0.09-0.12-0.07+0.03$

$= 0.68$

P 24 : (B) 50

Sol. We have $\left[\frac{1}{2}+\frac{x}{100}\right] = \begin{cases} 0, & \text{if } 0 < x < 49 \\ 1, & \text{if } 50 \le x \le 99 \end{cases}$

Thus

$$\left[\frac{1}{2}\right]+\left[\frac{1}{2}+\frac{1}{100}\right]+\left[\frac{1}{2}+\frac{2}{100}\right]+....+\left[\frac{1}{2}+\frac{99}{100}\right] = 50$$

P 25 : (B) $3\sqrt{3}$

Sol. The equation of given circle is $x^2 + y^2 = 1$

On differentiating w.r.t. t, we get

$$2x\frac{dx}{dt}+2y\frac{dy}{dt} = 0$$

But we have, $x = \frac{1}{2}, y = \frac{\sqrt{3}}{2}$ and $\frac{dy}{dt} = -3$,

then $\frac{1}{2}\frac{dx}{dt}+\frac{\sqrt{3}}{2}(-3) = 0$

$\Rightarrow \frac{dx}{dt} = 3\sqrt{3}$.

P 26 : (B) $y = x^2 - x$

Sol. The point on y-axis is $\left(0, y - x\frac{dy}{dx}\right)$.

According to given condition,

$$\frac{x}{2} = y - \frac{x}{2}\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = 2\frac{y}{x} - 1$$

Putting $\frac{y}{x} = v$

We get $x\frac{dv}{dx} = v - 1 \Rightarrow \ln\left|\frac{y}{x} - 1\right| = \ln|x| + c$

$\Rightarrow 1 - \frac{y}{x} = x = x^2 - y = x$

$\therefore y = x^2 - x$

P 27 : (D) $\vec{a}\cdot\vec{b}=\vec{b}\cdot\vec{c}=\vec{c}\cdot\vec{a}=0$

Sol. $|(\vec{a}\times\vec{b})\cdot\vec{c}|=|\vec{a}|\ |\vec{b}|\ |\vec{c}|$

or $\|\vec{a}||\vec{b}|\sin\theta\hat{n}.\vec{c}|=|\vec{a}||\vec{b}||\vec{c}|$

or $\|\vec{a}||\vec{b}||\vec{c}|\sin\theta\cos\alpha|=|\vec{a}||\vec{b}||\vec{c}|$

or $|\sin\theta||\cos\theta|=1$

$\Rightarrow$ $\theta=\pi/2$ and $\alpha=0$

$\Rightarrow$ $\vec{a}\perp\vec{b}$ and $\vec{c}\|\hat{n}$

Or perpendicular to both $\vec{a}$ and $\vec{b}$

$\Rightarrow$ $\vec{a}.\vec{b}=\vec{b}.\vec{c}=\vec{c}.\vec{a}=0$

P 28 : (A) no real solution

Sol. The given equation can be written as

$e^{2\sin x}-4e^{\sin x}-1=0$

$\Rightarrow$ $e^{\sin x}=\dfrac{4+\sqrt{16+4}}{2}=2+\sqrt{5}$

$\Rightarrow$ $\sin x=\ln(2+\sqrt{5})$

($\ell n(2-\sqrt{5})$ not defined as $(2-\sqrt{5})$ is negative)

Now, $2+\sqrt{5}>e \Rightarrow \ln(2+\sqrt{5})>1$

$\Rightarrow$ $\sin x>1$

Which is not possible. Hence no real solution.

P 29 : (C) $\cos^{-1}(4/5)$

Sol. Without loss of generality, let the right angled ΔOAB be such that OA = OB = 2 units. Along OA take unit vector as $\vec{i}$ and along OB take unit vector as $\vec{j}$

So that $\overrightarrow{OA}=2\vec{i};\overrightarrow{OE}=\vec{i}$

$\overrightarrow{OB}=2\vec{j};\overrightarrow{OD}=\vec{j}$

$\overrightarrow{AD}=\overrightarrow{OD}-\overrightarrow{OA}=\vec{j}-2\vec{i}$

$\overrightarrow{BE}=\overrightarrow{OE}-\overrightarrow{OB}=\vec{i}-2\vec{j}$

$\overrightarrow{AD}.\overrightarrow{BE}=\left|(\vec{i}-2\vec{j})\cdot(\vec{j}-2\vec{i})\right|$

$=\sqrt{5}\sqrt{5}\cos\theta=4\Rightarrow\cos\theta=4/5$

$\therefore\ \theta=\cos^{-1}4/5$

P 30 : (D) None of these

Sol. Since, the given equation is

$\sin^{-1}(3x-2-x^2)+\cos^{-1}(x^2-4x+3)=\dfrac{\pi}{4}$

$\Rightarrow \sin^{-1}(-(x-1)(x-2))+\cos^{-1}(x-3)(x-1)=\dfrac{\pi}{4}$

For $x\in[1,2]$, we have

$\sin^{-1}(-(x-1)(x-2))\in\left[0,\dfrac{\pi}{2}\right)$

and $\cos^{-1}(x-3)(x-1)\in\left[\dfrac{\pi}{2},\pi\right]$

Hence, in this range, the equation possesses no solution.

Also, $-1\le-(x-1)(x-2)\le1$

and $-1\le-x^2-4x+3\le1$

$\Rightarrow$ $2-\sqrt{2}\le x\le\dfrac{3+\sqrt{5}}{2}$

P 31 : (A) 1.5×10^{10} Hz

Sol. The detector receives direct as well as reflected waves. Distance moved between two consecutive positions of maxima = $\lambda/2$

For 14 successive maxima = $14\times\dfrac{\lambda}{2}$

Given that ; $14\times\dfrac{\lambda}{2}=0.014$ m

$\therefore$ $\lambda=2\times10^{-2}$ m

So, $v=\dfrac{c}{\lambda}=\dfrac{3\times10^8}{2\times10^{-2}}=1.5\times10^{10}$ Hz

P 32 : (B) 20%

Sol. When an observer moves towards a stationary source, the apparent frequency heard by the observer is,

$v'=v_0\left(\dfrac{v+v_0}{v}\right)$

where, v_0 = frequency of the source

v_0 = velocity of the observer

v = velocity of the sound

$\therefore$ $v'=v_0\left(\dfrac{v+v/5}{v}\right)=\dfrac{6v_0}{5}$

Percentage change in apparent frequency

$$= \frac{v'-v_0}{v_0}\times 100 = \frac{\left(\frac{6v_0}{5}-v_0\right)}{v_0}\times 100 = \frac{1}{5}\times 100$$

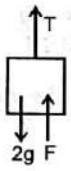
$= 20\%$

P 33 : (C) 52 N

Sol. Buoyancy force,

$$F = V\rho g = \frac{2}{10\times 10^3}\times 10^3 \times 10 = 2N$$

FBD of block is an shown



$\Rightarrow T + F_b \Rightarrow 20\ T = 18$ N

The reading of S_1 is equal to tension in string.

The effect of putting the mass inside the water is equivalent to adding an amount of water equal to volume of block.

So total force on S_1 is $5g + 2 = 52$ N

P 34 : (C) increase k times

Sol. Energy density $= \frac{1}{2}4k\epsilon_0 E^2$

Since the cell remains connected, V remains unchanged (and therefore E remains unchanged)

$\Rightarrow$ Energy density will increase k times.

P 35 : (B) 60°

Sol. Given :

Resistance R= 10Ω, impendance Z= 20Ω

The impedance in the LCR circuit is given by

$$Z = \sqrt{R^2 + (X_L + X_C)^2}$$

$$20 = \sqrt{(10)^2 + (X_L - X_C)^2}$$

Now, squaring both sides equation no. (1), we get

$$400 = 100 + (X_L - X_C)^2 \text{ or}$$

$$(X_L - X_C)^2 = 400 - 100$$

$$X_L - X_C = \sqrt{300} = 10\sqrt{3}$$

The relation for the phase difference between the current and voltage is given by

$$\tan\phi = \frac{X_L - X_C}{R} = \frac{10\sqrt{3}}{10} = \sqrt{3}$$

$$\phi = \tan^{-1}\sqrt{3} = 60°$$

P 36 : (A) 39.5×10^5

Sol. Radius of the area covered by T.V. telecast

$d = \sqrt{2hR_e}$

Total population covered = $\pi d^2 \times$ population density

$= 2\pi hR_e \times$ population density

$$= 2 \times 3.14 \times 100 \times 6.4 \times 10^6 \times \frac{1000}{10^6}$$

$= 39.503 \times 10^5$

P 37 : (A) zero

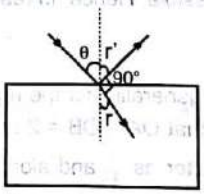
Sol. $30 = u + a \times 2$ ....(i)

$60 = u + a \times 4$ ....(ii)

Solve to get $u = 0$

P 38 : (B) $\tan^{-1}(\sin\phi)$

Sol. According to question, we can draw the following diagram.



From figure, $\theta = r'$ ...(i)

$r' + r + 90° = 180° \Rightarrow r + r' = 90°$

$r = 90° - r'$ ....(ii)

According to snell law, $\mu = \frac{\sin\theta}{\sin r}$

$\sin\theta = \mu \sin r \Rightarrow \sin\theta = \mu \sin(90° - r')$ [eq. (ii)]

$\sin\theta = \mu\cos r' \Rightarrow \sin\theta = \mu\cos\theta \Rightarrow \mu = \tan\theta$

$$\sin\phi = \frac{1}{\cot r} \Rightarrow \sin\phi = \tan r$$

$\Rightarrow r = \tan^{-1}(\sin\phi)$

P 39 : (C) 49 A

Sol. Given current in wire (x) $I_1 = 100$ A

Current in wire (y) $I_2 = I$

Distance (d) = 10 mm = 10×10^{-3}m

Linear mass density of wire (λ) = 10^{-2}kg/m

We know that for unit length if we balance

force $\Rightarrow \lambda g = \frac{\mu_0 I_1 I_2}{2\pi d}$

$$10^{-2} \times 9.8 = \frac{2\times10^{-7}\times100\times I}{10\times10^{-3}} \left[\frac{\mu_0}{2\pi} = 2\times10^{7}\right]$$

$$I = 10^2 \times \frac{9.8}{20} = \frac{980}{20} = 49A$$

P 40 : (A) 12

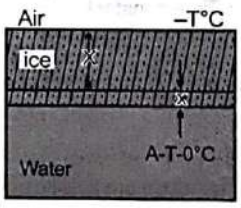
Sol. $m = \frac{A_m}{A_c} \Rightarrow A_m = 0.6 \times 20 = 12$

P 41 : (C) 19.1

Sol. The cold air (below 0°C) above the water in a lake takes heat from the water of the lake and hence, the water begins to freeze into ice layer. Here, we shall obtain an expression for the rate of growth of this layer.

As shown in figure, let a layer of ice x cm thick has already formed on a lake at 0°C. The air above lake is at –T°C. Let A be the area of the layer, L the latent heat of fusion of ice and ρ its density. The heat given up when the ice layer increase in thickness by dx,

= mass × latent heat = (A.dx.ρ)×L cal



Let this quantity of heat is conducted upwards through the layer in dt seconds.

Then

$$A(dx)\rho \times L = \frac{kA\{0-(-T)\}}{x}dt$$

where k = Thermal conductivity of ice.

$$\therefore \quad \frac{dx}{dt} = \frac{kT}{\rho L x}$$

or $$dt = \frac{\rho L x}{kT}dx$$

Now the time in which the thickness of ice will increase from x_1 to x_2 can be obtained by integrating eq. (i) within the limits of x from x_1 to x_2. Hence

$$t = \frac{\rho L}{kT}\int_{x_1}^{x_2} x\,dx = \frac{\rho L}{kT}\left[\frac{x^2}{2}\right]_{x_1}^{x_2}$$

or $$t = \frac{\rho L}{2kT}\left(x_2^2 - x_1^2\right)$$

In the given problem

$k = 0.004 \text{cal-cm}^{-1}\text{s}^{-1}{}^{\circ}\text{C}^{-1}$

L = 80 cal/g, – T = – 10°C, x_1 = 5 cm

x_2 = 10 cm (doubled)

Substituting these values, we get

$$t = \frac{0.92\times80}{2\times0.004\times10}(10^2 - 5^2)$$

= 69000 s = 19.1 h

P 42 : (C) 1 : 2

Sol. We know magnetic dipole moment $\mu = n\ell iA$

As in the question $\mu_1 = \mu_2$

$\Rightarrow n_1\ell_1 i_1 A_1 = n_2\ell_2 i_2 A_2$

$\Rightarrow n_1\ell_1 A_1 = n_2\ell_2 A_2$

As $\frac{\ell_1 A_1}{\ell_2 A_2} = 1:2 \Rightarrow n_1 : n_2 = 2 : 1$

for same magnetic field B

$\mu_0 m_1 i_1 = \mu_0 m_2 i_2$

$$\Rightarrow \frac{\ell_1}{\ell_2} = \frac{1}{2}$$

P 43 : (B) $\frac{1}{2}\left[\frac{P_1}{T_1}+\frac{P_2}{T_2}\right]$

Sol. For a closed system the total number of moles remains constant

$P_1V = n_1RT_1$, $P_2V = n_2RT_2$,

$P(2V) = (n_1 + n_2)RT$

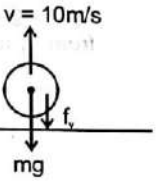
$$\therefore \quad \frac{P}{T}=\frac{n_1+n_2}{2}=\frac{1}{2}\left[\frac{P_1}{T_1}+\frac{P_2}{T_2}\right]$$

P 44 : (A) $\frac{10}{3}$m

Sol. $f_v + mg = ma$

$a = g + 3h$

$a = vdv/dh$



$$\int_{10}^{0} vdv = \int(-10-3h)dh$$

$$0 - 50 = -10h - \frac{3h^2}{2}$$

$3h^2 + 20h - 100 = 0$

$h = 10/3$

P 45 : (B) 2 sec

Sol. $F_{net} = ma_{cm}$

$a_{cm} = g\sin\theta + \mu g\cos\theta = 10\ m/s^2$

$\tau_{net} = I\alpha$

$\mu mg\cos\theta\,(R) = mR^2\,(\alpha) \Rightarrow \alpha = \mu g\cos\theta$

for pure rolling, $v = R\omega \Rightarrow 28 - 10T = \mu T$

$\Rightarrow t = 2sec$

P 46 : (B) $J_0\frac{A}{3}$

Sol. We consider a hollow cylinder of radius r and thickness dr.

The cross-sectional area of considered element is $dA = 2\pi rdr$

The current in considered element is

$$dI = JdA = J_0\left(1-\frac{r}{R}\right)2\pi rdr$$

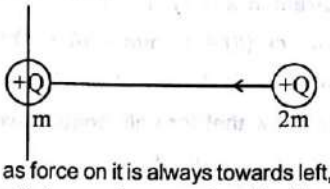
$$\Rightarrow \quad dI = 2\pi J_0\left(1-\frac{r}{R}\right)rdr$$

$$\Rightarrow \quad I = 2\pi J_0\int_0^R\left(1-\frac{r}{R}\right)rdr$$

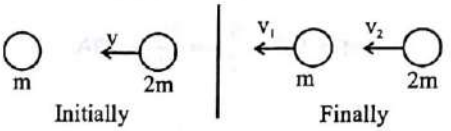
$$\Rightarrow \quad I = J_0\frac{\pi R^2}{3} = J_0\frac{A}{3}$$

P 47 : (A) 4v/3

Sol. The speed of particle at origin, keeps on increasing,



as force on it is always towards left, so its speed will be maximum, when it will reach infinity. There is no external force and the kinetic energy of system remains same, so it is similar to elastic collision between m and 2m.



$2mv = mv_1 + 2mv_2$

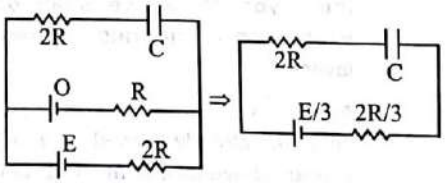
{By momentum conservation}

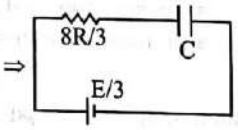
$v = v_1 - v_2$ {by e=1}

$$\Rightarrow v_1 = \frac{4v}{3}$$

P 48 : (D) $\frac{CE}{3}(1 - e^{-3t/8RC})$

Sol.





$$\Rightarrow \text{Time constant} = \frac{8RC}{3};\ -q_0 = 0 \text{ and } q_\infty = \frac{CE}{3}$$

$$\Rightarrow q = \frac{CE}{3}(1 - e^{-3t/8RC})$$

P 49 : (B) 1 : 4

Sol. The maximum kinetic energies of the ejected electrons from the two photons will be $1 - 0.5 = 0.eV$ and $2.5 - 0.5 = 2.0eV$ hence option (B) is correct.

P 50 : (B) $\frac{1}{6\pi\sqrt{LC}}$

Sol. Equivalent inductance $L_{eq} = L + 2L = 3L$

$C_{eq} = C + 2C = 3C$

Frequency $f = \frac{1}{2\pi\sqrt{L_{eq}C_{eq}}} = \frac{1}{6\pi\sqrt{LC}}$

P 51 : (A) 1m sec.

Sol. $i = i_0(1 - e^{-t/\tau})$

$$|e| = L\frac{di}{dt} = \frac{Li_0}{\tau}e^{-t/\tau} \quad .... (1)$$

also $|e| = e_{max}e^{-t/\tau}$ .... (2)

From (1) and (2)

$$e_{max} = \frac{Li_0}{\tau} \Rightarrow \tau = \frac{Li_0}{e_{max}} = \frac{(3mH)\times 2}{6} = 1m/sec.$$

P 52 : (C) $\frac{2B\pi R^2}{L}$

Sol. Flux can't change in a superconducting loop.

$\Delta\phi = 2\pi R^2.B$

Initially current was zero. So self flux was zero.

$\therefore$ Finally $Li = 2\pi R^2 \times B. \Rightarrow i = \frac{2\pi R^2 \times B}{L}$

P 53 : (A) t = 0

Sol. $E = -\frac{d\phi}{dt} = aB_0Ae^{-at}$

(where A is area of the loop)

It is maximum when t = 0

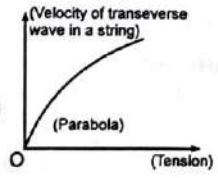
P 54 : (D) $\frac{2\ell}{\alpha\left(\sqrt{T_2}+\sqrt{T_1}\right)}$

Sol. $\frac{dx}{dt} = V = \alpha\sqrt{T_1 + \left(\frac{T_2 - T_1}{\ell}\right)x}$

$$\int_{x=0}^{x=\ell}\frac{dx}{\sqrt{T_1 + \left(\frac{T_2 - T_1}{\ell}\right)x}} = \int_0^t \alpha dt$$

on solving we get $t = \frac{2\ell}{\alpha(\sqrt{T_1} + \sqrt{T_2})}$

P 55 : (C)



Sol. Velocity of sound in air (V) = $\sqrt{\frac{\gamma RT}{M}}$

$\Rightarrow$ $V^2 \propto T$ (in kelvin)

not $V^2 \propto T$ (in ^0C)

Hence (B) is incorrect.

Velocity of transverse wave in a string :

$$V = \sqrt{\frac{T}{\mu}} = V^2 \propto T$$

Hence (C) is a correct graph.

P 56 : (A) 5000

Sol. A_v = The voltage given = $\frac{\beta R_C}{R_{in} + (\beta+1)R_E}$

$$\cong \frac{R_C}{R_E} = \frac{R_C}{2}$$

For proper biasing $v_C > v_B$

$$\Rightarrow 20 - \frac{2}{1000}A > 10 + \frac{2}{1000}$$

$$\Rightarrow \frac{2A}{1000} < 10 - \frac{2}{1000} \Rightarrow 2A < 10000$$

$$\Rightarrow A < 5000 \Rightarrow A_{vmax} = 5000$$

P 57 : (C) kx(x+L)

Sol. Rotational K.E. = Rotational degree of freedom x $\frac{1}{2}nRT$

$= 2 \text{ x } \frac{1}{2}nRT = nRT = PV$

$= PA.\frac{V}{A}$ = force on piston (L + x) = kx (L + x)

P 58 : (C) 12 cm

Sol. Let distance of S from lens is x and focal length is f.

So $\frac{1}{50-x} - \frac{1}{-x} = \frac{1}{f}$ ...(i)

If water is filled, shift = $8\left(1 - \frac{3}{4}\right) = 2$ cm

So u = –(x – 2), v = 50 – x + 6

So $\frac{1}{56-x} - \frac{1}{-(x-2)} = \frac{1}{f}$ ...(ii)

By solving x = 20 and f = 12 cm

P 59 : (C) $\left[\frac{Rt_{1/2}}{0.693}\right]$

Sol. As the element is being produced and disintegrating at same time.

$$\frac{dN}{dt} = R - \lambda N$$

after a long time,

$\frac{dN}{dt} \to 0$ as N increases so

Number N will become constant

$$\text{So } R-\lambda N = 0 \Rightarrow N = \frac{R}{\lambda} = \frac{Rt_{1/2}}{\ell n 2} = \frac{Rt_{1/2}}{0.693}$$

P 60 : (B) 1 : 1

Sol. The rate of heat loss by a thin hollow sphere of thickness 'Δx' , mean radius 'r' and made of density 'ρ' is given by

$$mS\frac{dT}{dt} = -\epsilon\sigma A(T^4 - T_0^4)$$

$$(\rho\, 4\pi r^2 \Delta x)\ S\ \frac{dT}{dt} = -\epsilon\sigma\, 4\pi r^2 (T^4 - T_0^4)$$

$$\Rightarrow \frac{dT}{dt} = -\frac{\epsilon\sigma(T^4 - T_0^4)}{S\Delta x}$$ is independent of radius

Hence rate of cooling is same for both spheres.

P 61 : (C) 8.0×10^{15} Hz

Sol. Given frequency (ν) = 3.2×10^{16} Hz

For calculation of threshold frequency (ν_0), we use photoelectric effect.

Hence, $\nu = \nu_0 + KE$

$3.2 \times 10^{16} = \nu_0 + KE_1$ ....(i)

If light frequency $\nu = 2.0 \times 10^{16}$ Hz

then, $2 \times 10^{16} = \nu_0 + \frac{KE_1}{2}$ ...(ii)

For Eqs. (i) and (ii)

$3.2 \times 10^{-16} HZ - \nu_0 = KE_1$ ..(iii)

$2 \times 10^{16} HZ - \nu_0 = KE_{1/2}$ ..(iv)

For calculation of ν_0, Eq. (iii) + Eq. (iv)

$$\frac{32\times10^{16} - \nu_0}{2\times10^{16} - \nu_0} = \frac{2}{1}$$

or $4 \times 10^{16} - 2\nu_0 = 32 \times 10^{16} - \nu_0$

$= 8.0 \times 10^{15}$ Hz

P 62 : (B) CH_3CH_2OH

Sol. $CH_3CH_2OH \xrightarrow[\text{Iodoform reaction}]{NaOI} CH_3-\underset{\underset{O}{\|}}{C}-ONa + CHI_3\downarrow$ Yellow ppt.

P 63 : (D) osmotic pressure

Sol. For the determination of molar mass of the polymer, osmotic pressure is used as

$\pi = CRT$

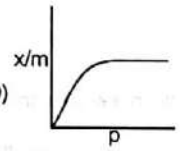
$$\pi = \frac{\text{weight}}{\text{Molecular weight}} \times RT$$

$$\text{Molecular weight} = \frac{wt \times R \times T}{\pi}$$

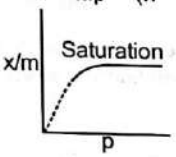
P 64 : (A) XeF_2, IF_2^-

Sol. Both are linear.

P 65 : (D)



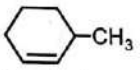
Sol. Freundlich adsorption isotherm explains the empirical relationship between the quantity of gas adsorbed by unit mass of solid adsorbent and pressure at particular temperature. Following equation explains the relationship, $X / m = k.p^{1/n}$ $(n > 1)$



P 66 : (B) I_2O_5

Sol. $5CO + I_2O_5 \longrightarrow 5CO_2 + I_2$

$I_2 + 2Na_2S_2O_3 \longrightarrow 2NaI + Na_2S_4O_6$

P 67 : (C) 

Sol. β-H is eliminated from least substituted β-carbon.

P 68 : (A) –54 J/K

Sol. $\Delta G^\circ = \Delta H^\circ - T\Delta S_{univ} = -373 \times 56$

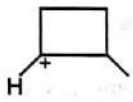
Also $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

$$= \Delta S^\circ = \frac{\Delta H^\circ - \Delta G^\circ}{T}$$

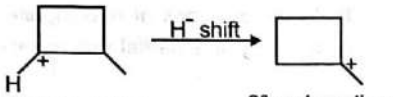
$$= \frac{-4.1\times10^4 + 373\times56}{373} = -53.9$$

P 69 : (B) TiO > VO > CrO > FeO

P 70 : (A)



Sol.



The carbocation 'b' will undergo methyl shift while 'c' will undergo ring expansion to produce more stable carbocation.

P 71 : (A) 1.58×10^{-4} mol L^{-1}

Sol. pH of $Mg(OH)_2$ solution is 10.5

$\therefore$ pOH = 14 – pH = 14 – 10.5 = 3.5

So, the $[OH^-] = 3.2 \times 10^{-4}$

$[OH^-] = 2 \times$ solubility

$$\therefore \quad \text{Solubility} = \frac{3.2\times10^{-4}}{2}$$

$= 1.6 \times 10^{-4}$ mol L^{-1}

P 72 : (B) 3

Sol.
$$\ln\left(\frac{k_{310}}{k_{300}}\right) = \frac{85000}{8.314}\left(\frac{10}{300\times310}\right)$$

$$\Rightarrow \frac{K_{310}}{K_{300}} = 3$$

P 73 : (A) –100

Sol. $\Delta_{vap}H = 35.3$ kJ/mol

$T_b = 273 + 80 = 353$ K

$$\Delta S = \frac{\Delta H}{T_b} = \frac{35.3\times1000\ \text{Jmol}^{-1}}{353\ \text{K}}$$

$= 100\ JK^{-1}\ mol^{-1}$

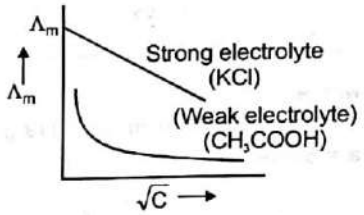
As the transition of vapour to liquid then the entropy becomes less i.e. – 100 J/molK

P 74 : (D) All of the above.

Sol. All these are properties of electrophilic addition reaction to an alkene.

P 75 : (B) II

Sol.



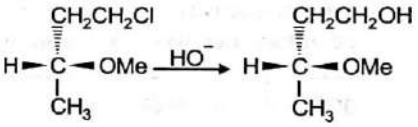
P 76 : (D) The ion flow through the salt bridge in galvanic cell is opposite to the flow in electrolytic cell.

Sol. There is no salt bridge in electrolytic cell.

P 77 : (B) Absence of bond polarity

P 78 : (A) The product will have same configuration

Sol. Substitution occur at non chiral carbon, hence configuration at chiral carbon will not be disturbed.



P 79 : (B) superoxide like KO_2

Sol. Li and Na do not form superoxide MO_2

P 80 : (C) XeO_3

Sol. $XeF_6 + 3H_2O \longrightarrow XeO_3 + 6HF$

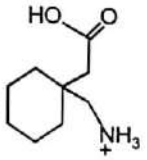
P 81 : (D) No isomerism

Sol. $\left.\begin{matrix}[CO(NH_3)_5SO_4]Br \\ [CO(NH_3)_5SO_4]Cl\end{matrix}\right]$ not isomer of each other.

P 82 : (D) A = HBO_2, B = B_2O_3

Sol. $4H_3BO_3 \xrightarrow[-4H_2O]{100^\circ C} 4HBO_2 \xrightarrow[-H_2O]{\text{Red hot}} B_2O_3$

P 83 : (C)



Sol. $-NH_2$ is the basic group which accepts H^+ to form the conjugate acid.

P 84 : (B) Schottky - defect

Sol. For calculation of density (m × crystal)

$$d_{ob} = \frac{Z \times \text{molar mass}}{Na \times (\text{celllength})^3}$$

where

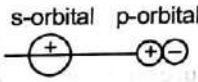
Z = 4 (FCC), molar mass = 119 g

a = 6.58Å = 6.58×10^{-8} cm

$$d_{ob} = \frac{4 \times 119g}{6.023 \times 10^{23} \times (6.58)^3 \times 10^{-24} cm^3}$$

d_{ob} = 2.78 g/cm^3

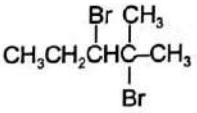
Calcualted density d_{ob} (2.78 g/cm^3) is greater than actual density (2.44 g/cm^3). Due to Schottky defect, density decreases.

P 85 : (B) 

Sol. Only symmetrical overlap leads to bonding.

P 86 : (A) $TeF_6 > SeF_6 > SF_6$

Sol. As a size of atom increases, the electronegativity decreases and the bond polarities increase leading to increase in reactivity or decrease in stability down the group. TeF_6 undergoes hydrolysis readily.

P 87 : (A) 

Sol. Vicinal dihalide undergo dehalogenation with Zn to give alkene.

P 88 : (D) There is steric hindrance to the approach of necleophile from the front side as some of the leaving group are not departed completely which favour inversion of configuration.

Sol. In S_N1 reaction, some solvent separated ion pair and some intimate ion pair are formed that restrict attack of nucleophile from one side giving only partial recemization.

P 89 : (B) 3

Sol. Piperazine is formed

(a) (+) cis (b)(–) cis(c) Meso form of trans.

P 90 : (D) $[Co(NH_3)_6]^{3+}$

Sol. Complex will be more stable if higher value of charge density is present.

$$\text{Charge density} = \frac{\text{Charge}}{\text{Radius}}$$

For stability : (i) More basic ligand then complex will be more stable.

(ii) Higher the oxidation state of central atom and its coordination number.

(iii) Higher the stability constant (K)

	$[Co(NH_3)_6]^{3+}$	$[Cu(NH_3)_4]^{2+}$
Stability	↓	↓
constant (K)	5×10^{33}	4×10^{11}
	$Ag(NH_3)_2]^+$	$[Ag(CN)_2]^{-1}$
Stability	↓	↓
constant (K)	1.6×10^7	1×10^{22}

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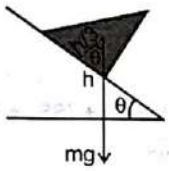
P 1 : (D) 3.00

Sol. Toppling occurs when the centre of mass of prism is above right corner of the base. This reduces the problem to trigonometry.



$\tan\theta = 3$

Sliding occurs if ; $\tan\theta > \mu$

We can equate further.

P 2 : (C) 1.4

Sol. $S = (\mu - 1)t\,(D/d)$

But $\beta = \frac{D\lambda}{d}$ or $\frac{D}{d} = \frac{\beta}{\lambda}$

$\therefore \quad S = (\mu - 1)r\,(\beta/\lambda)$

or $\quad 20\beta = (\mu - 1)(2.5 \times 10^{-3})$

$$\left[\frac{\beta}{5000\times10^{-8}}\right]$$

or $$(\mu - 1) = \frac{20\times5000\times10^{-8}}{2.5\times10^{3}} = 0.4$$

$\therefore \quad \mu = 1.4$

P 3 : (A) 1

Sol. We know that absorption coefficient of sound for a surface is the ratio of quantity of sound energy absorbed by the surface to the quantity absorbed by an equal area of a perfect absorber. Since, open window absorbs the total quantity of sound energy, therefore its absorption coefficient is 1.

P 4 : (B) 32 m/s

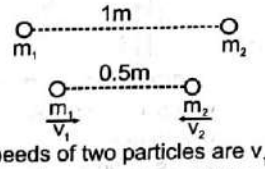
Sol. From $A_1v_1 = A_2v_2$

$\pi \times (0.5)^2 \times 2 = 25\pi \times (0.25)^2 \times v_2$

$\Rightarrow v_2 = 32$ m/s

P 5 : (D) $v_1 = \sqrt{\frac{160G}{3}}$ & $v_2 = \sqrt{\frac{40G}{3}}$

Sol. As no external force is acting on the system so it means momentum of system remains conserved.



Let speeds of two particles are v_1 and v_2 when they are at a separation of 0.5m.

From momentum conservation,

$0 = m_1v_1 - m_2v_2 \Rightarrow 20v_1 = 40\,v_2 \Rightarrow v_1 = 2v_2$

From work energy theorem,

$$\frac{20v_1^2}{2} + \frac{40v_2^2}{2} - 0 = -\left[\frac{-G\times800}{0.5} - \left(\frac{-G\times800}{1}\right)\right]$$

Solving above equation we get

$$v_1 = \sqrt{\frac{160G}{3}} \;\&\; v_2 = \sqrt{\frac{40G}{3}}$$

P 6 : (D) -1.67×10^{11} J

Sol. Orbital speed of satellite is, $v = \sqrt{\frac{GM}{r}}$

Initial KE, $K_i = \frac{mv^2}{2} = \frac{GMm}{2r}$

From work energy theorem,

$$\frac{mv_f^2}{2} - \frac{mv_i^2}{2} = -\left[-\frac{Gmm}{R_E} + \frac{GMm}{r}\right] + W_{\text{air friction}}$$

$$\Rightarrow W_{\text{air friction}} = \frac{mv_f^2}{2} - \frac{GMm}{2r} - \frac{GMm}{R_E} + \frac{GMm}{r}$$

$= -1.67 \times 10^{11}$ Joule

P 7 : (B) $\sqrt{3}\pi$

Sol. Pitch, $P = (V\cos 30°)\left(\frac{2\pi m}{qB}\right) = \left(1.\frac{\sqrt{3}}{2}\right)\left(\frac{2\pi.1}{1\times1}\right)$

$= \sqrt{3}\,\pi$

P 8 : (D) 60 Hz

Sol. In a transformer, frequency remains same for input and output.

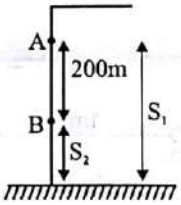
P 9 : (C) $\frac{\mu_0 I_0}{2\pi r}$

Sol. According to Ampere's law, when $r > R$,

$B = \frac{\mu_0 I_0}{2\pi r}$

P 10 : (A) 45 m

Sol. Velocity of Ist stone when passing at A →



$V^2 = 0 + 2 \times 10 \times 5$

$V = 10$ m/s

$S_1 - S_2 = 20$ m

$$\Rightarrow \quad \left(10 \times t + \frac{1}{2} 10 \times t^2\right) - \left(\frac{1}{2} \times 10 \times t^2\right) = 20$$

$t = 2s$

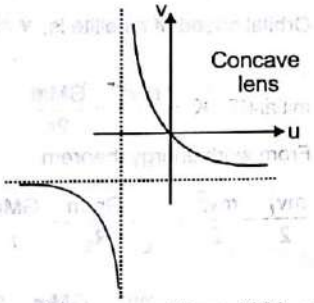
$$S_2 = \frac{1}{2} \times 10 \times 4 = 20m$$

$t = 25 + 20 = 45$ m

P 11 : (*D) 0.5

Sol. $$m = \frac{A_m}{A_C} = \frac{1}{2} = 0.5$$

P 12 : (*D)



Sol. Convex lens cannot form V_i/V_0 so U(+ve) and V(–ve) is not possible so graph cannot be in IVth quadrant. Similarly concave lens cannot form R_i/R_0 so U(–ve) and V(+ve) is not possible so graph cannot be in IInd quadrant. So answer is D.

P 14 : (*B) 4 Q

Sol. By Stefan's law, $Q = \sigma T^4$

The rate of dissipation of heat by a body of emissivity e is given by :

$Q' = e\sigma T'^4$

Here $T' = 2T$, $e = 0.25$

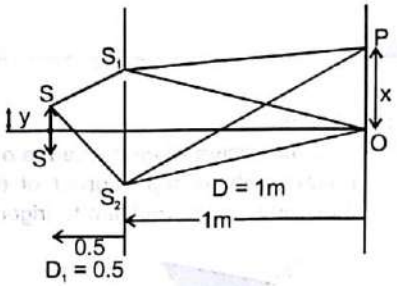
$\therefore \quad Q' = 0.25 \times \sigma(2T)^4 = 4\sigma T^4 = 4Q$

P 13 : (*D) $y = -10 \sin [\pi t + 30°]$

Sol. $d = 0.4$mm

$D = 1$ meter

at



OPD at P

OPD = $(SS_2 + S_2P) + (SS_1 + S_1P)$

$$OPD = \frac{xd}{D} + \frac{yd}{D_1}$$

for central maxima

$x = -2y$

$\Rightarrow x = -10\sin(\pi t + 30°)$

P 15 : (C) 120 m

Sol. The equation of trajectory

$$y = x\tan\theta - \frac{gx^2}{2u^2}\sec^2\theta$$

for $x = 50$

$$y = 50\tan\theta - \frac{10(50)^2}{2(50)^2}\sec^2\theta$$

$y = 50\tan\theta - 5\sec^2\theta$

for maximum y, $\frac{dy}{d\theta} = 0$

$$\frac{dy}{d\theta} = 50\sec^2\theta - 10\sec^2\theta\tan\theta$$

$\Rightarrow \tan\theta = 5$

$y_{max} = 250 - 5(26)$

$y = 120$ m

P 16 : (*C) $x_{cm} = 0$, $y_{cm} = 0$

Sol. $\lambda = \lambda_0\sin^2\theta$ so if we take a small element its mass will be $dm = \lambda_0\sin^2\theta \times Rd\theta$ so by definition,

$$x_{cm} = \frac{\int dmx}{\int dm} = \frac{\lambda_0 R\int \sin^2\theta\cos\theta d\theta}{\lambda_0 R\int \sin^2\theta d\theta} = 0$$

$$y_{cm} = \frac{\int dmy}{\int dm} = \frac{\lambda_0 R\int \sin^2\theta\sin\theta d\theta}{\lambda_0 R\int \sin^2\theta d\theta} = 0$$

P 17 : (*D) $2\sin^{-1}\left(\frac{1}{3}\right)\sqrt{\frac{Re}{g}}$

Sol. The particle will perform SHM in the tunnel with angular frequency $\omega = \sqrt{\frac{g}{Re}}$. Let amplitude of SHM = A

So, $V_e = \sqrt{\frac{g}{Re}}\sqrt{A^2 - Re^2}$

$\Rightarrow \quad \sqrt{3}\,Re \; \{V_e = \sqrt{3gRe}\}$

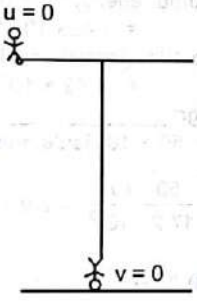
So, time taken by particle will be

$$t = \frac{2\sin^{-1}(1/3)}{\omega}$$

$$\Rightarrow \quad t = 2\sin^{-1}\left(\frac{1}{3}\right)\sqrt{\frac{Re}{g}}$$

P 18 : (B) 32 N/m

Sol. By energy conservation



$$0 + 80(10)(200) + 0 = 0 + 0 + \frac{1}{2}k(100)^2$$

$$k = \frac{2\times800\times200}{100\times100} = 32 \text{ N/m}$$

P 19 : (A) $2\pi \times 10^{-7}$ T

Sol. No. of turns (N) = $\frac{5 \text{ meter}}{1\times10^{-3}} = 5000$

Length of wire = $2\pi(5 \times 10^{-2})\times5000 = 500\pi$

Resistance of wire = $500\pi\ \Omega$

Current = $\frac{2}{500\pi}$A

magnetic field

$$= \frac{\mu_0 Ni}{\ell} = \frac{(4\pi\times10^{-7})(5000)\left(\frac{2}{500\pi}\right)}{5}$$

$= 8 \times 10^{-7}$ T

P 20 : (*B) will be same

Sol. Force between two plates will depend on charges of plate

$$F = \frac{Q^2}{2A\epsilon_0}$$

So, By increasing separation or filling with dielectric there will be no change in force between them.

P 21 : (B) 6.8×10^4 Pa

Sol. Let P_1 is the pressure on upper wing surface, then for vertical equilibrium of plane where P is pressure on lower surface of wing.

$2P_1A + mg = 2PA$

$$\Rightarrow P_1 = P - \frac{mg}{2A} = 7 \times 10^4 - \frac{1.6\times10^4\times10}{2\times40}$$

$= 6.8 \times 10^4$ Pa

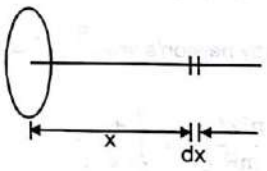
P 22 : (A) $\frac{GM\lambda}{R}$

Sol. Consider an elemental length dx of wire at distance x from ring centre as shown in figure.

Its mass = λ.dx

$\therefore$ Force on this length,

$$dF = E(\lambda dx) = \frac{GM\lambda x dx}{(R^2 + x^2)^{3/2}}$$

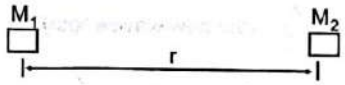


$\therefore$ Total force on the wire

$$F = GM\lambda \int_{x=0}^{x=\infty} \frac{x dx}{(R^2 + x^2)^{3/2}} = \frac{GM\lambda}{R}$$

P 23 : (C) $6\left(\frac{\mu_0}{4\pi}\right)\frac{M_1M_2}{r^4}$

Sol. Magnetic field due to M_1 = $2\left(\frac{\mu_0}{4\pi}\right)\frac{M_1}{r^3}$



So force on M_2

$$\vec{F} = \frac{2\mu_0}{4\pi} M_1 \left[\frac{m}{r^3} - \frac{m}{(r+d)^3}\right]$$ {m is pole strength and d is length of magnet}

$$\vec{F} = \frac{2\mu_0}{4\pi} M_1 \left[\frac{m(d)\left(r^2+(r+d)^2+r(r+d)\right)}{r^3(r+d)^3}\right]$$

{d is small}

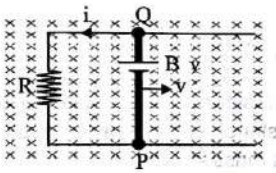
$$\vec{F} = 2\left(\frac{\mu_0}{4\pi}\right) M_1 \frac{\left[M_2\left(3r^2\right)\right]}{r^6} = 6\left(\frac{\mu_0}{4\pi}\right)\frac{M_1M_2}{r^4}$$

P 24 : (A) $v_0 e^{-\frac{B^2\ell^2 t}{mR}}$

Sol. Let velocity of slider is v at any time

So motional EMF = $B\ell v$

Which causes current , $i = \frac{B\ell v}{R}$ from P to Q



So force on slider , $F = \frac{B^2\ell^2 v}{R}$ leftwards

$(F = Bi\ell)$

So by newton's law, $\frac{B^2\ell^2 v}{R} = -\frac{mdv}{dt}$

$$\Rightarrow \frac{B^2\ell^2}{mR}\int_0^t dt = \int_{V_0}^{V}\frac{dV}{V}$$

$$\Rightarrow v = v_0 e^{-\frac{B^2\ell^2 t}{mR}}$$

P 25 : (A) reduce to $\frac{1}{\sqrt{2}}$

Sol. We can treat electron beam as wave and its wavelength can be given by De broglie relation

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2meV}}$$

if potential difference is doubled then :

$\lambda' = \frac{\lambda}{\sqrt{2}}$ {the new wavelength}

We know, fringe width = $\frac{\lambda D}{d}$

So, fringe will reduce to $\frac{1}{\sqrt{2}}$ times of the original width.

P 26 : (D) 39.503×10^5

Sol. Radius of the area covered by T.V. telecast

$d = \sqrt{2hR_e}$

Total population covered = pd^2 × population density

= $2phR_e$ × population density

$$= 2 \times 3.14 \times 100 \times 6.4 \times 10^6 \times \frac{1000}{10^6}$$

$= 39.503 \times 10^5$

P 27 : (A) 2.9×10^{-7} kg

Sol. $2_1^2H \rightarrow {}_2^4He + Q(\text{value})$

$\Rightarrow Q = [2 \text{ mass } ({}_1^2H) - \text{mass } ({}_2^4He)]C^2$

$\Rightarrow Q = [4.028204 - 4.002630] \times 931 \text{ MeV}$

$\Rightarrow Q = 23.8 \text{ MeV} = 23.8 \times 10^6 \times 1.6 \times 10^{-19} J$

$\Rightarrow Q = 38.08 \times 10^{-13}$

Output energy = $38.08 \times 10^{-13}(0.3)$

$= 11.42 \times 10^{-13}$

So 49m deuterium will give energy

$= 11.42 \times 10^{-13} \times 6.02 \times 10^{23}$ J

1 gm _____ $= 17.2 \times 10^{10}$ J

So 50×10^6 joule energy is given by

$$= \frac{50 \times 10^6}{17.2 \times 10^{10}} = 2.9 \times 10^{-7} \text{ kg}$$

P 28 : (B) 22 Hz.

Sol. The mechanical strain = $\frac{\Delta\ell}{\ell} = \alpha\,\Delta T$

$= 1.21 \times 10^{-5} \times 20 = 2.42 \times 10^{-5}$

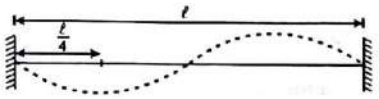
The tension in wire = $T = Y\frac{\Delta\ell}{\ell}A$

$= 2 \times 10^{11} \times 2.42 \times 10^{-5} \times 10^{-6} = 48.4$ N

$\therefore$ speed of wave in wire

$$V = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{48.4}{0.1}} = 22 \text{ m/s}$$

Since the wire is plucked at $\frac{\ell}{4}$ from one end



The wire shall oscillate in 1st overtone (for minimum number of loops)

$\lambda = \ell = 1$m

Now $V = f\lambda$ or $f = \frac{V}{\lambda} = 22$ Hz.

P 29 : (A) 70 J/m °C, 312.5 kg

Sol. Let at any instant temperature of water be T, then heat current,

$$i = \frac{kA}{x}.(T-0) \qquad \text{— (1)}$$

where $A = 6a^2 = 6\ m^2$; x = thickness = 1 mm $= 10^{-3}$ m

Rate of heat lost from water,

$$\frac{dQ}{dt} = +ms\frac{dT}{dt} \qquad \text{— (2)}$$

So, we get from (1) & (2), $-ms\frac{dT}{dt} = \frac{kAT}{x}$

$$\Rightarrow -\int_{50^\circ}^{25^\circ}\frac{dT}{T} = \frac{kA}{mSx}\int_0^{10\ell n2} dt$$

$$\Rightarrow \ell n\,(2) = \frac{kA}{mSx}.10\,(\ell n\,2)$$

$$\text{So, } \frac{kA(10)}{mSx} = 1$$

Putting values

$$\Rightarrow k = \frac{mSx}{10A} = \frac{(10^3 kg)(4.2\times10^3\ J/kg^\circ C)10^{-3}m}{10\times(6m^3)}$$

$\Rightarrow$ k = 70 J/m °C

$\Rightarrow$ Total heat transferred will be = total heat

$$Q = \int dQ = \int\frac{kA}{x}Tdt$$

Lost by water, $Q = m S\Delta T = 10^3\times4200\times25$ J $= (10^6\ gm)\ (1\ cal)\ (25) = m_{ice}\ L$

$$\Rightarrow m_{ice} = \left(\frac{10^6\times25}{80}\right)gm = \frac{25000}{80}kg = 312.5\ kg$$

$\Rightarrow$ mass of ice melted = 312.5 kg

P 30 : (B) 0.707

Sol. (B) $dB = 10\log\left(\frac{I}{I_0}\right) = 10\log\left(\frac{K/r^2}{I_0}\right) = 10$

[log (K') – 2 log r]

$dB_1 = 10\ (\log K' - 2\log r_1)$

$dB_2 = 10\ (\log K' - 2\log r_2)$

$$3 = dB_1 - dB_2 = 20\log\left(\frac{r_2}{r_1}\right)$$

$$(0.3) = \log\left(\frac{r_2}{r_1}\right)^2 \Rightarrow \left(\frac{r_1}{r_2}\right) = \frac{1}{\sqrt{2}} = 0.707$$

P 31 : (B) 5

Sol. $dU = nC_V dT = n\left[16.5 + \frac{T}{100}\right]dT$

$$\Rightarrow \Delta U = n\left[16.5\Delta T + \frac{T_2^2 - T_1^2}{200}\right]$$

$$= 2.5\left[16.5\times100 + \frac{400^2 - 300^2}{200}\right]$$

5000 J = 5 kJ

P 32 : (A) 63.5 V

Sol. From relationship between wavelength and potential difference,

$$eV = \frac{h^2}{2m}.\frac{1}{\lambda^2}$$

or $\lambda^2 = \frac{h^2}{2meV}$ [V = potential difference]

or $\lambda = \frac{h}{\sqrt{2me}}.\frac{1}{\sqrt{V}}$

or $\lambda \propto \frac{1}{\sqrt{V}}$

or $\lambda = \sqrt{\frac{150\times10^{-8}}{V}}$

or $\lambda = 0.154\ nm = 154\times10^{-12}\ m$

or $V = \frac{150\times10^{-8}}{\lambda^2}$

$$= \frac{150\times10^{-8}}{154\times10^{-12}} = 63.5$$

P 33 : (B) 2.5×10^{-5}

Sol. $\Lambda_m = \frac{\kappa\times1000}{M} = \frac{0.2\times10^{-3}\times1000}{10^{-2}}$

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\infty} = \frac{20}{400} = \frac{1}{20} = 5\times10^{-2}$$

	HA	$\rightleftharpoons$	H^+	+	A^-
Initially	1		0		0
At eq.	1-α		α		α

$$K = \frac{C\alpha\times C\alpha}{C(1-\alpha)} = \frac{C\alpha^2}{1-\alpha}$$

(for dilute solution $K = C\alpha^2$)

$K = C\alpha^2 = 10^{-2}\times(5\times10^{-2})^2 = 25\times10^{-6}$

$= 2.5\times10^{-5}$

P 34 : (A)

Sol. Only 1°-amine gives isocyanide test.

P 35 : (*B) Finkelstein's reaction

Sol. Finkelstein reaction

$$RCl \text{ or } R-Br \xrightarrow[Acetone]{NaI} R-I.$$

P 36 : (B) 169

Sol. $AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$;

$K_{sp_1} = 1.2\times10^{-10}$

$AgSCN(s) \rightleftharpoons Ag^+(aq) + SCN_{(aq)}$;

$K_{sp_2} = 7.1\times10^{-13}$

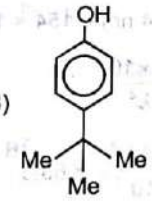
For $AgCl(s) + SCN(aq) \rightleftharpoons AgCNS(s) + Cl^-(aq)$;

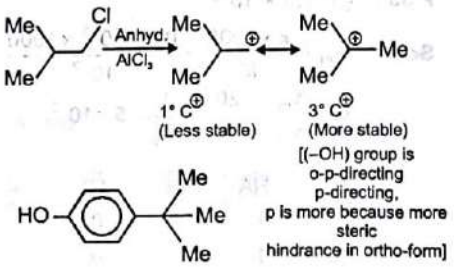
$$K_{eq} = K_{sp_1} \times \frac{1}{K_{sp_2}}$$

$$\therefore \quad K_{eq} = \frac{1.2\times10^{-10}}{7.1\times10^{-13}} = 169$$

P 37 : (*A) Raised by $(2/5)/\Delta_t$

Sol. t_{2g} orbitals have higher energy by $(2/5)/\Delta_t$.

P 38 : (*B) 

Sol. 

P 39 : (B) 9

Sol. $$\ln\frac{k_c}{k_{uc}} = \ln 36 = \frac{\Delta E}{8.314\times300}$$

$\Rightarrow \Delta E = 9$ kJ

P 40 : (C)

Sol. Compounds which are more acidic than H_2CO_3 gives $NaHCO_3$ test. and para-nitrophenol (C) is more acidic than H_2CO_3.

P 41 : (A) 1165 g

Sol. From depression in freezing point

$\Delta T_f = k_f.m$

Given, $\Delta T_f = 7°C$

$K_f = 1.86$ K/mole kg

Molar mass of ethylene glycol = 62 g/mol

Solvent weight = 5 kg

$$\text{So, } 7 = 1.86\times\frac{\text{weight of ethylene glycol}}{62}\times\frac{1}{5}$$

$$\text{Weight} = \frac{7\times62\times5}{1.86} = 1165 \text{ g}$$

P 42 : (*D) it allows ions to flow to prevent the build up of charge.

Sol. Salt bridge allows ions to flow into anode chamber and cathode chamber to prevent build up of charge and maintain electroneutrality, otherwise cell will not function.

P 43 : (A) 1.28×10^5 s

Sol. Given, rate constant for adsorption $= 1.8 \times 10^{-5}\ s^{-1}$

Time period = ?

From first order rate constant equation,

$$k = \frac{2.303}{t}\log\frac{A_0}{A}$$

$\therefore A_0$ (initial concentration) = 100

A (final concentration) = 10

$$\text{Now, } 1.8\times10^{-5}s^{-1} = \frac{2.303}{t}\log\frac{100}{10}$$

$$1.8\times10^{-5}\ s^{-1} = \frac{2.303}{t}\log10 \quad [\because \log 10 = 1]$$

$$\text{or } 1.8\times10^{-5}\ s^{-1} = \frac{2.303}{t}$$

$$\text{or} \quad t = \frac{2.303}{1.8\times10^{-5}s^{-1}} = 1.28\times10^5 s$$

P 44 : (B) Br_2

Sol. In non-polar molecule, the van der Waals force is most important in determining melting point and boiling point.

P 45 : (*C) p-Nitro bromobenzene

Sol. a. Vinyl halide does not undergo either S_N1 or S_N2 reaction.
b. Aryl halide does not undergo S_N reaction unless benzene ring is activated by EWG [eg., (NO_2) group].
c. It undergoes ArS_N reaction, since the benzene ring is activated by [eg., (NO_2) group].
d. Bridge-head halides are inert to both S_N1 and S_N2 reactions.

P 46 : (A) $\Delta G = 0$, ΔS = +ve

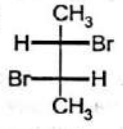
Sol. At 1 bar and 350 K temperature.

$$CCl_4(l) \longrightarrow CCl_4(g)$$

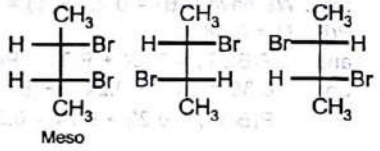
Here, equilibrium present between $CCl_4(l)$ and $CCl_4(g)$.
at equlibrium change in Gibbs free energy $\Delta G = 0$ $(G_1 = G_2)$
Conversion of liquid into gas increases their randomness, therefore ΔS = + ve.

P 47 : (*D) None of these

P 48 : (*A)



Sol.



Meso

(I) (II) (III)

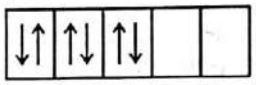
I and II and I and III are diastereomers, but II and III are enantiomers.

P 49 : (*A) I, III

Sol. Bond order = $\frac{1}{2}[N_b - N_a]$.

P 50 : (A) K^+ would be coloured ion

Sol. K(19) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^1$
$K^+(19)$ $1s^2 2s^2 2p^6 3s^2 3p^6$
colourless being no unpaired electron in d-orbital
Thus, (A) is incorrect.
(b) Na(11) $1s^2 2s^2 2p^6 3s^1$ s-block–ture
(c) Cu(29) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$ s-block–true
(d) Cr(24) $1s^2 2s^2 2p^6 3s^2 3p^6 3s^2 3p^6 3d^6$



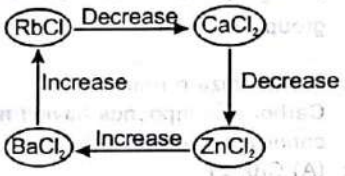
Hund's rule absent

P 51 : (D) All the above are true statements

P 52 : (C) both (A) and (B)

Sol. It is a fact.

P 53 : (A)



Sol. Ionic character = $RbCl > BaCl_2 > CaCl_2 > ZnCl_2$

P 54 : (D) (I), (II), (III) and (IV)

Sol. It is excited state of X_2 molecule.

P 55 : (C) Both (A) and (B)

Sol. $2MnO_4^- + 5SO_2 + 2H_2O \rightarrow 2Mn^{2+} + 4H^+ + 5SO_4^{2-}$
$Cr_2O_4^{2-} + 2H^+ + 3SO_2 \rightarrow 2Cr^{3+} + H_2O + 3SO_4^{2-}$

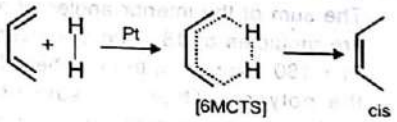
P 56 : (C) 0.414

Sol. Ionic salt AX such as NaCl crystallizes in the face centred cubic unit cell. The radius ratio of

$$\text{NaCl is } 0.414 - 0.732 \left(\frac{r_{Na^+}}{r_{Cl^-}} \text{ or } \frac{r_{A^+}}{r_{X^-}} \right)$$

P 57 : (B)

Sol.



P 58 : (*B)

Sol. More the EDG, faster is the SE reaction or F. C. reaction but ED power of ($-NH_2$) (by resonance) is greater than ($-CH_3$) (by hyperconjugation).
But aniline does not undergo F. C. reaction.
Therefore, in (a), three H. C. structures.
In (c), one H. C. structure.
In (d), ($-NO_2$) EWG, F.C. reaction does not take place,
SO in (b), F. C. reaction is fastest.

P 59 : (D) LAH

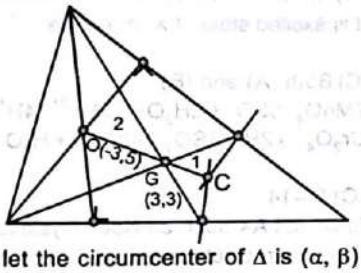
Sol. All these reagents convert $\left(\rangle=O\right)$ group to ($-CH_2-$) group, but LAH convert into $\rangle-OH$ group.

P 60 : (C) Cannizaro reaction

Sol. Carbonyl compounds having no α–H gives cannizaro reaction with base.

P 61 : (A) C(6, 2)

Sol. Since, we know that centroid divides the join of orthocenter and circumcenter in the ratio of 2 : 1.



let the circumcenter of Δ is (α, β)

$$\Rightarrow G(3, 3) = G\left(\frac{2\alpha-3}{3}, \frac{3\beta+5}{3}\right)$$

$$\therefore \frac{2\alpha-3}{3} = 3 \text{ and } \frac{3\beta+5}{3} = 3$$

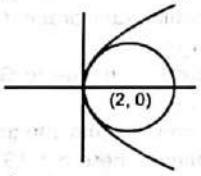
or $\alpha = 6$ $\beta = 2$
Therefore, C(6, 2)

P 62 : (A) 8

Sol. The sum of the interior angles of a polygon are multiples of 180 and are given by (n – 1) × 180 where n is the number of sides of the polygon. Thus, the sum of interior angles of a polygon would be a member of the series : 180, 360, 540, 720, 900, 1080, 1260. The sum of the series with first term 100 and common difference 10 would keep increasing when we take more and more terms of the series. In order to see the number of sides of the polygon, we should get situated where the sum of the series represented by 100 + 110 + 120..... should become a multiple of 180. The number of sides in the polygon would then be the number of terms in the series 100, 110, 120, at the point.
If we explore the sums of the series represented by 100 + 110 + 120 ....
We realize that the sum of the series becomes a multiple of 180 for 8 terms as well as for 9 terms. It can be seen in : 100 + 110 + 120 + 130 +140 + 150 + 160 + 170 = 1080
Or 100 + 110 + 120 + 130 + 140 + 150 + 160 + 170 + 180 = 1260.
but 180 angle make the two sides as single side. So answer is 8.

P 63 : (A) 0

Sol. Solving $y^2 = 4x$ and $x^2 + y^2 = 4x$



we get $x^2 + 4x = 4x$
Hence the two curve intersect only at (0, 0)
Number of common tangents = 1
which is the y–axis i.e. x = 0

P 64 : (D) 0.09

Sol. Let B be the event that a truck stopped at the roadblock will have faulty brakes and T be the event that it will have badly worn tires. We have P(B) = 0.23, P(T) = 0.24 and $P(B \cup T) = 0.38$.

and $P(B \cup T) = P(B) + P(T) - P(B \cap T)$

So $0.38 = 0.23 + 0.24 - P(B \cap T)$

$\Rightarrow$ $P(B \cap T) = 0.23 + 0.24 - 0.38 = 0.09$

P 65 : (°C) $n \sin\left(\frac{\pi}{n}\right)$

Sol. $\therefore 0 < x_i < \pi$, i = 1,2,3,....n
Therefore, $\sin x_i > 0$, i = 1,2,3,.....n

We know $\left(\frac{\sin x_1 + \sin x_2 + ... + \sin x_n}{n}\right)$

$$\leq \sin\left(\frac{x_1 + x_2 + ... + x_n}{n}\right)$$

$$= \sin\left(\frac{\pi}{n}\right)$$

Hence, $\sin x_1 + \sin x_2 + ...\sin x_n \leq n\sin\left(\frac{\pi}{n}\right)$

P 66 : (D) $\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=3$

Sol. We know that $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$ ...(1)

Centroid $\left(\frac{a}{3},\frac{b}{3},\frac{c}{3}\right)$ i.e., $\alpha = a/3$, $\beta = b/3$, $\gamma = c/3 \Rightarrow a = 3\alpha$, $b = 3\beta$, $c = 3\gamma$

From equation (1), $\frac{x}{3\alpha}+\frac{y}{3\beta}+\frac{z}{3\gamma}=1$

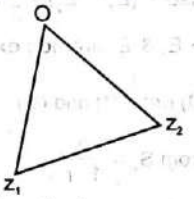
Therefore, $\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=3$.

P 67 : (B) $a^2 = 3b$

Sol. As z_1, z_2 are roots of $z^2 + az + b = 0$

Therefore, $z_1 + z_2 = -a$, $z_1z_2 + 0$

Again, O, z_1, z_2 are vertices of an equilateral triangle.



Thus, $0^2 + z_1^2 + z_2^2 = 0z_1 + z_1z_2 + z_20$

$z_1^2 + z_2^2 = z_1z_2$

$\Rightarrow$ $(z_1 + z_2)^2 = 3z_1z_2 \Rightarrow a^2 = 3b$

P 68 : (B) 4

Sol. $z = x + iy$

$z^{1/3} = a - ib \Rightarrow z = (a - ib)^3$

$z = a^3 - i^3b^3 - 3ab^2 - 3a^2bi$

Comparing Equations (1) and (2)

$(a^3 - 3ab^2) = x$

$-(-b^3 + 3a^2b) = y$

$\Rightarrow$ $\frac{x}{a} = (a^2 - 3b^2) \Rightarrow -\frac{y}{b} = -b^2 + 3a^2$

$\Rightarrow$ $\frac{x}{a}-\frac{y}{b} = 4(a^2 - b^2)$ So, k = 4.

P 69 : (B) $x^2 - 18x + 16 = 0$

Sol. A = 9, G = 4 are, respectively the AM and GM between two numbers, then the quadratic equation having its roots as the two numbers is given by $x^2 - 2Ax + G^2 = .0$, i.e., $x^2-18x+16 = 0$

P 70 : (C) $\left(\frac{12}{5},\frac{4}{5}\right)$

Sol. Let $(\alpha, 5 - 3\alpha)$ be any point on the line $3x + y = 5$

Equation of the chord of contact of $(\alpha, 5 - 3\alpha)$ to the circle $x^2 + y^2 = 4$ is

$\alpha x + (5 - 3\alpha) y = 4$

i.e. $a(x - 3y) + 5y - 4 = 0$

it is the family of straight lines passing through the point $\left(\frac{12}{5},\frac{4}{5}\right)$.

P 71 : (A) $y = x + 2a$

Sol. Since the line is tangent to the parabola $y^2 = 8ax$

$\therefore$ its equation is $y = mx + \frac{2a}{m}$. Also it is tangent to circle $x^2 + y^2 = 2a^2$

$$\therefore \left|\frac{0-0-\frac{2a}{m}}{\sqrt{1+m^2}}\right| = \sqrt{2}\,a \Rightarrow \frac{2}{m^2} = 1 + m^2$$

$\Rightarrow$ $m^2 = 1 \Rightarrow m = \pm 1$

hence equations of tangents are $y = x + 2a$ and $y = -x - 2a$

P 72 : (A) $\frac{9!\,7}{6}$

Sol. $\overline{x}\,S,\ \overline{y}\,S,\ \overline{z}\,S,\ \overline{w}$

$x + y + z + w = 7$

$x \geq 0, y \geq 0, z \geq 1, w \geq 0$

Let $z = 1 + t$

$x + y + t + w = 6$

Required number of ways $= {}^{6+4-1}C_6 \times 7!$

$= {}^9C_6 \times 7! = \frac{9!}{6:3!} \times 7! = \frac{9! \times 7}{6}$

P 73 : (A) $\pm\frac{\hat{i}+\hat{j}-2\hat{k}}{\sqrt{6}}$

Sol. Let $\vec{d} = x\hat{i} + y\hat{j} + z\hat{k}$

where $x^2 + y^2 + z^2 = 1$ ...(1)

($\vec{d}$ being a unit vector)

$\vec{a} \cdot \vec{d} = 0$

$x - y = 0$ or $x = y$ ...(2)

$\Rightarrow$ $[\vec{b}\ \vec{c}\ \vec{d}] = 0$

$$\Rightarrow \quad \begin{vmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ x & y & z \end{vmatrix} = 0$$

Or $x + y + z = 0$

Or $2x + z = 0$ [Using (2)]

Or $z = -2x$ ...(3)

From (1), (2) and (3) we have

$x^2 + x^2 + 4x^2 = 1$

$$x = \pm\frac{1}{\sqrt{6}}$$

Therefore, $\vec{d} = \pm\left(\frac{1}{\sqrt{6}}\hat{i} + \frac{1}{\sqrt{6}}\hat{j} - \frac{2}{\sqrt{6}}\hat{k}\right)$

$$= \pm\left(\frac{\hat{i} + \hat{j} - 2\hat{k}}{\sqrt{6}}\right)$$

P 74 : (B) $\sqrt{12\sqrt{3}}$

Sol. Vertices of the triangle are $(\lambda, 0), \left(\frac{\lambda}{2}, \frac{\lambda}{2\sqrt{3}}\right)$ and $\left(\frac{\lambda}{2}, \frac{-\lambda}{2\sqrt{3}}\right)$

$\therefore$ area $= \frac{\lambda^2}{4\sqrt{3}} = 3 \quad \therefore \lambda^2 = 12\sqrt{3}$

i.e. $\lambda = \pm\sqrt{12\sqrt{3}}$.

P 75 : (D) $2x + y - 4 = 0$

Sol. Let the equation of the line be $\frac{x}{a} + \frac{y}{b} = 1$. The coordinates of the mid-point of the intercept AB between the axes are $\left(\frac{a}{2}, \frac{b}{2}\right)$.

So, $\frac{a}{2} = 1$, $\frac{b}{2} = 2 \Rightarrow a = 2, b = 4$. Hence, the equation of the line is $\frac{x}{2} + \frac{y}{4} = 1$

or $2x + y = 4$.

P 76 : (A) 1

Sol. $P\left(\frac{A}{B}\right) = P(A) = \frac{1}{2}$

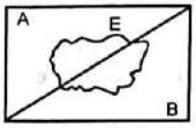
Also $1 = P(A \cup B) = P(A) + P(B) - P(A)$.

$P(B) = \frac{1}{2} + P(B) - \frac{1}{2}P(B)$

$\therefore \quad P(B) = 1$

P 77 : (B) E_1 and E_2 are exhaustive

Sol. $P(E_2/E_1) = \frac{P(E_1 \cap E_2)}{P(E_1)}$



$$\frac{1}{2} = \frac{P(E_1 \cap E_2)}{1/4}$$

$\Rightarrow P(E_1 \cap E_2) = \frac{1}{8} = P(E_2).\ P(E_1/E_2)$

$= P(E_2).\ \frac{1}{4} \Rightarrow P(E_2) = \frac{1}{2}$

Since $P(E_1 \cap E_2) = \frac{1}{8} = P(E_1).\ P(E_2)$

$\Rightarrow$ events are independent

Also $P(E_1 \cup E_2) = \frac{1}{2} + \frac{1}{4} - \frac{1}{8} = \frac{5}{8}$

$\Rightarrow E_1$ & E_2 are non exhaustive

P 78 : (D) Both (B) and (C)

Sol. From $S_\infty = \frac{a}{1-r}$

We get, $x = \frac{1}{1-\cos^2\phi} = \frac{1}{\sin^2\phi}, y\frac{1}{1-\sin^2\phi} = \frac{1}{\cos^2\phi}, z = \frac{1}{1-\cos^2\phi\sin^2\phi} = \frac{1}{1-\frac{1}{xy}} = \frac{xy}{xy-1}$

$\Rightarrow xyz - z = xy \Rightarrow xyz = xy + z$ (1)

Also, $\frac{1}{x} + \frac{1}{y} = \cos^2\phi + \sin^2\phi = 1 \Rightarrow x + y = xy$

From Equation (1), $xyz = x + y + z$

P 79 : (C) right angled

Sol. $\cot\frac{A}{2} = \frac{b+c}{a} \Rightarrow \frac{\cos A/2}{\sin A/2} = \frac{\sin B + \sin C}{\sin A}$

$$\Rightarrow \quad \frac{\cos A/2}{\sin A/2} = \frac{2\sin\left(\frac{B+C}{2}\right)\cos\left(\frac{B-C}{2}\right)}{2\sin\frac{A}{2}\cos\frac{A}{2}}$$

$$\Rightarrow \quad \cos\frac{A}{2} = \cos\left(\frac{B-C}{2}\right) \Rightarrow \frac{A}{2} = \frac{B-C}{2}$$

$\Rightarrow \quad A = B - C \quad \Rightarrow \quad A + C = B.$

But $A + B + C = \pi$. Therefore, $B = \frac{\pi}{2}$

P 80 : (A) 1, –6

Sol. $f(1) = f(3)$

$\Rightarrow a + b + 11 - 6 = 27a + 9b + 33 - 6$....(i)

$\Rightarrow 26a + 8b = -22 \Rightarrow 13a + 4b = -11$ ...(ii)

$\Rightarrow f'(x) = 3ax^2 + bx + 11 \Rightarrow f'\left(2+\frac{1}{3}\right)$

$= 3a\left(4+\frac{1}{3}+\frac{4}{\sqrt{3}}\right)+2b\left(2+\frac{1}{\sqrt{3}}\right)+11-6$

Solving (i) & (ii), we get the value of a & b is 1, –6 respectively.

P 81 : (A) 6

Sol. $\frac{2}{\sqrt{2x^2+1}+\sqrt{2x^2-1}} = \frac{2\left(\sqrt{2x^2+1}-\sqrt{2x^2-1}\right)}{(2x^2+1)-(2x^2-1)}$

$= \sqrt{2x^2+1}-\sqrt{2x^2-1}$

Thus, the given expression can be written as follows :

$\left(\sqrt{2x^2+1}+\sqrt{2x^2-1}\right)^6+\left(\sqrt{2x^2+1}-\sqrt{2x^2-1}\right)^6$

But $(a + b)^6 + (a - b)^6 =$

$2[a^6 + {}^6C_2a^4b^6 + {}^6C_4a^2b^4 + b^6]$

Therefore,

$\left(\sqrt{2x^2+1}+\sqrt{2x^2-1}\right)^6+\left(\sqrt{2x^2+1}-\sqrt{2x^2-1}\right)^6$

$= 2[(2x^2 + 1)^3 + 15 (2x^2 + 1)^2 (2x^2 - 1) + 15 (2x^2 - 1) + (2x^2 - 1)^2 + (2x^2 - 1)^3]$

which is a polynomial of degree 6.

P 82 : (B) $\frac{2}{\sqrt{3}}$

Sol. Since $c_1 c_2 (a_1a_2 + b_1b_2) < 0$

$\therefore$ origin lies in acute angle

P(1, 2) lies in abutes angle

acute angle between the asymptotes is $\frac{\pi}{3}$

$\therefore e = \sec\frac{\theta}{2} = \sec\frac{\pi}{6} = \frac{2}{\sqrt{3}}$

P 83 : (C) The lines $2x + 3y + 19 = 0$ and $9x + 6y - 17 = 0$ cut the coordinate axes in concyclic points .

Sol. (A) $\begin{vmatrix} 1 & \log_a b & \log_a c \\ \log_b a & 1 & \log_b c \\ \log_c a & \log_c b & 1 \end{vmatrix}$

$= \frac{1}{\ell n\, a \,.\, \ell n\, b}\begin{vmatrix} \ell n\, a & \ell n\, b & \ell n\, c \\ \ell n\, a & \ell n\, b & \ell n\, c \\ \log_c a & \log_c b & 1 \end{vmatrix} = 0$

$\therefore$ (A) is false

(B) Let $\tan\theta = .2$

$\therefore \tan 2\theta = \frac{2\times.2}{1-(.2)^2} = \frac{5}{12}$

$\therefore$ the lines is $y = \frac{5}{12}x$.

$\therefore$ (B) is false

(C) Intercepts made by the line $2x + 3y + 19 = 0$ on the coordinate axes are $-\frac{19}{2}$ and $-\frac{19}{3}$

Intercepts made by the line $9x + 6y - 17 = 0$ on the coordinate axes are $\frac{17}{9}$ and $\frac{17}{6}$

Now $-\frac{19}{2}\times\frac{17}{9} = -\frac{19}{3}\times\frac{17}{6}$

$\therefore$ (C) is true

(D) Equation of the line pair through the origin and perpendicular to the line pair $xy - 3y^2 + y - 2x + 10 = 0$ is $3x^2 + xy = 0$

$\therefore$ (D) is false

P 84 : (B) $\frac{1}{t^2}+\frac{1}{n^2} = \frac{1}{t'n'}$

Sol. t = length of tangent = $\frac{|y_1|}{\sin\psi}$

n = length of normal = $\frac{|y_1|}{\cos\psi}$

t' = sub tangent = $\frac{|y_1|}{\tan\psi}$

n' = sub normal = $\frac{|y_1|}{\cot\psi}$

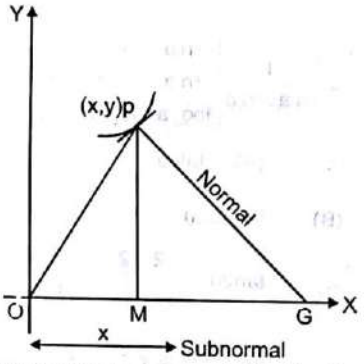
(i) $t'n' = \frac{|y_1|}{\tan\psi}\cdot\frac{|y_1|}{\cot\psi} = y_1^2$

(ii) $\frac{1}{t^2}+\frac{1}{n^2} = \frac{\cos^2\psi}{y_1^2}+\frac{\sin^2\psi}{y_1^2} = \frac{1}{y_1^2}$

(iii) $nt' = \frac{|y_1|}{\cos\psi}\cdot\frac{|y_1|}{\tan\psi} = \frac{|y_1|}{\sin\psi} = |y_1|\,t$

P 85 : (B) a rectangular hyperbola

Sol. It is given that the triangle OPG is an isosceles triangle.
Therefore, OM = MG = subnormal



On integration, we get $x^2 - y^2 = C$, which is a rectangular hyperbola.

P 86 : (C) Both A and B are true

Sol. $\cos(\beta-\gamma)+\cos(\gamma-\alpha)+\cos(\alpha-\beta)=-\frac{3}{2}$

$\Rightarrow 2[\cos(\beta-\gamma)+\cos(\gamma-\alpha)+\cos(\alpha-\beta)]+3=0$
$\Rightarrow 2[\cos(\beta-\gamma)+\cos(\gamma-\alpha)+\cos(\alpha-\beta)]$
$+\sin^2\alpha+\cos^2\alpha+\sin^2\beta+\cos^2\beta+\sin^2\gamma+\cos^2\gamma=0$
$\Rightarrow (\sin\alpha+\sin\beta+\sin\gamma)^2+$
$(\cos\alpha+\cos\beta+\cos\gamma)^2=0$

P 87 : (B) $\pi/2-\beta$

Sol. We have $\Sigma x_1 = \sin 2\beta$, $\Sigma x_1x_2 = \cos 2\beta$
$\Sigma x_1x_2x_3 = \cos\beta$, and $x_1x_2x_3x_4 = -\sin\beta$
so, $\tan^{-1}x_1 + \tan^{-1}x_2 + \tan^{-1}x_3 + \tan^{-1}x_4$

$$= \tan^{-1}\left(\frac{\sum x_1 - \sum x_1x_2x_3}{1-\sum x_1x_2 + x_1x_2x_3x_4}\right)$$

$$= \tan^{-1}\left(\frac{\sin2\beta-\cos\beta}{1-\cos2\beta-\sin\beta}\right) = \tan^{-1}(\cot\beta)$$

$= \tan^{-1}(\tan(\pi/2-\beta)) = \pi/2-\beta.$

P 88 : (*C) $\frac{3}{16}$

Sol. Indeed $\sin20°\sin40°\sin60°\sin80°$

$$= \frac{\sqrt3}{2}\sin20°\sin(60°-20°)\sin(60°+20°)$$

$$= \frac{\sqrt3}{2}\sin20°[\sin^260°-\sin^220°]$$

$$= \frac{\sqrt3}{2}\sin20°\left[\frac{3}{4}-\sin^220°\right]$$

$$= \frac{\sqrt3}{2}\times\frac{1}{4}\left[3\sin20°-4\sin^320°\right]$$

$$= \frac{\sqrt3}{2}\times\frac{1}{4}(\sin60°)$$

$$= \frac{\sqrt3}{2}\times\frac{1}{4}\times\frac{\sqrt3}{2} = \frac{3}{16}$$

P 89 : (B) $\frac{5\pi^2}{4}$

Sol. $I=(\sec^{-1}x)^2+(\operatorname{cosec}^{-1}x)^2$
$= (\sec^{-1}x+\operatorname{cosec}^{-1}x)^2 - 2\sec^{-1}x.\operatorname{cosec}^{-1}x$

$$= \frac{\pi^2}{4} - 2\sec^{-1}x\left(\frac{\pi}{2}-\sec^{-1}x\right)$$

$$= \frac{\pi^2}{4}+2\left((\sec^{-1}x)^2-\frac{\pi}{2}.(\sec^{-1}x)+\frac{\pi^2}{16}-\frac{\pi^2}{16}\right)$$

$$\Rightarrow \frac{\pi^2}{8}+2\left[\left(\sec^{-1}x-\frac{\pi}{4}\right)^2\right] \Rightarrow I_{max} = \frac{\pi^2}{8}+2\left[\frac{9\pi^2}{16}\right]$$

$$\Rightarrow \frac{\pi^2}{8}+\frac{9\pi^2}{8} = I_{max} \Rightarrow I_{max} = \frac{5\pi^2}{4}$$

P 90 : (B) $\alpha^2+\beta^2+\gamma^2 = (\alpha')^2+(\beta')^2+(\gamma')^2$

Sol. Since PQ is perpendicular to the plane and direction ratio of PQ are $\alpha-\alpha'$, $\beta-\beta'$, $\gamma-\gamma'$
hence equation of plane can be written as :
$(\alpha-\alpha')x+(\beta-\beta')y+(\gamma-\gamma')z=0$
also plane passes through

$$\left(\frac{\alpha+\alpha'}{2},\frac{\beta+\beta'}{2},\frac{\gamma+\gamma'}{2}\right), \text{ Hence}$$

$$(\alpha-\alpha')\left(\frac{\alpha+\alpha'}{2}\right)+(\beta-\beta')\left(\frac{\beta+\beta'}{2}\right)+(\gamma-\gamma')$$

$$\left(\frac{\gamma+\gamma'}{2}\right)=0$$

$\Rightarrow \alpha^2+\beta^2+\gamma^2 = (\alpha')^2+(\beta')^2+(\gamma')^2$

REPLICA 16

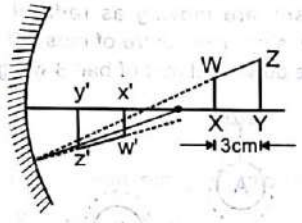
P 1 : (C) $7A^2$

Sol. $A_{res} = \sqrt{A^2 + 4A^2 + 2A(2A)\cos 60} = \sqrt{7}A$

$I \propto 7A^2$

P 2 : (C) will neither be congruent nor be similar but shape will be quadrilateral

Sol.



with x-axis, Angle of line WZ is not equal to Angle w'z', so image is not similar to object.

P 3 : (B) $2|\vec{b}|$

Sol. $\frac{a+b}{a-b} = \frac{3}{1}$ or $3a - 3b = a + b$

or $2a = 4b$ or $a = 2b$

P 4 : (B) 7.5 kg

Sol. Pulley is massless so spring force will be equal to 2T.

$F_{sping} = 2T$

Now as block will move so,

$50 - T = 5a$ ....(1)

$T - 30 = 3a$ ....(2)

$T = 37.5$ N

So $F_{sping} = 75$ N $= 7.5$ kg

P 5 : (D) all the above in series

P 6 : (B) $\sqrt{\frac{2d}{\mu g\cos\theta}}$

Sol. Given $\mu_A < \mu_B < \tan\theta$

Since, friction force acting on each block is less than pulling force Mg cosθ.

i.e., both blocks have some acceleration and also $\mu_A < \mu_B$ i.e., acceleration of block A is greater than block B.

Acceleration of block B,

$$a_B = \frac{Mg\sin\theta - \mu_B Mg\cos\theta}{M} = g(\sin\theta - \mu_B\cos\theta)$$

Similarly, $a_A = g(\sin\theta - \mu_A\cos\theta)$

So, Acceleration of block A, w.r.t. block B,

$a_{AB} = g(\mu_B - \mu_A)\cos\theta = \mu g\cos\theta$.

initially both are at rest so $u_{AB} = 0$, $s_{AB} = d$

So, $d = 0 + \frac{1}{2}\mu g\cos\theta \times t_1^2$

$$\Rightarrow t_1 = \sqrt{\frac{2d}{\mu g\cos\theta}}$$

P 7 : (D) $\frac{R}{2}\left(\frac{\gamma+1}{\gamma-1}\right)$

Sol. Let V be the initial volume of the gas. It is expanded to a volume ηV. The work done in the process is given by :

$$W = \int_V^{\eta V} pdV = \int_V^{\eta V} \alpha V dV = \alpha\left[\frac{V^2}{2}\right]_V^{\eta V}$$

$$= \frac{\alpha V^2}{2}\left[\eta^2 - 1\right]$$

The pressure of the gas varies with volume $p = \alpha V$. So, the initial and final pressures will be αV and $\eta\alpha V$. The change in internal energy is given by :

$$U = \frac{p_f V_f - p_i V_i}{\gamma - 1} = \frac{\eta\alpha V^2 - \alpha V^2}{\gamma - 1}$$

$$= \frac{\alpha V^2}{2}\left[\eta^2 - 1\right]\left[\frac{2}{\gamma - 1} + 1\right]$$

$$= \frac{\alpha V^2}{2}\left[\eta^2 - 1\right]\left[\frac{\gamma+1}{\gamma-1}\right]$$

Here, $T_i = \frac{p_i V_i}{nR} = \frac{\alpha V^2}{nR}$

and $T_f = \frac{p_f V_f}{nR} = \frac{\eta^2 \alpha V^2}{nR}$

Now, Heat capacity $C = \frac{Q}{T_f - T_i}$

$$C = \frac{Q}{T_f - T_i}\left[\frac{aV^2}{2}(\eta^2 - 1)\left\{\frac{\gamma+1}{\gamma-1}\right\}\right]$$

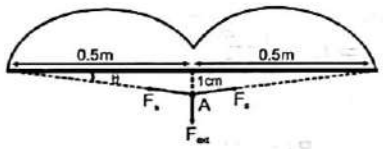
$$= \frac{nR}{\alpha V^2(\eta^2 - 1)}\left[\frac{aV^2}{2}(\eta^2 - 1)\left\{\frac{\gamma+1}{\gamma-1}\right\}\right]$$

Here, n = 1

$$\therefore \quad C = \frac{R}{2}\left(\frac{\gamma+1}{\gamma-1}\right)$$

P 8 : (A) 0.16N

Sol.



For point 'A' if we balance force

$2F_s\sin\theta = F_{ext}$

$F_s = kx$ and $x = 2(1\ \sin\theta)$ cm

$F_s = 10000 \times 2\sin\theta \times 10^{-2}$

$\Rightarrow F_{ext} = 400[\sin\theta]^2$

$\Rightarrow F_{ext} = 400[1/50]^2 = 0.16N$

P 9 : (D) $\frac{A_1}{A_2} = \frac{\alpha_2 Y_2}{\alpha_1 Y_1}$

Sol. $Y = \frac{\text{Stress}}{\text{Strain}} = \frac{T/A}{\Delta\ell/\ell}$

$$T = \frac{Y.\Delta\ell}{\ell}A = Y.A\,\alpha\,\Delta T$$

In both the rods tension will be same so

$T_1 = T_2$

Hence $Y_1 A_1 \alpha_1 \Delta T = Y_2 A_2 \alpha_2 \Delta T$

$$\frac{A_1}{A_2} = \frac{Y_2\alpha_2}{Y_1\alpha_1}$$

P 10 : (B) 23 cm

Sol. Using $\alpha = \frac{\Delta L}{L_0 \cdot \Delta T}$

$$\alpha_1 = \frac{0.0650}{30 \times 100}\ ;\ \alpha_2 = \frac{0.0350}{30 \times 100}$$

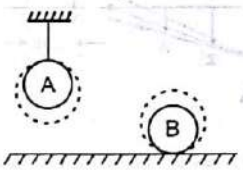
$L_{01} \cdot \alpha_1 \cdot \Delta T + L_{02} \cdot \alpha_2 \cdot \Delta T = 0.0580$

and $L_{01} + L_{02} = 30$

Solving $L_{01} = 23$ cm

P 11 : (A) $T_A > T_B$

Sol. When the balls are warmed, their centre of masses are moving as radii of balls are increasing. The centre of mass of ball A will come down and that of ball B will go up.



In case of ball A, the gravitational potential energy decreases.

This corresponds to additional heating of ball.

$\therefore T_A > T_B$

P 12 : (B) $\lambda_0 = \left(\frac{1}{\lambda} - \frac{eEd}{hc}\right)^{-1}$

Sol. The maximum kinetic energy of the electrons immediately upon ejection is the difference between the energy of the incident photon and the threshold energy.

$$K = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$$

This kinetic energy of ejected electron is converted to electrostatic potential energy, $\Delta U = eEd$, as electrons come to rest while moving in the direction of electric field. Therefore, K = Eed

and $\lambda_0 = \left(\frac{1}{\lambda} - \frac{eEd}{hc}\right)^{-1}$

P 13 : (C) 7.52×10^{18} Hz

Sol. By Mosley's law $\sqrt{\nu} = a(z-b) \Rightarrow \nu = a^2(z-1)^2$

So $1.88 \times 10^{18} = a^2(29-1)^2$

$\nu_{La} = a^2(57-1)^2$

$\nu_{La} = 4(1.88 \times 10^{18}) = 7.52 \times 10^{18}$ Htz

P 14 : (A) $\frac{5}{128}$

Sol. Let the body have temperatures T_1 and T_2 respectively at wavelength $\lambda_1 = 8000Å$ and $\lambda_2 = 4000Å$.

$\therefore$ From Wien's displacement law

λT = constant

$\Rightarrow \lambda_1 T_1 = \lambda_2 T_2$ or $8000 \times T_1 = 4000 T_2$

or $\frac{T_1}{T_2} = \frac{1}{2}$

Emissive power = $e \sigma A T^4$

$\therefore$ Ratio of emissive powers at these temperature is

$$= \frac{e_1 T_1^4}{e_2 T_2^4} = \frac{0.5}{0.8} \times \left(\frac{1}{2}\right)^4 = \frac{5}{128}$$

P 15 : (A) A

Sol. Rate of heat transfer is same through all walls

$$\frac{K_1 . A . (10)}{40\,cm} = \frac{K_2 A (20)}{20\,cm} = \frac{K_2 A (40)}{10\,cm}$$

$$\Rightarrow \frac{K_1}{4} = K_2 = 4K_3 \quad \Rightarrow K_1 = 4K_2 = 16\,K_3 .$$

P 16 : (A) $Q_A - Q_D = W_A - W_D$

Sol. Work done by gas in all four process is positive and in order

$W_A > W_B > W_C > W_D$

$\Rightarrow$ (C) is false

The change in initial energy U is same for all process.

$\therefore$ $Q_A = U + W_A$ ...(1) $Q_B = U + W_B$ ...(2)

$Q_C = U + W_C$...(3) $Q_D = U + W_D$...(4)

Hence $Q_A > Q_B > Q_C > Q_D$

$\Rightarrow$ (D) is false

from equation (1) and (4)

$Q_A - Q_D = W_A - W_D$

$\Rightarrow$ (A) is true

from equation (2) and (3) $Q_B - Q_C = W_B - W_C$

$\Rightarrow$ (B) is false.

P 17 : (D) 0.99

Sol. Current gain in common emitter

$\beta = 99$

For a transistor the current gain β is given by

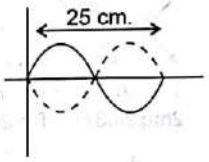
$$\beta = \frac{a}{1-a} \Rightarrow 99 = \frac{a}{1-a}$$

$99 - 99a = a$ or $100\,a = 99$ or $\lambda = 0.99$

P 18 : (B) 27.04 N

Sol. $\mu = \frac{2.5}{25} = 0.1\ g/cm = 10^{-2}\ Kg/m$

I[st] overtone

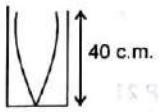


$\lambda_s = 25\,cm = 0.25\,m$

$f_s = \frac{1}{\lambda_s}\sqrt{\frac{T}{\mu}}$

pipe in fundamental freq

$\lambda_p = 160\,cm = 1.6\,m$



$f_p = \frac{V}{\lambda_p}$

$\because$ by decreasing the tension , beat freq is decreased

$\therefore$ $f_s > f_p \Rightarrow f_s - f_p = 8$

$$\Rightarrow \frac{1}{0.25}\sqrt{\frac{T}{10^{-2}}} - \frac{320}{1.6} = 8 \Rightarrow T = 27.04\ N$$

P 19 : (A) $\frac{9}{11}$

Sol. λ_i = wavelength of the incident sound =

$$\frac{10u - \frac{u}{2}}{f} = \frac{19u}{2f}$$

f_i = frequency of the incident sound = $\frac{10u - u}{10u - \frac{u}{2}} f$

$= \frac{18}{19} f = f_r$ = frequency of the reflected sound

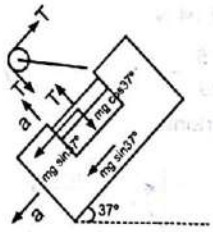
λ_r = wavelength of the reflected sound

$$= \frac{10u + u}{f_r} = \frac{11u}{18f} \times 19 = \frac{11 \times 19}{18} \cdot \frac{u}{f}$$

$$\frac{\lambda_i}{\lambda_r} = \frac{19u}{2f} \times \frac{18f}{11 \times 19u} = \frac{9}{11}$$

P 20 : (A) $a_A = \frac{4}{3}\sqrt{2}\ m/s^2$.

Sol.



$T - mg\cos37° = ma$

$2mg\sin37° - T = 2ma$

$\Rightarrow \quad a = \frac{4}{3}$ m/s

$\Rightarrow \quad a_B = \frac{4}{3}$ m/s

$\Rightarrow \quad a_A = \frac{4\sqrt{2}}{3}$ m/s.

P 21 : (C) g

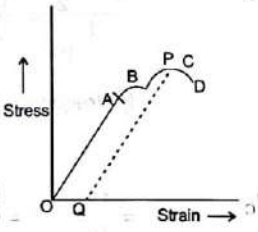
Sol. As the particle of mass m looses contact with the sphere, no normal reaction is exerted by sphere on the particle. The only force acting on the particle is its weight. Hence its acceleration is g(acceleration due to gravity)

P 22 : (B) – (1/2) k_o

Sol. Angular momentum will be conserved so on sticking, angular speed becomes half and moment of inertia gets doubled thus kinetic energy will be halved.

P 23 : (D) will contract but final length will be greater than original length

Sol. Stress-strain graph of a ductile material is shown in figure. Point A shows limit of proportionality. Hooke's law is valid up to this limit. Point B shows yield point. Material is elastic up to this point. If the material is strained up to this point, then on releasing it will regain its original shape and size. But the material is deformed beyond this limit, say upto point P; then on releasing, it will follow dotted line PQ. It means, a deformation OQ will remain permanently. Hence, final length of the wire will contract but final length will be greater than original length.



P 24 : (B) $\frac{1}{2}(\alpha_1 + \alpha_2)$

Sol. Consider a small element of length dx at a distance x from one end of the rod. Let L be the length of the rod. Now the increase in the coefficient of linear expansion per unit length of the rod is $(\alpha_2 - \alpha_1)/L$. Therefore, the value of α at the element located at x is

$$\alpha_x = \alpha_1 + \left(\frac{\alpha_2 - \alpha_1}{L}\right)x$$

Increase in the length of the element = $\alpha_x dx\,\Delta t$, where ΔT is the rise in temperature. Therefore, the increase in the length of rod is

$$\Delta L = \int_0^L \alpha_x dx \Delta T = \Delta T \int_0^L \left\{\alpha_1 + \left(\frac{\alpha_2 - \alpha_1}{L}\right)x\right\} dx$$

$$= \Delta T\left|\alpha_1 x + \left(\frac{\alpha_2 - \alpha_1}{L}\right)\frac{x^2}{2}\right|_0^L$$

$$= \Delta T\left[\alpha_1 L + \left(\frac{\alpha_2 - \alpha_1}{2}\right) L\right]$$

$$= \left(\frac{\alpha_2 + \alpha_1}{2}\right) L\Delta T = \alpha_{eff} L\Delta T$$

where $\alpha_{eff} = \frac{1}{2}(\alpha_1 + \alpha_2)$.

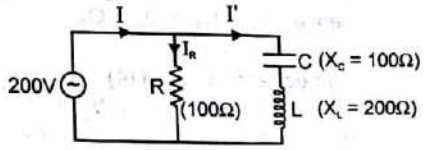
P 25 : (B) 0.02

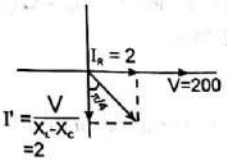
Sol. Dispersive power $(\omega) = \frac{n_b - n_r}{n_y - 1}$

$$= \frac{1.521 - 1.510}{1.550 - 1} = 0.02$$

P 26 : (B) $2\sqrt{2}$ A

Sol.





$$I_R = \frac{V}{R} = \frac{200}{100} = 2A$$

$$I' = \frac{V}{X_L - X_C} = \frac{200}{100} = 2A$$

$$I = \sqrt{I_R^2 + I'^2} = 2\sqrt{2}\ \text{Amp.}$$

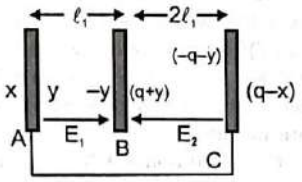
P 27 : (A) $q_{1(outer)} = q/2$, $q_{1(inner)} = -2q/3$
$q_{2(outer)} = q/2$, $q_{2(inner)} = -q/3$

Sol. Let the charge on outer surface of plates (1) be x and on the inner (interfacing) surface of plate (1) be y, then charge distribution on other plates is shown in figure (by using charge conservation and E = 0 inside the metallic plates).

For electric field to be zero inside the plate 1

$$\frac{x}{2A\varepsilon_0} = \frac{(q-x)}{2A\varepsilon_0}$$

$$\Rightarrow \quad x = \frac{q}{2} \qquad \& \qquad q - x = \frac{q}{2}$$



Since $V_A = V_C \quad \Rightarrow V_A - V_B = V_C - V_B$

$\Rightarrow \quad E_1 \ell_1 = E_2 \ell_2$

$$\Rightarrow \quad \frac{y}{A\varepsilon_0}\ell_1 = \frac{(-q-y)}{A\varepsilon_0}2\ell_1$$

$$y\ell_1 = -2q\ell_1 - 2y\ell_1$$

$$3y\ell_1 = -2q\ell_1$$

$$y = \frac{-2q}{3}$$

$$\text{So, } -q - y = -q + \frac{2q}{3} = \frac{-q}{3}$$

So, Charge on outer surface of plate 1 is $x = \frac{q}{2}$

Charge on inner (interfacing) surface of plate 1 is $y = \frac{-2q}{3}$

Charge on outer surface of plate 2 is $q - x = \frac{q}{2}$

Charge on inner (interfacing) surface of plate 2 is $(-q - y) = \frac{-q}{3}$

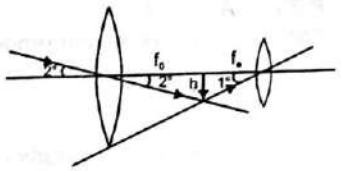
$$q_{1(outer)} = \frac{q}{2} \text{ , } q_{1(inner)} = \frac{-2q}{3}$$

$$q_{2(outer)} = \frac{q}{2} \text{ , } q_{2(inner)} = \frac{-q}{3}.$$

P 28 : (B) 24^0

Sol.

$$\frac{2^\circ = \frac{h}{f_0}}{\theta^\circ = \frac{h}{f_e}}$$



$$\theta = \frac{f_0}{f_e} \times 2 = \frac{60}{5} \times 2 = 24^\circ$$

P 29 : (C) directly as the distance between the slit and the screen

Sol. Width of the diffraction band, $\beta = \frac{D\lambda}{a}$

where , D = distance between slit and the screen
a = width of the slit
λ = wavelength

P 30 : (A) Triton energy is less than that of a He^3 nucleus.

Sol. Triton is more stable then He^3 because of higher n/Z ratio, so it will not have sufficient energy to convert in He^3.

P 31 : (B) {–3, 5}

Sol. $x^2 + y^2 = \lambda^2$

$x^2 + y^2 - 2\sqrt{2}\,x - 2\sqrt{2}\,y - \lambda = 0$

$(x - \sqrt{2})^2 + (y - \sqrt{2})^2 = \lambda + 4$

$c_1 \equiv (0, 0)$; $r_1 = |\lambda|$

$c_2 \equiv (\sqrt{2}, \sqrt{2})$, $r_2 = \sqrt{\lambda + 4}$

for exactly one common tangent, circles touch each other internally

$$c_1c_2 = |r_1 - r_2|$$

$$\sqrt{4} = |\,|\lambda| - \sqrt{\lambda+4}\,|$$

$$\Rightarrow \quad 4 = \lambda^2 + \lambda + 4 - 2|\lambda|\sqrt{\lambda+4}$$

$$\Rightarrow \quad \lambda(\lambda + 1) = 2|\lambda|\sqrt{\lambda+4}$$

$$\Rightarrow \quad \lambda^2 + 2\lambda + 1 = 4\lambda + 16$$

$$\Rightarrow \quad \lambda^2 - 2\lambda - 15 = 0$$

$$\lambda = 5, \; -3$$

$\lambda = 5, -3$ both satisfy

$$\sqrt{4} = |\lambda| - \sqrt{\lambda+4}\,|$$

$\therefore \quad \lambda = -3, 5$

P 32 : (B) $3(x^2 + y^2) - 25x = 0$

Sol. $(x - \alpha)^2 + y^2 = \alpha^2$

$x^2 + y^2 - 2\alpha x = 0$

since it passes through (3, 4)

$$25 - 6\alpha = 0 \quad \Rightarrow \quad \alpha = \frac{25}{6}$$

$\therefore$ equation is

$3(x^2 + y^2) - 25x = 0$

P 33 : (A) 0.1583

Sol. A PIN is a sequence of four symbols selected from 36 (26 letters + 10 digits) symbols.
By the fundamental principle of counting, there are $36\times36\times36\times36 = 36^4 = 1,679,616$ PINs in all. When repetition is not allowed the multiplication rule can be applied to conclude that there are
$36 \times 35 \times 34 \times 33 = 1,413,720$ different PINs
The number of PINs that contain at least one repeated symbol
$= 1,679,616 - 1,413,720 = 2,65,896$
Thus, the probability that a randomly chose PIN contains a repreated symbol is

$$\frac{265,896}{1,679,616} = 0.1583$$

P 34 : (C) town A

Sol. Let distance of the school form A = x
Let distance of the school from B = 60 – x
Total distance covered by 200 students
$= 2[150x + 50(60 - x)]$
$= 2[100x + 3000]$
This is minimum when x = 0
$\therefore$ School should be at town A.

P 35 : (B) $y = ce^{x^2-x}$

Sol. Differential equation $\frac{dy}{dx}+2xy = y$

The given equation may be written as :

$$\frac{dy}{dx} = (1-2x)y \Rightarrow \frac{dx}{dy} = (1-2x)dx$$

integrating, we get

$$\log y = x - x^2 + \log c \text{ or } \log \frac{y}{c} = x - x^2$$

or $y = ce^{x-x^2}$

P 36 : (C) x – 2y – 3 = 0

Sol. Let P(x', y') be the coordinates of the point P(x, y) after rotation of axes at an angle of 180°.

$$\Rightarrow \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Since, here $\theta = 180°$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -x \\ -y \end{bmatrix}$$

Therefore, x = –x' and y = –y'
Hence, the new equation of curve :
x – 2y + 3 = 0 is
(–x') – 2(–y') + 3 = 0
$\Rightarrow$ –x' + 2y' + 3 = 0
$\Rightarrow$ x' – 2y' – 3 = 0
Or x – 2y – 3 = 0 in general.

P 37 : (C) (2, –3)

Sol. Let the new coordinates of point P(x, y) after shifting the origin to (h, k) is P(x', y')
$\Rightarrow$ x' = x – h
and y' = y – k $\Rightarrow$ xx' + hx' + h
and y = y' + k
Since, (x, y) satisfy the equation, we have
$(x' + h)^2+(y' + k)^2 - 4(x' + h)+6(y' + k)-7= 0$
$\Rightarrow (x')^2 + (y')^2 + 2(h - 2)x'$
$+ 2(k + 3)y' + (h^2 + k^2 - 4h + 6k - 7) = 0$
Since, the terms containing linear powers are missing
$\therefore$ h – 2 = 0 and k + 3 = 0 $\Rightarrow$ (h, k)≡(2,– 3)

P 38 : (D) $(0!)^{0!}$

Sol. For looking at the zeroes in the expression we should be able to see that the number of zeroes in the third term onwards is going to be very high. Thus, the number of zeroes in the expression would be given by the number of zeroes in :
$4 + 24^{24}$. 24^{24} has a unit digit 6. Hence, the number of zeroes in the expression would be $1 = (0!)^{0!}$

P 39 : (A) $A_2\hat{i} - A_1\hat{j} + A_3\hat{k}$

Sol. Since, given vector has components A_1, A_2 and A_3 in the coordinate system OXYZ

$$\vec{A} = \hat{i}\,A_1 + \hat{j}\,A_2 + \hat{k}\,A_3$$

When given system is rotated through $\pi/2$, the new x-axis is along the old y-axis and the new y-axis is along the old negative x-axis, z remains same as before.
Hence, the components of A in the new system are A_2, $-A_1$ and A_3.

Therefore $\vec{A}$ becomes $A_2\hat{i} - A_1\hat{j} + A_3\hat{k}$.

P 40 : (A) 2

Sol. $np - npq = 1$ i.e. $np^2 = 1 \Rightarrow p^2 = \frac{1}{25}$

$$\therefore p = \frac{1}{5}$$

$\therefore$ npq = 4 and so $\sqrt{npq} = 2$

P 41 : (B) 210

Sol. The number of trains a day (the digits 1, 2,3) are three groups of like elements from which a sample must be formed. In the time-table for a week the number 1 is repeated twice, the number 2 is repeated 3 times and the number 3 is repeated twice. The number of different time-tables is equal to:

$$p(2, 3, 2) = \frac{7!}{2!3!2!} = 210$$

P 42 : (B) $^{15}C_5 - 3.\ ^{10}C_5 + 3.\ ^5C_5$

Sol. Number of ways = Total ways – (Number of ways of omitting exactly one section + number of ways of omitting exactly two sections)

$^{15}C_5 - [3\{^{10}C_5 - (^5C_5 + ^5C_5)\} + 3.\ ^5C_5]$

$= ^{15}C_5 - [3.\ (^{10}C_5 - ^{25}C_5] + 3.\ ^5C_5]$

$= ^{15}C_5 - 3.\ ^{10}C_5 + 3.\ ^5C_5$

P 43 : (A) Straight line

Sol. Equation of BE:

$$y = \frac{\lambda b}{\lambda a + c}\ x \text{ or } k(\lambda a + c) = h\lambda b$$

$$\Rightarrow \lambda = \frac{kc}{bh - ka}$$

Similarly equation of CD:

$$y - 0 = \frac{\frac{\lambda b}{1+\lambda} - 0}{\frac{\lambda a}{1+\lambda} - c}(x - c)$$

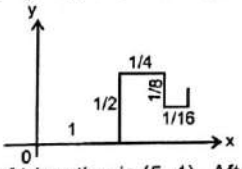
$$k\left(\frac{\lambda a}{1+\lambda} - c\right) = \frac{\lambda b}{1+\lambda}(h - c)$$

$$\Rightarrow \lambda = \frac{kc}{(a-c)k + bc - bh}$$

Equating the two values of λ we get the locus of P (h, k) as

$2bx - (2a - c)y = bc$. which is a straight line

P 44 : (C) $3x - y - 11 = 0$



Sol. Point of trisection is (5, 4) . After rotation the slope of the line

$$\tan\theta_1 = \frac{\tan 45° + \tan\theta_2}{1 - \tan 45°\tan\theta_2} = \frac{1 + \frac{1}{2}}{1 - \frac{1}{2}} = 3$$

P 45 : (C) obtuse angled

Sol. $3x + 4y = 0$ .....(i)

$4x + 3y = 4$ .....(ii)

$5x + 5y = 7$ .....(iii)

slopes of the lines are $-\frac{3}{4}, -\frac{4}{3}$ and -1

since $-\frac{3}{4} \times -\frac{4}{3} = 1$ and

slope of line (iii) is -1

$\therefore$ angle between (i) and (iii) = angle between (ii) and (iii)

Let this equal angle be θ. The θ is acute angle

$$\therefore \tan\theta = \left|\frac{-\frac{3}{4} - (-1)}{1 + \left(-\frac{3}{4}\right)(-1)}\right| = \frac{1}{7} < 1$$

$\therefore \theta < 45°$

$\therefore$ The third angle between the lines (i) and (ii) is obtuse.

P 46 : (C) $\left(\frac{9}{5}, \frac{8}{5}\right)$

Sol. Tangent at the vertex passes through the points (3, 4) and (4, 6)

$\therefore$ equation of this tangent is

$$y - 4 = \frac{2}{1}(x - 3)$$

$y = 2x - 2$

vertex is foot of perpendicular from the focus on tangent at vertex

$$\therefore \frac{x-1}{2} = \frac{y-2}{-1} = -\frac{2-2-2}{5}$$

$$\therefore x = \frac{9}{5}, \quad y = \frac{8}{5}$$

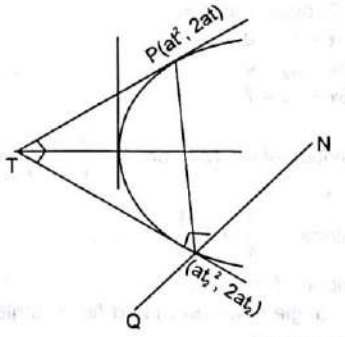
P 47 : (B) $d = \left|\frac{a(1+t^2)^{3/2}}{t^2}\right|$

Sol. Since QN and PT are parallel and $t_2 = -t - \frac{2}{t}$

equation of tangent at P is : $yt = x + at^2$

distance between PT and QN is equal to the

length of perpendicular from Q to PT = $\left|\frac{a(1+t^2)^{3/2}}{t^2}\right|$



P 48 : (C) 7

Sol. Equation of hyperbola is $\frac{x^2}{\frac{144}{25}} - \frac{y^2}{\frac{81}{25}} = 1$

$\therefore \quad \frac{81}{25} = \frac{144}{25}(e^2 - 1)$

$\therefore \quad e = \frac{5}{4}$ and so $ae = 3$

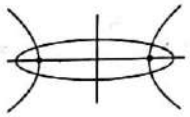
Now equation of the ellipse is $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$

$\therefore \quad ae = 3 \quad \Rightarrow \quad e = \frac{3}{4}$

$\therefore \quad b^2 = a^2(1 - e^2) = 16\left(1 - \frac{9}{16}\right) = 7$

P 49 : (C) $\sqrt{3}$

Sol. Let e_1 & e_2 are the eccentricity of ellipse & hyperbola.



$\because \quad b = ae_1 \quad \ldots (1)$

$b^2 = a^2(1 - e_1^2) \quad \ldots (2)$

$a^2 = b^2(e_2^2 - 1) \quad \ldots (3)$

From (1) & (2) $2e_1^2 = 1 \quad \Rightarrow \quad e_1 = \frac{1}{\sqrt{2}}$

From (1) & (3) $2 = e_2^2 - 1 \quad \Rightarrow \quad e_2 = \sqrt{3}$

P 50 : (D) –i

Sol. Let $z = r_1 e^{i\theta}$ and $\omega = r_1 e^{i\phi}$

$\Rightarrow \quad \bar{z} = r_1 e^{-i\theta}$

Given $|z\omega| = 1 \Rightarrow \left|r_1 e^{-i\theta} . r_2 e^{i\phi}\right| = 1$

$\Rightarrow r_1 r_2 = 1$

and $\arg(z) - \arg(\omega) = \frac{\pi}{2} \quad \Rightarrow \quad \theta - \phi = \frac{\pi}{2}$

Then $\bar{z}\omega = r_1 e^{-i\theta} . r_2 e^{i\phi}$

$= r_1 r_2 e^{-i(\theta - \phi)}$

Therefore, from Eq. (1) and Eq. (2), we get,

$\bar{z}\omega = 1.e^{-i\pi/2} = \cos\frac{\pi}{2} - i\sin\frac{\pi}{2} = \bar{z}\omega = -i$

P 51 : (B) $\frac{2}{5}$

Sol. B_1 – a bag is selected from first group.

B_2 – a bag is selected from second group.

A – a black ball is drawn.

Therefore, $P(B_1) = \frac{4}{7}$, $P(B_2) = \frac{3}{7}$

Also, $P\left(\frac{A}{B_1}\right) = \frac{3}{9} = \frac{1}{3}, P\left(\frac{A}{B_2}\right) = \frac{4}{6} = \frac{2}{3}$

Therefore, required probability = $P\left(\frac{B_1}{A}\right)$

$\Rightarrow P(E) = \frac{P(B_1)P\left(\frac{A}{B_1}\right)}{P(B_1).P\left(\frac{A}{B_1}\right) + P(B_2).P\left(\frac{A}{B_2}\right)}$

Therefore, $P(E) = \frac{\frac{4}{7} \times \frac{1}{3}}{\frac{4}{7} \times \frac{1}{3} + \frac{3}{7} \times \frac{2}{3}} = \frac{4}{10} = \frac{2}{5}$

P 52 : (B) $\frac{\pi}{4}$

Sol. $\tan(\alpha + \beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$

$\Rightarrow \tan(\alpha + \beta) = \frac{\frac{1}{1 + \frac{1}{2^x}} + \frac{1}{1 + 2^{x+1}}}{1 - \frac{1}{1 + \frac{1}{2^x}} \cdot \frac{1}{1 + 2^{x+1}}}$

$\Rightarrow \tan(\alpha + \beta) = \frac{2^x + 2.2^{x+x} + 2^x + 1}{1 + 2^x + 2.2^x + 2.2^{x+x} - 2^x}$

$\Rightarrow \tan(\alpha + \beta) = 1 = \tan\frac{\pi}{4} \Rightarrow \alpha + \beta = \frac{\pi}{4}$

P 53 : (D) 36

Sol. $\text{cosec}^2x + 25\sec^2x = 26 + \cot^2x + 25\tan^2 x$
$= 26 + 10 + (\cot x - 5\tan x)^2 \geq 36.$

P 54 : (A) $\frac{1}{8}$

Sol. $\cos\left(\frac{\pi}{2}-\frac{\pi}{18}\right).\cos\left(\frac{\pi}{2}-\frac{5\pi}{18}\right).\cos\left(\frac{\pi}{2}-\frac{7\pi}{18}\right)$
$= \cos\frac{4\pi}{9}\cos\frac{2\pi}{9}\cos\frac{\pi}{9}$
$= \frac{\sin\left(2^3\frac{\pi}{9}\right)}{2^3.\sin\frac{\pi}{9}} = \frac{\sin\frac{8\pi}{9}}{8\sin\frac{\pi}{9}} = \frac{1}{8}$

P 55 : (A) zero

Sol. Here, $\frac{1}{2}\sin(2e^x) = \frac{1}{4}(2^x + 2^{-x}) \geq \frac{1}{4}.2 = \frac{1}{2}$
so, $\sin(2e^x) \geq 1$
But $\sin(2e^x) \not< 1$
therefore, $\sin(2e^x) = 1$
but equality can hold when, $2^x = 2^{-x} = 1$
i.e., $x = 0$.
then $\sin(2.e^0) = 1$, which is not true.

P 56 : (C) $-\frac{\pi}{4}$

Sol. $\cos(4095°) = \cos(45 \times 90° + 45°)$
$= -\sin 45° = -\sin\frac{\pi}{4} = \sin\left(-\frac{\pi}{4}\right)$
Therefore, $\sin^{-1}\{\cos(4095°)\}$
$= \sin^{-1}\sin\left(-\frac{\pi}{4}\right) = -\frac{\pi}{4}$

P 57 : (A) all values of θ

Sol. since $2^{\sin\theta}$ and $2^{\cos\theta}$ are positive real numbers, so A.M. (Arithmetic Mean) of these two numbers is greater or equal to their G.M.(Geometric Mean) and hence

$$\frac{2^{\sin\theta}+2^{\cos\theta}}{2} \geq \sqrt{2^{\sin\theta}+2^{\cos\theta}} = \sqrt{2^{\sin\theta+\cos\theta}}$$

$$\geq 2^{\frac{\sin\theta+\cos\theta}{2}} = 2^{\frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}\sin\theta+\frac{1}{\sqrt{2}}\cos\theta\right)}$$

$$\geq 2^{\frac{1}{\sqrt{2}}\sin\left(\frac{\pi}{4}+\theta\right)}$$

$-1 \quad -1 \leq \sin\left(\frac{\pi}{4}+\theta\right) \leq 1$, we have

$$\frac{2^{\sin\theta}+2^{\cos\theta}}{2} \geq 2^{\frac{-1}{\sqrt{2}}}$$

$$\Rightarrow 2^{\sin\theta}+2^{\cos\theta} \geq 2^{1-\frac{1}{\sqrt{2}}}$$

So its true for all θ.

P 58 : (B) $4x + 3y + 5z = 50$

Sol. Let $(3\lambda+1, \lambda+2, 2\lambda+3)$ lie on line (I) and line (II)

Hence $\frac{3\lambda+1-3}{1} = \frac{\lambda+2-1}{2} \Rightarrow \lambda = 1$

Hence point of intersection P of the two lines can be given as (4, 3, 5)
Equation of plane perpendicular to OP where O is (0, 0, 0) and passing through P is
$4x + 3y + 5z = 50$

P 59 : (B) 4 square unit

Sol. Since $y = f(x)$ & $y = f^{-1}(x)$ are symmetrical about $y = x$
since $y = f(x)$ intersect $y = x$ at $x = \pi$ & 2π

so required area $= 2\int_{\pi}^{2\pi}(x-(x+\sin x)\,dx$

$= 2\int_{\pi}^{2\pi}(-\sin x)\,dx = 2(\cos x)_{\pi}^{2\pi} = 4$ sq. unit.

P 60 : (B) 4

Sol. $\Delta = \begin{vmatrix} [x]+1 & [y] & [z] \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix}$

$= ([x] + 1)(1 - 0) - [y](-1 - 0) + [z](0 + 1)$
$= [x] + 1 + [y] + [z]$
$= 1 + [x] + [y] + [z]$
max. $\Delta = 1 + 1 + 0 + 2 = 4$

P 61 : (B) $\overset{\oplus}{H_3N} - CH_2COOH$

Sol. At pH = 4 (acidic range), glycine reacts with H^+ ion

$$\underset{\text{Zwitter ion}}{\overset{\oplus}{H_3N}-CH_2-\overset{\ominus}{COO}} \xrightarrow{H^{\oplus}} \overset{\oplus}{H_3N}-CH_2-COOH$$

P 62 : (D) H_2SO_4, $Na_2Cr_2O_7$ and $K_2Cr_2O_7$

Sol. This reaction is the method of preparation of $K_2Cr_2O_7$.

$$2Na_2CrO_4 + H_2SO_4 \rightarrow \underset{\text{Sodium dichromate (orange)}}{Na_2Cr_2O_7} + Na_2SO_4 + H_2SO_4$$

$$2Na_2Cr_2O_7 + 2KCl \rightarrow \underset{\text{Potassium dichromate (Orange red)}}{K_2Cr_2O_7} + 2NaCl$$

P 63 : (A) 4

Sol. $\underset{\text{iron(III) hydroxides}}{Fe(OH)_3} \rightleftharpoons \underset{[S]}{Fe^{3+}} + \underset{[3S]\,mol/L}{3OH^-}$

(Solubility product) $K_{sp} = [Fe^{3+}][OH^-]^3$

$= [S][3S]^3 = 27\,S^4$

Given, $K_{sp} = 1.8 \times 10^{-37}$

Then solubility, $S = \sqrt[4]{\frac{1.8\times10^{-37}}{27}}$

or $S_{(OH)} = 2.85 \times 10^{-10}$ mol/L

pOH = –log OH

pOH = $-\log (2.85 \times 10^{-10})$

$= -\log 2.85 + 10$

pOH = –0.4548 + 10 = 9.54

$\therefore$ pH + pOH = 14

pH = 14 – 9.54 = 4.4 = 4

P 64 : (A) CH_3OH

Sol. Compounds which are more acidic than H_2O are soluble in aq. NaOH.

Acidic strength –

$CH_3OH > H_2O > CH_3CH_2OH >$ $> CH_3C{\equiv}CH.$

P 65 : (A) 9

Sol. $\Delta U = 0 \Rightarrow q = -w = \pi r^2$

$\Rightarrow q = 9\pi L.\ kp_a = 9\pi$ J

$\because$ (1L kp_a = 1J)

P 66 : (B) Resonance structures have equal potential energy.

Sol. Only equivalent resonance structures have equal potential energies.

P 67 : (B) $MgO > Al_2O_3 > SiO_2$; basic character

Sol. Acidic character increases with increase in electronegativity of central atom.

Ionic radius decreases with increase in positive charge on cation.

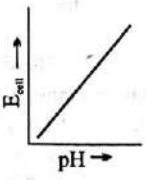
P 68 : (D) $[CuBr_4]^{2-}$

Sol. $Ni(CO)_4$; $\mu_{eff} = 0$

$[Co(NH_3)_4(NO_2)_2]$; $\mu_{eff} = 0$

$[Ag(CN)_2]^-$; $\mu_{eff} = 0$

$[CuBr_4]^{2-}$; $\mu_{eff} = 1.732$ BM

P 69 : (B) 

Sol. $E = E° - \frac{0.59}{2}\log\frac{[H^+]^2}{[Ag^+]^2}$

$= E° + 0.059 \log [Ag^+] + 0.059$ pH

$\Rightarrow E = C + 0.059pH (C = E° + 0.059 \log [Ag^+])$

E v/s pH is a straight line with positive slope. Remember the straight line is not origin.

P 70 : (C) 4.48 dm^3

Sol. When ehthyl amine react with $NaNO_2$ and HCl then alcohol is formed.

$\underset{(9g)}{C_2H_5NH_2} \xrightarrow{NaNO_2/HCl} C_2H_5OH + N_2 + H_2O$

Moles of ethyl amine = $\frac{9}{45} = \frac{1}{5}$ mole

[M of $C_2H_5NH_2$ = 45]

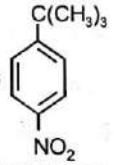
Number of moles of ethylamine equal to evolved N_2 gas.

Thus, moles of nitrogen (N_2) = $\frac{1}{5}$ mole

Volume of one mole gas = 22.4 dm^2 at 273K.

therefore, volume of released N_2 gas

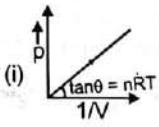
$= \frac{1}{5} \times 22.4\ dm^3 = 4.48\ dm^3$

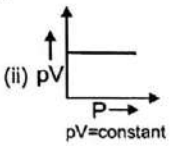
P 71 : (C) 

Sol. Nitration occurs at para position due to activating t-butyl group.

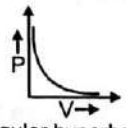
P 72 : (C) IV

Sol. $pV = nRT \Rightarrow p = (nRT)\frac{1}{V}$.





(iii) pV - constant i.e. xy = constant

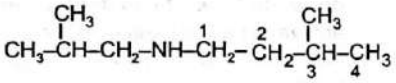


Rectangular hyperbola

Hence, (IV) i.e. option (c) wrongly represent Boyle's law.

P 73 : (B) 3-methyl-N-(2-methylpropyl)-1-butanamine

Sol.



3-methyl-N-(2-methylpropyl)-1-butanamine

P 74 : (D) All statements are true.

Sol. All of these favour S_N2 reaction.

P 75 : (D)

Sol. Benzene ring does not give addition reaction with Br_2/H_2O.

P 76 : (C) $k[NH_4^+][NO_2^-]$ and $M^{-1}s^{-1}$

Sol. $NH_4^+(aq) + NO_2^- \longrightarrow N_2(g) + 2H_2O(l)$

Rate of reaction directly proportional to concentration of reactant molecules.

Rate of reaction (r) $\propto$ Concetration of reaction (C)

or $\frac{r_1}{r_2} = \left[\frac{C_1}{C_2}\right]^n$ where, n = order of reaction.

order of reaction with respect to NH_4^+

or $\frac{7.2\times10^{-6}}{3.6\times10^{-6}} = \left(\frac{0.24}{0.12}\right)^n$

Order of reaction with respect to NO_2^-

$$\frac{3.6\times10^{-6}}{5.4\times10^{-6}} = \left(\frac{0.1}{0.15}\right)^n$$

$$n = 1$$

Thus, rate of reaction $\frac{dx}{dt} = -k\,[NH_4^+]\,[NO_2^-]$

Total order of reaction

n = 1 + 1 = 2 (second order of reaction)

Unit of k = $(mol/L)^{1-n}s^{-1}$

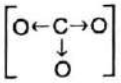
= $(mol/L)^{1-2}s^{-1}$

= $(mol/L)^{-1}s^{-1}$

= $mol^{-1}L\,s^{-1}$

P 77 : (D) $CO < CO_2 < CO_3^{2-}$

Sol. Bond-length is inversely proportional to bond-order



bond order = 4/3

3

2

P 78 : (D) all of the above

Sol. For correct statements,

(a) $Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2$ Thus, (a) is true	$Na_2CO_3 \xrightarrow{\Delta} X$
(b) LiCl soluble in ether thus covalent Thus, (b) is true	NaCl insoluble in ether but soluble in H_2O thus ionic
(c) Li forms nitride Thus, (c) is true	Na does not form nitride

P 79 : (A) CaC_2O_4 CO CO_2 CaO $Ca(OH)_2$

P 80 : (B) Below 1073 K CO is better reducing agent

Sol. Because below 1073 K Gibb's energy of formation of Fe_2O_3 is –500 which is more than Gibb's energy of formation of CO_2. Thus CO_2 is more stable below 1073 K. As a result Fe_2O_3 is reduced by CO.

$$Fe_2O_3 + 3CO \xrightarrow{1073K} 2Fe + CO_2$$

P 81 : (C) XYZ_3

Sol. $X = 8 \times \frac{1}{8} = 1$

$Y = 1$

$Z = 1$ on each edge $= \frac{12}{4} = 3$

the formula of the compound = XYZ_3.

P 82 : (A) $HOCl < HClO_2 < HClO_3 < HClO_4$

Sol. Acid strength $\propto$ oxidation number of Cl atom

$HClO < HClO_2 < HClO_3 < HClO_4$

↑ ↑ ↑ ↑

+1 +3 +5 +7

P 83 : (B) $CdCO_3$

Sol. Sulphides of Sb^{+3}, Sn^{+2}, Pb^{+2}, Cd^{+2}, Ni^{+2}, Co^{+2} do not give H_2S gas with dil. H_2SO_4. But when Zn pieces are added, H_2 gas is evolved.

$Zn + H_2SO_4 \longrightarrow ZnSO_4 + 2H$

$PbS + 2H \longrightarrow Pb^{+2} + H_2S\uparrow$

$S^{-2} \longrightarrow H_2S\uparrow$

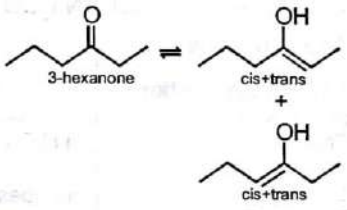
$SO_3^{-2} + CdCO_3 \longrightarrow Cds + CO_2\uparrow + 1/2O_2\uparrow$

$SO_3^{2-} + CdCO_3 \longrightarrow$ No reaction

P 84 : (B) sp^3d^3, distorted octahedral

P 85 : (C) 4

Sol.



P 86 : (C) $K_2Cr_2O_7$, H_2SO_4/H_2O

Sol. $K_2Cr_2O_7/H_2SO_4/H_2O$ oxidises a 2° alcohol to ketone.

P 87 : (D) 3.34 V

Sol. The standard potential (E°) of MnO_4^- is= 1.51V
the standard potential (E°) of MnO_2/Mn^{2+}= 1.23V
then we prepare two half cell reactions as follows :

$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O = 1.51V$...(i)

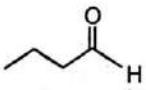
$2 \times [Mn^{2+} + 2H_2O \rightarrow MnO_2 + 4H^+ + 2e^-]$

$= 1.23\ V \times 2 = 2.46$ ...(ii)

$MnO_4^- \rightarrow 2MnO_2 - 1e^-$

$E° = nE = 1 \times 3.9 = 3.9 = 3.3$

P 88 : (D)



Sol. All alcohol gives victor mayer test but carbonyl compound does not.

P 89 : (C) 6:3:2

Sol. According to Faraday's second law, deposited weight of metal depends on equivalent weight and number of transferred electrons. Here, three solution $AgNO_3$, $NiSO_4$, $CrCl_3$ are used respectively

	$AgNO_3$ ↓	NSO_4 ↓	$CrCl_3$ ↓
Oxidation number	Ag^+	Ni^{2+}	Cr^{3+}
From balancing	$3Ag^+ + 3e^- \rightarrow 3Ag$	$3/2Ni^{2+} + 3e^- \rightarrow 3/2Ni$	$Cr^{+3} + 3e^- \rightarrow Cr$
Metal ion	3 3	3/2 1.5	1 1
From balancing or	3×2 6	1.5×2 3	1×2 2

Thus, deposited ratio of metal ion is 6:3:2.

P 90 : (A) 1:4

Sol. Energy of electron $= -13.6 \times \frac{n^2}{Z}$

For hydrogen atom (H), n = 1

E = –13.6 eV

For Be^{3+} ion, n = 4, Z = 4

$E = -13.4 \times \frac{16}{4}$

For Eqs. (i) and (ii)

$\frac{E_H}{E_{Be^{3+}}} = \frac{-13.6}{-54.4}$

$\frac{E_H}{E_{Be^{3+}}} = \frac{1}{4}$ or 1 : 4